

# Pricing the Sensing Budget in $\rho$ -POMDPs, with Expected Free Energy as the Canonical Information Weight

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**Abstract.** An agent acting under partial observability must decide how much effort to spend gathering information before committing to a decision, and standard POMDPs give it no direct way to state a target for that effort as an expected sensing usage. Information has value only through its downstream effect on reward. We address this by treating information acquisition in a  $\rho$ -POMDP (a POMDP extended with a belief-dependent utility  $\rho$ ) as an explicitly budgeted resource. Given a stated sensing budget  $B$ , the observation usage the agent is expected to spend before committing, stated as a target in expectation rather than a per-episode cap, the weight to place on an information-gain utility is the budget’s operational shadow price  $w^*(B)$ , the weight at which the agent’s expected sensing usage crosses  $B$ . That value is computable offline from a usage curve, trackable online by dual descent when the reward scale itself shifts mid-run, and scale equivariant for the receding-horizon planner, exactly in the mathematics and up to floating-point rounding in code, so it need not be re-tuned when rewards are rescaled. This budgeted formulation rests on an exact theoretical identification. Minimizing Expected Free Energy (EFE) under active inference is provably equivalent to solving a  $\rho$ -POMDP whose utility is expected information gain at the canonical weight  $w=1$ , exact under log scoring rules [4], for observe-then-commit POMDPs and, for the hidden-state-preserving observation and navigation actions, for factored observation POMDPs covering interleaved observe-act problems such as non-destructive testing and mobile sensing. The canonical weight  $w=1$  is one weight whose expected usage equals the budget the ambient reward scale happens to imply, a member of that budget’s level set rather than the price the procedure would report for it, making it a derived default under a shared reward convention rather than a universal, scale-invariant calibration. Across environments ranging from the classic Tiger problem through RockSample to a Structural Inspection benchmark with over 65,000 states, this default lands inside the reward-maximizing weight bracket on three of the five environments we sweep (and is beaten on the two exceptions, Testbed, where the reward-tied bracket sits at the low-weight boundary, and Tileworld, where the surface has a genuinely interior optimum) and is statistically equivalent to a near-optimal SARSOP reference, by TOST at the predeclared sensing-cost margin, on the three core observe-then-commit envi-

ronments. The budgeted formulation further reports a crossing bracket and an endpoint mixture at usage gaps rather than a misleading point estimate, reducing the practical design question from “what should the information weight be” to “what expected sensing usage do we intend.”

**Keywords:** Active inference · Expected free energy ·  $\rho$ -POMDPs · Epistemic value · Belief-dependent utility · Partial observability · Information gain · Sensing budgets · Shadow prices

## 1 Introduction

Decision-making under partial observability requires agents to balance exploiting current knowledge against gathering information to reduce uncertainty about hidden states. In a standard partially observable Markov decision process (POMDP), information gathering has no intrinsic value. It is useful only insofar as it leads to higher expected reward. This creates a well-known difficulty. A reward-only planner values an observation only if it looks far enough ahead to see the reward that the observation eventually buys. The exploration–exploitation trade-off must therefore be resolved either by a planning horizon deep enough to see that far or by heuristic exploration bonuses.

The  $\rho$ -POMDP framework [2] addresses this by extending POMDPs with a belief-dependent utility  $\rho(b)$  that allows the agent to derive value directly from properties of its belief state. This enables explicit optimization over uncertainty reduction, information gain, or other belief-state properties alongside task reward. However, the choice of  $\rho$  remains largely heuristic and environment-specific, and, when  $\rho$  is an information-gain term, so does the weight placed on it.

This paper treats that weight as a design problem with a resource-allocation answer. Most of the environments we study pose an observe-then-commit (OTC) problem. Each observation costs something, and the agent gathers as many observations as it chooses before committing to a terminal decision whose payoff depends on the hidden state. In that setting a natural design input is a sensing budget  $B$ , the amount of sensing the agent is expected to use before committing, stated as an expectation over episodes rather than as a hard per-episode limit. A target of that kind arises when sensing is itself a deliverable, as in a coverage, monitoring, or verification requirement that fixes how many inspections are to be made regardless of what the task reward alone would justify, and the question is then which weight delivers that expected usage. We measure that amount as sensing usage, the expected number of observation actions (tests, scans, or checks) the agent takes per episode, and  $B$  is stated in the same units (Section 4.1). The budget sections also allow usage to be denominated in cumulative sensing cost instead, but throughout this Introduction usage means the count. The proposal is to state that budget and to price the information-gain utility at the budget’s *operational shadow price*  $w^*(B)$ , the weight at which the agent’s expected usage crosses  $B$ . Where the usage curve jumps over  $B$ , so that no single weight meets it exactly, the price is still that one crossing weight, and

the procedure estimates it by a bracket of neighboring grid weights, as the next paragraph explains. In constrained optimization, a shadow price is the marginal value of one more unit of a scarce resource, obtained as the Lagrange multiplier on that resource’s constraint, and the adjective operational marks a deliberate departure from that meaning. Our price is read off a usage curve, the map from weight to the agent’s expected usage, rather than obtained as that multiplier, and Section 4 makes precise what  $w$  is and is not a multiplier of. The question “what weight should the information-gain utility carry” becomes “what expected sensing usage do we intend,” with the weight read off the stated usage target rather than searched per task.

Four properties make this price useful in practice. It can be estimated offline, tracked online, and rescaled with rewards. When the usage curve jumps over a budget, a crossing bracket estimates the price and an endpoint mixture can attain the budget in expectation. Offline, the price is computable from that usage curve (Section 4). Online, it is trackable by dual descent, a feedback rule that nudges the weight down whenever measured sensing usage runs above the target and up whenever it runs below it. For the projected, decaying-step form of that rule, stationary convergence is guaranteed under stated conditions (Proposition 6), and re-adaptation under mid-run reward-scale shift is demonstrated empirically (Section 6.3). Under rescaling, the price is scale equivariant for the receding-horizon planner the paper implements, one that re-plans from its current belief at every step and executes only the first action of each plan, exactly in the mathematics and up to floating-point rounding in code. Equivariance here means that multiplying every reward and cost by a constant  $\alpha > 0$  multiplies the price by the same constant, so that the count-usage curve at the new scale is the unscaled curve read against  $w/\alpha$  in place of  $w$ , while a cost-usage curve also rescales by  $\alpha$ . The price at a rescaled problem is therefore obtained by multiplying the old price by  $\alpha$ , not by searching for it again (Proposition 4). The fourth property follows from the shape of the usage curve. The agent chooses each action by comparing finitely many alternatives, so a small change in  $w$  either leaves its policy, and with it its sensing usage, unchanged or switches it. A switch can move usage by a whole observation at once. The usage curve is therefore a staircase. It is not always monotone, because a switch at a higher weight can, for example, trade several weak observations for fewer but individually more informative ones (Section 4.3 gives the full account). No single weight exactly meets a budget inside a staircase jump. At such a usage gap the crossing threshold is still a single number, the location of the jump, and the procedure reports it as a crossing bracket, the pair of neighboring grid weights that straddle the jump, together with a per-episode mixture of the two endpoint policies that attains the budget in expectation (Definition 2). The bracket’s width is grid resolution, not an interval of exact prices.

The active inference (AIF) framework [18, 42] identifies one canonical point on this price curve. AIF casts perception and action as approximate Bayesian inference, selecting policies that minimize Expected Free Energy (EFE). EFE naturally decomposes into a pragmatic term (goal-seeking) and an epistemic

term (information-seeking). Because both terms come from a single variational objective, their relative coefficient is fixed once pragmatic value is identified with task reward. In the discrete-state formulation we use, that coefficient is  $w=1$  in information units, one reward unit per unit of information (Proposition 1). The claim is exact under log scoring rules [4], rewards that pay the agent the log probability it assigned to the outcome that occurred, so that reward is already measured in information units. Under a shared reward convention, a fixed exchange rate between the units of task reward and the information units of the epistemic term, it is an empirically robust untuned default. That default is not scale-invariant. Multiplying every reward and cost by  $\alpha > 0$  leaves the bit-valued information term unchanged, so a fixed  $w=1$  implements a different trade-off at every scale. Rather than treating this as a limitation to caveat, we read it as confirmation that  $w$  is a price. In the budgeted formulation above,  $w=1$  is then one of the weights whose expected usage equals the budget it happens to imply at the ambient reward scale, not a universal, scale-invariant calibration. A convention-free comparison between a reward-denominated quantity and a bit-denominated one is not on offer to any fixed weight, ours included. The log-scoring derivation gives a principled anchor within this unavoidable choice of units, while leaving the reward convention itself to the designer.

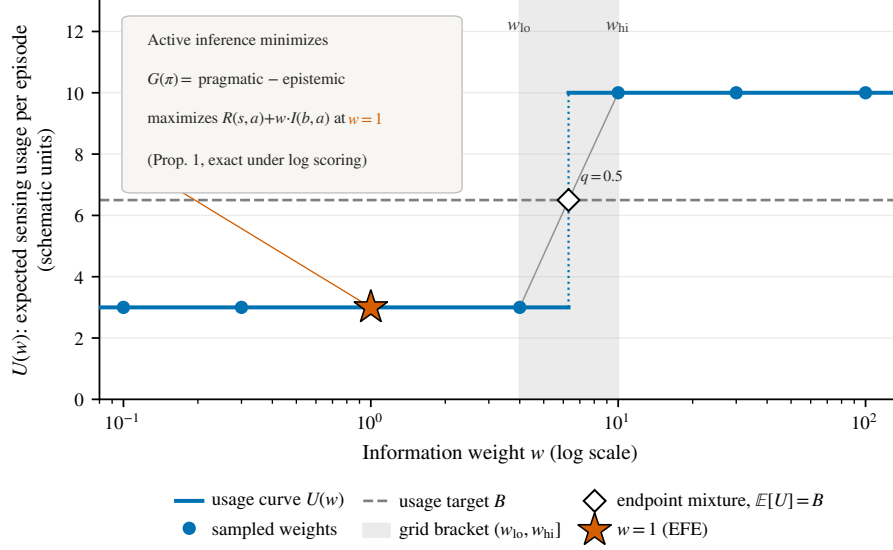
Figure 1 shows how a sensing target is read from the price curve. Figure 2 places that calculation in an engineer’s inspection workflow.

The two figures answer different practical questions. Figure 1 shows which weight a specified usage target selects, including what to do when the usage curve jumps over it. Figure 2 shows what an engineer must supply and check: a model of tests and outcomes, an expected-usage target when that is the design requirement, calibration runs to estimate the usage curve, and a separate assessment of reward. The agent still chooses each observation or commitment from its updated belief; the budget selects the policy family member rather than prescribing a fixed number of tests for every item.

Our contributions are as follows.<sup>1</sup>

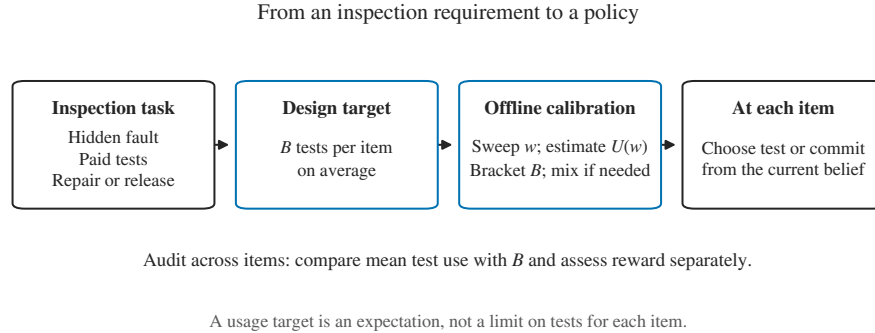
1. Budgeted reformulation. We recast the information weight as the operational shadow price  $w^*(B)$  of a sensing budget  $B$  (Section 4), prove exact scale equivariance of the resulting policy family for the receding-horizon planner under the scale-independent tie-break the implementation uses (Proposition 4), separate the level set, the crossing threshold, the finite-grid crossing bracket, and the endpoint mixture at usage gaps (Definition 2), and give an online dual controller, a projected decaying-step recursion, with a stationary convergence guarantee under stated conditions (Proposition 6). In experiments, that controller, whose step size shrinks over time, re-adapts after a mid-run reward-scale shift. With an added change-detection heuristic that restarts the step-size schedule when it declares a shift, it re-adapts on all ten controller seeds tested, and without that heuristic on nine of the ten inside the post-shift window, roughly three times more slowly on a recovery-time

<sup>1</sup> Code and experiment data are available at [https://github.com/PatrickAllenCooper/rho\\_aif](https://github.com/PatrickAllenCooper/rho_aif).



**Fig. 1.** A schematic price curve for an information-gain weight (illustrative, not measured data). The blue curve is a usage curve  $U(w)$ , the expected sensing an agent uses at information-gain weight  $w$ , drawn with one jump between two sampled grid weights (dots). Measured usage curves elsewhere in this paper carry further plateaus, jumps, and occasional dips and are not asserted to be monotone. A stated expected sensing budget  $B$  (dashed line) is a target for expected usage, not a per-episode cap. Here it falls inside the usage gap the jump opens, so no single weight attains it and the level set of Definition 2 is empty. The dotted riser at an unsampled weight is the crossing threshold, and the shaded band is the crossing bracket  $(w_{lo}, w_{hi}]$ , the grid’s estimate of where that threshold lies, a grid interval rather than a set of prices. The open diamond on the budget line is the endpoint mixture, a randomized policy that plays  $w_{hi}$  with probability  $q$  and  $w_{lo}$  otherwise, drawn afresh each episode, and so attains  $B$  in expectation (here  $q=0.5$ ), with leader lines to the two grid points it mixes. The boxed aside states the equivalence result in the notation of the axes. Minimizing the recursive, reward-based Expected Free Energy objective we specify is exactly equivalent to maximizing  $R(s, a) + w \cdot I(b, a)$  at  $w=1$  for observe-then-commit POMDPs and for the hidden-state-preserving actions of factored observation POMDPs, exact under log scoring rules (Propositions 1 and 3). The star marks that  $w=1$  on the curve, derived from the objective rather than tuned, and sitting outside this particular bracket. The price is scale equivariant for the receding-horizon planner (Proposition 4), so rescaling every reward and cost by a constant rescales  $w$  by the same constant and leaves this count-usage curve unchanged once it is read against the rescaled weight, and  $w=1$  is one weight whose expected usage equals the budget the ambient reward scale happens to imply, not a universal number.

estimand defined for every seed (Section 6.3). This is the paper’s primary



**Fig. 2.** From an inspection requirement to a running policy (schematic, not experimental data). For a state-preserving task with a hidden fault and paid tests, the engineer states a target  $B$  in expected tests per item. An offline sweep estimates usage  $U(w)$  and brackets  $B$ ; when a usage gap lies between the selected endpoint policies, mixing them across items can meet the target under Definition 2’s conditions. The deployed agent chooses tests and terminal actions from its current belief. Usage and reward are assessed separately: meeting  $B$  does not establish reward optimality under a sensing cap.

contribution. It turns a hyperparameter into the price of a stated expected sensing usage target.

2. Theory. A formal bridge between  $\rho$ -POMDPs and active inference, which is what licenses reading  $w=1$  as an anchor on the price curve rather than an arbitrary choice. We prove the equivalence for observe-then-commit POMDPs (Proposition 1), characterize when the information-unit weight  $w=1$  is near-optimal (Proposition 2), and extend the equivalence to factored observation POMDPs (Proposition 3). A factored observation POMDP (Definition 1) splits the state into a fully observable part and a hidden part. Its actions come in three kinds, namely observation actions that look at the hidden part, navigation actions that move the agent without touching the hidden part, and exploitation actions that collect reward that depends on the hidden part. Observing and moving may interleave with exploiting rather than precede a single commit, and the extension covers the case in which the observation and navigation actions preserve the hidden state.
3. Evidence. The empirical evidence comes in two batteries, one for each theoretical contribution. The first exercises the budgeted formulation of Contribution 1, and the second evaluates the equivalence results of Contribution 2. The second battery runs controlled comparisons across five OTC environments, namely Tiger, Diagnosis, Bandit, Tileworld, and the two-state Testbed, and the first battery adds a sixth, the distractor variant of Diagnosis (Section 6.10). Navigation, a further environment whose study is reported

in Appendix F, is counted separately because it has only move actions, with no observation actions and no terminal commit, so it falls outside the OTC class as defined above. The two principal baselines are same-horizon planning (hereafter Planning), a reward-only planner that looks ahead to the same depth as the EFE agent, and tuned information gain (Planning+IG), the same planner with a tuned information-gain bonus added to its objective. POMCP [55], an online Monte Carlo tree-search planner for POMDPs, is a further baseline on the first four of those environments. The suite also includes four instances of a cost-modified RockSample variant [58]. Section 5 states the six deviations from the original benchmark, and Appendix M reports a full tree-search POMCP with a tuned, frozen configuration on all four instances, distinguished there from a flat one-ply Monte Carlo reference. A Pareto analysis, which sweeps  $w$  and places each weight on a success-rate versus reward frontier, shows that on Diagnosis and Bandit,  $w=1$  Pareto-dominates Planning, reaching higher success with comparable or better reward, while remaining competitive with tuned Planning+IG across environments. The first battery (Section 6) exercises the budgeted formulation in five ways. It shows usage-curve collapse across reward scales, meaning that curves measured at different reward scales coincide once the weight is rescaled with them. It shows the closed-form onset thresholds, the weights at which sensing first turns on, recovered as the first jump of the measured usage staircase. It shows the online dual controller tracking a stated budget, across seeds, through a  $\times 10$  reward rescale. It tests the calibration procedure at target budgets set by a predeclared rule from the calibration curve’s range, independent of the weight-one usage though not application-specified, on held-out seeds, and reports where meeting a budget earns reward and where it costs reward relative to the best budget-feasible single weight and to a near-optimal constrained reference. And it shows a near-optimal reference computed by SARSOP, an offline point-based POMDP solver, statistically equivalent to EFE ( $w=1$ ) on the three core OTC environments (Tiger, Diagnosis, and Bandit) by the two one-sided tests (TOST) procedure at a fixed sensing-cost margin, an equivalence test that supports, rather than merely fails to reject, the conclusion that the two agents differ by less than the stated margin.

4. Practical guidance. We identify the tested conditions under which EFE-as- $\rho$  helps and the model structures in which information gain fails by construction. In the tested regimes, the advantage appears when the agent must choose among multiple observation actions. We isolate that dependence directly on the Diagnosis task, in which the agent must identify one of  $N$  hidden conditions using  $K$  available tests, by holding the environment fixed and varying only the number of tests (Appendix D.2). There, at  $N=8$ , the success-rate gap between EFE and reward-only planning is zero at  $K=1$ , within sampling error at  $K=2$ , and +13.2pp (percentage points) at  $K=3$ . The main battery’s Diagnosis instance, which has two tests at a smaller  $N$ , supplies the corresponding  $K=2$  evidence in Table 8.

The advantage also opens at the largest scale of the Tileworld sweep ( $69.4\% \pm 1.0\text{pp}$  vs.  $1.4\% \pm 0.3\text{pp}$  success, seed-level SE, at  $8 \times 8$ , Section 7). Tileworld’s

observation actions are scans, each a noisy yes/no question about the location of a hidden target tile. At that grid size and the sweep’s fixed lookahead depth, no scan repays its cost before the lookahead ends and the planner must value a commit. Reward-only Planning therefore stops scanning entirely, while EFE, searching to the same depth but crediting each scan’s expected information gain directly, does scan. The comparison there is therefore EFE against a non-scanning agent, a joint horizon-and-scale effect rather than a pure scale effect. That result invites the reading that the advantage grows with problem size, a reading the abridged workshop version of this article made and that we revise below. What our results actually support is narrower than a size law. The advantage tracks how much slack a problem’s structure leaves between what the reward-only planner does on its own and the information-optimal behavior that the agent’s search can actually represent. On Tileworld that slack is the room between never scanning and the scanning that the same depth-limited search can represent, and it correlates with size there but not in general.

RockSample makes the point. In the bracket notation `RockSample[·,·]`, the first entry is the side of the square grid and the second is the number of rocks whose hidden quality the agent may check. The second entry therefore counts the observation actions and plays the role of the  $K$  of the Diagnosis sweep. On RockSample, the advantage is largest with several rocks to check (+9.54 reward over reward-only Planning on `RockSample[7,8]`,  $K=8$ ) but does not persist to every instance we test, vanishing on `RockSample[11,11]` ( $K=11$ ) where every tree-search agent scores within sampling error of every other. On the two larger instances a hand-coded unbounded-horizon heuristic beats all of them, which bounds what these instances can be asked to show about weighting and locates the limitation in the depth of the search rather than in the epistemic term (Section 7.4).

Two structural limits accompany these results. Ordinary information gain is reward-blind, so it will spend a sensing budget reducing uncertainty that has no bearing on reward. A variant that weights information by its relevance to reward eliminates that wasted spending entirely at every weight in the sweep, and it does so using only quantities the agent already holds (Section 6.10). Naive information gain can also misvalue an action outright when observing changes the hidden state (Example 1).

An abridged version of this article was accepted as a poster and spotlight presentation at the 7th International Workshop on Active Inference (IWAI 2026) [10]. It comprised the equivalence results in Contribution 2 above (Propositions 1, 2, and 3) and the second experiment battery of Contribution 3. This article is a substantially extended version. The entire budgeted  $\rho$ -POMDP formulation of Section 4 (Definition 2 and Propositions 4, 5, 6, and Corollary 1), the first experiment battery of Section 6, and the distractor and destructive-sensing results of Contribution 4 are new material that was not part of the IWAI submission. Contribution 4 otherwise revises the IWAI version’s characterization, which claimed

the advantage grows with state space size and observation action count, a claim RockSample[11,11] does not support.

The remainder of the article is organized as follows. Section 3 sets up OTC  $\rho$ -POMDPs and EFE, states the equivalence and near-optimality results of Contribution 2 with proof sketches (the full proof of Proposition 1 is in Appendix A), and introduces the agent family used throughout. Section 4 develops the budgeted formulation of Contribution 1, including the crossing bracket and endpoint mixture, the scale-equivariance result, and the online dual controller. Section 5 describes the environments and the statistical protocol. Section 6 reports the first experiment battery, which evaluates the budgeted formulation, and Section 7 reports the second, which evaluates the equivalence results. Section 8 takes up scope and limitations before the conclusion. We first situate the work in the related literature.

## 2 Related Work

This section covers the literatures the article draws on, in the order the argument needs them. It begins with POMDP solvers and the  $\rho$ -POMDP framework that supplies the formal object of study, then turns to the classical value-of-information and experimental-design results on which the identification of reward with information rests. It then takes up active inference and its expected free energy objective, from which the information weight is derived, and the sophisticated-inference form the paper builds on. It closes with three neighboring lines, namely scaling active inference, exploration and intrinsic motivation, and resource-constrained information acquisition.

*POMDPs and solvers.* POMDPs formalize sequential decision-making under state uncertainty [29, 57]. Exact finite-horizon POMDP planning is PSPACE-complete. Point-based offline methods (PBVI [45], HSVI [58], SARSOP [32]) approximate the value function on reachable beliefs [54]. Online solvers plan from the current belief. POMCP [55] uses Monte Carlo tree search (MCTS) with UCB1 action selection and evaluates leaves by rollouts. UCB1 is the upper-confidence-bound rule that picks the action whose estimated return plus an exploration bonus is largest. DESPOT [68] searches a regularized sparse belief tree, sparse because each node branches on a small set of sampled scenarios rather than on every possible observation, and regularized because the search penalizes the size of the policy it builds so as not to overfit those samples. POMCPOW [61] extends POMCP with progressive widening, a technique that limits how many children a search node expands as a function of its visit count, so that the tree can be grown over continuous spaces. All explore implicitly through stochastic simulations rather than explicitly valuing information gain. A recent counterpoint reasons about value of information explicitly inside the tree search. Laouar et al. [34], sharing an author with Sunberg and Kochenderfer [61], propose Value of Information Monte Carlo planning, which selectively skips observation branches whose expected value of information is low rather than exploring them uniformly.

Our EFE agent sits on the same side of that divide but values information in a different way. It computes the expected information gain of each observation action in closed form from the current belief, whereas POMCP estimates the value of a branch from sampled rollouts.

The rest of this paragraph previews where the two valuations are compared and what controls accompany the comparison. Section 8 and Appendix S report where the closed-form information valuation outperforms POMCP with semi-informed rollouts. A semi-informed rollout, defined in Appendix S, picks observation actions at random but ends in a belief-optimal commit, the commit action that maximizes expected reward under the rollout’s Bayesian-updated belief. The same section and appendix also test whether that advantage can instead be attributed to solver configuration, using three controls, and state what those controls do and do not isolate. The first is a per-environment sweep of POMCP’s exploration constant, the coefficient on the UCB1 exploration bonus just described. The second is an information-gain rollout that, unlike the semi-informed one, picks the observation with the largest exact one-step information gain at every rollout step. The third is a component ablation of MCTS-EFE, the paper’s own tree-search variant, which the paragraph on scaling active inference below introduces and that section defines. The ablation switches two of MCTS-EFE’s components, the in-tree information-gain reward and the max-backup rule, on and off in every combination.

*$\rho$ -POMDPs.* Araya-López et al. [2] introduced  $\rho$ -POMDPs, replacing the state-dependent reward  $R(s, a)$  with a belief-dependent reward  $\rho : \Delta(S) \times \mathcal{A} \rightarrow \mathbb{R}$ , where  $\Delta(S)$  denotes the set of probability distributions over states, so the objective becomes  $\max_{\pi} \mathbb{E}_{\pi}[\sum_t \gamma^t \rho(b_t, a_t)]$ . In their notation  $\rho$  denotes the whole belief-dependent reward, and they note that it may itself combine the expected state-action reward with an uncertainty measure. This paper adopts that additive form but reserves the symbol  $\rho$  for the belief-dependent information term alone, so that the per-step utility is written  $R(s, a) + \rho(b, a)$ , with  $R$  the ordinary state-dependent reward. The information weight  $w$  of the Introduction lives inside  $\rho$ . The  $\rho$  this paper studies is  $w$  times the expected information gain of an observation action. At  $w=1$ , the EFE case, it is exactly the  $\rho_{\text{EFE}}$  of Proposition 1, the expected information gain of the chosen observation action and zero on commit actions. For general  $w$  the budgeted sections write the same utility as  $R + wI$ , with the information gain abbreviated to  $I$ , since there the weight itself is the object of study.

Whether standard solver machinery carries over depends on the shape of  $\rho$ , and Araya-López et al. [2] showed when it does. When  $\rho$  is convex in the belief, the value function is likewise convex. A convex  $\rho$  can moreover be approximated arbitrarily well by a piecewise linear and convex (PWLC) function. Once  $\rho$  is PWLC the value function itself is PWLC, so standard solvers can be reused with limited changes. Fehr et al. [15] extended this to Lipschitz-continuous non-convex  $\rho$ , and Benchetrit et al. [5] developed an anytime incremental  $\rho$ -POMDP planner for continuous spaces built on progressive widening. The  $\rho$  studied in this paper is not among the convex cases. It does not inherit the convex value-function

structure that convex  $\rho$  yields [2], nor the PWLC approximation that structure supports. Expected information gain is concave, not convex, in the belief, so  $\rho_{\text{EFE}}$ , the symbol Section 3 adopts for it, falls in the non-convex regime. For that regime Fehr et al. [15] give Lipschitz-continuous  $\epsilon$ -optimal value functions when  $\rho$  is itself Lipschitz in the belief. Expected information gain meets that hypothesis whenever every observation likelihood is strictly positive, since  $I_a(b) = H(O_a b) - \sum_s b(s) H(O_a(\cdot | s))$  with  $O_a b$  the predictive observation distribution, which then stays a fixed distance from the boundary of the observation simplex where the entropy’s slope is unbounded (for a binary distribution  $H(\epsilon, 1 - \epsilon)/\epsilon$  diverges as  $\epsilon \rightarrow 0$ ). It can fail for observation models with zero entries, such as a noiseless sensor, where that predictive distribution reaches a vertex. None of this article’s results relies on the guarantee either way. The exact tree enumeration of Section 3 evaluates the recursion directly and needs no approximation guarantee.

The same framework-level move, rewarding information explicitly rather than only through its effect on state-dependent reward, also appears under another name. POMDPs with information rewards (POMDP-IR) [59] augment planning with explicit information rewards. That framework was developed for active cooperative perception, in which sensing actions are chosen across cooperating sensors to serve a shared task. Satsangi et al. [51] exploit submodular value functions for scaling active perception and note the equivalence between POMDP-IR and  $\rho$ -POMDPs. The aim behind  $\rho$ -POMDPs predates their formalization. Boutilier [7] casts preference elicitation as a POMDP whose hidden state is the user’s utility function and whose belief is a distribution over those functions, selecting the next preference query by sequential rather than myopic value of information. Its reward remains a function of the state, so it is a predecessor in motivation rather than in construction. What it shares with  $\rho$ -POMDPs is the aim of choosing queries for what they reveal, with an information-gathering criterion fixed by that problem rather than derived from a variational objective.

Common choices for  $\rho$  (entropy reduction, KL divergence from a target belief, and information gain) each encode a different notion of epistemic value, but the choice and weight remain heuristic and environment-specific. We instead derive the information-unit weight of the Introduction from the variational objective of active inference, yielding an untuned default under a shared reward convention. That convention, one reward unit per unit of information, is the exchange rate that fixes the coefficient. Observation costs are stated on that same reward scale. That unit of information can be taken in bits or in nats, and the paper uses both. The Introduction’s information term is measured in bits, the base-two logarithm unit, while the EFE derivation is written in nats, the natural-logarithm unit, and the two differ by a constant factor. The paragraph on reward-to-nats calibration in Section 3 makes this precise and states why the default is tied to that scale rather than invariant to it.

*Value of information and experimental design.* The idea that information has quantifiable decision-theoretic value predates both POMDPs and active inference. Howard [28] formalized the value of information in decision analysis, and Lindley [36] introduced expected information gain as a criterion for optimal

Bayesian experimental design. One classical result grounds the identification of reward with information on which this paper rests. Under a proper log scoring rule, proper meaning that reporting one’s true belief maximizes the expected score, the agent is paid the logarithm of the probability it assigned to the outcome that occurred. Its expected payoff is therefore the negative entropy of its belief, and the expected improvement in that payoff from an observation is exactly the expected information gain. Bernardo [4] showed on this basis that expected information is expected utility when rewards are proper log scoring rules. In the bandit setting, information-directed sampling (IDS) [48] explicitly trades off instantaneous regret against information gain by minimizing the information ratio  $\Gamma_t = \delta_t^2/g_t$ , where  $\delta_t$  is the expected regret and  $g_t$  is the information gain. The key structural difference from EFE is that IDS minimizes the ratio at each step (a relative weighting that adapts to the current belief), while EFE fixes the weight at  $w=1$  (an absolute weighting in information units under a fixed reward convention). This paper includes an observe-then-commit IDS baseline (Section 3, with full results in Appendix N).

Of these classical lines, sequential Bayesian experimental design is the closest structural analogue to the problem class we study, since our observe-then-commit structure parallels its selection of experiments (observations) to maximize information about an unknown state before a terminal decision. The  $\rho$ -POMDP framework operationalizes this connection by setting the belief-dependent term of each observation action to the expected information gain that action yields. Section 3 defines that quantity formally and writes it  $I_a(b)$ , the expected information gain of observation action  $a$  at belief  $b$ , so that the operationalization reads  $\rho = I_a(b)$ . The  $I(b, a)$  of Figure 1 and the bare  $I$  of  $R + wI$  above denote the same quantity, with the action written in a different place or left implicit. Walraven et al. [66] approach the same connection from the active-inference side, connecting active inference to information gathering in POMDPs and situating EFE-style epistemic value alongside classical POMDP information-seeking objectives.

*Active inference and EFE.* The Active Inference (AIF) framework [18] casts perception and action as variational inference under the free-energy principle. Parr et al. [43] provide a comprehensive textbook treatment. Friston et al. [19] formalized the decomposition of policy value into extrinsic (goal-seeking, the component the Introduction called pragmatic) and epistemic (information-seeking) components, showing that curiosity-driven exploration arises automatically from expected free energy minimization. Da Costa et al. [11] synthesized discrete-state AIF from first principles, showing that the posterior over policies takes the form  $Q(\pi) \propto \exp(-\mathcal{G}(\pi))$  where  $\mathcal{G}$  is the Expected Free Energy. Parr and Friston [42] showed that EFE decomposes into pragmatic value (divergence from preferred observations) and epistemic value (expected information gain), with both arising from a single variational objective and requiring no tunable exploration weight. The same paper compared EFE against a second functional, the generalised free energy, which differs in construction. Where EFE averages a free energy over the observations a policy is predicted to produce, the generalised free energy treats

those future observations as unobserved variables to be inferred alongside hidden states, within one free-energy functional spanning past and future time points. Despite different constructions, the two functionals produced closely matching policy posteriors in that paper’s simulations. They are not identical in general, and the coefficient result of Proposition 1 is established here for EFE only.

The mathematical foundations of EFE have been critically examined. Milidge et al. [40] showed that naively extending the variational free energy, the perception-time objective that upper-bounds the surprise of observed data, into the future does not yield exploratory behavior, proposing the Free Energy of the Expected Future (FEEF) as an alternative with clearer mathematical grounding. The literature also contains four algebraically distinct expressions that are all called expected free energy. Champion et al. [9] addressed the “unification problem”, whether the four EFE formulations in the literature follow from a single root definition, by studying two candidate roots. The two roots trade coverage against justification. All four formulations follow from the first root, but at a price. Under that root only a restricted class of prior preferences over observations is compatible with the generative model’s likelihood mapping. The second root can be justified directly as a definition in its own right, but it recovers only two of the four formulations. de Vries et al. [13] recast EFE-based planning as variational inference on a generative model augmented with preference and epistemic priors, an alternative derivation intended to enable message-passing implementations that the paper itself leaves to future work. Our own results depend on one of these formulations only. The equivalence of Section 3 is established for the discrete-state EFE whose epistemic term is the expected information gain about the hidden state, evaluated recursively over belief trajectories (the sophisticated-inference form described next). Whether the same coefficient survives under FEEF, the generalised free energy, or the recasting just described is not assumed. Section 3 lists those alternatives, together with one further derivation not discussed in this section, as cases the result leaves open.

Two contemporaneous 2024–2026 papers examine EFE’s relationship to other decision-theoretic targets rather than its internal formulation. Wei [67] analyze EFE’s optimality gap against a reward-only Bayes-optimal policy within a belief-MDP casting, treating the epistemic term as an information-value correction rather than an exact identity with a specific target. Sweeney et al. [62] survey formal correspondences between EFE minimization and six decision-theoretic frameworks (Bayesian decision theory, resource rationality, optimal control, reinforcement learning, rate-distortion theory, and maximum-entropy inference), a broader unification effort distinct from our narrower exact identity with  $\rho$ -POMDPs under log scoring at  $w=1$  (Proposition 1).

*Sophisticated inference.* Standard AIF evaluates policies open-loop. It scores whole action sequences under the current belief without conditioning later actions on intermediate, counterfactual observations. Friston et al. [20] introduced sophisticated inference, a closed-loop recursive extension implementing deep tree search over belief trajectories rather than states. Sophistication (maintaining be-

liefs about future beliefs) enables counterfactual reasoning about the downstream epistemic consequences of actions. Da Costa et al. [12] proved an optimality result for this recursive scheme in a specific regime. In their setting the preferences over outcomes encode a reward scaled by an inverse temperature  $\beta$ , called a precision in active-inference vocabulary, so that larger  $\beta$  makes the preferences encode the reward more sharply. As  $\beta \rightarrow \infty$ , the zero-temperature limit of precise reward-encoding preferences, the reward term dominates, so that the epistemic terms only break ties among reward-maximizing action sequences. In that limit the recursive scheme recovers a policy that is Bellman-optimal for the reward, for any finite horizon. The result is stated for POMDPs and assumes exact state estimation. Standard AIF, by contrast, recovers such reward-Bellman-optimal policies only for single-step planning. Our recursive EFE agent (Equation 3) is derived from this framework, adapted to the  $\rho$ -POMDP commit-action structure. The recursive counterfactual reasoning over future beliefs is preserved. What changes is that our observe-then-commit setting does not involve state transitions between time steps, simplifying the belief-trajectory computation.

*Scaling active inference.* Discrete-state AIF with full policy enumeration is limited to small state-action spaces. Several lines of work address scaling. Fountas et al. [17] combined deep generative models with Monte Carlo tree search for EFE-optimal planning in continuous state spaces. Tschantz et al. [65] developed an RL-compatible objective (the FEEF objective introduced above) that inherits AIF’s exploration-exploitation balance while scaling to standard RL benchmarks. Maisto et al. [37] combined active inference with Monte-Carlo tree search for large POMDPs, reproducing state-of-the-art POMDP solutions on RockSample [58]. Our MCTS-EFE variant (Section 8, Appendix S) follows this direction, using EFE as the in-tree observe reward and leaf heuristic within MCTS to extend planning horizons beyond exact tree search.

*Exploration, control as inference, and intrinsic motivation.* The control-as-inference perspective [35, 63] casts maximum-entropy reinforcement learning as probabilistic inference. The correspondence is exact under deterministic dynamics and only variational under stochastic dynamics. The reason is that conditioning on optimality under stochastic transitions would credit the agent with control over chance outcomes, and the variational treatment removes that by holding the transition model fixed. Soft actor-critic [24] and KL-minimizing stochastic optimal control [46] are algorithmic instances. Millidge et al. [39] provided a formal comparison of AIF and control-as-inference, tracing their differences to how value is encoded in the generative model. Sajid et al. [50] showed that intrinsic motivation behaviors arise under EFE. Intrinsic motivation methods, including curiosity [44, 52], count-based exploration [3], VIME [27], and random network distillation [8], all require tunable bonus weights. Oudeyer and Kaplan [41] survey and classify the computational forms these intrinsic signals take. Bayes-Adaptive MDPs [14, 23] and Bayesian RL more broadly [21] address model uncertainty (unknown dynamics), distinct from our focus on state uncertainty under a known model. Our  $\rho$ -POMDP formalism makes the connection between EFE and these lines

of work precise while fixing a derived information-unit weight under a shared reward convention.

*Resource-constrained information acquisition.* A separate literature treats information acquisition itself as resource-constrained, without reference to  $\rho$ -POMDPs. Rational inattention [38, 56] takes an information-processing constraint as primitive, imposed as a modeling assumption rather than derived. In Sims [56] the constraint is a Shannon-capacity cap on how many bits per period the decision-maker’s chosen action may carry about the state, and in Matějka and McKay [38] it is instead a cost charged per unit of information acquired. Constrained MDPs dualize an expected-cost constraint via a Lagrange multiplier [1], and constrained POMDPs enforce the cumulative-cost bound directly by point-based dynamic programming over belief and admissible-cost pairs [30]. Soft Actor-Critic’s automatic entropy-temperature tuning [25] drives a resource (policy entropy) to a target by dual gradient descent. Sequential Bayesian experimental design [16, 36], met above as the closest structural analogue to our setting, adaptively selects the next experiment to maximize information about a latent parameter. Section 4.6 positions our budgeted  $\rho$ -POMDP formulation in detail against each of these four lines, namely rational inattention, constrained MDPs and POMDPs, automatic temperature tuning, and sequential experimental design. None of these approaches gives a designer an operational sensing budget specifically for  $\rho$ -POMDP planning, exact scale equivariance of the resulting policy for the receding-horizon planner, or a crossing bracket and endpoint mixture at usage gaps. The next section sets up the observe-then-commit  $\rho$ -POMDP class and the EFE objective that the rest of the paper, including those comparisons, builds on.

### 3 Methodology

This section fixes the objects that the rest of the article uses. It defines the general  $\rho$ -POMDP and the observe-then-commit class studied here, states the Expected Free Energy objective and shows how its recursive form maps onto that class, proves the equivalence between the two and extends it to factored observation POMDPs, and closes by specifying the agents the experiments compare.

#### 3.1 The $\rho$ -POMDP Framework

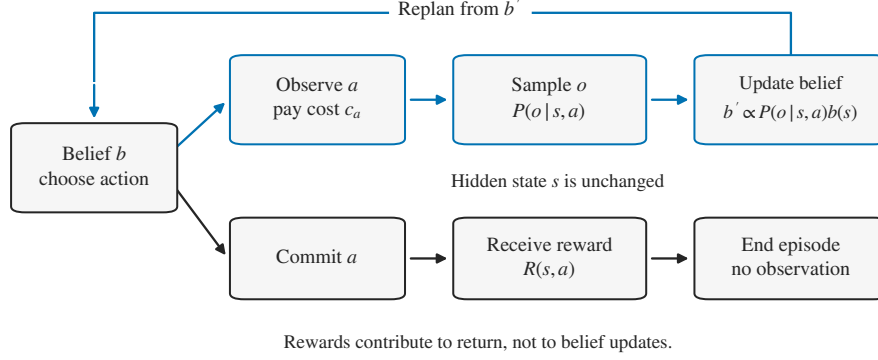
A  $\rho$ -POMDP extends the standard POMDP with a belief-dependent utility  $\rho : \Delta(S) \times \mathcal{A} \rightarrow \mathbb{R}$ , where  $S$  is the finite set of hidden states,  $\mathcal{A}$  is the action set, and  $\Delta(S)$  is again the set of probability distributions over  $S$ , also called the belief simplex. The agent’s objective becomes

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[ \sum_{t=0}^H \gamma^t (R(s_t, a_t) + \rho(b_t, a_t)) \right] \quad (1)$$

Here  $\pi$  is a policy, a rule that selects an action from the agent’s current belief,  $s_t$  is the (unobserved) hidden state,  $b_t$  is the agent’s belief over states at time  $t$ , a point of  $\Delta(S)$ ,  $H$  is the planning horizon, the number of steps the agent looks ahead, and  $\gamma$  is the discount factor, which weights a reward received  $t$  steps ahead by  $\gamma^t$ . The equivalence of Section 3.3 is stated for the undiscounted case  $\gamma=1$ . The outer expectation  $\mathbb{E}_\pi$  is over the joint trajectory distribution induced by  $\pi$ , i.e., over hidden-state transitions  $s_t \sim T(\cdot \mid s_{t-1}, a_{t-1})$  under the transition model  $T$ , observations  $o_t \sim P(\cdot \mid s_t, a_{t-1})$  under the observation model  $P$ , and the resulting deterministic Bayesian belief updates  $b_t = \text{update}(b_{t-1}, a_{t-1}, o_t)$ . This is the general framework. Under it the reward term  $R(s_t, a_t)$  depends on the sampled hidden state, while  $\rho(b_t, a_t)$  depends on the belief induced by the observations that state generates, together with the action taken from that belief. The two terms are therefore coupled through the same trajectory distribution. When  $\rho = 0$ , we recover the standard POMDP. When  $\rho$  encodes information gain, the agent is explicitly rewarded for reducing uncertainty.

This paper restricts attention to observe-then-commit  $\rho$ -POMDPs, in which the action set  $\mathcal{A}$  partitions into observation actions  $\mathcal{A}_{\text{obs}}$  (which update the belief at known cost but do not change the hidden state) and terminal commit actions  $\mathcal{A}_{\text{com}}$  (which end the episode with state-dependent reward). Commit actions emit no observation. Throughout, rewards are not observations, and the belief is conditioned on the observation symbol alone, an assumption Definition 1 restates for the interleaved class. An episode consists of a variable-length sequence of observation actions followed by a single commit action. The hidden state is fixed throughout, so  $s_t$  stays at its initial draw. The expectation in the objective above runs over that initial draw, the stream of observations, and the belief updates they drive. We start with this class for four reasons. First, it isolates the epistemic-value question from state-transition dynamics. Second, for a two-state model with one observation action, it admits a proved characterization of when  $w=1$  lies in a near-optimality interval (Proposition 2). The interval is a range of weights between an under-observing threshold and an over-observing one, where over-observing means paying for a second observation whose instrumental value, the improvement in expected commit reward that the observation is expected to buy, falls below its cost. Third, near-optimal SARSOP baselines are available for it, reference solutions computed offline by a point-based POMDP solver against which the agents can be compared. Fourth, it covers a useful range of applications (medical testing, sensor tasking) on its own. Proposition 3, the RockSample results of Section 7.4, the Structural Inspection results of Section 7.5, and the interleaved usage curves of Section 6.5 extend the equivalence and the budgeted formulation beyond this class to interleaved observe-act settings.

Figure 3 summarizes the two action types and the observation channel that drives belief updates.



**Fig. 3.** The observe-then-commit decision loop, shown schematically. An observation incurs a cost and updates the belief about a hidden state that remains fixed. The agent then chooses again from the updated belief. A commit instead pays a state-dependent reward and ends the episode without emitting an observation. Rewards are not inputs to the belief update. The loop describes the episode, while the planner’s finite horizon limits how far it looks ahead at each decision.

### 3.2 Expected Free Energy as $\rho$

In the standard AIF formulation, the EFE for a policy  $\pi$  at future time  $\tau$  is

$$\mathcal{G}(\pi) = \underbrace{-\mathbb{E}_{Q(o_\tau|\pi)}[\ln P(o_\tau|C)]}_{\text{Pragmatic value}} - \underbrace{\mathbb{E}_{Q(o_\tau|\pi)}[D_{\text{KL}}[Q(s_\tau|o_\tau, \pi) \| Q(s_\tau|\pi)]]}_{\text{Epistemic value}} \quad (2)$$

where  $Q(\cdot|\pi)$  denotes the agent’s predictive distribution under policy  $\pi$ , so that  $Q(o_\tau|\pi)$  is the distribution of observations it predicts at time  $\tau$ , and  $P(o_\tau|C)$  is a second distribution over the same observations, the prior preference distribution of the agent’s generative model. Here  $C$  stands for the agent’s prior preferences over outcomes, the parameters that fix which observations are preferred, and  $P(\cdot|C)$  is the distribution those preferences induce. The letter  $P$  is reused from the previous subsection, where  $P(\cdot | s_t, a_{t-1})$  was the observation model. The two are different objects distinguished by what they condition on, the observation model on a state and an action, the preference distribution on the preferences  $C$ . Both belong to the same generative model in the active-inference convention, which is why they share a letter. The two letters  $P$  and  $Q$  mark two different roles.  $Q$  says what the agent expects to observe if it follows  $\pi$ , and  $P(\cdot|C)$  says what it would like to observe. That is why the predicted observations are scored under  $P$  rather than under the  $Q$  that generates them. The agent minimizes  $\mathcal{G}$ , so lower is better, and the signs of the two terms encode what it seeks. The pragmatic term is the expected negative log-probability of the predicted observations under the preference distribution, so it is small when the outcomes  $\pi$  is predicted to produce are ones the agent prefers. The epistemic term is the divergence  $D_{\text{KL}}$  between

the posterior over hidden states after a predicted observation,  $Q(s_\tau|o_\tau, \pi)$ , and the prior  $Q(s_\tau|\pi)$ , averaged over the predicted observations. That average is the expected information gain about the hidden state. It enters with a minus sign, so policies whose observations are expected to sharpen the belief lower  $\mathcal{G}$ . The recursion below scores one action at a time from the current belief rather than a whole policy at a fixed future time, so the index  $\tau$  does not appear again.

In our observe-then-commit setting, the pragmatic and epistemic terms take concrete forms, and we take the two action types in turn, commit actions in this paragraph and observation actions in the next. Commit actions carry no epistemic term because, by construction of the observe-then-commit class, a commit emits no observation. Its payoff depends on the hidden state, but rewards are not observations in this article, the belief is conditioned on the observation symbol alone, and so there is no posterior belief for the epistemic term to score. That the episode ends on commitment says the same thing from the other direction, and Proposition 1 states it that way, since a terminal action emits no further observation and leaves no successor belief to score. Termination also removes the recursion’s continuation term, the expected best EFE from a posterior belief, so a commit is scored by its pragmatic term alone. The pragmatic term for a commit is still well defined, but it does not score a fresh observation. For a commit, the outcome is the reward the commit pays, which depends on the hidden state, so the expectation in the pragmatic term runs over the current belief  $b$  rather than over a predicted observation. The term is then fixed by identifying preferences with task reward, taking log-preference proportional to reward, an identification made precise in the reward-to-nats calibration paragraph of Section 3.3. Under it, the pragmatic term of  $\text{commit}_i$  reduces to minus the expected reward under the current belief,  $\mathcal{G}(\text{commit}_i) = -\mathbb{E}_b[R_i]$ , so that minimizing  $\mathcal{G}$  over commit actions maximizes expected reward. Here  $R_i$  denotes the state-dependent reward  $R(s, \text{commit}_i)$  of the  $i$ -th commit action and  $\mathbb{E}_b[R_i] = \sum_s b(s) R(s, \text{commit}_i)$ . The implementation applies this identification by reading the reward matrix, the table of values  $R(s, \text{commit}_i)$  indexed by state and commit action, directly, rather than by specifying a preferred outcome distribution  $P(o|C)$  separately and scoring commits against it. Within the implementation’s reward-based recursion, the two routes express the same identification up to the preference normalizer. They differ in whether the reward scale is stated explicitly. Equivalence to a separately normalized preference distribution holds only in the cases the reward-to-nats calibration paragraph of Section 3.3 sets apart, since over variable-length episodes the preference normalizer contributes a policy-dependent term that the recursion omits. Reading the reward matrix directly declares the scale explicitly, at  $\beta=1$  per reward unit as Section 3.3 writes out. A separately specified  $P(o|C)$  leaves that same scale implicit, because how sharply it concentrates on preferred outcomes acts as an undeclared inverse temperature, the scale  $\beta$  at which log-preference is set proportional to reward. That undeclared scale is the preference-calibration issue we avoid. Changing it shifts the balance between the pragmatic and epistemic terms, and prior AIF work has flagged this calibration as a source of hidden tuning.

Observation actions carry both terms. Writing  $\text{observe}_k$  for the  $k$ -th observation action (abbreviated  $\text{obs}_k$  in subscripts and in some later displays), its pragmatic term is the observation cost  $c_k > 0$  (the magnitude the agent pays to observe). This follows from the same identification used for commit actions, because the reward of an observation action is minus its cost, so minus its expected reward is plus the cost. Its epistemic term is the expected information gain  $I_k(b) = H(b) - \mathbb{E}_{o|\text{obs}_k}[H(b'_o)]$ . Here  $H(b) = -\sum_s b(s) \log b(s)$  is the Shannon entropy of the belief, a measure of how uncertain the agent is about the hidden state. The belief  $b'_o$  is obtained from  $b$  by Bayesian conditioning on observation  $o$  under  $\text{observe}_k$ , with the expectation taken over the observations the agent predicts from  $b$ . The entropy functional  $H(\cdot)$  is distinct from the planning horizon  $H$  of the objective above, and which is meant is always clear from whether it takes an argument. The entropy's unit, nats or bits, is fixed by the base of the logarithm (Section 3.3).

Following the sophisticated inference scheme of Friston et al. [20], the EFE agent evaluates actions via a recursive tree search over belief states. In words, at belief  $b$  each observation action is scored by its cost, minus the information it is expected to buy, plus the best EFE attainable from each posterior belief it could lead to, averaged over those posteriors. Commit actions are the leaves of the tree, scored by minus their expected reward as above. The recursion is cut off at the planning horizon  $H$ , where no further observation action is expanded and only commit actions are scored, so every branch of the tree ends in a commit. Writing  $\mathcal{G}(a | b)$  for the EFE of action  $a$  taken at belief  $b$ , and suppressing  $b$  when it is the current belief, the observe branch reads

$$\mathcal{G}(\text{observe}_k) = c_k - I_k(b) + \mathbb{E}_o \left[ \min_a \mathcal{G}(a | b'_o) \right] \quad (3)$$

The agent selects  $\arg \min_a \mathcal{G}(a)$ . Standard AIF instead draws its policy from a softmax distribution  $Q(\pi) \propto \exp(-\kappa \mathcal{G}(\pi))$ , which introduces a policy-precision parameter, an inverse temperature controlling how sharply the policy concentrates on low-EFE actions. We write it  $\kappa$  to keep it distinct from the reward inverse temperature  $\beta$  that calibrates preferences to reward. Our deterministic  $\arg \min$  corresponds to  $\kappa \rightarrow \infty$ , which eliminates this tunable knob.

Equation 3 contains no separate exploration weight, and within the reward-based recursion that Section 3.3 adopts as a convention, this is not a free choice. The reason is a matter of units. In EFE's native form, before pragmatic value is replaced by task reward, both terms are built from logarithms of probabilities, the pragmatic term as an expected log-probability of the predicted observations under the preference distribution and the epistemic term as an expected KL divergence, itself an expected difference of two log-probabilities. Written with the natural logarithm, both are therefore in nats. The coefficient one on expected information gain is not merely written into the display. It relates two quantities carried in the same unit, so it is dimensionless and no conversion factor is needed between them. By contrast, a coefficient placed between reward units and information units, which is what  $w$  is when a planner adds an information bonus to task reward, has to be set by a calibration of the two scales. The pragmatic term

keeps the shared-unit reading once task reward stands in for preference, because the identification used for commit actions above supplies exactly that calibration, setting log-preference proportional to reward at  $\beta=1$  per reward unit. This shared scale is why the reduction of Equation 3 to a  $\rho$ -POMDP carries  $w=1$  exactly (Proposition 1). Section 3.3 states that reduction, makes the calibration precise, and returns to the question of units, since the implementation measures entropy in bits.

### 3.3 Formal Equivalence with $\rho$ -POMDPs

This subsection makes the claim of the previous one precise. Negating the recursive EFE of Equation 3 turns it, line for line, into the Bellman recursion of a  $\rho$ -POMDP whose belief utility  $\rho$  is expected information gain and whose reward  $R$  on observation actions is minus the observation cost. Proposition 1 states that identity. Two scope remarks come first, both already made in the related-work discussion. First, the action argument in  $\rho(b, a)$  is admissible, since Araya-López et al. [2] state the framework for action-dependent belief utilities and permit the additive form  $R(s, a) + \rho(b, a)$ . Second, because expected information gain is concave rather than convex in the belief, the convexity and PWLC results of Araya-López et al. [2], which require  $\rho(\cdot, a)$  convex in the belief for each fixed  $a$ , do not apply, and the Lipschitz-continuity guarantee of Fehr et al. [15] for the non-convex case applies only where its Lipschitz-reward hypothesis holds, which expected information gain satisfies for observation models whose likelihoods are all strictly positive and can violate for models with zero entries, and the exact enumeration below does not rely on it either way (as the Related Work section notes).

The recursion in Equation 3 is indexed by the depth  $d$  of the current node in the belief tree, that is, the number of observations taken so far along the branch. We suppressed  $d$  there for readability and make it explicit now, because the horizon bound  $H$  is enforced through it. Writing  $\mathcal{G}(a, b, d)$  for the EFE of action  $a$  at belief  $b$  and depth  $d$ , define  $V(a, b, d) \triangleq -\mathcal{G}(a, b, d)$ . Then Equation 3 becomes

$$V(\text{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o \left[ \max_a V(a, b'_o, d+1) \right] \quad (4)$$

$$V(\text{commit}_i, b, d) = \mathbb{E}_b[R_i] \quad (5)$$

Once  $d$  reaches  $H$  the observe branch of Equation 4 is no longer available, so only commit actions remain. The recursion bottoms out at the best commit value  $\max_i \mathbb{E}_b[R_i]$ , which is how the implementation enforces the bound. With that convention in place, this is exactly the Bellman recursion for a  $\rho$ -POMDP with action-dependent belief utility, that is, the dynamic-programming form of the summed objective displayed at the start of Section 3 (Equation 1), with minus the observation cost playing the role of  $R$  and the information gain playing the role of  $\rho$ .

**Proposition 1.** Define  $\rho_{\text{EFE}}(b, a)$  as

$$\rho_{\text{EFE}}(b, a) = \begin{cases} I_a(b) & \text{if } a \text{ is an observation action} \\ 0 & \text{if } a \text{ is a commit action} \end{cases}$$

where  $I_a(b) = H(b) - \mathbb{E}_{o|a}[H(b'_o)]$  is the expected information gain from observation action  $a$  at belief  $b$ . Assume (i) pragmatic value is identified with task reward at  $\beta=1$  per reward unit, (ii) beliefs are updated by exact Bayesian conditioning, (iii) information is measured in the same units on both terms, and (iv) actions are selected by deterministic  $\arg \min$  over  $\mathcal{G}$ , the  $\kappa \rightarrow \infty$  limit of the standard softmax policy with precision  $\kappa$ . Commit actions carry  $\rho_{\text{EFE}}=0$  not by stipulation but because the episode terminates on commitment, so no further observation is emitted and the epistemic term has no successor belief to score. Then for undiscounted finite horizon ( $\gamma=1$ ), minimizing recursive EFE (Eq. 3) over horizon  $H$  produces the same policy as solving the  $\rho$ -POMDP Bellman equation  $V^*(b) = \max_a \{R(b, a) + \rho_{\text{EFE}}(b, a) + \mathbb{E}_o[V^*(b'_o)]\}$  over the same horizon, with terminal commit actions carrying no continuation term.

*Proof (Proof sketch).* The negation  $V = -\mathcal{G}$  converts  $\arg \min \mathcal{G}$  to  $\arg \max V$ . Substituting into Equation 3 gives, for observation actions,  $V = -c_k + I_k(b) + \mathbb{E}_o[\max_{a'} V(a', b'_o)]$ , which matches the  $\rho$ -POMDP Bellman backup with  $R(b, \text{obs}_k) = -c_k$  and  $\rho = I_k(b)$ . For commit actions,  $V = \mathbb{E}_b[R_i]$  with  $\rho = 0$ , matching a terminal  $\rho$ -POMDP action. The recursive structure is identical, so the policies agree at every belief node.

The result just stated covers one specific formulation of EFE, and we name what it does not cover before interpreting it. Equation 3 instantiates the recursive, closed-loop form of EFE associated with sophisticated inference [12, 20]. In Champion et al. [9]’s taxonomy of EFE formulations this is the variant whose epistemic term is the expected information gain about the hidden state, evaluated recursively over belief trajectories. We establish the  $w=1$  coefficient for that formulation only. Whether the same coefficient survives under the Free Energy of the Expected Future [40], generalised free energy [42], the epistemic-prior variational treatment of de Vries et al. [13], or the Bethe-Lagrangian stationary point of Kouw [31] is not established here. A negative answer for any of them would narrow the scope of Proposition 1 without touching the budgeted results of Section 4.

Operationally, Proposition 1 says that EFE fixes the information weight rather than removing it, and it makes precise what EFE-as- $\rho$  means. The epistemic term of EFE functions as an action-dependent belief utility. The operational payoff concerns Planning+IG, the tuned information-gain baseline of the Introduction, which runs the same tree search as the EFE agent and adds  $w$  times expected information gain to each observation action’s expected reward (the objective is written out in Proposition 2). The  $\rho$ -POMDP of Proposition 1 has exactly that objective with the weight fixed at one, so the EFE agent of Equation 3 is equivalent to Planning+IG with  $w=1$  in information units under

a fixed reward convention. What EFE supplies in place of the tuning is a relative coefficient derived from the variational objective itself, fixing  $w=1$  without per-environment search.

As for what Proposition 1 is and is not, this equivalence is a notational bridge rather than a new optimality theorem. Its content is the negation  $V = -\mathcal{G}$  together with the naming of the two terms, minus the observation cost as  $R$  and the information gain as  $\rho$ . It rests on the same substitution that underlies the Bellman-optimality result for sophisticated inference of Da Costa et al. [12]. That result takes the reward inverse temperature  $\beta$  to the limit  $\beta \rightarrow \infty$ , in which the epistemic term only breaks ties among reward-maximizing action sequences. Here the same substitution is applied with action-dependent  $\rho = I_a(b)$  at  $\beta=1$ , so the epistemic term is retained at full weight in the observe-then-commit action structure. Appendix A carries the self-contained proof.

*Reward-to-nats calibration.* The equivalence needs an exchange rate between reward and information. The pragmatic term of EFE is an expected log-probability, measured in nats, while task reward arrives in whatever units the environment specifies, so identifying the two means fixing how many nats one reward unit is worth. Identifying pragmatic value with task reward chooses  $P(o|C) \propto \exp(\beta R)$  with  $\beta=1$  per reward unit, which sets that rate at one nat per reward unit. This identification is a convention adopted in assumption (i) of Proposition 1, not a derivation, and the proposition is a statement about the recursion of Equation 3 as written, in which the pragmatic value of each scored outcome is its task reward and no normalizer appears. Deriving that recursion from a normalized preference distribution would add a term the recursion omits. Writing  $Z = \sum_{o'} \exp(\beta R(o'))$  for the normalizer over the outcome space,  $\ln P(o|C) = \beta R(o) - \ln Z$ , and the constant  $\ln Z$  is scored once per outcome, so a path that scores  $T$  outcomes before it ends accumulates  $-T \ln Z$ . Observe-then-commit paths of different lengths score different numbers of outcomes, so that accumulated constant is policy-dependent and can change which of observing and committing minimizes the objective. Apart from the log-scoring case, in which the reward is itself a log probability so that  $Z=1$  and the constant vanishes, it cancels only under a convention that scores the same number of outcomes on every path, for example a fixed-length construction in which an episode that has committed keeps emitting a null outcome of zero reward from an absorbing state at every remaining step, scored against one preference distribution over a common outcome space. We do not adopt such a construction, and the proposition’s scope does not extend to a general normalized-preference derivation for variable-length episodes. Three things are therefore kept apart in this article. The exact case is Bernardo’s, where the reward is itself a log probability and no normalizer arises. The proved case is the reward-based recursion, conditional on assumption (i). The empirical convention is one reward unit per bit, stated below. The choice has a consequence for scale. Rescaling all rewards by a positive factor  $k$  (the factor written  $\alpha$  in Proposition 4) at fixed  $w$  induces the same policy as weight  $w/k$  on the unscaled rewards, so  $w=1$  is not scale-invariant, and the reward-maximizing weight  $w_{\text{ret}}^*$  (the subscript abbreviates return) scales as  $w_{\text{ret}}^*(k) \propto k$  (Appendix Q). Following

Bernardo [4], the identification is exact when rewards are log scoring rules, that is, when the reward for an outcome is the log probability the agent assigned to that outcome, so that reward is already measured in nats and no unit conversion stands between the two terms. Our claim is weaker and operational. Within a fixed reward convention, the information-unit weight  $w=1$  is a robust untuned default.

The implementation measures information in bits rather than nats (`scipy.stats.entropy` with `base=2`), a constant-factor difference that we make explicit here. Every result reported as EFE ( $w=1$ ) in this article runs the agent at  $w=1$  in those bit units, that is, one reward unit per bit. Since one bit is  $\ln 2$  nats, one reward unit per bit is one reward unit per  $\ln 2$  nats, or  $1/\ln 2$  reward units per nat, which is an effective nat-denominated weight of  $1/\ln 2 \approx 1.44$ . Running the same conversion in reverse, the exactly canonical nat-unit identification of Proposition 1 corresponds to  $w = \ln 2 \approx 0.69$  in the implementation’s bit units.

Whether that constant factor changes any reported result is settled directly in Appendix R, which runs the exactly nat-canonical agent of Proposition 1,  $w = \ln 2$  in the implementation’s bit units, under the Pareto sweep protocol of Section 7.2 on all five swept environments (`results_nat_canonical_check.csv`). On Tiger, Diagnosis, Bandit, and Testbed it is bit-identical to the implemented  $w=1$  agent in reward, success rate, and observation count. Where that sweep’s mean reward is exactly tied across adjacent swept weights, its reward-maximizing weight is reported as the closed range of tied grid weights, the reward-maximizing tied bracket. On Tiger and Diagnosis both weights lie inside it, and on Bandit, whose bracket has grid endpoints  $[1, 2]$  and so contains  $w=1$  but not  $0.69$ , the directly run nat-canonical agent attains the bracket’s tied reward. Tileworld is the one swept environment where the base changes the policy, and there  $w=1$  remains Pareto-dominated by  $w=20$  under either convention.

Whether  $w=1$  is near-optimal for expected reward under the paper’s reward convention is an empirical question we address in Section 7.2. Before that empirical answer, a closed-form statement is available for the simplest instance, and the intuition behind it is that an information weight can fail in two directions. At too small a weight the agent commits without observing at all, and at too large a weight it pays for a second observation whose instrumental value falls below its cost. The following result computes, for a two-state model with a single observation action, the weight at which each failure switches on, a lower threshold for the first, derived at horizon  $H=1$ , and an upper threshold for the second, derived at  $H=2$ . The lower threshold can be negative, and when it is, the  $H=1$  agent observes at every nonnegative weight. The range between the two thresholds is what the proposition calls the near-optimality interval, and the result characterizes the conditions under which  $w=1$  lies in it for the two-state case.

**Proposition 2.** *Consider a two-state observe-then-commit  $\rho$ -POMDP with uniform prior  $b(s_0)=b(s_1)=\frac{1}{2}$ , a single observation action (accuracy  $\frac{1}{2} < p \leq 1$  in Part (a) and  $\frac{1}{2} < p < 1$  in Part (b), cost  $c > 0$ ), and two commit actions (correct reward  $R^+ > 0$ , incorrect penalty  $R^- < 0$  with  $|R^-| > R^+$ ). Define the reward*

asymmetry ratio  $\alpha = |R^-|/R^+$  and the informativeness ratio  $\eta = I_{\max}/c$  where  $I_{\max} = \ln 2 - H_{\text{post}}(p)$  is the maximum expected information gain in nats, with  $H_{\text{post}}(p) = -p \ln p - (1-p) \ln(1-p)$  the entropy in nats of the post-observation belief  $(p, 1-p)$ , so that  $I_{\max}$  is the uniform prior's entropy  $\ln 2$  less the entropy that one observation of accuracy  $p$  leaves behind. This proposition states two separate results at two separate horizons. Part (a) is a statement about  $H=1$ , a single decision to observe once or not and then commit, and Part (b) is a statement about  $H=2$  (whether a second observation is also worthwhile). They share notation but are logically independent claims, proved separately below.

**Part (a) ( $H=1$  lower threshold).** The minimum weight  $w_{\text{thresh}}^*$  above which the Planning+IG objective, expected reward plus  $w$  times expected information gain, prefers observing once to committing immediately is

$$w_{\text{thresh}}^* = \frac{c - (p - \frac{1}{2})(R^+ - R^-)}{I_{\max}} = \frac{c - (p - \frac{1}{2})(1 + \alpha)R^+}{I_{\max}}$$

For any  $w > w_{\text{thresh}}^*$  the agent observes before committing at  $H=1$ . The threshold  $w_{\text{thresh}}^*$  is negative, making  $w=1$  sufficient for at least one observation, whenever  $\alpha > c/[(p - \frac{1}{2})R^+] - 1$ . In particular,  $w_{\text{thresh}}^* \rightarrow -\infty$  as  $\alpha \rightarrow \infty$  for any fixed  $p > \frac{1}{2}$  and  $c > 0$ .

**Part (b) ( $H=2$  near-optimality interval).** At the post-first-observation belief  $b=(p, 1-p)$ , define the instrumental value of a second observation  $\text{VoI}_2(b)$  and the expected information gain  $I(b)$  of one further observation. The upper (over-observation) threshold is

$$w_{\text{over}}^* = \begin{cases} (c - \text{VoI}_2(b))/I(b) & \text{if } \text{VoI}_2(b) < c, \\ +\infty & \text{otherwise.} \end{cases}$$

At  $H=2$  for this two-state case, when  $w_{\text{thresh}}^* \leq 0$ , so that the first observation is instrumentally justified, every  $w \in (w_{\text{thresh}}^*, w_{\text{over}}^*) \cap [0, \infty)$  reproduces the instrumentally optimal single observation, and we call this set the near-optimality interval. When  $\text{VoI}_2(b) \geq c$  a second observation is instrumentally justified at any weight, so no over-observation threshold exists and the interval is unbounded above. In this uniform-prior symmetric two-state model  $\text{VoI}_2(b) = 0$  identically, since the disconfirming second observation returns the belief exactly to the tie point, so the finite branch always applies and all four rows of Table 1 fall in it. The reading is sign-conditional. When  $w_{\text{thresh}}^* \leq 0$ , weights inside the interval reproduce the instrumentally optimal single observation, while weights above a finite  $w_{\text{over}}^*$  buy a second observation whose instrumental VoI is below cost. When  $w_{\text{thresh}}^* > 0$  committing immediately is reward-optimal, weights below the threshold are exactly reward-optimal, and weights inside the interval buy one instrumentally unprofitable observation at a reward loss of  $c - (p - \frac{1}{2})(R^+ - R^-)$ , positive in this regime and bounded above by the observation cost  $c$ . Part (b) presupposes Part (a)'s threshold as its lower endpoint but is otherwise a distinct claim. Perfect sensing,  $p=1$ , is excluded from Part (b), since a second observation then carries no information,  $I(b)=0$ , and the over-observation ratio is undefined,

while Part (a) holds at  $p=1$  with  $H_{\text{post}}(1) = 0$  under the convention  $0 \ln 0 = 0$ , so that  $I_{\text{max}} = \ln 2$ .

*Proof (Proof sketch). Part (a).* At  $H=1$ , the agent observes once and commits. A Planning+IG agent with weight  $w$  observes iff  $-c + w \cdot I(b) + \mathbb{E}_o[\max_i \mathbb{E}_{b'_o}[R_i]] > \max_i \mathbb{E}_b[R_i]$ , where the left side is the observe-then-commit value and the right side is the immediate commit value. For the uniform prior, the immediate commit value is  $(R^+ + R^-)/2$ . After one observation, the posterior concentrates. With probability  $p$  the agent is correct, yielding expected commit value  $p \cdot R^+ + (1-p) \cdot R^-$ . The net gain from observing is  $(p - \frac{1}{2})(R^+ - R^-) - c + w \cdot I_{\text{max}}$ . Setting this to zero gives  $w_{\text{thresh}}^* = [c - (p - \frac{1}{2})(R^+ - R^-)] / I_{\text{max}}$ . Substituting  $R^- = -\alpha R^+$  yields the second form. When  $\alpha \gg 1$ , the marginal reward improvement  $(p - \frac{1}{2})(1 + \alpha)R^+$  dominates the cost  $c$ , making  $w_{\text{thresh}}^*$  negative, so the agent should observe at any  $w \geq 0$ , including  $w=1$ .

*Part (b).* Fix the post-first-observation belief  $b=(p, 1-p)$ . The agent prefers a second observation over committing when  $-c + w \cdot I(b) + \mathbb{E}_{o'}[\max_i \mathbb{E}_{b'_o}[R_i]] > \max_i \mathbb{E}_b[R_i]$ , i.e., when  $w > (c - \text{Vol}_2(b)) / I(b)$  whenever  $\text{Vol}_2(b) < c$ . That cutoff is  $w_{\text{over}}^*$ . Combining both inequalities yields the  $H=2$  near-optimality interval.

Under the paper’s reward convention,  $w=1$  lies in this interval for Tiger, the one tabulated instance that is genuinely two-state with  $|R^-| > R^+$  and so falls under the proposition as stated. For Diagnosis and Tileworld, which are multi-state, multi-action environments outside the proposition’s stated hypothesis, the same thresholds are applied heuristically to a two-state reduction, for which no formal correspondence is proved. The reduction takes each environment’s observation accuracy as  $p$ , its per-observation cost as  $c$  (a test on Diagnosis, a scan on Tileworld), and its correct and incorrect commit rewards as  $R^+$  and  $R^-$ , and it keeps nothing of the state and observation-action counts. Diagnosis and Tileworld share all four of those values (Appendix E), which is why their two rows in the table coincide. Table 1 provides a descriptive comparison against Proposition 2 on this basis. It reports  $\alpha$ ,  $\eta$ , the lower and upper thresholds  $w_{\text{lo}}^*$  and  $w_{\text{hi}}^*$ , and the observed reward-maximizing weight from the Pareto sweep (Section 7.2). The table writes Part (a)’s  $w_{\text{thresh}}^*$  as  $w_{\text{lo}}^*$  and Part (b)’s  $w_{\text{over}}^*$  as  $w_{\text{hi}}^*$ .

Table 1 shows two patterns. On the three high-asymmetry environments (Tiger, Diagnosis, Tileworld)  $w=1$  sits comfortably inside the interval in either unit convention, whereas Testbed sits on the interval’s upper edge and on the boundary of the proposition’s hypothesis, and we take Testbed first. On Testbed, the lower threshold is negative and the upper bound is tight ( $w_{\text{hi}}^* \approx 1.01$ ). Nat-denominated  $w=1$  sits just inside, while the implemented agent’s bit-denominated  $w=1$  corresponds to 1.44 nats and falls outside the interval. This is consistent with the over-exploration observed relative to the reward-maximizing bracket  $w_{\text{ret}}^* \in [0.01, 0.5]$  (Appendix C). Under the sweep protocol at  $H=4$ , however, the nat-canonical agent of Appendix R over-explores exactly as the bit-denominated agent does, matching it in reward, success rate, and observation count (`results_nat_canonical_check.csv`), so the interval’s tight upper endpoint separates the two conventions on paper but not in measured behavior at

**Table 1.** Reward asymmetry ( $\alpha$ ), informativeness ( $\eta$  in nats), lower observation threshold  $w_{\text{lo}}^*$ , upper over-observation threshold  $w_{\text{hi}}^*$ , and the observed reward-maximizing weight (or bracket)  $w_{\text{ret}}^*$  from a fresh Pareto sweep (Section 7.2, the eleven-point grid  $w \in \{0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200\}$ , 5 seeds  $\times$  500 episodes, `results_pareto_sweep.csv`). Where the sweep’s reward is exactly tied across several adjacent grid weights,  $w_{\text{ret}}^*$  is reported as the closed range of tied grid weights rather than an arbitrary point within it. That range is a flat set of maximizers, the analogue of the level set of Definition 2, not a crossing bracket. Only Tileworld has a genuinely unique argmax in this grid. Thresholds in nats. The  $w=1$  column asks about the canonical nat-denominated weight, while the implemented bit-denominated agent corresponds to 1.44 nats (see text). Bandit omitted as multi-arm, so Prop. 2 does not apply directly.

| Environment | $\alpha$ | $\eta$ | $w_{\text{lo}}^*$ | $w_{\text{hi}}^*$ | $w=1$ in $(w_{\text{lo}}^*, w_{\text{hi}}^*)$ ? | $w_{\text{ret}}^*$ |
|-------------|----------|--------|-------------------|-------------------|-------------------------------------------------|--------------------|
| Testbed     | 1.0      | 1.31   | −3.06             | 1.01              | Nat units only (see text)                       | [0.01, 0.5] (tied) |
| Tiger       | 10.0     | 0.27   | −138.66           | 6.89              | Yes                                             | [0.01, 20] (tied)  |
| Diagnosis   | 5.0      | 0.19   | −88.20            | 7.91              | Yes                                             | [0.5, 20] (tied)   |
| Tileworld   | 5.0      | 0.19   | −88.20            | 7.91              | Yes                                             | 20 (unique)        |

that horizon. Testbed’s  $\alpha=1.0$  sits at the boundary of Proposition 2’s strict assumption  $|R^-| > R^+$ . Its row is included descriptively rather than as an instance of the proposition.

The second pattern is the high-asymmetry one. Tiger, Diagnosis, and Tileworld have  $w_{\text{lo}}^* \ll 0$  and  $w_{\text{hi}}^* \approx 7\text{--}8$ . The near-optimality interval between those thresholds is derived at  $H=2$ , and the multi-step case beyond  $H=2$  is harder to analyze because thresholds shift with belief. Read as a prediction about the sweep, the interval says that  $w=1$  lands among the reward-maximizing weights wherever it lies inside the interval. On Tiger and Diagnosis the prediction is borne out, since  $w=1$  sits inside the reward-maximizing tied bracket. On Tileworld it is not. Tileworld’s reward-maximizing weight is uniquely  $w=20$  in the implementation’s bit units (28.9 nats), which lies above the interval’s nat-denominated upper endpoint of 7.91, and  $w=1$  is Pareto-dominated there.

*Extension to discounting.* Discounting preserves the  $\rho$ -POMDP equivalence with  $\rho(b, a) = I_a(b)$ , but it changes the balance between the two terms along a branch. The measured net effect runs toward fewer observations rather than more. With  $\gamma < 1$ , the recursive EFE becomes  $\mathcal{G}(\text{obs}_k) = c_k - I_k(b) + \gamma \mathbb{E}_o[\min_a \mathcal{G}(a, b'_o)]$ , giving  $V(\text{obs}_k) = -c_k + I_k(b) + \gamma \mathbb{E}_o[\max_a V(a, b'_o)]$ . The information gain term  $I_k(b)$  appears undiscounted at the current step, while future values are discounted. The intuition is that the information gain at a node is credited immediately, while the reward that information eventually buys arrives only at the commit action and is discounted by  $\gamma$  raised to the number of steps remaining until then. In the undiscounted case ( $\gamma=1$ ), the immediate information gain and the commit reward it later buys are weighted equally at every depth. With  $\gamma < 1$  the epistemic share of an observe branch’s value therefore rises with the length of the branch. This does not make the agent more exploratory overall, because discounting also shrinks the whole observe branch relative to committing now, and the measured

effect runs the other way, with EFE’s observation count on Diagnosis falling from 9.68 at  $\gamma=1$  to 5.83 at  $\gamma=0.95$  (Table 24). Appendix K compares EFE with Planning, the reward-only planner of the same horizon, across discount factors, and Section 8 takes up the consequence. On environments with several observation actions EFE’s advantage requires  $\gamma \geq 0.99$ , because heavier discounting truncates the effective horizon.

*Extension to factored observation POMDPs.* Proposition 1 assumes the observe-then-commit structure. We now extend the equivalence to a broader class that includes interleaved observe-act POMDPs.

**Definition 1 (Factored observation POMDP).** *A POMDP is a factored observation POMDP if its state decomposes as  $s = (s_{\text{vis}}, s_{\text{hid}})$  where  $s_{\text{vis}}$  is fully observable and  $s_{\text{hid}}$  is hidden, and the action set partitions into (i) observation actions  $\mathcal{A}_{\text{obs}}$  that produce observations about  $s_{\text{hid}}$  without changing  $s_{\text{hid}}$  (though they may change  $s_{\text{vis}}$ ), (ii) navigation actions  $\mathcal{A}_{\text{nav}}$  that change  $s_{\text{vis}}$  deterministically without changing  $s_{\text{hid}}$  and produce no informative observation, and (iii) exploitation actions  $\mathcal{A}_{\text{exp}}$  that yield reward that may depend on  $s_{\text{hid}}$  and emit no informative observation. Throughout, rewards are not observations. The belief is conditioned on the observation symbol alone, so an exploitation action’s payoff, however informative about  $s_{\text{hid}}$  it would be to an agent that conditioned on it, does not update the belief.*

Observe-then-commit POMDPs are the special case where  $\mathcal{A}_{\text{nav}} = \emptyset$  and  $s_{\text{vis}}$  is trivial. Our RockSample variant [58] is an instance. In RockSample, an agent moves on a grid among rocks whose quality, good or bad, is hidden, checks rocks with a noisy sensor, samples them for a reward that depends on the hidden quality, and has an exit action that ends the episode (Section 5 specifies our variant in full). In the terms of Definition 1,  $s_{\text{vis}}$  is the agent’s grid position together with the record of which rocks have been sampled (both fully observable), and  $s_{\text{hid}}$  is the vector of rock qualities, which our variant holds fixed. Check actions are observation actions, moves are navigation actions, and sample/exit are exploitation actions. In the original, sampling a good rock turns it bad, so the hidden vector is not fixed there.

The extension rests on comparing two ways of updating the belief after an action, one that ignores any change the action makes to the hidden state and one that accounts for it, and it turns on whether the two agree. For an action  $a$  at belief  $b$ , write  $b'_o$  for the observation-only posterior,  $b'_o(s_{\text{hid}}) \propto O(o \mid s_{\text{hid}}, a) b(s_{\text{hid}})$ , and  $b'_{o,T}$  for the joint transition–observation posterior that also propagates the hidden-state dynamics,  $b'_{o,T}(s'_{\text{hid}}) \propto \sum_{s_{\text{hid}}} T_{\text{hid}}(s'_{\text{hid}} \mid s_{\text{hid}}, a) O(o \mid s_{\text{hid}}, a) b(s_{\text{hid}})$ . The observation likelihood may additionally depend on the current visible state (RockSample’s check accuracy depends on the agent’s distance to the rock). Since  $s_{\text{vis}}$  is known exactly at every decision point, we suppress it from the notation and evaluate every likelihood at the realized visible state. Define the *transition–observation coupling term* as the expected divergence between the two,  $\Delta_T(b, a) := \mathbb{E}_o D_{\text{KL}}[b'_{o,T} \parallel b'_o]$ . It measures how much the state-preserving belief update misrepresents the correct one when action  $a$  moves the hidden state.

Two conventions sit behind this definition, one about when the observation is drawn relative to the transition and one about the variable on which the two posteriors are compared. The first fixes the timing inside an action. Throughout this article the observation is generated from the hidden state as it stands before the action’s transition applies, and the dynamics then act, a convention that Example 1 below relies on. The second puts both posteriors on the same variable. A KL divergence needs both of its arguments to be distributions over the same variable, and as written  $b'_o$  is a distribution over the hidden state before the action while  $b'_{o,T}$  is a distribution over the hidden state after it. We therefore regard  $b'_o$  as a distribution over the post-transition variable  $s'_{\text{hid}}$  via the identity coupling  $s'_{\text{hid}} = s_{\text{hid}}$ , which is exactly what the state-preserving assumption asserts, so both posteriors live on the same space.

**Proposition 3.** *In a factored observation POMDP (Definition 1), let  $b$  denote the belief over  $s_{\text{hid}}$ . For any action  $a$  that preserves  $s_{\text{hid}}$  (i.e.,  $a \in \mathcal{A}_{\text{obs}} \cup \mathcal{A}_{\text{nav}}$ ), the transition–observation coupling term vanishes,  $\Delta_T(b, a) = 0$ , and  $\rho_{\text{EFE}}(b, a) = I_a(b)$  for observation actions,  $\rho_{\text{EFE}}(b, a) = 0$  for navigation actions.*

*Proof (Proof sketch).* When action  $a$  preserves  $s_{\text{hid}}$ , the transition on the hidden component is  $T_{\text{hid}}(s'_{\text{hid}} | s_{\text{hid}}, a) = \delta(s'_{\text{hid}} = s_{\text{hid}})$ . The belief update over  $s_{\text{hid}}$  is then  $b'(s_{\text{hid}}) \propto O(o | s_{\text{hid}}, a) b(s_{\text{hid}})$ , depending only on the observation likelihood, identical to the observe-then-commit case. The posterior that would incorporate transitions,  $b'_{o,T}$ , coincides with the observation-only posterior  $b'_o$ , so  $D_{\text{KL}}[b'_{o,T} \| b'_o] = 0$ . For observation actions, EFE reduces to cost minus information gain plus expected continuation, matching the  $\rho$ -POMDP Bellman equation with  $\rho = I_a(b)$ . For navigation actions, the observation is uninformative ( $I_a(b) = 0$ ), giving  $\rho = 0$ .

Proposition 3 extends the formal bridge from observe-then-commit to any POMDP where the hidden state is preserved by information-gathering and navigation actions. Where the observe-then-commit argument had a single run of observations followed by one commit, the agent here may interleave observation, navigation, and exploitation in any order. The planner enumerates action sequences as a tree over belief states, using the recursive tree search of Equation 3 extended with navigation and exploitation branches. At each decision point, the EFE-as- $\rho$  equivalence holds on every branch of that tree made up of observation and navigation actions. Because navigation actions never touch the hidden state, interleaving them with observation actions does not disturb the equivalence, and  $\rho_{\text{EFE}}$  is well defined along any sequence of actions drawn from  $\mathcal{A}_{\text{obs}} \cup \mathcal{A}_{\text{nav}}$ , regardless of how many such actions precede a given decision.

Exploitation actions are not covered by Proposition 3, and they carry no belief-dependent utility term for two separate reasons, depending on whether they end the episode. An exploitation action that ends the episode, such as exit, carries  $\rho_{\text{EFE}} = 0$  for the reason Proposition 1 gives for commit actions, that termination leaves no successor belief to score. An exploitation action that does not end the episode, such as sampling a rock, yields a reward that depends on

the hidden state. That its reward already enters the objective is not by itself a reason for it to carry no epistemic term, because a payoff that depends on  $s_{\text{hid}}$  can reveal  $s_{\text{hid}}$ . The reason is instead an assumption of the formal model that the benchmark shares. Rewards are not observations. The belief is conditioned on the observation symbol alone, a non-terminal exploitation action emits an uninformative observation, and its payoff is never fed back into the belief, so its expected information gain under the model’s observation channel is zero and  $\rho_{\text{EFE}} = 0$  follows from the definition of  $I_a(b)$  rather than from a separate stipulation. In RockSample the implemented agents mark a sampled rock as sampled, which removes its quality from every later decision because re-sampling earns nothing, and they do not infer the quality from the sampling payoff (`rho_aif/agents/rocksample_agents.py`). A model in which the payoff of an exploitation action is itself observed and conditioned on would give that action a positive information gain, and the equivalence would then have to include an epistemic term for it. We do not treat that case. Whatever a non-terminal exploitation action does to the visible state is handled at the next decision point, where replanning starts from the resulting belief and recomputes  $\rho_{\text{EFE}}$  fresh. The coupling term  $\Delta_T$  only reenters if an action nominally classified as observation or navigation in fact changes  $s_{\text{hid}}$ , which Definition 1 rules out by construction rather than by approximation.

The extension therefore covers RockSample’s check and navigation actions. Beyond that benchmark, it covers mobile sensor placement and sequential testing with spatial access costs, domains where the agent must navigate to observation locations before gathering information. Section 7.4 exercises this extension empirically across four RockSample instances, and empirical agreement exercises the algebraic result but does not prove it.

The factored observation structure is common in practice whenever the quantity being measured is static or slow-changing relative to the decision horizon (Table 2). The key requirement, that  $T_{\text{hid}}(s'_{\text{hid}} | s_{\text{hid}}, a) = \delta(s'_{\text{hid}} = s_{\text{hid}})$  for observation and navigation actions, breaks when information-gathering itself alters the hidden state. In such settings the coupling term may be nonzero,  $\Delta_T \neq 0$  at some beliefs, and the identity  $\rho_{\text{EFE}}(b, a) = I_a(b)$  behind the information-unit-weight equivalence (the EFE-as- $\rho$  equivalence at  $w=1$ ) then fails in general. We discuss this further in the limitations paragraph of Section 8. The following example makes precise what fails and how badly.

*Example 1 (Destructive sensing can make naive information gain actively harmful).* Let  $s \in \{0, 1\}$  with uniform prior  $b = (0.5, 0.5)$  and the two-state commit reward  $r(\text{Commit}_i, s) = +1$  if  $i = s$ ,  $-1$  otherwise (the Proposition 2 testbed). Add one destructive test action  $a_D$  with cost  $c > 0$  that perfectly reveals the pre-transition state,  $O(o=s | s, a_D) = 1$ , but always leaves the hidden state depleted afterward,  $T(s'=0 | s, a_D) = 1$  for every  $s$ . Drilling a core sample destroys whatever ore was there, rich or poor.

The state-preserving assumption behind Propositions 1–3 treats  $a_D$  as if it left  $s$  untouched, giving the naive posterior  $b'_o \propto O(o | s) b(s)$  (a point mass at

the revealed value  $o$ ) and the naive value

$$V_{\text{naive}}(a_D) = -c + w I(b) + \mathbb{E}_o[\max_i r(\text{Commit}_i, o)] = -c + w \cdot 1 \text{ bit} + 1,$$

which at  $w=1$  is  $2 - c$ , exceeding  $V_{\text{naive}}(\text{Commit}_i) = \mathbb{E}_b[r(\text{Commit}_i, s)] = 0$  for any  $c < 2$ .

The correct joint transition–observation posterior,  $b'_{o,T}(s') \propto \sum_s T(s' | s, a_D) O(o | s, a_D) b(s)$ , evaluates to  $b'_{o,T} = (1, 0)$  for either observed outcome. The observation describes ore that drilling has already destroyed, so the correct post-action belief is certain the state is now 0 no matter what was observed. The coupling term is not merely nonzero but maximal here. The posteriors  $b'_{o,T}$  and  $b'_o$  are disjoint point masses whenever  $o=1$ , so  $D_{\text{KL}}[b'_{o,T} \| b'_o] = \infty$ . The correct Bellman value is

$$V_{\text{true}}(a_D) = -c + \max_i r(\text{Commit}_i, 0) = 1 - c,$$

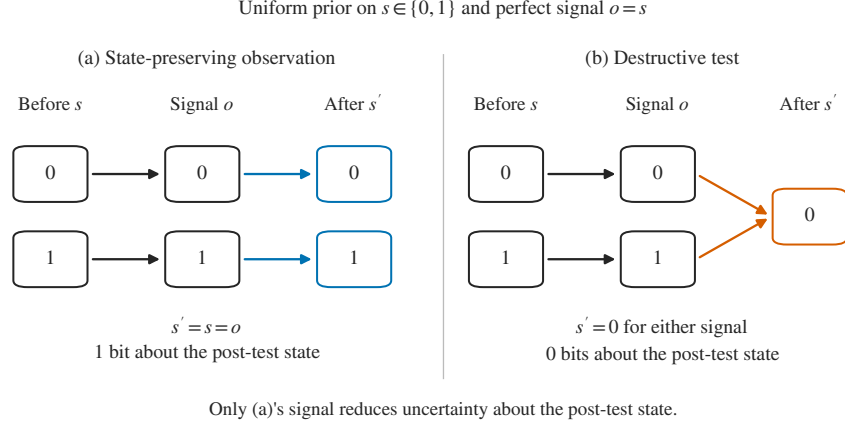
exactly one bit of illusory credit below the naive estimate, for both outcomes.

A transition-aware EFE evaluation, one that uses the correct joint posterior  $b'_{o,T}$  above, assigns the drill zero epistemic value about the post-drill state and recovers  $V_{\text{true}}$ .

For  $c \in (1, 2)$  the two evaluations disagree in sign.  $V_{\text{naive}}(a_D) = 2 - c > 0 = V_{\text{naive}}(\text{Commit}_i)$  recommends drilling, while  $V_{\text{true}}(a_D) = 1 - c < 0 = V_{\text{true}}(\text{Commit}_i)$  recommends committing immediately. That is a full ranking flip from a two-state, one-step example. The failure is worse than a mispriced tradeoff. An agent that drills and then acts on its now-obsolete observation realizes expected reward  $-c + 0.5 r(\text{Commit}_0, 0) + 0.5 r(\text{Commit}_1, 0) = -c$  in the true environment, strictly below the immediate-commit value of 0 for every  $c > 0$ , not only within the sign-flip range. Verified numerically in `tests/test_destructive_boundary.py`.

Figure 4 isolates the timing distinction behind Example 1.

*Remark 1 (The scope restriction is on the reduction, not on what EFE can price).* Example 1 shows a failure of the factored  $\rho$ -POMDP reduction, not a failure of Expected Free Energy. For any action  $a$  that violates Definition 1’s state-preserving hypothesis ( $\Delta_T(b, a) \neq 0$ ), replacing the naive epistemic term  $I_a(b) = \mathbb{E}_{o|a} D_{\text{KL}}[Q(s_{\text{hid}} | o, a) \| Q(s_{\text{hid}} | a)]$ , computed from the pre-transition posterior  $b'_o$ , with the transition-aware term computed from the correct joint posterior  $b'_{o,T}(s') \propto \sum_s T(s' | s, a) O(o | s, a) b(s)$ , under this article’s observe-then-transition timing convention where the observation is emitted from the pre-transition hidden state and the dynamics then act, removes the illusory information credit. In Example 1 the post-transition state is degenerate, the transition-aware epistemic term vanishes, and the substitution recovers the reward-only Bellman value  $V_{\text{true}}(a_D)$  exactly. In general it restores the correct belief dynamics while still crediting whatever genuine post-transition information the action carries, so it prices destructive sensing correctly rather than reducing to reward-only planning. Propositions 1–3 therefore delimit which  $\rho$ -POMDPs the factored reduction targets (those with  $\Delta_T \equiv 0$  on information-gathering



**Fig. 4.** An observation can describe a state that no longer exists. These schematic timelines use a perfect sensor to contrast state-preserving sensing with the destructive test of Example 1. The sensor reports the pre-action state in both cases. Under preservation that report identifies the state used by the next decision. Under destruction both possible initial states become 0, so the correct post-action belief is  $(1, 0)$  for either report. Treating a report of 1 as the current state gives an obsolete posterior. The example limits the state-preserving reduction, not transition-aware EFE.

actions), not which decision problems Expected Free Energy can correctly evaluate. EFE’s epistemic term is defined over post-transition beliefs by construction, so it prices destructive sensing correctly once the epistemic term is recomputed from transition-aware predictive and posterior distributions as above. The coupling term  $\Delta_T$  itself is a diagnostic of when the factored hypothesis fails, and we have not shown it to be an additive correction to the EFE value or a bound on decision error. We leave a transition-aware treatment of the budgeted machinery of Section 4, and any perturbation bound in terms of  $\Delta_T$ , as future work (Section 8).

### 3.4 Agents

With the objective now fixed, this subsection specifies the agents that the experiments compare and how the one free parameter among them, the information weight  $w$ , is set. We compare six core agents (Table 3), all built on the same exact Bayesian belief update over a known generative model and differing only in objective function and planning depth. Myopic and Planning plan on reward alone, at horizons  $H=1$  and  $H > 1$ . Info Gain and Planning+IG add a weighted information-gain bonus at those same two horizons. EFE is the agent of Proposition 1, and Epistemic-only is its ablation. The Planning and Planning+IG baselines use the same recursive tree search as the EFE agent, isolating the effect of

**Table 2.** Taxonomy of real-world POMDPs by factored observation structure. Factored settings preserve the hidden state under observation and navigation actions, while non-factored settings do not.

| Factored ( $\Delta_T=0$ ) | Non-factored ( $\Delta_T \neq 0$ ) |
|---------------------------|------------------------------------|
| Non-destructive testing   | Destructive testing (drilling)     |
| Medical imaging (CT, MRI) | Biopsy / tissue sampling           |
| Structural inspection     | Active interventions               |
| Environmental monitoring  | Predator-prey (target moves)       |
| Security screening        | Chemical testing (consumes sample) |
| Mineral exploration       | Quantum measurement                |
| Mobile sensor networks    | Adversarial surveillance           |

the  $\rho$  function from planning depth. The paragraphs below take the critical baseline, the ablation, and the weakest baseline in turn, then the baselines outside this family, and finally the tuning of  $w$ . Every  $H > 1$  agent is receding-horizon. At each real decision point it recomputes the full depth- $H$  tree from the current belief and executes only the resulting root action, replanning from scratch at the next step rather than committing to a fixed open-loop  $H$ -step policy.

**Table 3.** Agent specifications. All use exact Bayesian belief updates over a known generative model.

| Agent          | $\rho$ function  | Horizon | Controls for                |
|----------------|------------------|---------|-----------------------------|
| Myopic         | $\rho = 0$       | $H=1$   | Weakest baseline            |
| Planning       | $\rho = 0$       | $H > 1$ | Planning depth              |
| Info Gain      | $w \cdot I(b)$   | $H=1$   | Epistemic bonus (myopic)    |
| Planning+IG    | $w \cdot I(b)$   | $H > 1$ | IG + planning depth         |
| EFE            | $I_a(b)$ via EFE | $H > 1$ | Joint objective (Prop. 1)   |
| Epistemic-only | $I_a(b)$ only    | $H > 1$ | Ablation: no pragmatic term |

Planning+IG is the critical baseline. It uses the same tree search as the EFE agent with an additive information-gain (IG) bonus at the same horizon, meaning that at every node of the tree the value of an observation action is its reward-only value plus  $w$  times the expected information gain  $I_a(b)$  of that observation. Planning+IG is also this article’s instance of a construction that two lines of prior work share, namely POMDPs with information rewards (POMDP-IR) and  $\rho$ -POMDPs with an information-gain reward [2, 59]. Satsangi et al. [51] show that these two formulations are equivalent, so Planning+IG stands in for both. A reader from the active-perception literature should therefore read Planning+IG at a tuned weight as the familiar baseline from that tradition rather than as a

construction specific to this article. By Proposition 1, the EFE agent is exactly Planning+IG with  $w=1$ , so any difference between EFE and Planning+IG at a tuned weight, in either direction, is attributable to the weight choice rather than a different mechanism. That weight-only reading presumes the EFE implementation computes the epistemic term it claims to, so the EFE computation itself is checked for consistency against `pymdp` [26], a reference Python library for active inference (Appendix H). This is a qualitative agreement check on the epistemic term, not a certified numerical equivalence.

The Epistemic-only agent is the ablation that removes the pragmatic term. It sets  $\mathcal{G}(\text{commit}) = 0$ , removing reward awareness from the decision of when to commit, so that the comparison between observing and committing sees only observation costs and information gain. Once that comparison favors committing, the agent takes the commit action for the state its belief rates most probable, the belief-argmax commit, which is how it can still succeed once its belief is sharp. Success, for an observe-then-commit environment, is the fraction of episodes in which the agent’s commit is correct. On Tiger, Diagnosis, Bandit, and Tileworld no observation’s information gain alone clears its cost, so the agent commits immediately from its prior and succeeds at the chance rate, the accuracy the prior alone affords (Section 7, per-agent numbers in Appendix B). Testbed is the exception, since its observation is cheap enough that information gain alone clears the cost, and there the ablation matches reward-only Planning exactly (Appendix C). The pragmatic term is what makes commit timing reward-aware, and the four chance-level environments show what its removal costs.

Four of the remaining baselines need a brief description here. Myopic ( $\rho=0$ ,  $H=1$ ) has no epistemic term at all, yet it observes on many environments. It does so because its one-step lookahead compares committing now against observing once and then committing, so it can see when a commit made after an observation is worth more than one made now. That difference is the instrumental value of information (VoI) of the observation, and Myopic observes for that reason rather than because of an epistemic bonus. The Greedy baseline provides a no-information lower bound in RockSample and Structural Inspection, both specified in Section 5. In RockSample it moves to rocks and samples them without first checking whether they are good. In Structural Inspection it declares a diagnosis for each component without running any test, so its declarations rest on the prior alone. We also include an observe-then-commit baseline built on information-directed sampling (IDS) [48], which picks the observation action that minimizes its information ratio, the squared expected regret of taking that action divided by the information the action yields about the optimal commit (results in Appendix N). The fourth baseline warrants a naming note. The posterior-vote baseline draws 100 states from the belief, finds for each sampled state the action that would be optimal if that state were the true state, and takes the majority vote over those actions. As the sample count grows, this vote approaches the most-likely-state rule, which acts as if the belief’s most probable state were the true state. It is not Thompson sampling, and we do not claim it as such. Thompson sampling draws a single state and acts optimally for that

one draw, so the frequency with which it takes each action matches the belief’s probability that the action is optimal, whereas the hundred-sample majority vote collapses toward a single action. The committed CSVs record it under the legacy label `ThompsonSamplingAgent`, retained so the artifacts stay stable.

Two of the six core agents carry a weight  $w$  that must be set. They are Info Gain and Planning+IG (Table 3). We report two tuning criteria for that weight, a success-maximizing criterion used in the main tables and a reward-maximizing criterion used in the Pareto analysis. In the main tables, the tuned weight for Info Gain and Planning+IG is the success-maximizing weight chosen by grid search over  $\{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$ . It is written  $w_{\text{succ}}^*$  in the results tables and  $w^*$  where the criterion is clear from context. Each grid point is evaluated on 200 episodes, except on *Tileworld*, where 100 episodes are used because its computational cost per episode is an order of magnitude higher than that of the other observe-then-commit environments. Every grid point is evaluated on one common tuning stream, seed 7, disjoint from the five evaluation seeds and reset for each candidate so that candidates face the same episodes, and ties go to the smallest weight reaching the maximum (`tune_info_gain_weight` in `experiments/run_experiment.py`). Each tuned weight is therefore a selection on one stream of 200 or 100 episodes and carries that stream’s sampling noise. Seed 7 is also the first held-out seed of Section 6.9, which evaluates only the calibrated family and never a tuned baseline. This tuned baseline weight is a different quantity from the shadow price  $w^*(B)$  of Section 4 and from the thresholds of Proposition 2, which share the star notation. The Pareto analysis (Section 7.2), which traces mean reward against success as  $w$  varies, sweeps a wider eleven-weight grid, listed in that section, and reports the reward-maximizing weight  $w_{\text{ret}}^*$ . The two criteria can select different weights, and both are reported so that a reader can see which one a comparison uses. Section 4 then offers a third way to fix the weight, reading it off a sensing budget as the bracket of adjacent grid weights between which the agent’s induced sensing usage crosses that budget.

## 4 Budgeted $\rho$ -POMDPs and the Price of Information

Section 3 fixes the information-unit weight at  $w=1$  and Section 7 shows this default is empirically competitive under a shared reward convention, the common scale at which rewards and observation costs are read. The competitiveness of the default is tied to that scale, since a fixed  $w=1$  implements a different pragmatic–epistemic trade-off at every reward scale (Section 3.3). Rather than asking the designer for a weight  $w$  and tuning it per task, we ask for a *sensing budget*, a target for the agent’s expected sensing usage over episodes, and derive from that budget the weight that delivers it. The budget is the design input and the weight is the output. That weight is an operational shadow price, computable offline and trackable online when the reward scale shifts mid-run. One caveat delimits the formulation’s scope. When the true economic cost of each sensing action is already known in the same units as the task reward, that cost belongs directly in  $R$  and no budget is needed. The budgeted formulation is for the common case

where the reward convention is arbitrary or the acceptable level of sensing is most naturally stated operationally rather than priced.

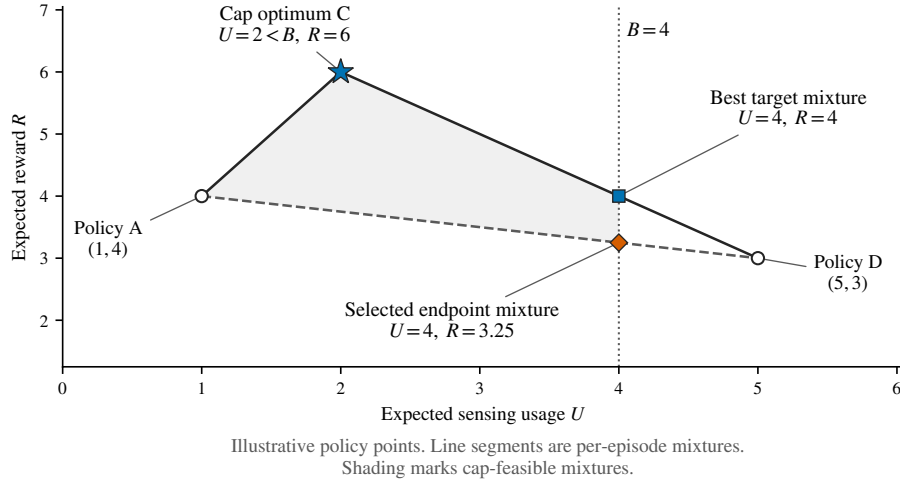
#### 4.1 The Budgeted Problem and the Usage Curve

Let  $\pi$  range over a fixed policy class  $\Pi$  (the receding-horizon Planning+IG family of Section 3), and write  $R(\pi)$  for expected episodic return and  $U(\pi)$  for expected sensing usage (observation/test count, or cumulative sensing cost). Two problems share a stated budget  $B$ . The calibration problem, which the rest of this article solves in practice, asks within the fixed family for the weight, or the per-episode mixture of two weights, whose expected usage equals  $B$ . The constrained problem asks for the most reward attainable under the usage cap  $B$ , over the same family here and over all policies for the near-optimal references of Section 6, and it is the reference against which calibration is measured (Section 6.9). The *budgeted  $\rho$ -POMDP problem* is

$$\max_{\pi} R(\pi) \quad \text{subject to} \quad U(\pi) \leq B. \quad (6)$$

The literal Lagrangian for (6), with  $\lambda \geq 0$  the multiplier on the usage cap, penalizes usage,  $R(\pi) - \lambda U(\pi)$ . Planning+IG instead maximizes  $R(\pi) + w \mathbb{E}_{\pi}[I]$ , with  $I$  the information gain accumulated over a trajectory and  $\mathbb{E}_{\pi}[I]$  its expectation over the trajectories that  $\pi$  generates, the cumulative expected information gain of  $\pi$ . We write this expectation  $I(\pi)$  from here on, so the objective reads  $R(\pi) + w I(\pi)$ . To see what kind of multiplier  $w$  is, consider a different constrained problem, which we call the floor problem. It maximizes return subject to collecting at least a given amount of information, a lower bound on  $I(\pi)$ . Dropping that lower bound and instead rewarding information in the objective at the fixed rate  $w$  yields  $R(\pi) + w I(\pi)$  again, so the Planning+IG objective is the Lagrangian relaxation of the floor problem. We therefore call this objective an *information-floor* dual. For a fixed environment and agent family, write  $U(w) := U(\pi_w^*)$  for the usage induced by the Planning+IG-optimal policy at weight  $w$ . This map from weight to induced usage is the usage curve that the Introduction sketched. Locating the weight at which  $U(w)$  crosses  $B$ , in the sense Definition 2 makes precise, gives the operational shadow price  $w^*(B)$ , the weight at which the family's sensing passes from below  $B$  to at or above it.

Two statements delimit precisely what  $w$  prices. First,  $w$  is an exact Lagrange multiplier of an information floor rather than of the usage cap in (6). If a policy  $\pi_w$  maximizes the weighted objective  $R(\pi) + w I(\pi)$  exactly over  $\Pi$ , for some  $w \geq 0$ , then  $\pi_w$  solves  $\max_{\pi \in \Pi} R(\pi)$  subject to  $I(\pi) \geq I(\pi_w)$ . The argument is two lines. Call  $\pi$  feasible for this floor problem when  $I(\pi) \geq I(\pi_w)$ . Any feasible  $\pi$  satisfies  $R(\pi) \leq R(\pi_w) + w (I(\pi_w) - I(\pi)) \leq R(\pi_w)$ , where the first inequality rearranges the optimality of  $\pi_w$  for the weighted objective and the second holds because feasibility makes the bracketed term nonpositive and  $w \geq 0$  keeps the product nonpositive. The if-clause of this statement is an idealization. It describes an exact one-shot maximizer  $\pi_w$  over a static policy class, the setting of Proposition 5 below, whereas the  $\pi_w^*$  of the usage curve is the implemented



**Fig. 5.** An expected-usage target and a usage cap answer different questions. This schematic uses three illustrative policies and their per-episode mixtures, not benchmark data. The target  $U = B$  is the vertical line, while the shaded portion of the mixture hull satisfies the cap  $U \leq B$ . Suppose increasing grid weights select policies C, A, then D, so A and D form the selected crossing pair. A selected endpoint mixture can meet the target yet earn less than another mixture at the same usage. The highest-return cap-feasible policy can also leave part of the budget unused. These possibilities explain why calibration alone supplies neither an equality-constrained nor a cap-constrained reward optimum. They do not assert that every instance has either shortfall.

receding-horizon agent, which replans at every step. Section 4.3 takes up the distinction.

Second, the operational shadow price  $w^*(B)$  is a calibration of this policy family to a usage budget, not the Lagrange multiplier  $\lambda^*(B)$  of problem (6) itself. We do not claim that maximizing  $R + wI$  solves (6) optimally, only that solving  $U(w) = B$  selects, within the Planning+IG family, the member whose induced usage meets the budget. The two notions align in practice because usage is the vehicle by which this family acquires information, but they are not formally identified. Section 4.6 states the resulting limitation against exact constrained-POMDP solvers.

Figure 5 separates meeting an expected-usage target from maximizing return under a cap, and also separates a selected endpoint mixture from the best mixture at that target.

Before the formal definition, four practical facts shape its form. First, the usage curve  $U(w)$  is not known in closed form. It is estimated by simulation on a finite, increasing, log-spaced grid of weights (Section 4.4 gives the grid and the episode counts). The true curve, the population quantity  $U(w)$  that unlimited episodes would reveal, is therefore available only through its grid estimate. The

words grid weight and grid point below refer to that grid. Second, the curve is a step function, noisy when estimated, because a small change in  $w$  either leaves the agent’s discrete policy unchanged or switches it, and a switch can move usage by a whole observation at once. We call a jump of this kind a usage gap, a range of usage values that no single weight attains. A budget  $B$  inside a gap has no weight with  $U(w) = B$ . Even when the curve does cross  $B$ , the weight at which it does so lies between two neighboring grid weights, so it is known only to grid resolution. The equation  $U(w) = B$  is accordingly solved as a bracket between two neighboring grid weights, one whose usage falls below  $B$  and the next whose usage reaches it, rather than as a single number. Third, usage is not monotone in  $w$ , a fact the definition records and Section 4.3 explains, so the curve can cross  $B$  more than once. In that case the bracket reports the crossing at the largest weight, the one beyond which no swept grid weight falls back below  $B$ . Finally, the family attains only a bounded range of usages, from the smallest usage any weight induces to the largest, so the definition also speaks of the endpoints of that attainable range, not only of the endpoints of the bracket. The largest attainable usage is bounded by belief saturation (once the belief is nearly certain, a further observation yields almost no information gain, so even a large weight stops buying observations) and by the episode step cap. With these facts in place, the definition below fixes the convention.

**Definition 2 (Operational shadow price, crossing bracket, and end-point mixture).** *Fix the usage curve  $U(w)$  of the implemented family on  $w \geq 0$  and a budget  $B$ . Four objects are involved, and they are distinct. First, the level set  $U^{-1}(B) = \{w \geq 0 : U(w) = B\}$  is the set of weights at which usage equals the budget exactly. It is empty when  $B$  falls inside a usage gap, and it can be a nondegenerate set when  $U$  is flat at level  $B$ . Second, for a budget that the curve crosses from below, the crossing threshold  $w^\dagger(B)$  is the single weight at which the curve last passes from below  $B$  to at or above  $B$ , namely the supremum of the weights  $w$  with  $U(w) < B$  that precede a weight with usage at or above  $B$ . It is undefined for a budget the curve never crosses from below, whether because the curve sits at or above  $B$  at every weight, as Tiger’s does at  $B=4$  in Table 4, or because it never reaches  $B$ , and the solver then reports a non-straddling fall-back flagged as unbracketed. Under the local non-monotonicity documented in Section 4.3 the curve can cross  $B$  more than once, and taking the last crossing is a convention adopted here, not a property of the problem. At a jump of the curve the threshold is a single number, the location of the jump, even though the level set is empty there. The operational shadow price of this article is this threshold,  $w^*(B) := w^\dagger(B)$ , defined wherever the threshold is. Third, the crossing bracket  $(w_{\text{lo}}, w_{\text{hi}}]$  is the finite-grid estimate of the threshold that the released solver reports (`rho_aif/budget.py`). Its lower endpoint is the largest grid weight with  $U(w_{\text{lo}}) < B$  and its upper endpoint is the first subsequent grid weight with  $U(w_{\text{hi}}) \geq B$ , so the bracket straddles the budget on the grid whenever a grid point at or above  $B$  follows the last grid weight with usage below  $B$ , and its closed enclosure  $[w_{\text{lo}}, w_{\text{hi}}]$  contains the threshold whenever the curve has no crossing of  $B$  that the grid fails to resolve, between grid weights or beyond the last one (a curve*

whose final grid point falls back below  $B$  would instead yield a non-straddling fallback flagged as such even though its threshold can exist, a shape no swept curve exhibits). The enclosure is stated closed because the threshold can coincide with  $w_{lo}$  itself, when the curve rises immediately after that grid weight, and the half-open reported bracket excludes that one point by construction. The width of the bracket is grid resolution. The weights inside it do not each attain the budget, and the bracket is not a set of exact prices. Offline solving additionally accepts the nearest grid point as achieving the budget when  $|U(w) - B| \leq \max(0.3, 0.1B)$ , a numerical tolerance that affects achievability calls only, never the bracket. Fourth, the endpoint mixture is the randomized policy that, when  $B$  falls inside a usage gap so that no single weight attains it, plays the policy at  $w_{hi}$  with probability  $q = (B - U(w_{lo})) / (U(w_{hi}) - U(w_{lo}))$  and the policy at  $w_{lo}$  otherwise, choosing afresh each episode. By linearity its expected usage is exactly  $B$ . This is one randomized policy attaining the budget in expectation, not an interval of scalar prices, and it mirrors the classical constrained-MDP fact that optimal budget-constrained policies may need to be randomized [1, 30].

A budget is attainable by some per-episode mixture of two members of the family iff  $\inf_{w \geq 0} U(w) \leq B \leq \sup_{w \geq 0} U(w)$ , up to the closure of the attainable usage set, since mixing two members whose usages straddle  $B$  attains it in expectation. The endpoint mixture of the fourth object attains it only when the budget is straddled by a last crossing from below, that is, when the solver returns a straddling bracket, which holds at every budget for which this article reports a mixture, while the unbracketed cases (Tiger at  $B=4$  in Table 4 and the declared-unattainable budgets of Section 6.9) are reported as such. A curve that only descends through  $B$ , with usage 2 below some weight and 1 above it and  $B=1.5$ , is attainable by mixing the two plateaus but has no crossing from below, so its crossing bracket and endpoint mixture are undefined and the solver flags the fallback. The lower endpoint need not be  $U(0)$ . Usage is not monotone in  $w$ , and the implemented family can spend less at a positive weight than at  $w=0$  (on Tileworld  $6 \times 6$ , reward-only Planning at  $w=0$  scans 15.64 times per episode while  $w=1$  scans 14.75, Section 7.3). Within the suite,  $U(0) > 0$  reflects instrumental sensing (observing pays for itself on reward alone at some instances) and the upper endpoint reflects belief saturation and the episode step cap.

## 4.2 Exact Scale Equivariance

The first structural result says that rescaling every reward and cost by a common positive factor is undone, as far as the agent’s choices go, by rescaling the weight by the same factor. The intuition is one line. Multiplying every reward and cost by  $\alpha$  multiplies the pragmatic part of every plan’s value by  $\alpha$ , while the information part carries no reward units and is untouched, so multiplying  $w$  by  $\alpha$  as well restores the original ranking of plans exactly. Let the  $\alpha$ -rescaled environment multiply every pragmatic quantity (commit rewards, penalties, observation costs) by  $\alpha > 0$ , leaving observation models and belief updates untouched, so information gain is unchanged. Assume argmax ties are broken by a fixed, scale-independent rule. The implementation’s tie-break satisfies that assumption by

construction. It compares candidates by strict improvement, favoring the incumbent commit action, with a tolerance equal to  $10^{-12}$  times the larger magnitude of the two values being compared (`rho_aif/agents/planning_infogain.py`, `efe.py`). An  $\alpha$ -rescale multiplies both sides of every such comparison by  $\alpha$ , so the rule is homogeneous of degree one and every decision is the same at every scale. An absolute tolerance would not be, since a fixed epsilon corresponds to  $\epsilon/\alpha$  in the unscaled units, and an earlier version of the implementation used one. Two qualifications accompany the proposition. Floating-point rounding is not exactly homogeneous when  $\alpha$  is not a power of two, so agreement across scales in the implementation is exact up to rounding rather than exact in the mathematical sense, and it is checked by measurement rather than assumed. And the proposition speaks of the agent's actions, so it fixes the observation count directly and fixes the cumulative sensing cost only up to the factor  $\alpha$  that the rescale applies to every cost, which is the cost identity it states. The identity of count-usage curves across scales that the proposition implies is the curve collapse, the scaled and unscaled usage curves landing on top of each other once the weight is rescaled with the rewards (the cost-usage curve also needs division by  $\alpha$ ), defined after the proposition and measured in Section 6.1.

**Proposition 4 (Exact scale equivariance).** *For every  $\alpha > 0$  and  $w \geq 0$ , the receding-horizon Planning+IG agent facing the  $\alpha$ -scaled environment with weight  $\alpha w$  selects the same action as the agent facing the unscaled environment with weight  $w$ , at every belief, provided ties are broken by a scale-independent rule. Consequently the closed-loop policies coincide, and so do the induced distributions of every quantity that depends on the actions alone. For usage denominated in observation count this gives*

$$U_{\text{count},\alpha}(\alpha w) = U_{\text{count}}(w), \quad (7)$$

and for usage denominated in cumulative sensing cost, which sums the rescaled costs of the same actions, it gives  $U_{\text{cost},\alpha}(\alpha w) = \alpha U_{\text{cost}}(w)$ .

*Proof.* For any candidate plan, the  $\alpha$ -scaled value is  $\alpha \cdot (\text{pragmatic sum}) + \alpha w \cdot (\text{IG sum}) = \alpha \cdot [\text{pragmatic sum} + w \cdot \text{IG sum}]$ , i.e. exactly  $\alpha$  times the unscaled value. Multiplying every candidate's value by the same positive constant preserves the ordering of the candidates, and the scale-independent rule compares values and a tolerance that have both been multiplied by  $\alpha$ , so it returns the same action. By induction over decision points, the closed-loop policies agree at every belief. Since the environment's stochastic dynamics are otherwise identical, the induced trajectory distributions agree, and with them the distribution of the observation count. Each action's cost is  $\alpha$  times its unscaled cost, so the cumulative sensing cost of every trajectory is  $\alpha$  times its unscaled value, which gives the cost identity.

This is stronger than an empirical scale-invariance check. It holds for the implemented receding-horizon agent, not only for the exact one-shot maximizers of Proposition 5 below, subject to the rounding qualification stated before Proposition 4. Three corollaries follow immediately. First, curve collapse is a theorem

for count usage. By equation (7), the scaled count-usage curve, plotted against  $w/\alpha$  rather than against  $w$ , coincides with the unscaled curve at every scale, so a measured deviation across scales can only be Monte Carlo noise or floating-point rounding, and Section 6.1 finds neither at the scales swept. The cost-usage curve collapses only after it is also divided by  $\alpha$ . Second, the shadow price and its bracket rescale together. For a count budget,  $w_{\text{count}}^*(B; \alpha) = \alpha \cdot w_{\text{count}}^*(B; 1)$  with no grid condition, including at gap budgets, and the reported bracket at scale  $\alpha$  is the bracket at scale 1 multiplied by  $\alpha$  when the search grid at scale  $\alpha$  is the base grid multiplied by  $\alpha$ , as it is in Section 6.1. For a cost budget held at a fixed number  $B$  while the costs are rescaled, the relation is  $w_{\text{cost}}^*(B; \alpha) = \alpha \cdot w_{\text{cost}}^*(B/\alpha; 1)$ , the bracket following on a grid rescaled the same way, and the count form is recovered only if the cost budget is rescaled to  $\alpha B$  along with the costs. Third, a fixed  $w=1$  at scale  $\alpha$  is behaviorally identical to  $w = 1/\alpha$  at scale 1. The sensing that the untuned  $w=1$  agent happens to spend at reward scale  $\alpha$ , which we call the *implicit EFE budget*, therefore drifts with scale as  $B_{\text{EFE}}(\alpha) := U(1/\alpha)$ . This is the precise sense in which the canonical weight of Proposition 1 is not scale invariant.

### 4.3 Monotone Comparative Statics and the Usage Staircase

Exact scale equivariance says nothing about whether  $U(w)$  is monotone. Definition 2 deliberately reports a crossing bracket instead of the level set  $U^{-1}(B)$ , because the level set can be empty, when  $B$  falls inside a usage gap, and because when  $U$  fails to be monotone it can contain several weights. Comparative statics, in the economist’s sense, asks how an optimizer’s chosen plan moves when a parameter of its objective moves, and the parameter here is the weight  $w$ . This subsection sorts out what monotonicity the theory does and does not guarantee, in three parts. The first part establishes the one general monotone property that does hold. For an exact maximizer over a fixed plan set, raising  $w$  can never lower the information the chosen plan collects and can never raise the return it earns. The second part explains why that property does not transfer to the curve the experiments plot. That curve is the grid estimate of the count-usage curve  $U(w)$ , usage being denominated here in observation count rather than in cumulative sensing cost (Section 4.4 takes up the other denomination). The curve is a step function (Section 4.1), and we call it the usage staircase. The third part shows that in the single observe-or-commit decision, where the staircase has at most one step, any step it has lands exactly at the closed-form onset threshold of Proposition 2. Throughout,  $R(\pi)$  and  $I(\pi)$  are the expected return and cumulative expected information gain of Section 4.1, and a plan  $\pi$  is a fixed course of action evaluated by those two quantities. The proposition takes  $\Pi$  to be any fixed set of such plans, the candidates among which the maximizer chooses once. That is a wider use of the symbol than in Section 4.1, where  $\Pi$  named the Planning+IG policy class, and the paragraph after the proof says what the difference costs when the result is carried to the implemented re-planning agent.

**Proposition 5 (Monotone comparative statics for exact maximizers).**  
*Let  $\Pi$  be any fixed set of plans and, for  $w \geq 0$ , let  $\pi_w \in \arg \max_{\pi \in \Pi} R(\pi) +$*

$w I(\pi)$ . Then for  $w_2 > w_1 \geq 0$ ,

$$I(\pi_{w_2}) \geq I(\pi_{w_1}) \quad \text{and} \quad R(\pi_{w_2}) \leq R(\pi_{w_1}). \quad (8)$$

*Proof.* Optimality of  $\pi_{w_1}$  at  $w_1$  gives  $R(\pi_{w_1}) + w_1 I(\pi_{w_1}) \geq R(\pi_{w_2}) + w_1 I(\pi_{w_2})$ , and optimality of  $\pi_{w_2}$  at  $w_2$  gives  $R(\pi_{w_2}) + w_2 I(\pi_{w_2}) \geq R(\pi_{w_1}) + w_2 I(\pi_{w_1})$ . Adding the two inequalities and simplifying yields  $(w_2 - w_1)(I(\pi_{w_2}) - I(\pi_{w_1})) \geq 0$ , and since  $w_2 > w_1$ ,  $I(\pi_{w_2}) \geq I(\pi_{w_1})$ . Substituting this back into the first inequality gives  $R(\pi_{w_1}) - R(\pi_{w_2}) \geq w_1(I(\pi_{w_2}) - I(\pi_{w_1})) \geq 0$ .

The proposition is exact, but two obstacles separate it from clean monotonicity of the usage staircase. Clean monotonicity of the staircase would mean that every step goes up, so that a higher weight never buys fewer observations. The first obstacle is a mismatch of quantities. The monotone quantity in Proposition 5 is cumulative expected information gain, not the observation count, and the two can locally disagree. At a higher weight the maximizer may switch to a plan with fewer but individually more informative observations, so information gain does not fall, as the proposition guarantees, while the observation count that the staircase plots falls. The second obstacle is a mismatch of decision problems. The proposition concerns a single maximization over one fixed plan set  $\Pi$ , whereas the implemented receding-horizon agent re-plans at every step from the belief it has reached, so the maximization it solves is rooted at a different,  $w$ -dependent belief at every step. Proposition 5 therefore applies to the argmax at each individual decision, but the sequence of decisions along an episode need not inherit its monotonicity. Trajectory-level usage can dip as  $w$  rises. The consequence of these two obstacles is visible empirically. The usage staircases on Bandit and Tileworld are only roughly monotone (Section 6.6).

At the opposite extreme from a long receding-horizon episode sits the single observe-or-commit decision, and there the staircase is characterized exactly. Recall that Part (a) of Proposition 2 gave a closed-form weight above which a two-state agent with planning horizon  $H=1$  observes once rather than committing immediately. In that setting the usage curve is a single step from 0 to 1. We call a weight at which the staircase steps up a knot, so the first knot is the onset of any sensing at all. The corollary and the remainder of the article write  $w_{\text{thresh}}$ , without the star, for the threshold that Proposition 2 denotes  $w_{\text{thresh}}^*$ . The two symbols name the same quantity, and the star is dropped so that it does not collide with the starred symbol this section is about, the shadow price  $w^*(B)$  of Definition 2. The corollary states that the onset knot and the closed-form threshold coincide.

**Corollary 1 (Proposition 2’s onset thresholds are usage-staircase knots).**

*In the two-state  $H=1$  setting of Proposition 2 with  $w_{\text{thresh}} \geq 0$ , equivalently  $c \geq (p - \frac{1}{2})(R^+ - R^-)$ , for any budget  $B \in (0, 1)$  the crossing threshold of Definition 2 is exactly  $w_{\text{thresh}}$ , the weight at which the staircase leaves zero, and the closed enclosure  $[w_{\text{lo}}, w_{\text{hi}}]$  of a finite-grid crossing bracket contains that threshold whenever the sampled usages straddle  $B$ , locating it to grid resolution rather than equalling it. When  $w_{\text{thresh}} < 0$  every nonnegative weight already observes,*

so  $U \equiv 1$  on  $[0, \infty)$  and no onset knot lies in the nonnegative range. The onset boundary of the budgeted problem recovers Proposition 2’s closed form as the first knot of the usage staircase.

*Proof.* At  $H=1$  the only two plans are commit now and observe once then commit, and Proposition 2 establishes the observe plan is preferred iff  $w > w_{\text{thresh}}$ . Consequently  $U(w) = \mathbb{K}[w > w_{\text{thresh}}]$  exactly for this single observe-or-commit decision, so  $U$  is 0 below  $w_{\text{thresh}}$  and 1 above it. For any  $B \in (0, 1)$  the supremum of the weights with  $U(w) < B$  that precede a weight with  $U(w) \geq B$  is therefore  $w_{\text{thresh}}$ , which is the stated crossing threshold, and a grid whose sampled usages straddle  $B$  returns a bracket whose closed enclosure contains  $w_{\text{thresh}}$ . The claim is about this  $H=1$  decision, not about the receding-horizon usage curves of Section 6, which take many values and need not be monotone.

The corollary is verified numerically (`TestProp2OnsetExact` in `tests/test_budget.py`). The check runs the implemented  $H=1$  planner over whole episodes rather than over the single decision the corollary analyzes. Within an episode that planner re-plans after each observation and is free to observe again before committing, so the test asserts the location of the onset rather than a usage of exactly 1. Usage is exactly 0 at  $0.9w_{\text{thresh}}$  and at least 1 at  $1.2w_{\text{thresh}}$  on a positive-threshold Testbed with observation accuracy  $p=0.6$ , observation cost  $c=0.3$ , and correct and incorrect commit rewards  $R = \pm 1$ , whose threshold is  $w_{\text{thresh}} \approx 3.44$  in the implementation’s bit units. Those bit units are the convention of the swept weight grid, and the calibration paragraph following Proposition 1 gives the conversion to the nat units of the proposition. Section 6.2 extends the check to a refined weight grid on two such environments.

#### 4.4 Solving for the Shadow Price

For each environment we estimate  $U(w)$  on a log-spaced grid by multi-seed Monte Carlo (five seeds, 100 episodes per  $(w, \text{seed})$  cell on the discrete observe-then-commit domains, 40 on Tileworld-6×6, lighter settings on tree-search domains, Section 6). Per-seed means give a standard error at each grid point. Because discrete policy switches make the count-usage curve a noisy step function, we estimate  $w^*(B)$  on the grid and report the crossing bracket rather than a bisection root, together with the grid weight minimizing  $|U - B|$ , written  $\hat{w}(B)$ , which is the point the staircase figures mark and which can sit on the bracket’s lower edge or, at a slack budget, below it. Usage can be denominated either in observation count or in cumulative sensing cost. The two agree only when all observation actions share a cost, and Section 6.4 demonstrates a case where they disagree in an interpretable way.

#### 4.5 Online Dual Control of the Shadow Price

The grid solve of Section 4.4 is an offline procedure, and it becomes unavailable when the reward scale can change mid-deployment. The reason is the second

corollary noted after Proposition 4, that the crossing bracket of a count budget, the kind the controller below tracks, rescales with the reward scale. A bracket for  $w^*(B)$  computed at one reward scale is therefore the bracket for that scale only, and an unannounced mid-run rescale silently moves the target weight. An agent in deployment cannot stop to re-sweep the grid. What it can do is adjust  $w$  from the usage it observes.

We adapt the automatic entropy-temperature update of Soft Actor-Critic (SAC) [25] to sensing usage in place of policy entropy, the temperature being the coefficient SAC places on the policy-entropy bonus in its objective. SAC raises its temperature when the policy’s entropy falls below a target and lowers it when entropy rises above the target, so that entropy settles at the target. Here  $w$  plays the temperature’s role, sensing usage plays entropy’s, and the budget  $B$  is the target.

Written out, the update is a one-line recursion. After each episode with observed usage  $U_t$ ,

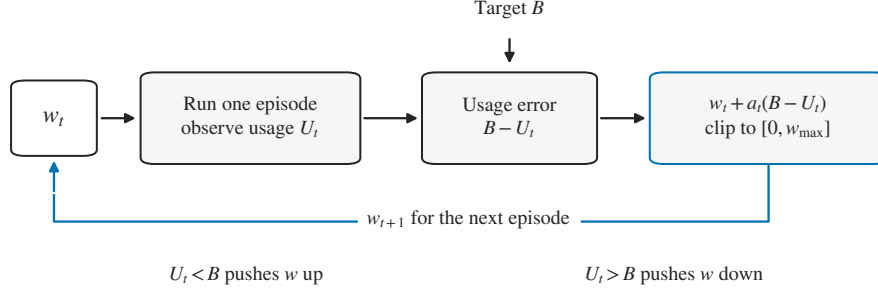
$$w_{t+1} = \text{Proj}_{[0, w_{\max}]}(w_t + a_t (B - U_t)), \quad a_t = \frac{\eta_0}{1 + \delta t}, \quad (9)$$

with  $\mathbb{E}[U_t \mid w_t] = U(w_t)$ . Here  $a_t$  is the step size at episode  $t$ ,  $\eta_0$  is its initial value,  $\delta$  is the decay rate that shrinks the step as episodes accumulate (at  $\delta=0$  the step is constant),  $\text{Proj}_{[0, w_{\max}]}$  clips the result into the interval  $[0, w_{\max}]$ , and  $w_{\max}$  is a fixed upper bound on the weight. The condition on  $U_t$  says in words that each episode’s observed usage is an unbiased but noisy sample of the usage curve at the current weight, so the controller sees the curve only through noise. The update is the Robbins–Monro recursion [47], the classical stochastic-approximation scheme for locating the root of a function observed only through noisy samples, here projected onto a bounded interval. Its target is the weight at which  $U(w) - B$  changes sign, a root when one exists and the crossing point of a jump otherwise. The sign convention is that overspending usage decreases  $w$  and underspending increases it, the direction that moves usage back toward the budget when a larger weight buys more sensing.

Figure 6 makes the feedback sign and the role of projection explicit.

Three versions of this controller appear in the article, and it matters which one each statement is about. The first is the projected decaying-step controller, the recursion (9) with a finite  $w_{\max}$  and  $\delta > 0$ . It is the algorithm Proposition 6 covers, and it is the algorithm every reported run executes, with  $w_{\max}=100$ , the top of the swept weight grid, and  $\delta=0.02$  (`DualWeightAgent` in `rho_aif/agents/dual_descent.py` with `max_weight` set, wrapping `dual_update` in `rho_aif/budget.py`). No reported iterate exceeded 3.1, more than an order of magnitude below  $w_{\max}$ , so the upper projection never bound, but it is in force. The second is the released library default, which sets no upper projection and uses a constant step ( $\delta=0$ ). The proposition covers neither of those two departures, and no result in this article is reported for that default. The third is the reset-on-shift heuristic described below, which the proposition also does not cover.

**Proposition 6 (Stationary convergence of the dual controller).** *Consider the projected decaying-step controller, the recursion (9) with a finite  $w_{\max}$*



Stationary guarantee requires a single sign-consistent crossing and the stated assumptions.

Decaying step  $a_t = \eta_0 / (1 + \delta t)$ ,  $\delta > 0$ . Reset-on-shift is a heuristic.

**Fig. 6.** One episode of the projected weight controller, shown schematically. The current weight produces an observed usage  $U_t$ . The controller adds a step proportional to  $B - U_t$  and clips the result to  $[0, w_{\max}]$ . Underspending raises the weight and overspending lowers it, subject to clipping. This direction corrects usage under the sign-consistent crossing assumptions of Proposition 6. The proposition covers the projected decaying-step update in a stationary setting under its four stated conditions, not arbitrary usage curves or the reset-on-shift heuristic.

and step sizes satisfying  $\sum_t a_t = \infty$  and  $\sum_t a_t^2 < \infty$  (the schedule in (9) with  $\delta > 0$ ). Suppose four conditions hold. (i) The environment, agent family, and budget  $B$  are stationary, so that  $\mathbb{E}[U_t | w_t] = U(w_t)$  for a fixed function  $U$ . (ii) Per-episode usage is bounded, which holds here because usage is bounded by the episode step cap, and each episode is run on an environment reset and freshly seeded independently of the past, so that the noise  $U_t - U(w_t)$  is a martingale difference with bounded conditional second moment. (iii) There is a single crossing  $w^* \in (0, w_{\max})$  with  $U(w) < B$  for  $w < w^*$  and  $U(w) > B$  for  $w > w^*$ . (iv) The curve stays away from the budget outside every neighborhood of the crossing, that is, for every  $0 < \epsilon_1 < \epsilon_2$ ,  $\inf\{|U(w) - B| : \epsilon_1 \leq |w - w^*| \leq \epsilon_2, w \in [0, w_{\max}]\} > 0$ . Then  $w_t \rightarrow w^*$  with probability one, and hence in probability. Condition (iii) does not require  $U(w^*) = B$ , and condition (iv) does not require continuity, so the conclusion covers a gap budget in the sense of Definition 2, where  $U$  jumps over  $B$  at  $w^*$  without touching it. What converges in that case is the weight. The usage at the limit is  $U(w^*)$ , one side of the jump, so the expected usage at  $w_t$  need not converge to  $B$ . The running average of observed usage under the schedule in (9) does converge to  $B$ ,  $\frac{1}{n} \sum_{t \leq n} U_t \rightarrow B$  with probability one, gap budgets included, as the proof below shows. Attaining  $B$  in expectation with one stationary policy at a gap budget is the job of the endpoint mixture of Definition 2, not of this recursion.

The recursion originates with Robbins and Monro [47], with classical almost-sure extensions due to Blum [6]. Kushner and Clark [33] treats constrained

stochastic approximation. We prove the stated bounded scalar case directly, including discontinuous usage curves. Let  $\mathcal{F}_t$  be the history before episode  $t$ ,  $e_t = w_t - w^*$ , and  $M(w) = U(w) - B$ . By (ii),  $\mathbb{E}[U_t | \mathcal{F}_t] = U(w_t)$  and  $(B - U_t)^2$  is bounded by a constant  $C$ . Projection onto an interval containing  $w^*$  cannot increase distance from  $w^*$ , so

$$\mathbb{E}[e_{t+1}^2 | \mathcal{F}_t] \leq e_t^2 - 2a_t e_t M(w_t) + C a_t^2.$$

Condition (iii) gives  $e_t M(w_t) \geq 0$ , also when  $w_t = w^*$  without requiring  $M(w^*) = 0$ . Thus  $Y_t = e_t^2 + C \sum_{k>t} a_k^2$  is a nonnegative supermartingale and converges almost surely. Summing the expected drift inequality gives  $2\mathbb{E}[\sum_t a_t e_t M(w_t)] \leq \mathbb{E}[Y_0] < \infty$ , so that nonnegative series is finite almost surely. Since the tail  $\sum_{k\geq t} a_k^2$  vanishes,  $e_t^2$  has an almost-sure limit. If it were positive, the bounded interval and condition (iv) would keep  $e_t M(w_t)$  bounded away from zero eventually, contradicting  $\sum_t a_t = \infty$ . Hence  $w_t \rightarrow w^*$  almost surely.

The running-average claim follows from that convergence. Because  $w^*$  is interior and the increments  $a_t(B - U_t)$  vanish, usage being bounded by (ii), the projection is inactive from some episode on, so  $\sum_t a_t(B - U_t)$  converges, its partial sums being differences of the convergent iterates. Since  $1/a_t = (1 + \delta t)/\eta_0$  increases to infinity, Kronecker's lemma gives  $a_n \sum_{t<n} (B - U_t) \rightarrow 0$ , and because  $n a_n \rightarrow \eta_0/\delta > 0$  under that schedule, the one property of it the running-average claim uses, this is  $\frac{1}{n} \sum_{t<n} (B - U_t) \rightarrow 0$  almost surely. Under a constant step, which no reported run uses, the recursion need not converge to a point. The standard constant-step stochastic-approximation intuition is that the iterate is instead held near  $w^*$  in a neighborhood whose width shrinks with the step size, and that this residual motion is the mechanism that lets the controller re-adapt after the budget or environment shifts. Making it precise requires constant-step tracking arguments (stationary-distribution or moment bounds under drift and noise assumptions) that we do not develop here. Section 6.3 assesses re-adaptation under reward-scale shift empirically on the decaying schedule, to which the same intuition applies through whatever step size is left at the moment of the shift. That section also runs the reset-on-shift variant, described next.

The *reset-on-shift* variant of the controller watches the usage of the most recent episodes. When that usage persistently misses the budget after the step size has already decayed below half its initial value, it restarts the episode counter  $t$  in the step schedule of (9) at zero, so that the step size jumps back to  $\eta_0$  and the controller can move quickly again. Resetting the decay clock mid-run in this way breaks the trajectory into a sequence of Robbins–Monro-style segments stitched together by a change-detection heuristic, rather than a single non-increasing step schedule. The proposition therefore does not cover the reset-on-shift variant when that variant is treated as a single run.

The proposition's scope has one further limit. It is conditional on a single, sign-consistent crossing of  $B$ . It is silent when  $U$  crosses  $B$  more than once, which the discussion after Proposition 5 shows to be a real possibility, since the usage staircase need not be monotone.

#### 4.6 Positioning against Prior Art

Our weight  $w$  is calibrated within the Planning+IG family so that expected sensing usage meets a stated target, and it is not the Lagrange multiplier of the usage cap in problem (6) (Section 4.1). We do not claim novelty for the underlying idea of adjusting a scalar price until a resource quantity meets a target. That idea is classical in constrained MDPs [1], rational inattention [38, 56], and SAC temperature auto-tuning [25]. Sims [56] models agents with finite Shannon capacity, and Matějka and McKay [38] derive discrete-choice logits from costly information acquisition. Both take an information-processing constraint as the primitive. We take an operational sensing budget as the primitive and recover the Planning+IG weight that implements it. Kim et al. [30] extend constrained dynamic programming to partial observability directly, showing that optimal constrained POMDP policies can require randomization and giving point-based dynamic programming over belief and admissible-cost pairs. Their method is not a Lagrangian one. Point-based solvers represent the value function by a set of vectors over beliefs and discard the dominated ones at each backup, and their method enforces the cost bound inside that pruning step rather than pricing it. The dominated vectors are pruned exactly with a minimax quadratically constrained program and approximately with a linear program at each collected belief point. Their constraints are exogenous costs enforced offline. The usage target here is also stated on a non-epistemic quantity, the count of sensing actions or their cumulative cost, rather than on information gained. What differs is that the price  $w$  on belief-computed information gain is set offline by the grid solve of Section 4.4 or adjusted online by the controller of Section 4.5 until expected sensing usage meets the target.

The single-price receding-horizon family is the set of policies that run one scalar weight  $w$  throughout an episode. It includes the per-episode mixture of Definition 2, which randomizes between the two bracket endpoints from one episode to the next. This family attains the usage budget in expectation. We do not claim it is reward-optimal among all budget-feasible policies. An exact solver in the style of Kim et al. [30] can randomize belief-dependently and may attain strictly higher constrained reward, particularly at gap budgets. We do not have such a solver, so Section 6.8 benchmarks against a near-optimal constrained-POMDP reference on the three core observe-then-commit environments (Tiger, Diagnosis, and Bandit). The reference is built as a Lagrangian-penalized sweep. For each value of a penalty  $\lambda$  on a grid, the POMDP is re-solved by SARSOP, the offline point-based solver that already serves as this paper’s near-optimal reference (Section 6.7), with every observation action charged an extra  $\lambda$ . The reward and usage of the resulting policy are then estimated by Monte Carlo simulation. SARSOP is solved to precision  $10^{-3}$ , and the reference is a simulation estimate rather than a certified exact solution. Because a near-optimal reference is a feasible policy rather than a certified optimum, it can only estimate the constrained optimum from below, up to its own sampling error and the upward selection bias of taking a maximum over noisy sampled points, so any measured gap to it understates the gap to the optimum up to those two effects. The

estimated optimality gap, the gap between the EFE agent at  $w=1$  and this reference, is small, zero on Tiger and within Monte Carlo noise of zero on Bandit and Diagnosis.

The RockSample[7,8] reference calculation instead exposes a limitation. We apply the usage-cap Lagrangian construction of Section 6.8, a per-observation penalty of  $\lambda$ , to this project’s own depth-3 tree search (the reference planner for RockSample, standing in for SARSOP) and sweep the same log-spaced 12-point  $\lambda$  grid used there for the three core environments (`experiments/run_rocksample_cpomdp_reference.py`). The result is a degenerate frontier, the reward-versus-usage curve that the sweep traces collapsing to two points. Across the entire lambda range the penalized planner checks zero or one rock, reward 9.5 or 12.3, and never approaches EFE’s actual operating point of 9.84 checks, so no meaningful reference reward is available there. The failure is structural rather than a tuning artifact. A literal usage-cap penalty rewards a check only through what a 3-step lookahead can see beyond it. No such short horizon can justify checking eight rocks in sequence, whereas the information-floor dual that Planning+IG and EFE actually use credits each check’s information gain immediately at the node where it occurs, independent of horizon. The usage-cap Lagrangian that gives a near-optimal reference on Tiger, Diagnosis, and Bandit relies on SARSOP’s effectively unbounded horizon, and it has no depth-limited analogue that reaches RockSample’s operating region. Closing this gap remains future work with a sharper target, either a genuinely deeper RockSample solver or a usage-cap construction that values information locally the way the information-floor dual already does.

Among the classical dualizations listed above, SAC is the one from which the update rule itself, as distinct from the budgeting idea, is adapted. Soft Actor-Critic’s applications paper [25] automatically tunes the entropy temperature by dual gradient descent on a target-entropy constraint, and our update (9) mirrors that construction with sensing usage in place of policy entropy.

Adaptive test selection is also the subject of sequential Bayesian experimental design (SBED), which chooses the next experiment to maximize expected information gain about a latent parameter, then updates and repeats [36]. Recent work amortizes this loop with a learned design policy so that near-optimal adaptive experiments run in real time [16]. SBED’s design objective is expected information gain about the latent alone. Ours is the scalarized  $R + wI$  objective, task reward plus information gain at weight  $w$ , with  $w$  chosen inside the single-price receding-horizon family so that expected sensing usage equals a stated budget. Information gain is the quantity the weight prices, and sensing usage, in observation count or cumulative cost, is the quantity the budget controls. Reward at the calibrated weight is then a measured outcome, not a design target. The two objectives coincide only in the limit of a weight large enough that information gain dominates both sensing cost and task reward in the scalarized objective. In that limit the test chosen is the one expected information gain alone would select. Otherwise the budgeted objective declines an informative test that

an SBED policy would take, because the decision variable here is how much to test, not only which test to run next.

Two contemporaneous 2024–2026 papers make an even closer structural comparison available. Kouw [31] derive expected free energy itself as the stationary point of a constrained Bethe-Lagrangian optimization, the Bethe free energy being a variational free energy defined on a factor graph. In that optimization a Karush–Kuhn–Tucker multiplier, the multiplier that constrained optimization attaches to a constraint, sits on an information constraint. Standard EFE is recovered at one specific value of that multiplier as the constraint moves through inactive (not binding), interior, and saturated (binding at its limit) regimes. This is a Lagrangian reading of EFE’s information term, and it is the Bethe-side counterpart of this paper’s own information-floor reading of  $w$  in Section 4.1. The two floors bound different quantities. Kouw’s floor is a constraint internal to the Bethe free energy, inside the constrained optimization whose stationary point is EFE, whereas the floor that  $w$  prices here is on  $I(\pi)$ , the cumulative expected information gain over the trajectories a policy generates, a constraint on the planner’s choice of policy. Neither floor is the usage cap in (6), and Kouw’s treatment does not address scale equivariance, crossing brackets and endpoint mixtures at usage gaps, or online tracking under budget or reward-scale shift. Stocco et al. [60] apply Lagrangian-guided Monte Carlo tree search with recursive, history-dependent dual ascent inside constrained POMDP planning itself, the same dual-ascent mechanism as our online controller (Proposition 6), but tracking per-node multipliers within a single tree search rather than a single scalar  $w$  across episodes toward a stated sensing budget.

The budgeted formulation carries one cost. Estimating the usage curve  $U(w)$  well enough to solve  $U(w)=B$  offline requires sweeping  $w$  over a grid and running episodes at each point, which is at least as much sampling as the direct per-task weight tuning this paper criticizes. The budgeted formulation does not save that sampling. What it buys instead is twofold. The design primitive becomes a quantity a practitioner can actually state (a sensing budget in observation count or cost) rather than one they must guess. The online dual controller (Proposition 6) skips the offline curve entirely, converging from interaction alone to the budget crossing point. Scale equivariance (Proposition 4) lets a count-usage curve computed at one reward scale transfer across pure reward-and-cost rescalings without recomputation, and lets a cost-usage curve transfer after rescaling by the same factor. That is a property of the planner and its weight rather than of how the weight was chosen. The same transformation carries a directly tuned weight  $w$  to  $\alpha w$  under the same rescaling and tie-break assumptions, on a weight grid rescaled with the rewards, as the scaling of the reward-optimal tuned weights in Section 3.3 already records. The contribution is thus not generic Lagrangian duality, which is prior art in all of these literatures, but its concrete operationalization for budgeted  $\rho$ -POMDP sensing.

## 5 Experiments

The six environments are Tiger, Diagnosis, Bandit, Tileworld, RockSample, and Structural Inspection, and they form a progression in what the agent must decide about sensing. Tiger has a single sensor, so the only question is whether to observe again. Diagnosis and Bandit add a choice among several sensors, so the agent must also decide which observation to buy. Tileworld makes that choice spatial, with sensors that each partition a grid. RockSample and Structural Inspection interleave sensing with navigation and exploitation, so the agent must additionally move to where information is available. Four of the six (Tiger, Diagnosis, Bandit, Tileworld) follow the observe-then-commit structure of Section 3, in which a variable-length run of observation actions ends in a single terminal commit action and the hidden state never changes. The remaining two (RockSample and Structural Inspection) are factored observation POMDPs in the sense of Proposition 3. Their observation, navigation, and exploitation actions interleave within an episode, which is why the paper also calls them interleaved observe-act settings.

We call Tiger, Diagnosis, and Bandit the core environments. They run at the full protocol of 1,000 episodes per seed across 5 random seeds ( $\{42, 123, 456, 789, 1024\}$ ) for a total of 5,000 episodes. They anchor the main results table, and they are the three environments on which the near-optimal SARSOP reference of Section 6.7 is computed. Tileworld and Structural Inspection use 500 episodes per seed, and each RockSample instance uses the protocol stated in its table caption. The same agent on the same environment is also evaluated in other batteries, among them the Pareto sweep, the SARSOP and CPOMDP studies, and the shadow-price atlas, each under the episode count and seeding rule its own section states, so its usage and reward differ slightly from one battery to the next. Tiger EFE, for example, averages 4.20 observations in Table 8 and 4.32 in Table 5. Where a comparison does cross batteries, as the committed-reference column of Table 7 does, the text says so.

The rest of the protocol concerns how results are aggregated and tested. Each reported reward, success rate, and observation count is the mean over all episodes of all seeds. Pairwise statistical comparisons are seed-level Welch  $t$ -tests, which aggregate each seed’s episodes to one mean and then test across the 5 seeds, with Holm–Bonferroni correction for multiple comparisons (Appendix I). Bootstrap confidence intervals for the core-environment means, computed under the protocol given in the same appendix, are in the file `results_bootstrap_ci.csv` of the code repository. Comparison against POMCP is in Appendix S. That appendix includes a scaling analysis over POMCP’s simulation budget, meaning the number of simulated rollouts it runs per decision, a quantity distinct from the sensing budget  $B$  of Section 4, with wall-clock timing at 500–5,000 simulations per decision. Full specifications are in Appendix E, and all environments are implemented as Gymnasium [64] environments, Gymnasium being a widely used Python interface for reinforcement-learning environments.

*Tiger* [29]. A tiger waits behind one of two doors, so the hidden state takes two values. The single observation action, listen, reports the tiger’s door correctly with probability 0.85. The two commit actions open one door or the other and end the episode. A correct commit pays +10, an incorrect one −100, and each listen costs −1. Because there is only one sensor, Tiger poses no choice of what to observe, only whether to observe again.

*Sequential diagnosis* (Diagnosis in tables and figures). Exactly one of  $N=4$  hidden conditions is present, and the agent’s task is to name it. There are  $K=2$  binary tests (accuracy 0.80). Each test splits the four conditions into two groups along its own partition and reports, with that accuracy, which group contains the true condition, so the two tests together can identify any condition. There are  $N$  diagnose actions, one per condition, each of which commits and ends the episode. A correct diagnosis pays +10, an incorrect one −50, and each test costs −1. Unlike Tiger, the agent must choose which test to run.

*Structured bandit* (Bandit in tables and figures). One of  $K=4$  arms is the best, and which one it is forms the hidden state. The bandit is structured in the sense that this single hidden variable, rather than an independent unknown payoff per arm, ties the arms together, so evidence about one arm is evidence about the others. Each of the  $K$  inspect actions (accuracy 0.80, cost −0.5) targets one arm and returns a noisy yes/no signal about whether that arm is the best. Each of the  $K$  pull actions commits to an arm and ends the episode. Pulling the best arm pays +10, and pulling any other arm pays +1. As in Diagnosis, the agent must choose which arm to inspect.

*Tileworld*. Tileworld is a spatial generalization of Diagnosis. An  $N \times N$  grid ( $N=6$ ,  $|S|=36$ ) hides a single target tile, and the hidden state is the cell that tile occupies. Each scan action asks a yes/no question about one binary digit of the target’s row index or of its column index, and that question splits the grid in two, into the cells whose digit is 0 and the cells whose digit is 1. Three binary digits suffice to write any of the six row indices, and likewise for the six column indices. Three scans for the row digits and three for the column digits therefore give the  $K=6$  scan actions, which together pin down the cell. Each scan alone tells the agent only which side of its own split the target lies on, and each returns its answer as a noisy binary signal (accuracy 0.80, cost −1). Each of the  $N^2$  commit actions declares one cell as the target’s location and ends the episode (correct +10, incorrect −50). Appendix E and Figure 21 label this action collect, following the implementation. Because the belief lives on the grid itself, its evolution over an episode can be drawn directly, as shown later in Figure 15.

*RockSample* [58]. RockSample is the first environment in the suite where sensing is interleaved with movement. An  $N \times N$  grid holds  $K$  rocks at known positions, each with a hidden binary quality (good or bad), and the agent’s own position is fully observable. Move actions change the agent’s position. Check actions produce noisy observations of a chosen rock’s quality, with an accuracy that decays

with the agent’s distance from that rock. Sampling collects the rock at the current position (good +10, bad −10). Exiting ends the episode from wherever the agent stands and gives +10. All actions cost −0.5. We evaluate four instances, written  $RS[N, K]$  for an  $N \times N$  grid with  $K$  rocks, namely  $RS[5, 3]$ ,  $RS[7, 4]$ ,  $RS[7, 8]$ , and  $RS[11, 11]$ . Unlike the above environments, RockSample has interleaved observe-act dynamics with state transitions, the setting addressed by Proposition 3.

This is a cost-modified variant of Smith and Simmons [58]’s benchmark, not the original, and it differs in six respects that a reader comparing against published RockSample numbers must know. First, every action including checks carries the −0.5 cost. The original charges for neither moves nor checks. It pays +10 for sampling a good rock, −10 for sampling a bad one, and +10 for reaching the exit, with all other moves free, so the only thing penalizing a long path there is the discount factor. Second, rock positions are fixed per instance, as in the original, but are our own choice rather than the specific maps used in the published instances. Third, returns are reported undiscounted, with episodes truncating at the  $N^2 + 10K$  step cap stated in Appendix E, whereas the original relies on its discount factor instead.

The next two differences concern the dynamics. Fourth, the exit action grants its reward from any cell rather than only at the east edge. This makes exit an unconditional outside option, that is, a fallback of known value the agent can take at any moment from wherever it stands. The POMCP analysis in Appendix M turns on this fact. POMCP’s search tree values each action by the mean of the returns of the simulations that pass through it (a mean backup). A guaranteed exit payoff available at every cell is the fixed alternative against which every one of those averaged estimates is measured. Fifth, the hidden rock-quality vector stays fixed when a rock is sampled. In the original, sampling a good rock turns it bad. Here, which rocks have already been sampled is tracked outside the hidden state, and the agent can recover that list from its own action history and its fully observable position. A re-sample is therefore recognized without any change to the hidden state. Re-sampling a collected rock therefore earns zero sampling reward rather than the bad-rock penalty, although the action still pays the −0.5 cost that every action pays. This fixity is what Proposition 3’s application to RockSample relies on.

Sixth, the check sensor differs in both its close-range accuracy and its tail. Efficiency is the original paper’s name for the sensor’s reliability as a function of range. The original sensor is exact at range zero, blends toward chance as its efficiency decays with Euclidean distance, and never reaches chance at any finite range. The accuracy of our check sensor starts at 0.95, and it is the accuracy value itself that halves with every 4 cells of distance and is then floored at 0.5 (`rho_aif/environments/rocksample.py`). The decay meets that floor at a range of 3.7 cells, a little short of one full halving distance because the accuracy starts at 0.95 rather than at 1. Since a binary observation with accuracy 0.5 is a coin flip, our sensor is exactly uninformative beyond 3.7 cells rather than asymptotically so.

The undiscounted, check-charging reward model is what makes sensing usage directly interpretable as a budget in Section 4, which is why we adopt it. But it means the absolute returns reported here are not comparable to RockSample values in the literature. We use  $|S|$  throughout for the number of hidden-state configurations ( $2^K$  rock-quality vectors), excluding the fully observable agent position, so our  $|S|=2,048$  for RS[11,11] is not the convention under which the literature quotes larger state counts for the same instance.

*Structural inspection.* In Structural Inspection,  $N$  components sit at known spatial locations on a grid, and each has a hidden binary state (nominal/faulty, prior  $p_{\text{fault}}=0.3$ ) that the agent is tasked with declaring. There are  $K=2$  non-destructive test types, visual (accuracy 0.70, cost  $-0.5$ ) and detailed (accuracy 0.90, cost  $-2$ ). The agent navigates between components, runs tests, and declares a diagnosis for one component at a time, in whatever order its own navigation takes it, so a completed episode carries one declaration per component rather than a single terminal commit. An episode ends once every component has been declared, or earlier at a step cap. A correct nominal declaration pays  $+2$ , a correct fault declaration pays  $+5$ , a missed fault costs  $-50$ , a false alarm costs  $-5$ , and each move costs  $-0.5$ . Neither tests nor moves change fault states, so this is a factored observation POMDP. The environment maps directly to industrial inspection, medical screening, and fault detection domains. We evaluate  $N=8$  ( $|S|=256$ ) and  $N=16$  ( $|S|=65,536$ ).

Two additional environments, reported in the appendices, test the EFE agent in mild-penalty and small-state-space settings and mark the boundary of the settings in which the main results apply. The two-state testbed (Appendix C) pairs the smallest possible hidden state with a penalty for a wrong commit that is only as large as the reward for a correct one, so that a further observation has little reward to buy. Navigation (Appendix F) has no separate menu of observation actions at all, since information arrives only as a side effect of moving.

Section 6 reports the first battery. Section 7 then reports the second battery one environment class at a time, in the order core environments, Tileworld, RockSample, and Structural Inspection, with a Pareto analysis of the information-unit weight placed between the core results and Tileworld.

## 6 Experiments on Shadow Prices and Sensing Budgets

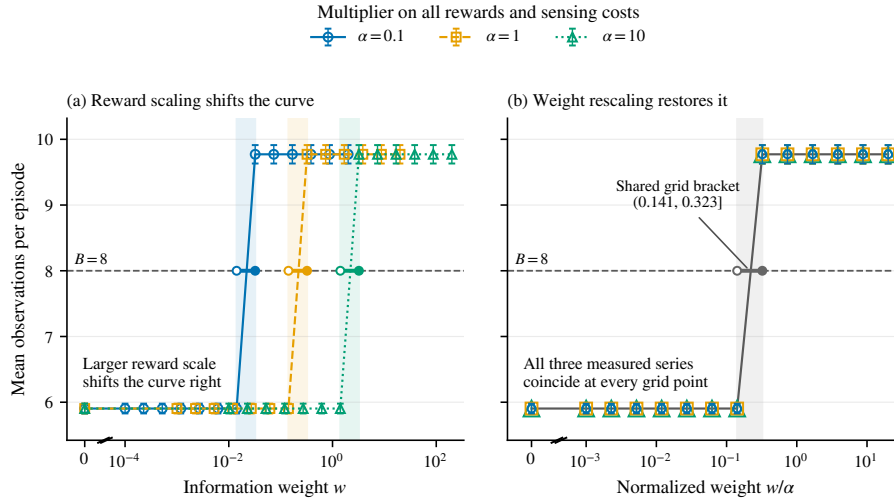
This section evaluates the budgeted reformulation of Section 4. Section 7, which follows it, evaluates the equivalence and near-optimality claims of Section 3 under the protocol of Section 5. The subsections of this section fall into three groups. The first three exercise the theory of Section 4 directly. They test usage-curve collapse across reward scales (Section 6.1), recovery of the closed-form onset threshold as the first knot of the usage staircase (Section 6.2), and online tracking through a mid-run reward rescale (Section 6.3). The next three widen the settings in which the shadow price is solved for. They cover budgets denominated in sensing cost rather than observation count (Section 6.4), interleaved

settings in which sensing and acting alternate within an episode rather than following the observe-then-commit structure (Section 6.5), and the shadow-price staircases read off across five domains (Section 6.6). The last five place the results in context. They provide two near-optimal offline references, SARSOP and a constrained-POMDP reference derived from it (Sections 6.7 and 6.8), a test of the calibration procedure at calibration-derived target budgets on held-out seeds, which separates meeting an expected usage from maximizing reward under a cap (Section 6.9), a distractor test that marks what a budget does and does not control, since a budget fixes how much sensing is bought but not what that sensing is about (Section 6.10), and an atlas of implicit budgets recording where each benchmark sits on its own usage curve (Section 6.11). All headline runs use five seeds  $\{42, 123, 456, 789, 1024\}$  and 100 episodes per evaluation cell, an evaluation cell being one weight at one seed, unless noted. The tree-search domains are the exceptions, and they run fewer episodes per cell because each of their episodes costs an order of magnitude more than an episode in the observe-then-commit domains. Tileworld-6 $\times$ 6 runs 40 episodes per usage-curve cell and 30 per scale-collapse cell, and Inspection- $N=8$  runs 30 per usage-curve cell. The code and the result CSVs behind every number in this section are in the accompanying repository.

### 6.1 Curve Collapse across Reward Scales

This subsection tests the first corollary of Proposition 4, that usage plotted against  $w/\alpha$  is one and the same curve at every reward scale. Here  $\alpha$  is the reward scale of Section 4.2. On Diagnosis and Bandit we estimate the usage curve  $U_\alpha$  of Equation 7 at reward scales  $\alpha \in \{0.1, 1, 10\}$ . The weight grid at scale  $\alpha$  is the base grid multiplied by  $\alpha$ , so each grid point at one scale has a partner with the same  $w/\alpha$  at the other two. The curves can be compared point by point. Figure 7 first shows how the Diagnosis usage curves shift along the raw weight axis as rewards and costs are rescaled, then shows how dividing the weight by the same scale restores their agreement. Every matched  $w/\alpha$  point coincides exactly across scales, bit-exact rather than merely within  $2 \cdot \text{SE}$ . The agreement can be exact rather than statistical because the pipeline seeds every episode’s environment stream, so an agent that takes the same actions sees the same outcomes. The crossing bracket of budget  $B=8$  in  $w/\alpha$  units likewise coincides across  $\alpha$  ( $[0.141, 0.323]$  at every scale). We extend the check to Tiger and Tileworld-6 $\times$ 6 to avoid resting the headline claim on two environments (Table 4).

Every environment shows bit-exact collapse, and every crossing bracket coincides across scales down to floating-point noise ( $\leq 10^{-15}$ ). This is the strongest form the check can take, and it is available only because environment streams are seeded per episode. The collapse CSVs were regenerated under the relative comparison tolerance of Section 4.2 and are byte-identical to those produced under the earlier absolute tolerance, which did not bind at any swept point. Tiger passes trivially. Instrumental listening already saturates usage (constant 4.37 observations at every grid point) across the entire  $w/\alpha \in [0, 20]$  range tested, so



**Fig. 7.** Rescaling the information weight compensates for rescaling rewards and sensing costs on Diagnosis. Both panels show the same separately evaluated conditions, with all rewards and costs multiplied by  $\alpha \in \{0.1, 1, 10\}$ . In (a), the usage curves and the grid brackets for  $B=8$  shift along the raw weight axis. In (b), dividing that axis by  $\alpha$  makes the three curves coincide and the brackets agree at  $(0.141, 0.323]$ . The plotted means and seed-level SEs are identical at every matched normalized weight. Each grid point uses 100 episodes per seed and five seeds, with shared per-episode random streams across conditions. Open series markers and their error bars show mean usage  $\pm 1$  SE. The segments joining sampled means guide the eye. Shaded bands and open-closed endpoint pairs on the budget line mark grid brackets, not uncertainty intervals or measured policies attaining  $B$ . The true crossing location between grid points remains unresolved.

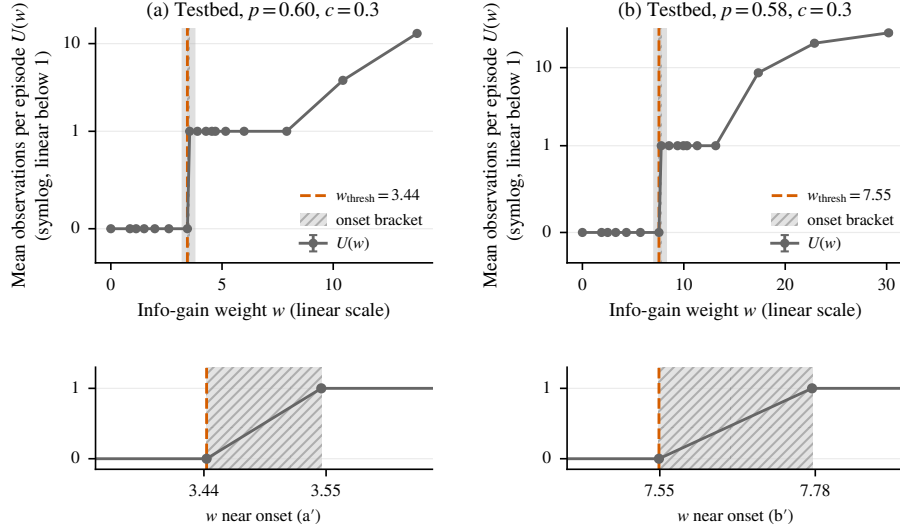
**Table 4.** Curve-collapse breadth. Fraction of matched  $w/\alpha$  points across  $\alpha \in \{0.1, 1, 10\}$  within  $2 \cdot \text{SE}$ , and whether the  $B$ -crossing bracket coincides across scales (in  $w/\alpha$  units).

| Environment                          | Within<br>$2 \cdot \text{SE}$ | Max spread               | Bracket at $B$                                              |
|--------------------------------------|-------------------------------|--------------------------|-------------------------------------------------------------|
| Diagnosis<br>( $B=8$ )               | 100%                          | 0.00 obs.<br>(bit-exact) | (0.141, 0.323] at every $\alpha$                            |
| Bandit<br>( $B=8$ )                  | 100%                          | 0.00 obs.<br>(bit-exact) |                                                             |
| Tiger<br>( $B=4$ )                   | 100% (trivial)                | 0.00 obs.<br>(bit-exact) | unbracketed ( $B$ below<br>observed $U$ at every $\alpha$ ) |
| Tileworld-6 $\times$ 6<br>( $B=15$ ) | 100%                          | 0.00 obs.<br>(bit-exact) | (5.80, 20.0] at every $\alpha$                              |

the budget formulation has no usable dial in this window. Proposition 4 guarantees collapse of whatever curve there is, not that every environment has an informative curve to collapse.

## 6.2 Proposition 2 Onset on Positive-Threshold Two-State Instances

This subsection checks that the closed-form onset threshold  $w_{\text{thresh}}$  of Proposition 2, the weight above which observing first switches on, is where the measured usage staircase takes its first upward step, as Corollary 1 predicts. The check needs instances whose threshold is positive. Standard Tiger and Testbed parameters yield negative  $w_{\text{thresh}}$ , so any  $w \geq 0$  observes. The check of Corollary 1 is vacuous there. We therefore construct two positive-threshold Testbed instances. Each is the two-state Testbed environment of Appendix C (the `InfoSeekingEnv` class of the implementation), whose standard parameters are tabulated in Appendix E. At each step the agent chooses between committing at once and paying cost  $c$  for one more observation of accuracy  $p$ . The two instances differ from the standard Testbed only in a costlier, less accurate observation ( $p \in \{0.60, 0.58\}$ ,  $c=0.3$ ,  $R_{\pm} = \pm 1$ , where  $R_{\pm}$  are the correct-commit reward and the incorrect-commit penalty). One point of notation before the numbers. The  $\alpha$  of Proposition 2 is its reward asymmetry ratio  $|R^-|/R^+$ , a different quantity from the reward scale that Section 6.1 writes with the same letter. With symmetric rewards these instances have  $\alpha=1$ , the boundary of the proposition’s strict asymmetry hypothesis, but the threshold formula itself does not require that hypothesis. Their closed-form thresholds are  $w_{\text{thresh}} \approx 3.44$  and 7.55 in the implementation’s bit units, the convention the swept weight grid uses, equivalently 4.97 and 10.89 nats. A refined weight grid around the threshold shows  $U=0$  at and below  $w_{\text{thresh}}$ , with onset in  $(w_{\text{thresh}}, 1.03 \cdot w_{\text{thresh}}]$  on both environments (upper relative error 3%, Figure 8), matching Corollary 1. In words, the measured onset never falls below the closed-form threshold. The upper endpoint of the onset bracket sits within three percent of the threshold, three percent being the



**Fig. 8.** Onset of observing on positive-threshold testbeds. The vermilion dashed vertical line marks the  $H=1$  closed-form threshold (Proposition 2) and the hatched gray band is the empirical onset bracket after grid refinement, magnified in the zoom row beneath each main panel, with the shaded strip on the main panel marking exactly the region the zoom row below it shows.

spacing of the first refined grid point above it, so the bracket is one grid step wide. Negative-threshold sanity checks on standard Testbed and Tiger remain consistent ( $U(0) > 0$  when  $w_{\text{thresh}} < 0$ ).

### 6.3 Multi-Seed Online Dual Control under Reward Rescale

Section 4.5 argued that an offline bracket is valid only at the reward scale it was solved at, so a deployed controller must re-adapt online after an unannounced rescale, and it left the speed of that re-adaptation to be measured. This subsection measures it across controller seeds and asks whether resetting the step-size clock after a detected shift makes it faster. The step-size clock is the decay clock of Section 4.5. The setting is Diagnosis with target  $B=8$ , learning rate  $\eta_0=0.05$ , decay  $\delta=0.02$ , and a  $\times 10$  reward-and-cost rescale at episode 200 of a 400-episode run, and we sweep 10 independent controller seeds for each of two variants. The decay-only variant runs the update of Equation 9 with its step decaying from the first episode and never intervenes. The reset-on-shift variant runs the same update and adds the change-detection heuristic of Section 4.5. It watches a rolling window of recent per-episode usage (window 20), and once the step has decayed below half its initial value it declares a shift when the mean usage over that window departs from  $B$  by more than a threshold. That threshold is the larger of  $k$  standard errors of the window mean ( $k=3$ ) and an absolute floor of 0.5

observations, so a departure smaller than 0.5 observations never counts as a shift however small the window’s standard error happens to be. A declared shift restarts the step-size clock, so the step is large again.

Re-adaptation time is measured on a rolling mean of usage that, at each post-rescale episode, averages per-episode usage over the previous 20 episodes and is computed from post-rescale episodes only. Re-adaptation time is the number of episodes after the rescale until that rolling mean stays within  $\pm 1$  of  $B$  for 20 consecutive episodes. Restricting the window to post-rescale episodes matters. A window allowed to straddle the rescale point reports a spuriously fast recovery by mixing in already-converged pre-rescale usage (`experiments/run_price_of_information.py:readaptation_episodes`).

Resetting the step-size clock re-adapts roughly  $3\times$  faster than decay alone. The comparison of record uses one estimand for every seed, the restricted recovery time  $\min(T, 180)$ , where  $T$  is the post-rescale-only recovery time and 180 is the last index at which the 20-episode hold can still be checked inside the 200 post-rescale episodes, with a seed that never completes the hold scored at 180. On that estimand decay-only averages 157.3 episodes and reset-on-shift 51.2. Because the two variants share their ten controller seeds, the paired difference is 106.1 episodes with a Student- $t$  95% interval of  $[96.8, 115.4]$ . Every one of the ten paired differences is positive, the smallest being 89 (`results_price_dual_multiseed_metrics.csv`, column `readapt_restricted`, and the summary JSON’s `dual_ms_readapt_restricted` block). For descriptive context, we also report mean recovery times among the seeds that recovered within the window. Under the post-rescale-only metric, which corrects the straddling one, decay-only recovery averages 154.8 episodes, 95% CI  $[142.7, 166.9]$ , over the 9/10 seeds that recover inside the window, one seed not recovering within it. Two of those nine seeds recover only at episode 180, the last index at which the 20-episode hold can still be checked, and the tenth seed’s rolling mean lies within the tolerance over the final episodes without having held there for 20 consecutive episodes, so the recorded decay-only sample is right-censored at 180, with the tenth seed’s time lying beyond the window. For these observed runs the restricted decay-only mean is therefore a lower bound on that variant’s recovery time, which is a statement about the ten runs measured, not a population inference about recovery times in general. Reset-on-shift recovery averages 51.2 episodes, CI  $[46.1, 56.3]$ , with 10/10 seeds recovering and every seed triggering exactly one step-size reset, so its restricted and conditional means coincide. Post-rescale steady-state  $|U - B|$ , the absolute gap between the budget and mean usage over a fixed block of episodes at the end of the run, is 0.06  $([0.024, 0.096])$  under reset versus 0.39  $([0.118, 0.666])$  under decay-only. Pre-rescale steady-state error, the same quantity over the block of episodes immediately before the rescale, is 0.10  $([0.051, 0.149])$  for both variants (identical, since decay and reset only diverge after the rescale), well inside the  $\pm 1$  tolerance used to declare recovery. Every CI in this analysis other than the paired interval above is a normal-approximation CI over the controller seeds that enter it, and the separate intervals of the two conditional means are not an interval for their ratio. These numbers come from

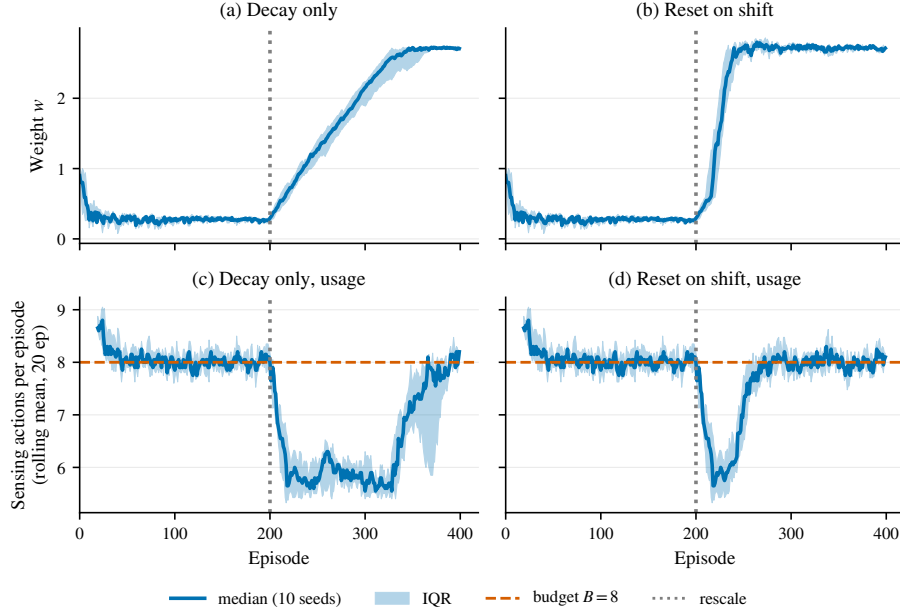
a rerun of the battery under per-episode seeding, after a review found that the environment constructed at the rescale had not been seeded (`results_price_dual_multiseed_metrics.csv`). The earlier run’s headline ratio was 2.65 with every decay-only seed recovering, so the correction moved the estimate in the direction of a larger advantage for reset-on-shift and a slower decay-only recovery that one seed did not complete inside the window, and it did not change the qualitative finding. The ratio of the two restricted means is 3.07. Figure 9 shows the median trajectory with interquartile band for both variants.

The multi-seed result strengthens rather than merely replicates the single-trajectory workshop finding, in which each variant was run once. Under the corrected pipeline, reset-on-shift recovers on all 10 seeds tested within the 200-episode post-rescale window and decay-only on 9 of them. Decay-only takes roughly  $3\times$  longer on average where it recovers and reaches a looser post-rescale steady state, whereas reset-on-shift recovers faster on every seed and reaches a tighter post-rescale steady state on five of the ten seeds, ties on four, and is looser on one. Both variants run the projected decaying-step controller of Section 4.5, but the mid-run rescale makes the environment nonstationary and the reset-on-shift variant restarts its schedule, so neither variant is covered by Proposition 6’s stationary guarantee and what is measured here is nonstationary re-adaptation. The mechanism is the residual step size that the discussion after Proposition 6 identifies. At the rescale episode the decayed step is still large enough to move the iterate, the controlled weight  $w_t$ , which is why decay-only recovers at all. But by then the step has decayed to a fraction of its initial value, so the iterate moves slowly. Reset-on-shift restarts the schedule as soon as the detector declares the shift, so its step is large again immediately afterward. Since the reset is the only thing that distinguishes the two variants, that restored step is where its speed advantage comes from.

#### 6.4 Cost-Denominated Budgets

The count-usage curves of this section so far treat every observation as one unit regardless of price. Section 4.4 noted that usage can instead be denominated in cumulative sensing cost, and that the two denominations agree only when every observation action costs the same. This subsection demonstrates end to end that the denomination matters. A budget stated in test count and a budget stated in sensing cost are not interchangeable, because the agent’s mix of cheap and expensive tests can change with  $w$ . We introduce a Diagnosis variant with heterogeneous per-test costs. Its two test types are priced differently, the cheaper test at 0.5 and the more expensive test at 2.5 (the cost vector  $[0.5, 2.5]$  in the implementation), while both tests keep the base environment’s accuracy. On this variant we estimate both the count-usage curve  $U_{\text{count}}(w)$  and the cost-usage curve  $U_{\text{cost}}(w)$  on the same agent family.

The mechanism is a shift in the agent’s test mix. Recall from Section 5 that each Diagnosis test reports, noisily, which side of its own fixed split of the hidden conditions the true condition lies on, and that the two tests use different splits. The two tests are equally accurate, but because their splits differ, how much



**Fig. 9.** Multi-seed ( $n=10$ ) dual control of  $w$  on Diagnosis through a mid-run  $\times 10$  reward rescale (vertical line). Decay-only (left column) vs. reset-on-shift (right column). The top row plots the controlled weight  $w$  itself, median trajectory with interquartile band, falling from its initial value to about 0.3 within the first few dozen episodes, holding there until the rescale invalidates it, and then climbing about tenfold, to about 2.7, as the controller re-adapts. The bottom row plots the resulting sensing usage (rolling mean, 20 episodes) against the dashed budget line  $B=8$  the controller targets. On the restricted recovery time  $\min(T, 180)$  defined for every seed, where  $T$  is the post-rescale-only recovery time and 180 the last index at which the 20-episode hold can be checked, reset-on-shift re-adapts in a mean of 51 episodes against 157 for decay alone (post-shift-only rolling window), a paired difference of 106 episodes with Student- $t$  95% interval  $[97, 115]$  over the ten shared controller seeds.

each reveals depends on how the current belief is spread across its split. At a given belief one test can therefore reveal more than the other. What the swept curves show is a step. The mean cost paid per test,  $U_{\text{cost}}(w)/U_{\text{count}}(w)$ , holds at 1.26 from  $w=0$  through  $w=10$  and rises to 1.37 and then 1.41 only at the two largest weights, the same two weights at which the number of tests also jumps (Figure 10), a 12.0% spread relative to the grid-mean cost per test. Our reading of the step is an interpretation rather than a measurement. While information gain is weighted lightly, the price holds the agent to its low-weight test mix, and once  $w$  is large enough to buy extra tests, test choice is driven more by what a test reveals at the current belief than by its price, which on this variant means using the expensive test more often. The direction is measured on the swept curves rather than implied by the cost vector, which by itself says nothing about which test reveals more. A count budget cannot see that shift, while a cost budget prices it. This confirms directly that the two denominations are not interchangeable.

Whether the two denominations also select different weights at a given budget is a separate question, and answering it requires a rule for pairing a count budget with a cost budget. A cost budget here is not a count budget converted through a cost per test. Instead each curve is measured on its own scale, and the two budgets of a pair are set at the same fractional height inside each curve's own usage range. The usage range of a curve is the span from the smallest to the largest mean usage on the tested grid. A margin is held back at both ends of that span before budgets are placed, so that each budget has a crossing bracket on the grid. We call two budgets paired in this way comparable operating points. Since the two denominations are not interchangeable, one might expect them to select different brackets at such a pair. At the two comparable operating points we solve for, however, they happen to land in the same crossing bracket at this grid's resolution. A count budget  $B=8.86$  has crossing bracket  $(10, 31.6]$ , and a cost budget  $B_{\text{cost}}=11.50$  (the comparable operating point on the cost curve, a budget on total sensing cost rather than on test count) also has crossing bracket  $(10, 31.6]$  (Figure 10). The same coincidence holds at the higher pair of budgets we test ( $B=13.71$  count,  $B_{\text{cost}}=19.20$  cost, both with crossing bracket  $(31.6, 100]$ ).

The coincidence has a simple explanation. The cost curve is the count curve multiplied by the mean cost per test. If that multiplier were the same at every weight, the pairing rule would make the two crossings coincide exactly, because a budget at a given fractional height of the count curve and the budget at the same fractional height of the cost curve would then be one and the same budget written in two units. The two curves would then cross them at the same weight. Only the variation of the multiplier across the grid can pull the two crossings apart, by shifting where along the weight axis the cost curve crosses its budget. That multiplier moves by at most the 12% spread just reported, while adjacent positive grid weights are log-spaced a factor of  $\sqrt{10} \approx 3.2$  apart. A bracket changes only if that shift carries the crossing past one of those grid weights. A rescaling that small need not carry a crossing into a neighboring grid cell, and at

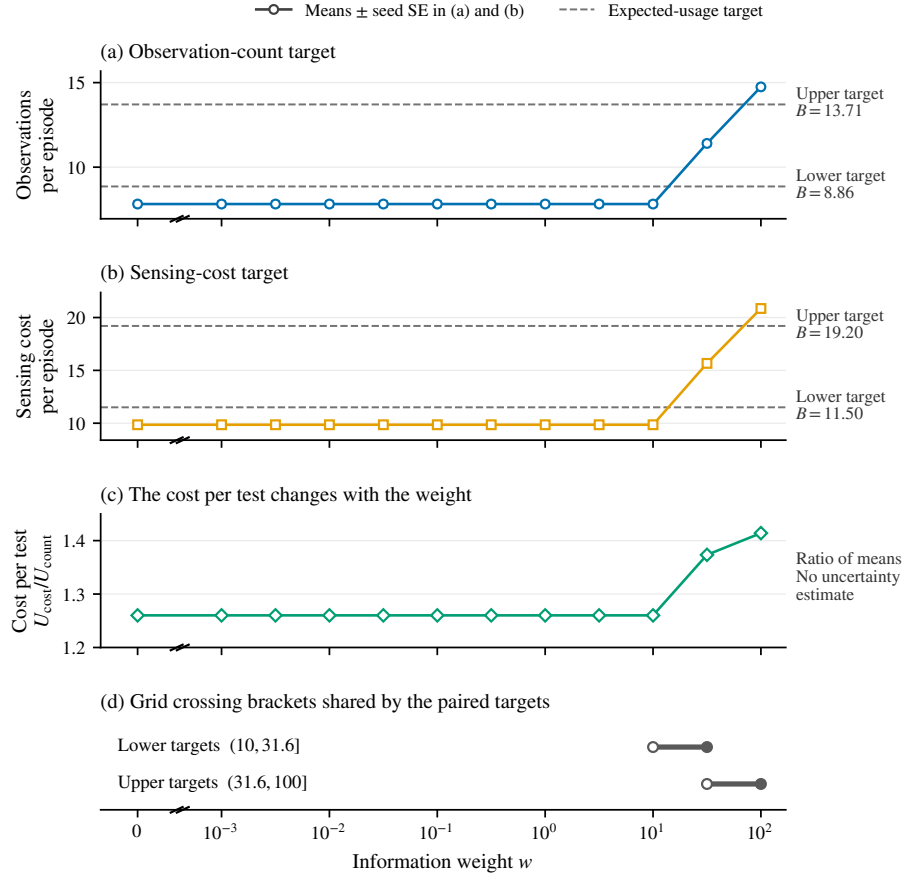
these two pairs it does not. The two denominations are therefore not guaranteed to disagree at every tested budget pair. They land in different brackets only when the variation in the cost per test is large enough, relative to the grid’s resolution, to move a crossing past a grid weight.

## 6.5 RockSample and Structural Inspection as Interleaved Settings

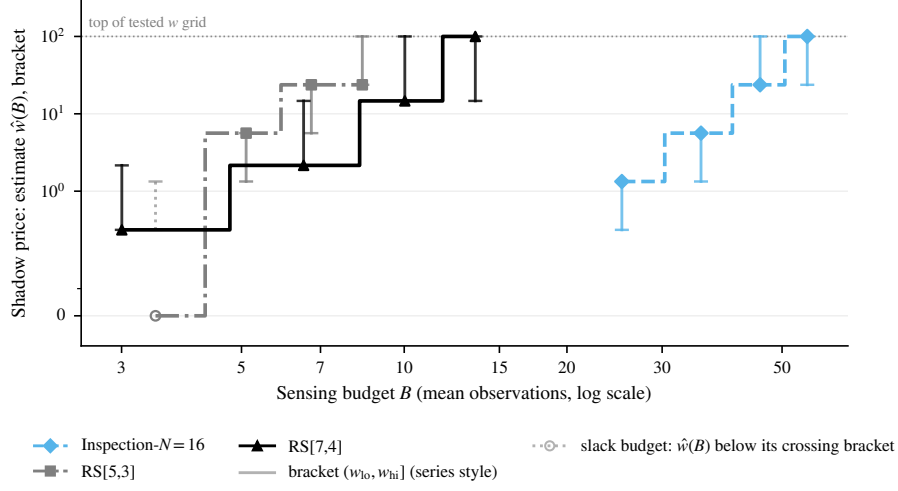
We now estimate usage curves and shadow prices in settings where sensing and acting alternate within an episode, the interleaved observe-act settings of Section 7.4 and Section 7.5. The extension is needed because Proposition 3’s factored-observation class admits navigation actions between observations, so that sensing and acting can alternate, whereas the observe-then-commit environments above separate observation from commitment by construction and so cannot exercise that case. We use the interleaved tree-search agents that those two sections specify, counting check/test actions as sensing usage. The three instances are RockSample[5,3], RockSample[7,4], and Inspection- $N=16$ . All three share one set of five seeds, and every episode count below is per seed at each grid weight. RockSample[5,3] uses a 10-point weight grid with 5 seeds  $\times$  50 episodes, and its usage range (Section 6.4) is [3.00, 8.83]. RockSample[7,4] uses an 8-point grid with 30 episodes and spans [2.00, 14.54]. Inspection- $N=16$  uses a 10-point grid with 50 episodes and spans [22.38, 58.56]. Budgets are placed inside each usage range by the rule of Section 6.4, at evenly spaced positions, with a narrower margin held back at each end than that section used. Inspection appears at  $N=16$ , the larger of the two instances evaluated in Section 7.5, while the staircases of Section 6.6 use the smaller  $N=8$  instance. The two subsections together therefore cover both sizes. Figure 11 shows the resulting shadow-price staircases  $w^*(B)$ , one per instance. At least four budgets are bracketed with SEs per environment, so each staircase has at least four steps, each carrying a crossing bracket and a standard error.

The usage curves  $U(w)$  from which these staircases are read off are nondecreasing between every pair of consecutive sampled grid points on RockSample[7,4] and Inspection- $N=16$ . That is cleaner than the count-usage curves of Bandit and Tileworld, which the discussion after Proposition 5 noted are only roughly monotone and whose shadow-price staircases follow in Section 6.6. RockSample[5,3] is the exception to this clean monotonicity. Its instrumental floor, the usage the agent spends at  $w=0$  because reward alone already motivates that much sensing, is  $U(0)=3.70 \pm 0.11$  (SE), and usage dips from there down to  $3.00 \pm 0.00$  at  $w=0.316$  before rising, a gap of roughly six SEs. This is the same kind of non-monotone dip the observe-then-commit environments show, so the interleaved class does not uniformly enjoy the cleaner monotonicity of the other two instances.

The instrumental floors also bound which budgets any weight can meet. On RockSample[7,4] and Inspection- $N=16$  an agent cannot buy less sensing than reward alone already motivates. No tested weight therefore spends less than the floor, and a budget below it is unachievable in the sense of Definition 2.



**Fig. 10.** Observation-count and sensing-cost targets on heterogeneous-cost Diagnosis. Panels (a) and (b) separate the two usage units, showing means  $\pm 1$  seed-level SE and the tested targets as dashed lines. The lower count/cost targets are 8.86/11.50, and the upper targets are 13.71/19.20. Panel (c) shows the ratio of mean sensing cost to mean observation count, which rises as the test mix changes. No uncertainty estimate is shown for this ratio. Panel (d) displays the paired targets' shared crossing brackets, (10, 31.6] and (31.6, 100]. These intervals measure grid resolution, not statistical uncertainty. Lines joining sampled means guide the eye.



**Fig. 11.** Shadow-price staircases  $w^*(B)$  on three interleaved (non-observe-then-commit) settings, RockSample[5,3], RockSample[7,4], and Inspection- $N=16$ , with crossing brackets marked as vertical bars, capped where they reach the dotted rule marking the top of the tested  $w$  grid, and an open-circle marker where a budget is slack (the grid weight whose usage comes nearest to the budget sits strictly below the bracket’s lower edge, here  $w=0$  on the lowest RockSample[5,3] budget). Every marker plots the nearest-point solution  $\hat{w}(B)$ , which can sit on a bracket’s lower edge. The budget axis is logarithmic and the weight axis uses a symmetric-log scale, linear near zero. The underlying usage curves  $U(w)$  are monotone at every sampled grid point on RockSample[7,4] and Inspection- $N=16$ , while RockSample[5,3] dips once near  $w=0.316$  before rising (see text).

The instrumental floors are large on all three instances ( $U(0) = 3.70$  on RockSample[5,3], 2.00 on RockSample[7,4], and 22.38 on Inspection- $N=16$ ), which confirms that this unachievable regime, the one that left Tiger’s budget unbracketed in Table 4, matters in interleaved settings too. RockSample[5,3] is again the exception, because there a positive weight buys less sensing than  $w=0$  does, so budgets between 3.00 and 3.70 are achievable even though they lie below  $U(0)$ . The lowest RockSample[5,3] budget in Figure 11 lies in this band. The figure draws it as slack because the grid weight whose usage comes nearest to that budget is  $w=0$  itself, which sits below the budget’s crossing bracket, not because the budget is unachievable.

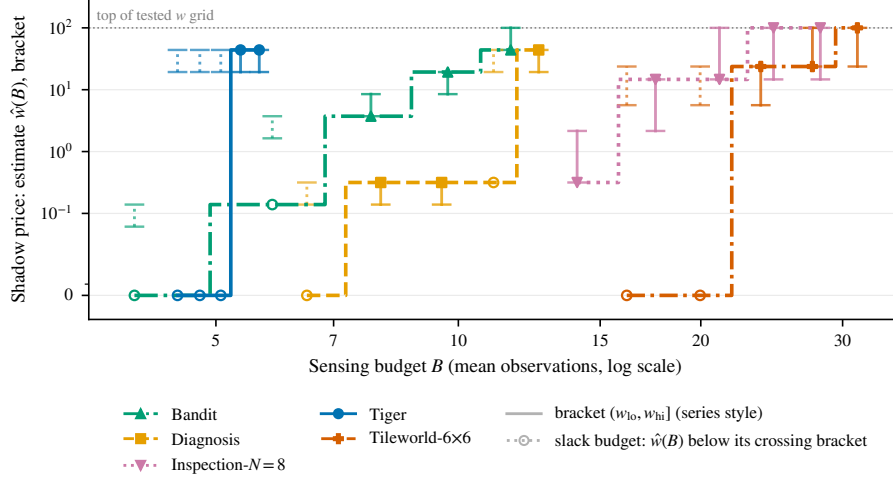
## 6.6 Shadow-Price Staircases across Five Domains

The observe-then-commit domains and the smaller Inspection instance now receive the same treatment as the interleaved instances. Figure 12 reports  $w^*(B)$  over identifiable budgets inside  $[U_{\min}, U_{\max}]$  for Tiger, Diagnosis, Bandit, Tileworld- $6 \times 6$ , and Inspection- $N=8$ , where  $U_{\min}$  and  $U_{\max}$  are the endpoints of each

curve’s usage range (Section 6.4). An identifiable budget is one placed strictly inside that range by the rule of that section, evenly spaced with the same margin held back at each end as in Section 6.5, so that the grid brackets it. The monotonicity at issue is that of the underlying usage curve  $U(w)$  from which each staircase is read off, taken between consecutive sampled grid weights. On Tiger, Diagnosis, and Inspection that curve is nondecreasing between every pair of consecutive grid weights. On Bandit and Tileworld it is not, showing local non-monotonicity from discrete policy switches, the first of the two obstacles noted after Proposition 5. Crossing brackets with standard errors are what we report for every domain (Definition 2). The standard error attached to a bracket is that of usage at a fixed weight, and the bracket’s width is set by the grid, so neither says how often a different sample of seeds would select a different bracket. We measured that directly. Resampling the five seeds with replacement 2000 times and re-solving each of the 25 budgets of Figure 12 on the resampled curve (`run_bracket_stability.py`, `results_price_bracket_stability.csv`, with the count of every bracket outcome in `results_price_bracket_stability_counts.csv`), the reported bracket is recovered in every resample at 19 budgets and in at least 99 percent of resamples at 21. The lowest agreement is 0.789, at Inspection’s lowest budget. In every disagreeing resample but one, the alternative is a neighbouring step of the staircase or, at Tiger’s lowest and highest budgets, which lie about one standard error of usage from the two ends of its usage range, no bracket at all. On Bandit and Tileworld a budget can be crossed more than once, and the reported bracket estimates the last crossing by the convention that definition adopts, so it names one threshold at grid resolution rather than displaying the reversals. Bandit’s budget  $B=5.87$  is the one tested case. Usage first reaches it at  $w=0.139$  (6.19 observations), falls back to 5.11 at  $w=0.72$ , and reaches it for the last time inside the reported bracket  $(1.64, 3.73]$  (`results_price_shadow_curves.csv`), which does not show the earlier crossing. Tiger’s price is weakly identified, consistent with Table 4, in the sense that the budget pins the price down only loosely. Instrumental listening already saturates most of the usage range at  $w=0$ , so the curve is flat at its floor across the low-weight range and rises in a single step at high weight. Every tested budget lands in that one step’s bracket.

## 6.7 SARSOP as a Near-Optimal Offline Reference

The staircases of the previous subsection price sensing, but they say nothing about how the reward the family attains compares with the best reward available. This subsection makes that comparison on the three core observe-then-commit environments, setting the canonical  $w=1$  agent’s reward and sensing usage against those of SARSOP [32], the offline point-based solver named in the related-work survey, which serves here as a near-optimal reference. The mechanics come first. We build the original APPL C++ SARSOP toolkit from source (Appendix T.1) and export the three benchmarks to the Cassandra `.pomdp` format, the plain-text model file that offline POMDP solvers read, through a single



**Fig. 12.** Operational shadow-price staircases  $w^*(B)$  across five observe-then-commit and inspection domains (Tiger, Bandit, Diagnosis, Inspection- $N=8$ , Tileworld- $6\times 6$ ), each in its own color, with a logarithmic budget axis and a symmetric-log weight axis that is linear near zero. Each staircase steps up as the sensing budget  $B$  rises, and the vertical bar at each step marks that budget’s crossing bracket  $(w_{lo}, w_{hi})$  (Definition 2), capped where the bracket’s upper edge coincides with the top of the tested  $w$  grid (dotted horizontal rule at 100). An open circle with a dotted bar marks a slack budget, one for which the grid weight whose usage comes nearest to the budget lies strictly below the lower edge of the crossing bracket ( $w=0$  at seven of the nine marked budgets,  $w=0.316$  at Diagnosis  $B=11.06$ , and  $w=0.139$  at Bandit  $B=5.87$ , the twice-crossed budget discussed in the text), so the nearest-point solution returns a weight below the bracket while the dotted bar still marks the crossing. Every marker plots that nearest-point solution  $\hat{w}(B)$ , which can also sit on a bracket’s lower edge. The usage curves behind Tiger, Diagnosis, and Inspection never step downward between consecutive sampled weights, while Bandit and Tileworld show local non-monotonicity from discrete policy switches, so their brackets mark the last crossing of each budget by the convention of Definition 2.

**Table 5.** SARSOP near-optimal reference vs. EFE ( $w=1$ ) on the three core OTC environments (mean  $\pm$  SE reward, mean usage  $\pm$  SE in observation count,  $\gamma=0.999$ ).

| Environment | SARSOP reward      | EFE reward         | SARSOP usage      | EFE usage         |
|-------------|--------------------|--------------------|-------------------|-------------------|
| Tiger       | $5.061 \pm 0.158$  | $5.061 \pm 0.158$  | $4.323 \pm 0.045$ | $4.323 \pm 0.045$ |
| Diagnosis   | $-1.452 \pm 0.336$ | $-1.217 \pm 0.184$ | $9.68 \pm 0.12$   | $9.73 \pm 0.13$   |
| Bandit      | $6.280 \pm 0.134$  | $6.261 \pm 0.112$  | $5.16 \pm 0.09$   | $5.09 \pm 0.07$   |

export path that works for any of the three because it reads the same hidden-state, action-layout, and observation-model definitions the agents already use. We then solve to precision  $10^{-3}$ , meaning the solver stops once the gap between its upper and lower bounds on the value of the initial belief falls below that threshold (under one second per environment at  $\gamma=0.999$ ).

What the solver returns is an alpha-vector policy, a finite set of vectors over states, each of which defines a linear function of the belief. The value of a belief is the largest of these linear functions evaluated at that belief, and the vector that attains the maximum names the action to take. We evaluate that policy in-simulator through the same episode runner used for the other observe-then-commit agents, so the agents compared in Table 5 share mechanics and metrics exactly. Table 5 reports the full protocol (5 seeds  $\times$  500 episodes).

On Tiger, SARSOP and EFE select identical actions on every episode. Tiger offers only one observation action, so both policies face the same listen-or-commit choice at every belief. On that choice the near-optimal policy and the EFE policy coincide exactly. On Diagnosis and Bandit the two show no statistically significant reward difference at our sample sizes, with the Diagnosis means differing by about 0.6 SE, and their usage differs by less than 0.1 observations (Table 5).

A non-significant difference is not evidence of equivalence, since a small sample can fail to reject any difference at all. We therefore formalize the comparison with a fixed-margin two one-sided tests (TOST) procedure [53]. TOST reverses the burden of proof. It posits a margin and tests two null hypotheses in parallel. One null says the true reward difference is at least as large as the margin in the positive direction, and the other says it is at least as large as the margin in the negative direction. Rejecting both nulls places the true difference inside the margin. The margin was set equal to the cost of one sensing action in each environment, which is  $\pm 1.0$  reward for Tiger and Diagnosis and  $\pm 0.5$  for Bandit, a value fixed by each environment’s own cost parameter. The test itself was added after an earlier comparison of the same two policies on the same canonical seeds, one without any equivalence test, had already been reported, so the margin was chosen with that comparison known and the five-seed test is retrospective rather than prospectively predeclared. The fifteen seeds that the  $n=20$  robustness study below adds to the canonical five were fixed before that study was run, after the margin, and are the prospective part of this evidence. It is an operational margin for this domain, not a universal equivalence standard.

Two choices in the test mechanics need stating, the unit of analysis and the decision not to pair. Each of the two one-sided tests is an unpaired Welch  $t$ -test on per-seed means ( $n=5$  seeds, 500 episodes per seed) rather than on pooled episode-level values, so degrees of freedom reflect independent seeds. We do not pair by seed index even though EFE and SARSOP share a nominal seed list. The reason is not that shared seeds leave the trajectories uncorrelated, which does not follow in general, since  $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y)$  and a shared seed can induce positive covariance. On Tiger the two policies take identical actions on every episode, so their trajectories are perfectly correlated and a paired test would report a zero difference with zero variance. On Diagnosis and Bandit the two policies diverge on many episodes and consume the environment’s random draws in different orders, so the covariance is partial and unknown. Treating the two samples as independent is the conservative choice when the covariance is positive, since it can only overstate the variance of the difference and so make the equivalence test harder to pass, and it is what the reported  $p$ -values assume. Pairing would be the sharper design where the randomness is genuinely shared, and a paired analysis of the per-seed differences is reported below as a sensitivity check, next to the  $n=20$  robustness study.

All three environments reject both one-sided nulls at  $\alpha=0.05$ , where the reported TOST  $p$ -value, written  $p_{\text{TOST}}$  below, is the larger of the two one-sided  $p$ -values (Tiger  $p=0.0010$ , Diagnosis  $p=0.0458$ , Bandit  $p=0.0130$ ). The same fact viewed as an interval is that the 90% CI of each mean difference lies entirely inside the margin (Tiger  $[-0.41, +0.41]$ , Diagnosis  $[-0.51, +0.98]$ , Bandit  $[-0.35, +0.31]$ ). The interval is 90% because two one-sided tests at level  $\alpha$  together correspond to a two-sided interval at confidence  $1 - 2\alpha$ . The three tests survive Holm–Bonferroni correction as a family. EFE at the canonical weight is therefore statistically equivalent to the SARSOP reference within the fixed margin on this suite (`results_tost_sarsop.csv`).

Five seeds is thin. We reran the identical procedure at  $n=20$  seeds, fixed before running rather than added post hoc, reusing the same SARSOP policies (`results_tost_sarsop_n20_robustness.csv`, per-seed means in `results_tost_sarsop_n20_robustness_per_seed.csv`). All three environments remain equivalent, and  $p_{\text{TOST}}$  tightens by roughly six, five, and eight orders of magnitude (Tiger  $4.5 \times 10^{-10}$ , Diagnosis  $6.8 \times 10^{-7}$ , Bandit  $6.0 \times 10^{-11}$ ). The intervals narrow accordingly, with Tiger’s 90% CI shrinking from  $[-0.41, +0.41]$  to  $[-0.21, +0.21]$ . The paired sensitivity check promised above runs the same two one-sided tests on the five per-seed differences (`results_tost_sarsop.csv`, columns `p_tost_paired` and companions, per-seed means in `results_tost_sarsop_per_seed.csv`). On Tiger the differences are identically zero, so the paired test is degenerate and equivalence holds trivially. On Diagnosis the paired  $p_{\text{TOST}}$  is 0.016 against the unpaired 0.046, and on Bandit it is  $3.6 \times 10^{-5}$  against 0.013, with the paired standard errors of the difference (0.24 and 0.03) well below the unpaired ones (0.38 and 0.18). The shared-seed covariance is therefore positive on both environments, so the unpaired analysis reported above (`results_`

`tost_sarsop.csv`) is the conservative one and the equivalence conclusion does not depend on the choice.

Two negative findings about alternative reference comparisons belong here, one a comparison that turns out to be ill-posed on Tiger and one a solver that was considered and declined because it targets a problem class this suite does not contain. Table 5 compares the two policies at  $w=1$ , where they need not spend the same amount of sensing, so a comparison at equal usage would put them on the same footing. A usage-matched comparison would pick the weight at which the Planning+IG family spends as much sensing as SARSOP does, which means solving  $U(w)$  equal to SARSOP’s usage and comparing there instead of at  $w=1$ . Inverting the offline usage curve in this way is ill-posed on Tiger because of the curve’s shape (Table 4). The curve is flat at its floor across the whole low-weight range and rises only once on the tested grid, in a single wide step at high weight, which leaves two cases, a target usage below the floor and a target usage above it. In the first case, which is where SARSOP’s own target lands, SARSOP’s usage of 4.32 sits just below the curve’s floor of 4.37 (within one seed-level SE). The inversion therefore finds no crossing at all and falls back to the low end of the tested weight grid,  $w=0$ . In the second case, any target the curve can reach above that floor is bracketed only by that single step,  $w \in (19.3, 43.9]$ , and the flat curve cannot discriminate any further within that bracket. The fallback in the first case is why the usage-matched row in `results_sarsop_baseline.json` is labeled Planning+IG ( $w=0$ ), a note on data lineage for readers of the released artifacts rather than part of the finding. We also considered POMCPOW [61] and declined it, because it exists to handle continuous state, action, and observation spaces via progressive widening, and every benchmark here is a discrete POMDP. Discrete POMCP (Appendix S) is already in hand on Tiger, Diagnosis, Bandit, Tileworld, and RockSample.

## 6.8 A Near-Optimal Constrained-POMDP Reference

Section 6.7 compared the family against a reference that is free to spend whatever sensing it likes. The budgeted framing of Section 4 calls for a second reference, one that is optimal subject to a charge on sensing. A natural objection to the single-price family is that no exact constrained-POMDP reference bounds it from above. A constrained POMDP is a POMDP solved for the best reward attainable under a bound on an expected cost, here sensing usage. An exact solution of one would give a certified ceiling on the reward attainable at the family’s own sensing level, against which the family’s reward could be measured. We provide a near-optimal one for the three environments where SARSOP already serves as our near-optimal reference (Tiger, Diagnosis, Bandit, Section 6.7), by extending the same SARSOP export with a literal per-unit usage penalty in the reward model. Observation action  $j$  already carries its own cost  $\text{cost}_j$  in the reward model, and the penalty adds  $\lambda$  on top of it. Each observation action’s reward becomes  $-(\text{cost}_j + \lambda)$  for  $\lambda \geq 0$ . Solving the resulting POMDP to SARSOP’s precision ( $10^{-3}$ ) then gives, for that  $\lambda$ , a near-optimal solution to  $\max_{\pi} \mathbb{E}_{\pi}[R] - \lambda \mathbb{E}_{\pi}[U]$  over the full space of POMDP policies (not the additive Planning+IG

family). The charge stands in for the cap because this penalized problem is the literal Lagrangian of the budgeted problem (6) (Section 4.1), the relaxation that penalizes usage rather than rewarding information. An exact maximizer of the penalized objective at rate  $\lambda$  cannot be beaten in reward by any policy that spends no more sensing than it does, by the mirror image of the two-line argument Section 4.1 gives for the information floor. Hence it is optimal under a cap set at its own usage. Each penalty rate therefore yields a constrained reference at one budget, the usage that rate induces, and sweeping  $\lambda$  collects such references across a range of budgets. The reference is near-optimal rather than certified exact, because SARSOP is a point-based approximate solver and the reported reward and usage are themselves Monte Carlo estimates from simulated episodes, not closed-form quantities.

Before presenting the protocol, two definitions clarify what a penalty sweep can and cannot certify. Call the achievable region the set of pairs of expected usage and expected reward that some POMDP policy attains, and call its frontier the curve giving the best reward attainable at each usage level. For a finite discounted constrained MDP, strong Lagrangian duality holds [1]. That result covers finite state spaces, and the belief MDP here (the Markov decision process over beliefs that the POMDP is equivalent to) has a continuous state space. We therefore do not assume a zero duality gap carries over. The distinction matters for how much of the frontier a sweep can reach. Strong duality would mean that every budget’s constrained optimum is attained by the penalized problem at some  $\lambda$ , so a fine enough  $\lambda$  sweep would recover the whole frontier. A duality gap means that some budgets have a constrained optimum that no  $\lambda$  picks out, and a sweep over  $\lambda$  misses those points. What the sweep delivers holds either way. The swept points are supported points of the achievable region, where a supported point is one that a linear trade-off between reward and usage picks out of the achievable set. Each swept point is the usage and reward that an actually solved policy attains. They therefore estimate the frontier from below, each point up to the sampling error of its own reward and usage rather than as a certified population bound, and any duality gap makes the reference conservative rather than optimistic about what is achievable. Conservative here refers to frontier achievability alone. For an optimality comparison, however, an underestimated reference has the opposite effect. A reference that undershoots the constrained optimum makes our agents look closer to optimal than they are, so the measured gap to the reference is a lower bound on the true gap.

With the caveat stated, the protocol is a grid sweep over the penalty. Sweeping  $\lambda$  over a log-spaced 12-point grid ( $\lambda \in [0, 100]$ ,  $\gamma=0.999$ , 5 seeds  $\times$  300 episodes per  $\lambda$ ) traces an estimated upper concave envelope of the achievable ( $\mathbb{E}[U], \mathbb{E}[R]$ ) region, the frontier of the previous paragraph once mixtures of policies are allowed, sampled at these 12 supported points rather than continuously. A per-episode mixture of two adjacent swept policies approximates any intermediate budget, the same mixture argument used for the Planning+IG shadow price (Definition 2), though the approximation inherits whatever slack the constituent

near-optimal solutions carry.  $\lambda=0$  recovers the ordinary reward-only SARSOP solve of Table 5 as a sanity check.

**Table 6.** Estimated CPOMDP reference gap at the budget  $B_{\text{EFE}} = U(\text{EFE}, w=1)$ .  $R_{\text{ref}}$  is the estimated feasible envelope of the Lagrangian-sweep reference at that usage, the largest reward attainable by a per-episode mixture of the sampled near-optimal SARSOP-Lagrangian policies whose expected usage, the mixture’s usage-weighted average, is at most  $B_{\text{EFE}}$ , the same definition Section 6.9 uses (not a certified bound, column **R\_ref** in **results\_cpomdp\_baseline.csv**, with the earlier equal-usage interpolation kept as the diagnostic column **R\_interp\_legacy**). The gap is  $R_{\text{ref}} - R_{\text{EFE}}$ . Planning+IG at the usage-matched grid weight actually run (not  $w=1$ , but 0.00, 0.45, and 1.07 for Tiger, Diagnosis, and Bandit respectively, each the nearest grid point on a lighter usage curve rather than a bracket, and each lying inside a run of consecutive sampled weights on which the shadow-price usage curve of **results\_price\_usage\_curves.csv** is constant, a run that also straddles  $w=1$ , namely  $[0, 19.3]$  on Tiger,  $[0.316, 19.3]$  on Diagnosis, and  $[0.72, 1.64]$  on Bandit) attained reward bit-identical to EFE’s on all three environments in this run (**results\_cpomdp\_baseline.csv**, where **R\_PIG** matches **R\_EFE** to the precision stored), an empirical finding about this policy family and this suite, not a consequence of Proposition 1, which is stated for  $w=1$  specifically. A single gap column covers both.

| Environment | $B_{\text{EFE}}$ | $R_{\text{ref}}$ | $R_{\text{EFE}}$ | Gap    | Gap %  |
|-------------|------------------|------------------|------------------|--------|--------|
| Tiger       | 4.32             | 5.016            | 5.016            | 0.000  | 0.00%  |
| Diagnosis   | 9.82             | -1.265           | -1.224           | -0.041 | -3.27% |
| Bandit      | 5.10             | 6.314            | 6.320            | -0.005 | -0.08% |

Table 6 reports the comparison. The reference at  $B_{\text{EFE}}$  is the estimated feasible envelope of the sampled reference policies, the best reward attainable by a per-episode mixture of sampled policies whose expected usage, the mixture’s usage-weighted average, is at most  $B_{\text{EFE}}$ , so every sampled policy that uses less than EFE does stays a feasible comparator (**rho\_aif/budget.py**, **feasible\_envelope**, shared with Section 6.9). On all three environments the envelope selects a single sampled policy rather than a mixture: the  $\lambda=0$  policy on Tiger and Bandit, and the  $\lambda=0.05$  policy on Diagnosis. In each case the highest-reward sampled policy on the frontier is itself feasible at  $B_{\text{EFE}}$  (its usage equals  $B_{\text{EFE}}$  on Tiger and lies below it on Bandit and Diagnosis, where  $B_{\text{EFE}}$  exceeds every sampled usage), so the usage cap does not bind and the envelope is the largest sampled reward. The estimated gap is zero on Tiger and  $-0.08\%$  on Bandit, smaller in magnitude than the reward change between adjacent swept points. Diagnosis shows a small negative gap ( $-3.27\%$ , i.e., the reference value is slightly below  $R_{\text{EFE}}$ ). The population constrained optimum is at least the expected reward of any feasible policy. A negative estimated gap can reflect sampling error on either side or approximation error in the reference, rather than a violation of that bound. The reference reward  $R_{\text{ref}}$  and EFE’s reward  $R_{\text{EFE}}$  are 1,500-episode Monte Carlo estimates with standard errors ( $\pm 0.12$ – $0.39$ ) several times

the gap, and the best feasible sampled Diagnosis policy uses 9.785 observations against EFE’s 9.824, so a finer  $\lambda$  grid near 0 and more episodes per point would sharpen the estimate. We report the current resolution rather than tuning the grid to close the gap. An earlier version of this table read the reference by equal-usage interpolation between the two sampled points bracketing  $B_{\text{EFE}}$ , clamped to the boundary value where  $B_{\text{EFE}}$  fell outside the sampled range, which is not the reference the constrained problem defines. That lookup put the Diagnosis reference at  $-1.297$  (a gap of  $-5.65\%$ ) and Bandit’s at  $6.313$  ( $-0.11\%$ ), and it is kept in the CSV as a diagnostic column only.

Read together, the TOST result in Table 5 and the constrained reference illuminate two distinct aspects of the comparison. The SARSOP comparison shows that  $w=1$  is statistically equivalent, within the fixed margin, to a near-optimal policy’s reward when sensing is left unconstrained, which is a single reward-maximizing operating point. The constrained reference compares rewards at the sensing level EFE actually uses,  $B_{\text{EFE}}$ . There the near-optimal reference estimates the best reward available at that usage, and EFE’s measured reward is empirically close to it. Because Planning+IG calibrated to that budget attains the same reward as EFE in this run (Table 6), the statement covers the calibrated Planning+IG family as well as  $w=1$ , on this three-environment suite (`results_cpomdp_baseline.csv`, `results_cpomdp_frontier.csv`, `experiments/run_cpomdp_baseline.py`).

The reference’s scope is deliberately narrow. It does not extend to RockSample or Structural Inspection’s larger factored state spaces, where SARSOP itself is no longer a practical solver at the sizes we study (Section 4.6 and Appendix O), so even an approximate optimality-gap estimate there remains future work. The comparison at  $B_{\text{EFE}}$  measures the family’s distance from near-optimal reward at the endogenous weight-one usage. The next subsection moves to budgets set by a predeclared rule from the calibration curve, independently of weight-one usage. It asks whether calibration meets those expected-usage targets on held-out seeds and what reward it earns. The subsection after it turns from how much reward the family attains to what the information term’s sensing is about.

## 6.9 Calibration-Derived Target Budgets and What Calibration Buys

The constrained reference above is compared at the endogenous budget  $B_{\text{EFE}}$ , the usage the  $w=1$  agent happens to spend. In an application, the designer would set the budget independently of the weight-one policy. This subsection tests the calibration procedure at target budgets that stand in for such a statement, set by a predeclared rule from each calibration curve’s range and largest jump, independent of the weight-one usage though not application-specified, on seeds it never saw, and it asks two separate questions that the earlier comparison ran together. Does the procedure meet the stated expected usage on held-out seeds, and what reward does meeting it cost or earn relative to the alternatives. The study was predeclared in the producer’s docstring before any result was inspected (`experiments/run_budget_frontier.py`), and Table 7 reports it.

The design is as follows. The three core environments are calibrated on the canonical five seeds, 100 episodes per seed, on the same 16-point weight grid and at the same horizons as the shadow-price curves of Section 6.6, recording usage and reward at every grid weight. Budgets are then set from each calibration curve’s attainable range  $[U_{\min}, U_{\max}]$  at 0.25, 0.5, and 0.75 of the range, at the midpoint of the curve’s largest jump (a gap budget that no single tested weight attains), and at half of  $U_{\min}$  (an unattainable budget, which the procedure must declare rather than approximate). Every selection is made on the calibration data alone. The crossing bracket and its mixture probability  $q$  come from Definition 2, and the best feasible single member is the grid weight with the highest calibration reward among those whose calibration usage is at or below  $B$ . Two further policies were added after the first results had been inspected, in response to a re-review, and are disclosed as such in the producer’s docstring. The best target mixture is the reward-maximizing mixture of calibration-grid weights whose calibration usage equals  $B$  exactly, and the best feasible mixture is the same program with calibration usage at most  $B$ . Each is a linear program over the calibration grid with at most two support weights, each is selected on the calibration data alone, and each is then run on the held-out seeds like the others. Neither says anything about weights or mixtures the grid does not sample. Each selected policy is then run on five held-out seeds,  $\{7, 8, 9, 10, 11\}$ , for 100 episodes each. The reference is the near-optimal constrained-POMDP reference of Section 6.8, taken as its estimated feasible envelope at  $B$ . Under the reference problem, maximize reward subject to expected usage at most  $B$ , every sampled reference policy whose usage is at most  $B$  remains feasible, so the envelope is the largest reward attainable by mixing the sampled reference points under that constraint, a linear program over the committed Lagrangian frontier whose solution mixes at most two points. It is defined at every attainable budget, and once  $B$  exceeds the largest sampled reference usage the constraint is slack and the envelope is flat at the largest sampled reward. On Tiger every attainable budget lies above the reference’s largest sampled usage of 4.32, so the envelope is 5.02 at all four budgets. On Bandit the three interior budgets are slack, with envelope 6.31, and the gap budget mixes two reference points to 6.23. On Diagnosis the envelope is  $-2.00$ ,  $-1.35$ ,  $-1.27$ , and  $-1.95$  at the 0.25, 0.5, 0.75, and gap budgets. An earlier version of this comparison read the reference by linear interpolation between sampled points at equal usage and left it blank outside the sampled usage range, which omitted eight feasible comparators and interpolated through points below the envelope. That interpolation is kept in the CSV as a diagnostic column (`frontier_interp_at_B`) and is no longer the comparator. The reference is at once the constrained reference and the direct usage-penalty method for these domains, since it is built from a Lagrangian penalty on SAR-SOP. All three unattainable budgets were declared. Tiger’s gap budget coincides with its 0.5 budget, since its curve has a single jump. Bandit’s calibration curve is not monotone, so the last-crossing convention of Definition 2 applies there. It changes the reported bracket at the 0.25 budget of 5.51, which the curve crosses twice, once between  $w=0.06$  and  $w=0.14$  and again between  $w=1.64$  and

$w=3.73$  after dipping to a usage of 5.11 on the plateau  $w \in [0.72, 1.64]$ , and the reported bracket is the later one,  $(1.64, 3.73]$ . At the gap budget of 4.71 the curve crosses only once, since the plateau’s usage stays above the budget, so the bracket  $(0.06, 0.14]$  is the only candidate.

The held-out usage results answer the first question directly. At all twelve attainable budgets, the endpoint mixture meets its expected-usage target on the held-out seeds to within 0.13 observations, with errors of 0.03 to 0.08 on Tiger, 0.02 to 0.12 on Diagnosis, and 0.09 to 0.12 on Bandit, each within one held-out standard error except at Bandit’s gap budget, where the error of 0.09 is about 1.3 standard errors. No single grid weight comes close. The bracket endpoints miss the budget by 0.3 to 1.1 observations on Tiger and by 0.2 to 3.6 elsewhere, and the best feasible single member, whose calibration usage is at most the budget by selection and whose held-out usage stays below the budget on all twelve rows here, a result rather than a guarantee, leaves 0.2 to 4.7 observations of it unspent. An expected budget is not a per-episode cap. The mixture’s held-out mean usage exceeds  $B$  on three of the twelve budgets, by at most 0.13 observations. Such deviations are possible when targeting a population expectation with a finite sample. A designer who needs a hard per-episode cap needs a different mechanism.

The reward comparison shows the limits of usage calibration. Where extra sensing pays in reward, meeting the budget earns more than leaving it unspent. On Diagnosis at  $B=7.72, 9.53$ , and the gap budget 7.84, the mixture earns  $-2.14, -2.11$ , and  $-2.06$  against the best feasible member’s  $-2.86$ , and on Bandit’s gap budget 4.71 it earns 5.62 against 5.51. Where the budget exceeds the usage that reward alone justifies, meeting it costs reward, because the family spends the budget on observations that lower expected return. On Tiger the best feasible member, which is reward-only planning at 4.32 observations, earns 5.68 at every budget while the mixture earns 5.36, 4.96, and 4.65 as the budget rises, and on Bandit at  $B=7.79$  and 10.07 the best feasible member earns 6.16 against the mixture’s 5.70 and 4.95. The remaining two budgets, Diagnosis at  $B=11.35$  and Bandit at  $B=5.51$ , also lie above the usage that reward alone justifies, and there the mixture sits nominally below the best feasible member ( $-2.31$  against  $-2.22$ , and 6.06 against 6.16) by less than one held-out standard error. The best feasible mixture over the calibration grid coincides with the best feasible member wherever the usage constraint is slack, which is every Tiger budget, the three interior Bandit budgets, and Diagnosis at  $B=11.35$ , and with the selected endpoint mixture at the other three Diagnosis budgets, where the constraint binds. At Bandit’s gap budget, where it also binds, it coincides with the best target mixture described next. The best target mixture is the selected endpoint mixture itself on eleven of the twelve budgets, up to endpoints that sit on the same usage plateau and so run the same policy. The exception is Bandit’s gap budget, where the reward-maximizing mixture that meets the target on the calibration grid mixes  $w=0$  and  $w=0.72$  rather than the two bracket endpoints, and on the held-out seeds it earns  $5.94 \pm 0.19$  against the selected mixture’s  $5.62 \pm 0.21$  while missing the usage target by 0.18 observations where the se-

lected mixture misses it by 0.09. The selected endpoint mixture is therefore not always the best mixture that meets the target even on the tested grid, and the family’s shortfall at that budget cannot be assigned to its planning depth on this evidence. Against the reference envelope, the mixture’s reward sits 0.14, 0.76, 1.05, and 0.10 below it on Diagnosis at the 0.25, 0.5, 0.75, and gap budgets, which is 0.3, 1.1, 2.3, and 0.2 held-out standard errors of the mixture, and 0.26, 0.62, 1.36, and 0.61 below it on Bandit, about 4%, 10%, 22%, and 10% of the reference reward. On Tiger the mixture’s reward is 0.34 above the envelope at the smallest budget and 0.05 and 0.36 below it at the two larger ones. These comparisons are descriptive rather than tests, and they compare estimates from different evaluation streams, since the reference was evaluated on the canonical seeds with once-per-seed seeding and the family on the held-out seeds with per-episode seeding. The stream effect is not small. Re-evaluated on the held-out stream, Tiger’s unpenalized reference policy scores 5.68 rather than 5.02, the same reward as the best feasible member, which runs the reward-only policy. A same-stream comparison, added after the results above had been inspected and in response to a referee, therefore re-evaluated every sampled reference policy on the held-out seeds under the family’s per-episode seeding, after checking that each saved policy reproduces the committed reference to  $10^{-9}$  on the canonical stream, and recomputed the envelope there, which pairs each seed’s family reward with a reference reward on the same episode seeds (the last two columns of Table 7, `experiments/run_frontier_reference_heldout.py`, `results_budget_frontier_heldout_reference.csv`). The reference support and mixture weights are fitted on these same held-out seed means and held fixed when computing the paired standard errors. The resulting intervals are nominal, pointwise plug-in summaries. They do not propagate support reselection, uncertainty in mixture weights or cap feasibility, or multiplicity across budgets. They describe this fitted comparison and do not establish coverage for the population constrained frontier. On that comparison the mixture falls short of the envelope with a nominal pointwise plug-in 95%  $t$  interval that excludes zero on nine of the twelve attainable budgets, eight of the eleven distinct ones since Tiger’s gap budget coincides with its 0.5 budget. The shortfalls are 0.33, 0.72, and 1.03 at Tiger’s 0.25, 0.5, and 0.75 budgets, 0.76 and 1.10 at Diagnosis’s 0.5 and 0.75 budgets, and 0.57, 1.32, and 0.59 at Bandit’s 0.5, 0.75, and gap budgets, the largest being about a fifth of Bandit’s reference reward. The intervals at the other three, Diagnosis’s 0.25 and gap budgets and Bandit’s 0.25 budget, include zero ( $-0.03 \pm 0.28$ ,  $+0.00 \pm 0.22$ , and  $-0.21 \pm 0.12$ ). The Tiger shortfalls carry a rare-event caveat. Neither the family nor any reference policy in the envelope’s support commits wrongly in its 500 held-out Tiger episodes, so every per-seed reward there is 10 minus the observation count and the paired Tiger gaps measure only the extra listening the family spends (`results_budget_frontier.csv`, `results_cpomdp_frontier_heldout.csv`). A wrong commit costs 110 reward units against a correct one, and the canonical stream records the reference committing wrongly at a rate of 0.6 percent (success 0.994, `results_cpomdp_frontier.csv`), worth about 0.66 in expected reward, nearly all of the 0.67 difference between the refer-

ence’s canonical and held-out rewards. In the extreme case where the reference erred at that rate and the family never did, the 0.25 shortfall would reverse and the 0.5 shortfall would nearly vanish, so on Tiger only the 0.75 shortfall is robust to the rare event. Tiger’s nominal excess at its smallest budget was therefore the stream effect, not a family reward above the reference. The same-stream envelope is still a maximum over noisy sampled points, so it removes the stream mismatch but not the envelope’s selection bias, and the reference remains near-optimal and estimated rather than exact.

The reading is therefore this. The procedure calibrates expected sensing usage within a fixed policy family, and it does that accurately on held-out data, at gap budgets and interior budgets alike, while declaring unattainable budgets. It does not solve the constrained problem (6) of maximizing reward subject to a usage cap, as Section 4.1 already stated, and this study puts a number on the difference. When a stated budget is a resource the designer wants spent, for coverage, monitoring, or verification reasons that the reward does not capture, the calibrated mixture spends it to within 0.13 observations. When the budget is only an upper limit and reward is the objective, the best budget-feasible mixture over the grid, or the constrained reference’s feasible envelope where one can be solved, is the right comparison, and the family’s reward at the budget can fall short of both. The budget-first framing of this article is the first case. The second case is the constrained-POMDP problem, which this article does not claim to solve.

## 6.10 Distractor Robustness under Reward-Irrelevant Sensing

A recurring question about information-gain objectives is how they behave when uncertainty exists over state components that do not affect reward. The question belongs in this section because it marks the boundary of what the sensing budget of Section 4 can do on its own, since a budget fixes how much sensing is bought and leaves open what that sensing is about. The experiment returns two findings, developed in turn below. Ordinary information gain does spend part of its budget on a reward-irrelevant test once the weight is large enough. A variant that scores information gain only on the reward-relevant part of the state takes that spending to exactly zero at every swept weight, using nothing the agent does not already hold. To measure the effect we construct `DistractorDiagnosisEnv`, an 8-joint-state Diagnosis variant. Its hidden state pairs a condition (the variable the commit action must identify, as in Diagnosis) with an independent binary nuisance bit, so the joint state space is 4 conditions  $\times$  a binary nuisance bit, 8 joint states in all. Commit rewards depend only on the 4-way condition, and the environment adds one test (an observation action) that is informative only about the nuisance bit, which we call the distractor test. Every observation model in the environment factors through exactly one of the two independent components, so the distractor test carries exactly zero information about the condition marginal (the agent’s belief over the condition alone), a property that Appendix T.2 states and verifies by unit test.

**Table 7.** The calibrated family at calibration-derived target budgets, set by a predeclared rule from each calibration curve and evaluated on five held-out seeds it was not calibrated on (100 episodes per seed, seed-level SE). Budgets are set from each calibration curve’s attainable range  $[U_{\min}, U_{\max}]$  at three fractions of the range, at the midpoint of the largest jump (gap), and below  $U_{\min}$  (declared unattainable, not run). The mixture plays the bracket’s upper endpoint with probability  $q$  and its lower endpoint otherwise, drawn afresh each episode. Usage is the observation count. See `results_budget_frontier.csv` and `results_budget_frontier_heldout_reference.csv`.

| Env.      | Budget           | $B$   | $q$  | Mixture usage         | Mixture reward   |
|-----------|------------------|-------|------|-----------------------|------------------|
| Tiger     | 0.25 of range    | 4.72  | 0.25 | $4.64 \pm 0.10$       | $5.36 \pm 0.10$  |
| Tiger     | 0.50 of range    | 5.07  | 0.50 | $5.04 \pm 0.14$       | $4.96 \pm 0.14$  |
| Tiger     | 0.75 of range    | 5.42  | 0.75 | $5.35 \pm 0.14$       | $4.65 \pm 0.14$  |
| Tiger     | gap              | 5.07  | 0.50 | $5.04 \pm 0.14$       | $4.96 \pm 0.14$  |
| Tiger     | below $U_{\min}$ | 2.18  | –    | declared unattainable |                  |
| Diagnosis | 0.25 of range    | 7.72  | 0.47 | $7.70 \pm 0.25$       | $-2.14 \pm 0.48$ |
| Diagnosis | 0.50 of range    | 9.53  | 0.94 | $9.47 \pm 0.20$       | $-2.11 \pm 0.67$ |
| Diagnosis | 0.75 of range    | 11.35 | 0.46 | $11.47 \pm 0.37$      | $-2.31 \pm 0.45$ |
| Diagnosis | gap              | 7.84  | 0.50 | $7.86 \pm 0.25$       | $-2.06 \pm 0.47$ |
| Diagnosis | below $U_{\min}$ | 2.95  | –    | declared unattainable |                  |
| Bandit    | 0.25 of range    | 5.51  | 0.18 | $5.62 \pm 0.21$       | $6.06 \pm 0.26$  |
| Bandit    | 0.50 of range    | 7.79  | 0.29 | $7.67 \pm 0.37$       | $5.70 \pm 0.25$  |
| Bandit    | 0.75 of range    | 10.07 | 0.17 | $9.95 \pm 0.36$       | $4.95 \pm 0.22$  |
| Bandit    | gap              | 4.71  | 0.50 | $4.62 \pm 0.07$       | $5.62 \pm 0.21$  |
| Bandit    | below $U_{\min}$ | 1.62  | –    | declared unattainable |                  |

We first measure how much of ordinary Planning+IG’s sensing goes to the distractor test as the weight grows, and the headline is that the distractor spend switches on at a weight we can bracket tightly. Planning+IG is swept over a sixteen-point log weight grid (5 seeds  $\times$  100 episodes). The quantity the grid is built to locate is the onset, the first weight at which the agent buys any distractor test. This is a switch-on point of the same kind as a knot of the usage staircase, here for distractor spending rather than for sensing overall. The grid itself is a coarse log grid with extra points inserted inside the one coarse step on which distractor spending first appears. Those extra points let the onset be located to within a fraction of that step rather than only to within the coarse step. The quantity tracked is the distractor fraction, the share of total sensing usage (all observation actions taken in an episode) that goes to the distractor test. That fraction is exactly 0 for  $w \leq 3.5$ , is  $0.103 \pm 0.001$  at  $w=4$ , reaches  $0.233 \pm 0.003$  by  $w=5$ , and saturates in the 0.330–0.332 range for  $w \geq 31.6$  (seed-level SE 0.004–0.005, Figure 13). The onset therefore lies in the bracket  $(3.5, 4.0]$ , where the agent buys its first distractor test.

Why ordinary information gain is drawn to the distractor, and what repairs it, both follow from how the epistemic term is scored. Ordinary information gain

**Table 7.** The calibrated family (continued). Held-out reward comparisons within the calibration grid. Best member is the calibration-grid weight with the highest calibration reward among those whose calibration usage is at or below  $B$ . Target mixture is the reward-maximizing mixture of calibration-grid weights whose calibration usage equals  $B$ , and feasible mixture the same with usage at most  $B$ , each a linear program with at most two support weights, selected on the calibration data and run on the held-out seeds.

| Env.      | Budget           | Best member           | Target mixture   | Feasible mixture |
|-----------|------------------|-----------------------|------------------|------------------|
| Tiger     | 0.25 of range    | $5.68 \pm 0.08$       | $5.36 \pm 0.10$  | $5.68 \pm 0.08$  |
| Tiger     | 0.50 of range    | $5.68 \pm 0.08$       | $4.96 \pm 0.14$  | $5.68 \pm 0.08$  |
| Tiger     | 0.75 of range    | $5.68 \pm 0.08$       | $4.65 \pm 0.14$  | $5.68 \pm 0.08$  |
| Tiger     | gap              | $5.68 \pm 0.08$       | $4.96 \pm 0.14$  | $5.68 \pm 0.08$  |
| Tiger     | below $U_{\min}$ | declared unattainable |                  |                  |
| Diagnosis | 0.25 of range    | $-2.86 \pm 0.43$      | $-2.14 \pm 0.48$ | $-2.14 \pm 0.48$ |
| Diagnosis | 0.50 of range    | $-2.86 \pm 0.43$      | $-2.11 \pm 0.67$ | $-2.11 \pm 0.67$ |
| Diagnosis | 0.75 of range    | $-2.22 \pm 0.68$      | $-2.31 \pm 0.45$ | $-2.22 \pm 0.68$ |
| Diagnosis | gap              | $-2.86 \pm 0.43$      | $-2.06 \pm 0.47$ | $-2.06 \pm 0.47$ |
| Diagnosis | below $U_{\min}$ | declared unattainable |                  |                  |
| Bandit    | 0.25 of range    | $6.16 \pm 0.23$       | $6.06 \pm 0.26$  | $6.16 \pm 0.23$  |
| Bandit    | 0.50 of range    | $6.16 \pm 0.23$       | $5.70 \pm 0.25$  | $6.16 \pm 0.23$  |
| Bandit    | 0.75 of range    | $6.16 \pm 0.23$       | $4.95 \pm 0.22$  | $6.16 \pm 0.23$  |
| Bandit    | gap              | $5.51 \pm 0.10$       | $5.94 \pm 0.19$  | $5.94 \pm 0.19$  |
| Bandit    | below $U_{\min}$ | declared unattainable |                  |                  |

does not distinguish reward-relevant from reward-irrelevant uncertainty, so once the weight is large enough to make the distractor’s nonzero (but task-useless) information gain worth its cost, Planning+IG spends budget on it. The sensing budget of Section 4 sets a target for expected sensing usage but does not by itself redirect that usage toward reward-relevant tests. EFE at the canonical  $w=1$  happens to stay at zero distractor fraction on this instance, but only because the onset bracket  $(3.5, 4.0]$  lies above 1 here, not because EFE is structurally immune to the effect. The root cause is that  $I(b)$  is computed over the full joint belief rather than over the quantities the pending commit decision actually depends on. The repair is to score information gain on the reward-relevant marginal,  $\tilde{I}_a(b) = D_{\text{KL}}[Q(s_{\text{rel}} \mid o, a) \parallel Q(s_{\text{rel}} \mid a)]$ , restricted to the state components  $s_{\text{rel}}$  the commit reward depends on. This is not a new objective. It is the ordinary expected information gain  $I_a(b)$  of Section 3, here written as the divergence between the posterior and prior over  $s_{\text{rel}}$  and evaluated on the belief over  $s_{\text{rel}}$  instead of on the full joint belief. The only change is therefore which belief the epistemic term is scored on.

Identifying  $s_{\text{rel}}$  turns out to need no information the agent does not already hold. In the implementation,  $s_{\text{rel}}$  is the reward-equivalence class of the state, defined as follows. Two states are reward-equivalent when every commit action

**Table 7.** The calibrated family (continued). Constrained-reference comparisons. The reference is the estimated feasible envelope of the near-optimal constrained-POMDP reference of Section 6.8, the largest reward attainable by mixing its sampled points subject to expected usage at most  $B$ , flat at the largest sampled reward where  $B$  exceeds the largest sampled usage. That reference was evaluated on the canonical seeds with once-per-seed seeding. Same-stream reference is the same envelope with every sampled reference policy re-evaluated on the held-out seeds under the family’s per-episode seeding, and paired gap is the mixture’s reward minus that envelope, per seed, with its seed-level SE. Reference support and mixture weights are fitted on those same held-out seed means and held fixed for the SE. The nominal pointwise plug-in intervals discussed in the text omit uncertainty from that fit and cap feasibility, and make no multiplicity adjustment.

| Env.      | Budget           | Reference             | Same-stream<br>reference | Paired gap       |
|-----------|------------------|-----------------------|--------------------------|------------------|
| Tiger     | 0.25 of range    | 5.02                  | 5.68                     | $-0.33 \pm 0.04$ |
| Tiger     | 0.50 of range    | 5.02                  | 5.68                     | $-0.72 \pm 0.06$ |
| Tiger     | 0.75 of range    | 5.02                  | 5.68                     | $-1.03 \pm 0.06$ |
| Tiger     | gap              | 5.02                  | 5.68                     | $-0.72 \pm 0.06$ |
| Tiger     | below $U_{\min}$ | declared unattainable |                          |                  |
| Diagnosis | 0.25 of range    | -2.00                 | -2.11                    | $-0.03 \pm 0.28$ |
| Diagnosis | 0.50 of range    | -1.35                 | -1.35                    | $-0.76 \pm 0.15$ |
| Diagnosis | 0.75 of range    | -1.27                 | -1.21                    | $-1.10 \pm 0.22$ |
| Diagnosis | gap              | -1.95                 | -2.06                    | $0.00 \pm 0.22$  |
| Diagnosis | below $U_{\min}$ | declared unattainable |                          |                  |
| Bandit    | 0.25 of range    | 6.31                  | 6.27                     | $-0.21 \pm 0.12$ |
| Bandit    | 0.50 of range    | 6.31                  | 6.27                     | $-0.57 \pm 0.09$ |
| Bandit    | 0.75 of range    | 6.31                  | 6.27                     | $-1.32 \pm 0.08$ |
| Bandit    | gap              | 6.23                  | 6.21                     | $-0.59 \pm 0.20$ |
| Bandit    | below $U_{\min}$ | declared unattainable |                          |                  |

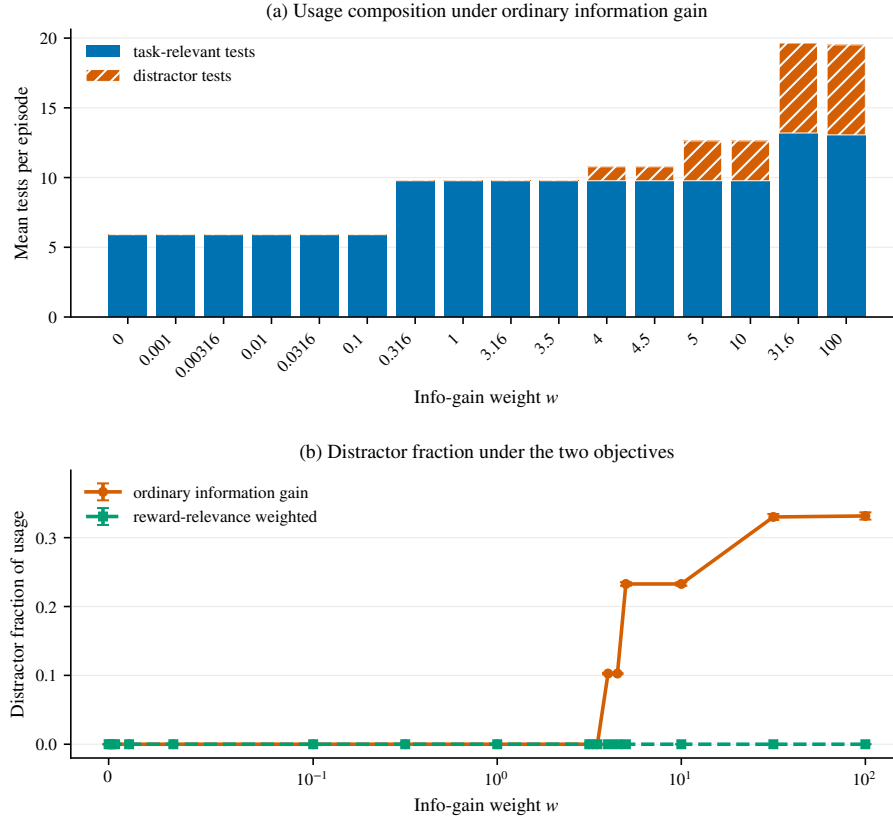
pays the same in both, so no belief mass moved between them can change which commit action is optimal. Reward-equivalence groups the states into classes, and that grouping (the partition) is read directly off the commit reward matrix the agent is given (Section 3). We implement the variant that way, as a partition into reward-equivalence classes with information gain scored on the class marginal, the belief over which class the state falls in. The classes play the role of  $s_{\text{rel}}$  in the formula above. On **DistractorDiagnosisEnv** the identification is concrete. The commit reward matrix has the same column for the two states that share a condition and differ only in the nuisance bit, and different columns for different conditions. The four reward-equivalence classes are therefore exactly the four conditions, and the class marginal is the condition marginal. Belief dynamics are untouched, since the finite-horizon lookahead (the continuation value the planner recurses over) always propagates the full joint posterior and only the term that scores information gain at each step changes. On an environment where no two

states are reward-equivalent the variant reduces exactly to ordinary information gain.

Sweeping ordinary information gain and the relevance-weighted variant over the same refined weight grid makes the effect a paired curve rather than a single point (Figure 13). The relevance-weighted agent takes exactly zero distractor tests at every one of the sixteen swept weights, against 6.46 tests and 33.2% of total usage for ordinary information gain at  $w=100$ . It also recovers the full reward cost of that spending, the reward that ordinary information gain gives up by buying distractor tests. Listing ordinary information gain’s mean reward first and the variant’s mean reward second at each weight, the pairs are  $-2.45$  against  $-1.45$  at  $w=4$ ,  $-4.31$  against  $-1.45$  at  $w=5$  and  $w=10$  (the two weights share the same pair of values), and  $-10.48 \pm 0.11$  against  $-3.64 \pm 0.36$  at  $w=100$  (seed-level SE). The variant’s refusal to buy distractor tests costs nothing in commit accuracy. Success rates for the two agents are identical through  $w=31.6$ , and the variant’s is nominally higher at  $w=100$  (99.2% against 98.4%, not significant). Seed-level Welch tests with Holm–Bonferroni correction over the weight-by-metric family (one test for each of the sixteen swept weights and each reported metric, corrected together as a single family) confirm the usage and distractor-count differences at every weight past onset, and the reward difference at  $w=31.6$  and  $w=100$  ( $p=1.2 \times 10^{-5}$  and  $1.5 \times 10^{-5}$ ). The smaller reward differences listed above at  $w=4$ ,  $w=5$ , and  $w=10$  do not survive the correction.

Relevance weighting also has a limit that matters here. At the two largest weights the relevance-weighted agent still runs 13.16 task tests, the tests that are informative about the condition (every test except the distractor test), where the same agent at  $w=10$  runs 9.77. It has therefore stopped buying reward-irrelevant information and has not stopped over-buying reward-relevant information. A sensing budget sets a target for expected sensing usage, how many tests are run. Relevance weighting decides what those tests are about. Neither substitutes for the other, and this environment needs both.

We also checked whether information-directed sampling (IDS) [48], which is explicitly reward-aware through its information-ratio numerator, is immune. That numerator is the squared expected regret of taking the observation action (Section 3). It is not immune, for an implementation reason worth reporting exactly. The denominator, information about the optimal commit (`info_astar` in the implementation), is correctly zero for the distractor action. The ratio is undefined when that denominator vanishes, and the implementation handles the case by falling back to a raw state-entropy reduction (`info_state`) whenever `info_astar` is at or near zero. That fallback lets the distractor’s genuine nuisance-bit informativeness through. Instrumented directly, IDS’s distractor fraction is  $0.305 \pm 0.005$  (seed-level SE), close to but measurably below Planning+IG’s saturated 0.330–0.332 at large  $w$  (seed-level SE 0.004–0.005, roughly a 3-SE gap, so not a clean tie, though both sit well above zero). This is not evidence that reward-aware information measures are useless against distractors in general. It is evidence that a specific, common implementation pattern (falling back to unconditional entropy reduction when the reward-conditional information measure



**Fig. 13.** Reward-irrelevant sensing on `DistractorDiagnosisEnv`. The upper panel decomposes ordinary Planning+IG’s usage into task-relevant and distractor tests as  $w$  grows. Distractor spend is zero below the onset bracket  $(3.5, 4.0]$  and saturates at roughly a third of total usage. The lower panel plots the distractor fraction for ordinary information gain against the reward-relevance-weighted variant on the same grid. The weighted variant is flat at exactly zero everywhere. Error bars are seed-level SE over 5 seeds.

vanishes) reintroduces exactly the reward-blindness the reward-aware measure was meant to avoid. Neither ordinary information gain nor the IDS adaptation implemented here is therefore a clean immune baseline against reward-irrelevant sensing on this benchmark.

The reward-relevance-weighted variant above is the clean case, and the contrast with IDS shows why. It never computes a reward-conditional quantity that can vanish and then need replacing, so it has no raw-entropy fallback through which reward-blindness could be reintroduced. It scores ordinary information gain on a coarsened state space, so a test that is uninformative about the reward-equivalence classes earns exactly zero rather than a small positive number that a fallback then rescues. Appendix N reports the IDS baseline in full on the observe-then-commit environments.

### 6.11 The $w^*$ Atlas

The distractor result marks where the budget instrument stops, and the last piece of evidence in this section records where each benchmark sits on its own usage curve. Table 35 (Appendix T.3) collects, for the eight benchmark instances whose usage curves we estimate, the usage range each curve spans from its floor to its ceiling, the implicit EFE budget  $B_{\text{EFE}} = U(w=1)$ , and two crossing brackets (Definition 2) at two canonical budgets, aggregating the usage curves of Sections 6.5 and 6.6 (`results_price_usage_curves.csv`, `results_price_interleaved_curves.csv`). The two canonical budgets are placed inside each instance’s own usage range, one a fixed fraction of that range up from its floor and one the same fraction down from its ceiling, so that every instance is bracketed at comparable relative positions on its curve. Read across instances, the table shows where the implicit EFE budget lands relative to each usage range, at or near the floor of the range on some instances and part of the way up it on others. This is an atlas of measured operating points, not a predictive meta-model relating  $B_{\text{EFE}}$  to environment parameters (reward scale, state count, horizon). With the family’s operating points recorded, the EFE benchmark battery (Section 7) follows. The Discussion after it draws the results together, beginning with the reward convention on which every claim about  $w=1$  rests.

## 7 Results

This section reports the second experiment battery, the comparison of the EFE agent against reward-only planning, tuned information gain, and POMCP. It opens with the three core observe-then-commit environments, then reads the weight sweep as a Pareto analysis, then turns to Tileworld, RockSample, and Structural Inspection, and closes with a summary of what the battery does and does not show.

## 7.1 Core Environments

Table 8 compares four of the six core agents of Table 3 on Tiger, Diagnosis, and Bandit. The four are the EFE agent at its derived weight  $w=1$ , the two reward-only baselines Myopic ( $H=1$ ) and same-horizon Planning, and Planning+IG at its success-tuned weight  $w_{\text{succ}}^*$ . Those four isolate the two questions this subsection answers, whether EFE’s untuned epistemic term buys anything over reward-only planning at the same depth, and whether it gives anything up against an information bonus whose weight was tuned. The paragraphs below follow the table’s order. Tiger, where EFE and Planning tie exactly, comes first, followed by the Epistemic-only ablation, then Diagnosis and Bandit, where EFE Pareto-dominates Planning, then the closest baseline that does not tie EFE. A panel-by-panel reading of the weight sweep of Section 7.2 then places the  $w=1$  default against the best weight one could have chosen, and the subsection closes with effect sizes and the two statistical conventions under which those numbers are read.

On Tiger (single observation action), EFE and same-horizon Planning coincide exactly. This is a measured fact rather than a structural necessity, and the arithmetic that makes it unsurprising is a matter of units. At  $w=1$  the epistemic term is measured in the same units as reward, one reward unit per bit (Section 3.3), and because Tiger’s hidden state takes only two values the information gain of one listen is bounded by one bit. A single listen can therefore move an action’s value by at most one reward unit through the epistemic term. At Tiger’s stakes (+10, −100) a margin of that size is small next to the payoffs a commit decides, and in the runs the epistemic term never flips a decision that reward does not already dictate. Both agents run the same finite-horizon recursion over the same belief and break ties between equal-valued actions by the same fixed rule, so once the epistemic term never changes which action wins, the two recursions select the same actions. The agents produce bit-identical trajectories (obs, success, and reward all tie to the displayed precision, Table 8, confirmed by `results_tiger_stats.csv`, where the difference is 0 with  $p=1$  and  $d=0$  on every metric). The exact tie with Planning does not itself describe EFE’s relation to the tuned Planning+IG alternative. With no weight selection at all, EFE reaches within a fraction of a percentage point of the success rate that Planning+IG attains only after a weight search.

The Epistemic-only ablation (Table 3) commits immediately rather than over-exploring. On Tiger, Diagnosis, and Bandit alike it takes zero observations and succeeds at each environment’s chance level (49.7%, 24.7%, and 24.7% respectively), and on Tileworld 6×6 it likewise commits immediately and matches Myopic’s chance-level performance exactly (2.9%, Appendix B). Pure information gain without reward alignment produces degenerate non-exploration here, not the catastrophic over-exploration a naive information-seeking ablation might suggest. Either way, the pragmatic term is essential.

On Diagnosis and Bandit, EFE Pareto-dominates same-horizon Planning, reaching substantially higher success with comparable or better reward and no weight search. On Diagnosis, EFE outperforms Planning in success rate (97.2%

**Table 8.** Results across three core environments (5,000 total episodes at 1,000 per seed  $\times$  5 seeds, {42, 123, 456, 789, 1024}). Here  $w^*$  is  $w_{\text{succ}}^*$ , the per-environment success-maximizing tuned weight (Section 7.2 reports the reward-maximizing  $w_{\text{ret}}^*$  separately). Reward shown as mean  $\pm$  seed-level SE (SE of the 5 per-seed means, the honest error bar for the number of independent replications actually run). The pooled episode-level SE, which treats all 5,000 episodes as independent, is the error bar reported in Appendix B, and the committed CSVs carry both columns. The two are not uniformly ordered, with pooled SE smaller for some agent/environment cells and larger for others, depending on within-seed versus between-seed variance. Appendix B adds the remaining agents, except the untuned Info Gain rows, which are bit-identical to Myopic, and Tiger’s pymdp-AIF row, which is bit-identical to Posterior-vote on every reported metric (`results_tiger.csv`). Effect sizes on reward (Cohen’s  $d$ ), including seed-level  $t$ -tests, are in Appendix I. Bold marks EFE and any agent bit-identical to it (Tiger’s Planning), not the single best value per column. Planning+IG’s tuned success rate exceeds EFE’s on Diagnosis and Bandit but at a reward cost, per the Pareto discussion below.

| Environment                   | Agent           | Obs.        | Success      | Reward                             |
|-------------------------------|-----------------|-------------|--------------|------------------------------------|
| Tiger<br>$H=6$ , $w^*=50$     | Myopic          | 1.00        | 85.3%        | $-7.15 \pm 0.41$                   |
|                               | <b>Planning</b> | <b>4.20</b> | <b>99.4%</b> | <b><math>+5.19 \pm 0.21</math></b> |
|                               | Planning+IG     | 5.66        | 99.9%        | $+4.21 \pm 0.08$                   |
|                               | <b>EFE</b>      | <b>4.20</b> | <b>99.4%</b> | <b><math>+5.19 \pm 0.21</math></b> |
| Diagnosis<br>$H=3$ , $w^*=50$ | Myopic          | 2.00        | 64.7%        | $-13.17 \pm 0.58$                  |
|                               | Planning        | 5.86        | 88.8%        | $-2.57 \pm 0.20$                   |
|                               | Planning+IG     | 13.16       | 99.2%        | $-3.66 \pm 0.12$                   |
|                               | <b>EFE</b>      | <b>9.66</b> | <b>97.2%</b> | <b><math>-1.37 \pm 0.18</math></b> |
| Bandit<br>$H=2$ , $w^*=20$    | Myopic          | 2.06        | 62.2%        | $+5.56 \pm 0.09$                   |
|                               | Planning        | 3.29        | 71.0%        | $+5.75 \pm 0.08$                   |
|                               | Planning+IG     | 9.57        | 98.9%        | $+5.11 \pm 0.04$                   |
|                               | <b>EFE</b>      | <b>5.11</b> | <b>86.9%</b> | <b><math>+6.27 \pm 0.04</math></b> |

against 88.8%, +8.3 pp, seed-level Welch  $p=4.0 \times 10^{-8}$ ) while achieving better reward ( $-1.37$  against  $-2.57$ ,  $p=0.002$ ). On Bandit it achieves both higher success (86.9% against 71.0%, +15.9 pp,  $p=2.1 \times 10^{-5}$ ) and higher reward ( $+6.27$  against  $+5.75$ ,  $p=0.001$ ). Tuned Planning+IG, success-maximizing on the tuning grid at  $w^*=50$  on Diagnosis and  $w^*=20$  on Bandit, reaches near-ceiling success of 99.2% and 98.9%. On Diagnosis that still costs substantial reward from over-exploration (13.16 tests,  $-3.66$  reward). On Bandit the correctly tuned weight leaves reward much closer to Planning and EFE than a mistuned weight would suggest (9.57 inspections,  $+5.11$  reward, positive and only modestly below EFE’s  $+6.27$ ). Against tuned Planning+IG as well as  $w=0$  Planning, EFE is the competitive untuned trade-off rather than uniquely best on every metric.

One baseline deserves explicit mention because it is the closest competitor in the suite among agents that do not tie EFE exactly. Posterior-vote draws 100 states from the belief and takes the majority vote over each sample’s optimal action, with no information-gain term at all. On Bandit it is not significantly

different from EFE on any of the three metrics (reward +6.32 against +6.27, seed-level Welch  $p=0.58$ , success 87.0% against 86.9%,  $p=0.97$ , inspections 5.01 against 5.11,  $p=0.30$ ), and it nominally leads on reward. On Testbed it ties reward-only Planning exactly and also beats EFE. It does not transfer. On Diagnosis it tracks Planning rather than EFE ( $-2.78$  against EFE’s  $-1.37$ ) and on Tiger it is well behind ( $+3.79$  against  $+5.19$ ). This bounds what the Bandit and Testbed results can be taken to show about the epistemic term specifically. Bootstrap 95% CIs (10,000 resamples over 5,000 episodes, Appendix I) confirm non-overlapping reward intervals between EFE ( $[6.18, 6.37]$ ) and Planning ( $[5.64, 5.86]$ ) on Bandit.

Viewed in the success-reward plane (Figure 14), the EFE agent sits inside the reward-maximizing tied bracket on three of the five swept environments and off it on two, while Planning under-explores and success-tuned Planning+IG over-explores. The regenerated sweep (`results_pareto_sweep.csv`) makes the exceptions precise. Reward is exactly tied across a contiguous run of sampled grid weights on four environments, so the reward-maximizing weight is a tied bracket in the sense of Section 3.3. The weight  $w=1$  lies inside that run on Tiger (every sampled weight from 0.01 to 20), Diagnosis (every sampled weight from 0.5 to 20), and Bandit (1 and 2). On Testbed,  $w=1$  falls outside the tied run of sampled weights from 0.01 to 0.5 and is a genuine frontier trade-off, buying +6.4pp success (96.4% against 89.9%) for  $-0.11$  reward. On Tileworld, the one environment with a unique argmax,  $w=1$  is Pareto-dominated outright, because  $w=20$  attains both higher reward ( $-18.95$  against  $-21.53$ ) and higher success (91.8% against 72.0%). The untuned default is reward-competitive where the reward surface is flat in  $w$ . It is beaten on Testbed, where the tied bracket sits at the sweep’s low-weight boundary and  $w=0$  Planning matches it, and on Tileworld.

Effect sizes clarify where the differences lie. We read Cohen’s  $d$  on the conventional bands that Table 22 lists, negligible, small, medium, and large. On reward,  $d$  between EFE and same-horizon Planning is negligible on Tiger, Diagnosis, and Bandit (Table 22). On Tiger it is exactly 0, since the two agents tie. On Diagnosis and Bandit it is  $|d| < 0.15$  at the pooled episode level. On Diagnosis and Bandit both agents already achieve reasonable returns once they explore enough, and on Tiger there is no gap to measure. The gap that remains is whether an agent gathers enough evidence before it commits, and it shows in success rate rather than in reward. On success rate  $d$  is larger, around 0.33 on Diagnosis and 0.40 on Bandit (Appendix I), values in the small band. EFE raises success rate, the objective that Planning under-serves, without sacrificing reward, which is the Pareto dominance reported above in miniature.

Two statistical conventions govern how we read those numbers. First, with a balanced design, meaning equal episodes per seed for both agents in a comparison, the seed-level and pooled point estimates of the mean difference coincide exactly, so agreement in sign is guaranteed by construction rather than an independent check. What the seed-level test adds is a significance decision computed from only 5 independent replications instead of 5,000 pseudo-replications. On

that stricter test, among the EFE against Planning comparisons on the three core environments, every pooled comparison that was significant after Holm-Bonferroni correction, which is Diagnosis and Bandit on both metrics, remains significant at the seed level. The one negligible pooled comparison in that set, Tiger EFE against Planning, is non-significant at both levels.

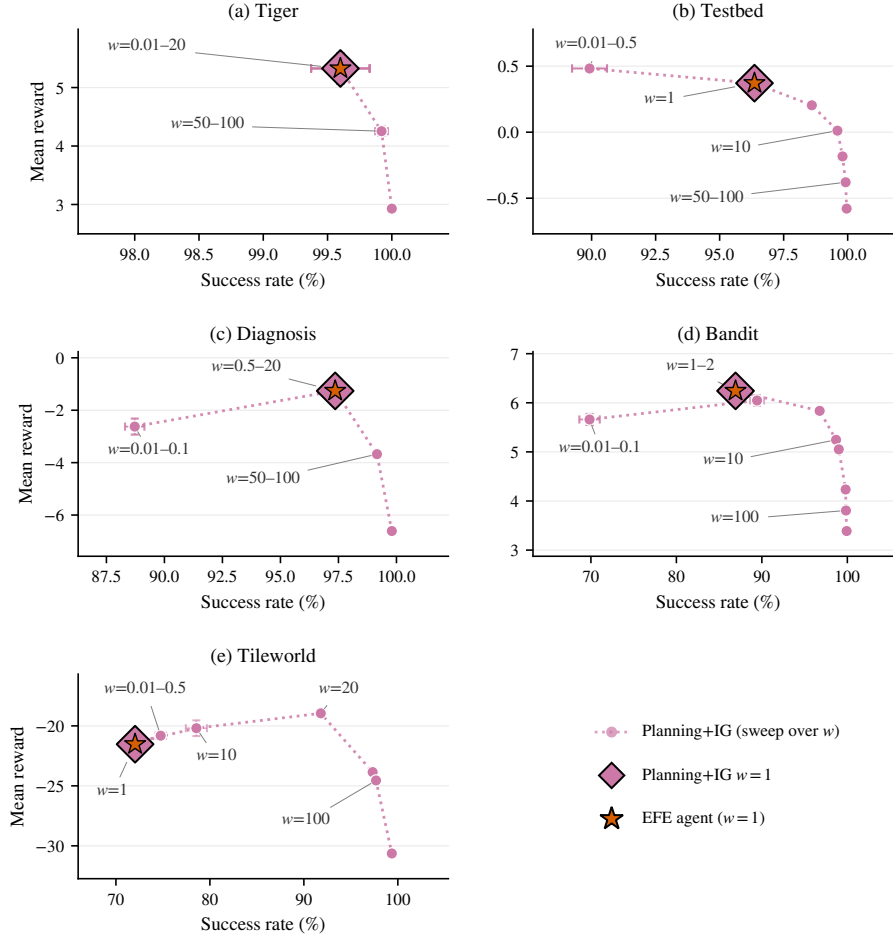
Second, seed-level Cohen’s  $d$  magnitudes are not comparable to the pooled values, since they are computed from the standard deviation of only 5 seed means per group. That much smaller and noisier denominator mechanically inflates  $|d|$  by roughly one to two orders of magnitude, factors of about 6 to 90 across the committed comparisons (Appendix I and the committed `*_stats.csv` files, which carry a seed-level  $d$  column for completeness). We therefore do not interpret seed-level  $d$  magnitudes and rely on the seed-level  $p$ -values and confidence intervals above.

Large reward effects ( $|d| > 0.8$ ) appear where Planning fails to explore sufficiently. The clearest case is the  $8 \times 8$  Tileworld grid of Section 7.3, where at the lookahead horizon used there reward-only Planning takes no scans at all. Tileworld  $8 \times 8$  gives pooled episode-level  $|d| = 1.13$  (`results_tileworld_8x8_stats.csv`). Against the over-exploring success-tuned baselines, Info Gain and Planning+IG at their tuned weights, the largest reward effect across the tested environments is medium,  $d=0.69$  on Testbed (Table 22). At the  $6 \times 6$  grid that Table 9 reports, the same EFE against Planning comparison is negligible at  $d \approx 0.01$ , since Planning and EFE scan similarly often at that smaller scale.

## 7.2 Pareto Analysis and the Information-Unit Weight $w=1$

By Proposition 1, EFE is exactly Planning+IG with  $w=1$ . That equivalence makes  $w=1$  a derived default rather than a tuned choice, and the question this subsection answers is where that default lands relative to the best weight one could have chosen. To place it on the success-reward frontier rather than relying on a single operating point, we sweep  $w$  over the grid  $\{0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200\}$ , eleven weights spanning four orders of magnitude at uneven spacing, on five observe-then-commit environments, the four of Section 5, Tiger, Diagnosis, Bandit, and Tileworld, together with the two-state Testbed of Appendix C (Figure 14). The previous subsection already read this figure panel by panel to make the tied brackets and the two exceptions precise, so here we draw only the general conclusion from it.

On three of the five environments in the Pareto sweep,  $w=1$  sits inside the reward-maximizing tied bracket, while the success-maximizing weight in the sweep lies at the sweep’s upper endpoint,  $w=200$ , on all five environments. Rock-Sample and Structural Inspection are not part of the sweep and are discussed separately in Sections 7.4 and 7.5. We emphasize that  $w=1$  is near-optimal for reward on three of the five swept environments, not for success rate. Agents requiring near-certain accuracy (e.g., safety-critical applications) would benefit from higher weights at the cost of reduced reward. On Tiger and Diagnosis, moving to  $w=50$ , the success-tuned Planning+IG weight of Table 8, buys at most 1.8pp of accuracy while costing 1.07 and 2.41 reward. The shape of that trade



**Fig. 14.** Pareto analysis. Mean reward vs. success rate as  $w$  varies from 0.01 to 200. The diamond marks Planning+IG at  $w=1$  and the star marks the EFE agent. By Proposition 1 with shared deterministic tie-breaking, the markers coincide. Error bars are seed-level SE on both axes, visible where larger than the marker, and weight labels give the tied range of  $w$  values sharing a plotted point. On Tiger, Diagnosis, and Bandit the markers lie inside the reward-maximizing tied bracket. On Testbed they sit off it as a success-for-reward trade, and on Tileworld they are Pareto-dominated by  $w=20$  on both axes (see text).

matters. Reward is unchanged at every swept weight from  $w=1$  up to the grid point just below  $w=50$ , and the entire reward cost arrives at the single step to  $w=50$ . No tested weight between them buys the extra accuracy at a lower reward cost, and the grid does not sample the interval between adjacent tested weights, so a cheaper untested weight is not ruled out. Bandit trades more evenly, gaining 9.8pp success for 0.41 reward at  $w=5$ , and Tileworld’s reward peaks at  $w=20$ , a weight inside the swept range rather than at either end of it, and falls again at larger weights. The contribution of EFE is to derive from the variational objective a principled weight that attains the swept grid’s reward maximum on three of the five swept environments, replacing a grid search whose reward-maximizing bracket moves, within the same eleven-weight grid, from at most 0.5 on Testbed to 20 on Tileworld.

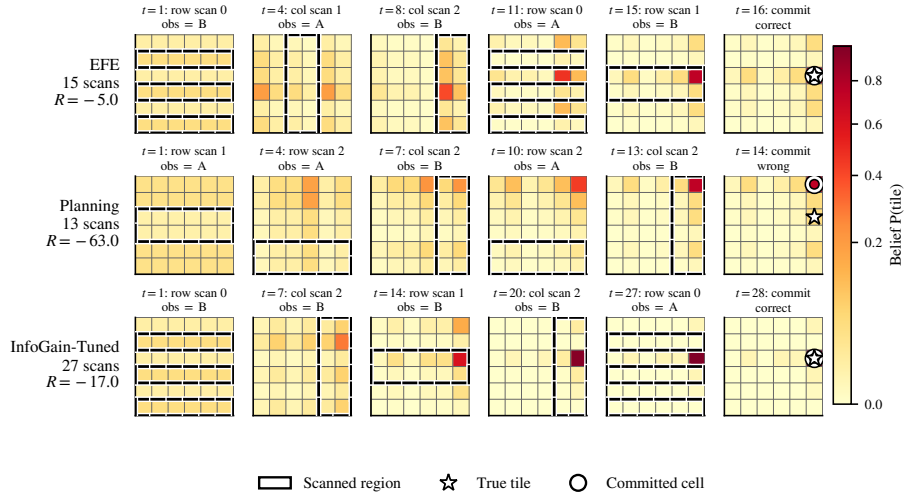
### 7.3 Spatial Epistemic Foraging on Tileworld

To test whether EFE’s advantage extends to larger state spaces, we turn to Tileworld, the spatial generalization of Diagnosis defined in Section 5, and vary its grid size  $N$  below. Tileworld projects the Diagnosis partition structure onto an  $N \times N$  grid ( $|S|=N^2$ ), producing spatially interpretable belief evolution (Figure 15, with step-by-step belief strips in Appendix D.4).

**Table 9.** Tileworld  $6 \times 6$  (2,500 episodes at 500 per seed  $\times$  5 seeds,  $H=2$ ). Tuned weight  $w_{\text{succ}}^*=50$ . Reward uncertainty is seed-level SE. Bold marks EFE and any agent not significantly different from it at the seed level (Welch  $p > 0.05$ , uncorrected), not the single best value per column.

| Agent                         | Scans        | Success      | Reward                              |
|-------------------------------|--------------|--------------|-------------------------------------|
| Myopic                        | 0.00         | 2.9%         | $-48.25 \pm 0.33$                   |
| Planning ( $H=2$ )            | 15.64        | <b>73.8%</b> | <b><math>-21.39 \pm 0.14</math></b> |
| Planning+IG ( $w=50$ )        | 32.25        | 97.3%        | $-23.85 \pm 0.19$                   |
| <b>EFE (<math>H=2</math>)</b> | <b>14.75</b> | <b>72.0%</b> | <b><math>-21.53 \pm 0.57</math></b> |

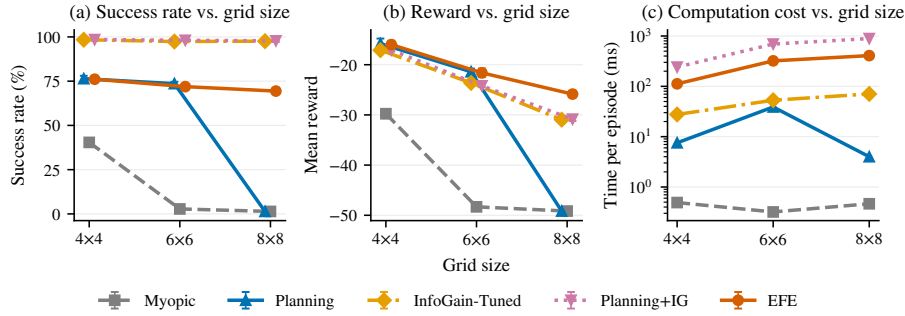
EFE and reward-only Planning are not significantly different on either reward ( $-21.53$  vs.  $-21.39$ , seed-level Welch  $p=0.82$ , a failure to reject, not a TOST-backed equivalence claim as in Section 6.7) or success (72.0% vs. 73.8%,  $p=0.17$ , both bolded in Table 9). EFE nevertheless reaches that outcome with significantly fewer scans (14.75 vs. 15.64, seed-level  $p=8.8 \times 10^{-5}$ , pooled  $p < 10^{-13}$ ). At this scale the epistemic bonus shows up in the number of scans rather than in raw reward or success. Planning+IG’s 32.25 scans, more than double EFE’s, erode reward despite reaching 97.3% success. The same over-exploration pattern from Diagnosis and Bandit recurs at this larger scale. Figure 15 visualizes the mechanism. EFE concentrates belief efficiently via partition-based narrowing and commits once confident, while Planning lacks scan-selection guidance and InfoGain-Tuned continues scanning past the point of diminishing returns.



**Fig. 15.** Agent comparison on the same  $6 \times 6$  Tileworld episode. All agents start from the identical uniform prior, stated here rather than drawn once per row. Belief probability is encoded by the shared colorbar on a square-root color scale. Black outlines mark the cell group queried by each scan, each panel’s title reports that scan’s noisy binary readout as  $\text{obs}=\text{A}$  or  $\text{obs}=\text{B}$  naming which half of the queried group the signal favors, the black-edged white star marks the true tile, and the black-stroked white ring marks the committed cell. Scan panels are subsampled evenly within each row, so columns are not time-aligned across rows and each panel’s  $t$  gives the true step. EFE (top) commits correctly with efficient scanning. Planning (middle) commits on the wrong cell after 13 scans. InfoGain-Tuned (bottom) over-scans well past the point of diminishing returns.

InfoGain-Tuned, the figure’s bottom row, is the myopic Info Gain agent of Table 3 run at its own success-tuned weight, a different agent from the tree-search Planning+IG row of Table 9, which adds the same bonus at every node of its lookahead tree.

*Spatial scaling.* We scale the grid from  $4 \times 4$  to  $8 \times 8$  with five agents, Myopic, Planning, InfoGain-Tuned, Planning+IG, and EFE (Figure 16). The sweep is a separate run from the full seven-agent Tileworld battery, of which Table 9 shows four rows, at the smaller episode count stated in the figure’s caption, and it omits that battery’s untuned Info Gain ( $w=1$ ) and Epistemic-only agents. The full battery’s per-agent tables, which exist at  $6 \times 6$  and  $8 \times 8$  and list all seven agents, are in Appendix B. At  $8 \times 8$  ( $|S|=64$ ), reward-only Planning collapses to  $1.4\% \pm 0.3\text{pp}$  success while EFE maintains  $69.4\% \pm 1.0\text{pp}$  (seed-level SE). The mechanism behind that contrast matters, because the headline number is easy to over-read. At  $8 \times 8$  reward-only Planning takes zero scans and is bit-identical to Myopic on every outcome metric (0.00 scans, 1.4% success,  $-49.16$  reward for both). A single binary scan of a 64-cell belief cannot repay its cost within an  $H=2$



**Fig. 16.** Tileworld scaling ( $H=2$ , 200 episodes per seed  $\times$  5 seeds) across Myopic, Planning, InfoGain-Tuned, Planning+IG, and EFE. Planning collapses at  $8 \times 8$  ( $|S|=64$ ), while EFE maintains 69.4%. Planning+IG at a fixed  $w=100$ , not tuned in class, achieves 97.8% but at higher reward cost. Error bars in (a) and (b) are seed-level SE, tied series are offset slightly along the categorical axis for legibility, and per-episode time in (c) is a pooled mean over all episodes and seeds, shown without error bars on a log scale.

lookahead. A depth-2 lookahead can see at most two scans before it must value a commit at its leaf, and after two binary scans a belief that began uniform over 64 cells is still spread over many cells. The commit valued at that leaf therefore remains close to chance, and its expected improvement over committing at once is smaller than what the scans cost. Planning therefore never scans at all, and the comparison at this scale is really EFE against a non-scanning agent. This is a joint horizon-and-scale effect at the fixed  $H=2$  of this sweep, not evidence about reward-only planning at deeper horizons, where the scan would begin to pay for itself.

Across scales, the gap between EFE and Planning is a threshold at  $8 \times 8$  rather than a gradual widening. At  $4 \times 4$  and  $6 \times 6$ , EFE’s reward and success are within sampling error of reward-only Planning’s, within 1 SE at both scales, with Planning nominally slightly ahead on both. No formal test was run for this specific sweep. EFE’s advantage opens up only at  $8 \times 8$ , exactly where Planning’s success collapses.

One further comparison in Figure 16 needs its own disclosure, because its two weighted series are not tuned the way the main tables’ weighted agents are. Planning+IG at  $w=100$  achieves 97.8% success but at substantial reward cost due to extensive scanning. The scaling sweep fixes  $w=100$  for both weighted agents, InfoGain-Tuned and Planning+IG, rather than tuning each separately as the main tables do, where the myopic Info Gain agent and the tree-search Planning+IG agent are each tuned within their own agent class. The figure is therefore a fixed-weight comparison and not a tuned one, and its InfoGain-Tuned series keeps that label from the main battery only as a series name.

*Observation structure sensitivity.* A natural concern is whether EFE’s advantage on Tileworld depends on the highly structured bit-level partition of scan

actions. We test this by replacing the deterministic row/column splits with two alternative observation structures, random partitions and overlapping partitions. We run the test at the  $6\times 6$  scale of Table 9. At that scale the advantage is the scan-count saving rather than a reward or success gap, so scan counts are the primary readout below, with reward and success reported alongside them. Under random partitions, each scan assigns the cells at random to two equal halves. The six splits therefore no longer correspond to separate index bits, and their answers are no longer guaranteed to combine into a unique cell, which is what the bitwise design’s orthogonality provided. Under overlapping partitions, each scan draws a random weight for the row index and another for the column index, scores every cell by the weighted sum of its two normalized coordinates, and assigns the cell to one group or the other according to whether that score exceeds a random threshold. Each scan therefore splits the grid along a random tilted line, and different scans cut off overlapping regions, producing correlated and partially redundant observations (construction in `rho_aif/environments/tileworld.py`).

The finding is that no partition mode separates EFE from Planning on reward or on success, while EFE uses fewer scans than Planning under all three. This battery is a separate run at 200 episodes per seed rather than the larger count of Table 9, so its bitwise figures differ slightly from that table’s. On  $6\times 6$  Tileworld ( $H=2$ , 200 episodes  $\times$  5 seeds, `results_partition_sensitivity_6x6.csv`, `experiments/run_tileworld.py partition`), we report EFE against Planning in reward  $\pm$  seed-level SE, with success in parentheses. Under the bitwise partition the pair is  $-21.62 \pm 0.87$  (71.9%) vs.  $-21.41 \pm 0.64$  (73.6%). Under the random partition it is  $-29.31 \pm 0.43$  (56.1%) vs.  $-30.32 \pm 0.66$  (55.1%). Under the overlapping partition it is  $-47.53 \pm 0.73$  (8.5%) vs.  $-47.66 \pm 0.68$  (8.4%). No partition mode separates the two agents on reward at the seed level, the largest gap being 1.01 reward under the random partition at 1.28 seed-level SE of the difference (Welch  $p=0.24$ ). The nominal ordering itself changes direction across modes, with Planning nominally ahead under the bitwise partition and EFE nominally ahead under the random and overlapping ones. Planning is nominally ahead on success under the bitwise partition, but on success the two are within sampling error of each other under all three partition modes. On scans, where the main  $6\times 6$  result lives, EFE stays ahead of Planning under all three modes, meaning that it uses fewer scans (14.76 vs. 15.57 bitwise, 12.97 vs. 13.38 random, 2.63 vs. 2.70 overlapping).

An earlier unseeded version of this experiment showed EFE nominally ahead of Planning on reward under the bitwise partition, and the seeded rerun reverses that reward ordering. That reversal is itself the clearest evidence that the nominal reward gaps here are sampling noise rather than signal. What the battery does establish is a large and consistent effect of observation structure on every agent. Both alternative modes cut absolute performance sharply, from roughly  $-21$  reward under the bitwise partition to  $-30$  under random scans and  $-48$  under overlapping ones, because less complementary scans leave less recoverable information for any agent to exploit. We therefore treat the Tileworld result as

scoped to the structured partition it was measured on, and we do not claim this experiment establishes EFE’s advantage is model-independent.

#### 7.4 RockSample as an Interleaved Observe-Act Setting

To exercise Proposition 3’s scope beyond observe-then-commit settings, we evaluate on a cost-modified variant of RockSample [58], a widely used POMDP benchmark with interleaved observe-act dynamics (Section 5 states the six deviations from the original). The agent moves on a grid among rocks of hidden quality, and the actions are the four cardinal moves (N/S/E/W), one check per rock, sample, and exit. Proposition 3 covers the check and navigation actions, and Remark 1 states its boundary for state-altering sensing.

The agents under comparison fall into two groups. The first is the weighted family, meaning the Planning, Planning+IG, and EFE agents (Table 10 abbreviates Planning+IG as Plan+IG), which use depth-limited belief-space tree search over factored beliefs (independent per-rock). The search depth is the  $d$  in each row label of Table 10, depth 3 or 4 on the three smaller instances and depth 2 on RS[11,11]. All three share one hand-coded leaf evaluation (`rho_aif/agents/rocksample_agents.py`) that, at the depth limit, greedily plans the remainder of the episode. That leaf rule scores each rock not yet sampled by its belief-weighted sample value, the belief that the rock is good times the good-sample payoff plus the belief that it is bad times the bad-sample penalty. It considers only the rocks whose value is positive, moves to the one whose value net of the move cost of reaching it is highest and samples it, repeats, and exits when no such rock remains. Because these three agents share that leaf value, they differ only in the information gain weight  $w$ .

The second group holds the reference agents. POMCP is a genuine search-tree solver, run at a configuration tuned once on separate seeds and then held fixed (Appendix M). The row labeled “Rollout only” runs POMCP’s approach-then-check rule standalone, with no search tree. That rule is POMCP’s rollout policy, the default policy that plays out each simulation once it leaves the search tree, and it is a hand-coded, unbounded-horizon rule rather than a depth-limited search. It walks to the unresolved rock with the highest belief per unit of distance (the nearest one at a uniform prior), checks it at close range where accuracy is 0.95, samples if the belief crosses a threshold, writes off any rock whose belief falls below 0.15, and exits when nothing is left. The prose below calls this standalone run the heuristic. Greedy, the remaining row, samples without checking.

Across the three smaller instances the reading is the same. EFE at  $w=1$  is never significantly behind the  $w=5$  comparator of Table 10, is ahead of it once, and is statistically tied with or below the heuristic. On RS[5,3], EFE and Planning+IG ( $w=5$ ) are not significantly different (+16.47 vs. +16.41, seed-level Welch  $p=0.74$ , a failure to reject, not a TOST-backed equivalence claim). Both are bolded in Table 10 as statistical ties with the best row, which is the heuristic at +16.67. On RS[7,4], EFE leads the weighted family outright (+15.96 vs. +14.56 for Planning+IG,  $p < 10^{-5}$ ) and is statistically tied with the heuristic’s +15.85 ( $p=0.65$ , again a failure to reject rather than demonstrated equivalence).

On RS[7,8], EFE and Planning+IG ( $w=5$ ) are again not significantly different from each other (+21.85 vs. +23.76,  $p=0.09$  uncorrected, not significant after Holm–Bonferroni), while both substantially outperform Planning (+12.31,  $p < 10^{-4}$ ). On this instance the heuristic reaches  $+28.33 \pm 0.82$  on the identical protocol. Both deficits against it survive the per-metric Holm correction (Planning+IG  $p=2.4 \times 10^{-3}$ , EFE  $p=4.6 \times 10^{-4}$ ), so the table holds the heuristic alone there. Planning, Planning+IG, and EFE all sit below that +28.33, so on RS[7,8] these are relative comparisons within a family that is jointly suboptimal.

More checking beyond  $w=5$  never significantly helps across these three instances, and twice significantly hurts. Planning+IG at  $w=10$  (Appendix M) falls to +20.55 on RS[7,8] ( $p=0.003$ , significant under the per-metric Holm family), so among the three nonzero weights we ran on RS[7,8],  $w=5$  attains the highest reward. It is not significantly different from  $w=1$ . Raising Planning+IG’s weight from 5 to 10 on RS[5,3] produces a decisive drop (+14.60 vs. +16.41,  $p=3.5 \times 10^{-9}$ ). The same step from 5 to 10 on RS[7,4] is flat ( $p=0.18$ ). On that instance EFE at  $w=1$  had already beaten both tuned weights outright, so neither tuned weight is the one to beat there. We ran only  $w \in \{5, 10\}$  as tuned comparators on RockSample, not the nine-point grid used for the observe-then-commit environments (Section 3), so this identifies the best of three sampled weights rather than a grid-searched argmax. We make no claim that  $w=5$  is the reward-maximizing weight over a denser sweep.

The Bad column of Table 10, the mean number of bad rocks sampled per episode, shows the information gain term at work, and the Good column beside it is the corresponding mean number of good rocks sampled. On RS[5,3], RS[7,4], and RS[7,8], EFE samples dramatically fewer bad rocks than Greedy, between 0.01 and 0.06 per episode (seed-level SE between 0.001 and 0.013) against Greedy’s 1.51 to 3.98 (seed-level SE between 0.01 and 0.05), with seed-level Welch  $p < 5 \times 10^{-8}$  at every instance, confirming that the information gain term drives checking behavior.

EFE’s gap over  $w=0$  Planning, largest on RS[7,8] among the three smaller instances, does not extend to RS[11,11] ( $|S|=2,048$ ), where EFE and Planning achieve identical reward (+13.23 each) despite that instance having the most rocks to check. Every tree-search agent vastly outperforms Greedy (+12.9 to +13.2 against −18.9), but nothing here differentiates EFE from reward-only planning. Every tree-search agent at its reported configuration converges to the same conservative pattern of checking one or two nearby rocks, sampling if a rock looks good, and exiting, so the weight on information gain moves reward by at most 0.06. This includes POMCP at its frozen setting (Appendix M). Moving from the table’s depth 2 to depth 3 does not change this. Rerunning at depth 3 (`results_rocksample_11x11_depth3_check.csv`) costs seconds and reproduces the same pattern within 1 SE, with Planning and EFE at +13.23 and Planning+IG at +13.17.

Two explanations fit that collapse. Either the instance leaves no exploitable slack, so every agent is already near the ceiling, or it leaves a great deal of slack that no depth-limited search can reach. The heuristic already in Table 10,

originally added for the POMCP comparison, settles which one holds, and the answer is not the flattering one. Recall that it is bound by no search depth and walks up to a rock before spending a check on it. Run on the same 5 seeds and 100 episodes as the table above, it scores  $+28.66 \pm 0.83$  on RS[11,11], where every tree-search agent, EFE and POMCP included, sits in the  $+12.94$  to  $+13.23$  band. On RS[7,8] it scores  $+28.33 \pm 0.82$  against the best tree-search agent's  $+23.76 \pm 0.58$ .

So the first explanation fails. There is a great deal of slack at 11 widely spaced rocks, and the depth-limited belief-space search cannot reach it. The Good column of Table 10 shows that slack in behavioral terms, since on RS[11,11] the heuristic samples 5.22 good rocks per episode while every tree-search agent samples about half a rock.

The mechanism is now identifiable. Recall that check accuracy decays with the agent's distance from the rock (Section 5), so a check is informative only when taken from close range. Reaching close range means walking to the rock first. At this instance the checks that could be informative are far from the agent, and a depth-2 or depth-3 tree cannot represent the walk that would make them informative. Within its horizon only a single check taken from the agent's own starting cell is worth making, so the weight on information gain has almost nothing to act on. The shared greedy leaf rule values a rock at its belief-weighted sample reward, which is exactly zero at the uniform prior because at belief 0.5 the  $+10$  good-sample payoff and the  $-10$  bad-sample penalty cancel. So no rock passes the rule's positive-value filter, and it recommends the exit. Because the informative checks lie beyond the depths we run, depth 3 leaves reward unchanged and every weight gives the same reward to within 0.06, with the higher weights buying only about half an extra check per episode.

One further detail of that leaf deserves disclosure. Its completion estimate charges move costs for a walk to the east edge before collecting the exit reward (`rocksample_agents.py`), while the environment pays the exit from any cell. This phantom travel cost deflates every depth-capped continuation relative to the exact value of exiting immediately, a bias toward earlier exits. The immediate exit escapes the deflation because an exit taken inside the tree at the current cell is valued at its true payoff, so only continuations that run on to the leaf carry the charge. Every agent in the family shares the phantom cost identically, so within-family comparisons are unaffected. But it means the leaf's exit recommendation is overdetermined here rather than resting on the zero prior expectation alone.

POMCP corroborates the mechanism from the other side. The heuristic profits from the far rocks because it walks up to each one before checking it, so that every check it spends is taken at close range where the check is accurate. A deeper POMCP tree gains the lookahead to plan a walk out to those rocks, but its in-tree check decisions are still taken from a distance, where accuracy is low. We read that from the behavioral trace in Appendix M rather than deriving it from the search procedure. At the deepest horizon we ran, the tree spends many times more checks per episode than its own rollout policy does and still collects fewer good rocks. Given horizon enough to reach the distant checks, its

reward falls monotonically, from +12.94 at  $H=10$  to -54.78 at  $H=40$ , because the tours it assembles from noisy distant checks do not pay (Appendix M). Both sides of the mechanism therefore point the same way. The heuristic walks up before checking and is bound by no depth, so it collects the slack. POMCP with a deeper horizon has the reach but still checks from a distance, so it does not. Reach alone, without walking up before checking, is not enough.

This is a limitation of the tree-search family at feasible depths on this instance, not evidence about information-gain weighting, and it bounds what RockSample[11,11] can be asked to show about EFE. Per-instance detail, including the Flat-MC reference (a flat one-ply Monte Carlo baseline with no search tree, kept in the appendix as a reference) and the standalone heuristic on every instance, is in Appendix M.

The RS[11,11] collapse, the RS[7,8] advantage, and the Tileworld scaling result are one mechanism seen at three points rather than three unrelated per-instance results. EFE’s advantage over reward-only Planning tracks how far the reward-only policy’s default behavior already sits from the information-optimal one. On Tileworld, reward-only Planning scans too little to disambiguate a large grid, so directed information gain buys a large success-rate margin. On RS[7,8], eight rocks spread across a mid-sized grid leave real ambiguity about which are worth sampling, so choosing which rock to check pays off. On RS[11,11] the rocks sit far enough apart that no check beyond the agent’s own cell is worth taking within the depths we run, so the weight has almost nothing to act on. The prediction this yields is that EFE’s advantage should track how much slack a problem’s structure leaves between the reward-only policy and the information-optimal one that the agent’s search can actually represent, not problem size or observation-action count in themselves. We did not design or run instances specifically to stress-test that prediction, so it remains a mechanistic reading of the four RockSample instances and three Tileworld scales already reported rather than an independently validated scaling law.

## 7.5 Structural Inspection

To exercise Proposition 3’s scope on a synthetic benchmark with an inspection structure and a larger state space than the environments above, we implement a Structural Inspection POMDP (Section 5). In that environment, the agent must declare each of the  $N$  components nominal or faulty, a declaration that cannot be revised once made. A declaration is made while the agent stands on that component, and declarations are interleaved with the moves and tests. The instance names  $N=8$  and  $N=16$  below give the two instance sizes by their component count. Tests do not alter component states, so this is a factored observation POMDP and Proposition 3 applies. In Table 11, Missed is the mean number of faulty components per episode declared nominal, and Tests is the mean number of tests run per episode. As on RockSample, all tree-search agents here share one hand-coded leaf evaluation (`rho_aif/agents/inspection_agents.py`) that greedily completes the episode from the depth limit, so agents in this section differ only in the information-gain weight and not in their leaf value.

EFE ( $w=1$ ) is a competitive untuned trade-off rather than uniquely best. On  $N=8$  ( $|S|=256$ ), Planning wins reward ( $-17.85$ ), while EFE achieves 91.1% accuracy ( $-20.95 \pm 0.30$ ), gaining +18.0pp accuracy over Planning (73.0%, seed-level Welch  $p < 10^{-10}$ ) at a moderate reward cost. Planning+IG ( $w=5$ ) reaches 96.3% accuracy but at  $-27.98$  reward, over-testing relative to reward. On  $N=16$  ( $|S|=65,536$ ), Planning posts the best mean reward ( $-45.42$ ), with EFE a close second at  $-45.80$ . EFE’s deficit is not significant (seed-level  $p=0.57$ ), so the two reward cells are bolded together. Planning+IG ( $w=5$ ) wins accuracy (97.3%) at substantial reward cost ( $-54.74$ , the worst of the three informed agents). That cost comes from extensive over-testing,  $38.3 \pm 0.05$  tests vs. EFE’s  $28.1 \pm 0.09$  (seed-level SE), roughly 36% more (seed-level Welch  $p=1.7 \times 10^{-11}$ ). EFE outperforms Planning by +12.6pp accuracy (90.8% vs. 78.2%, seed-level  $p < 10^{-7}$ ) at a reward cost of only 0.39 that is itself not significant, a far better accuracy-per-reward trade than Planning+IG’s. The  $N=16$  instance is the largest state space in our evaluation and shows that the factored belief tree search remains feasible at that size on this synthetic benchmark, with a known model and a hand-coded leaf rule. Validation on a deployed inspection problem is future work.

Inspection’s payoffs also let us read off what Proposition 2 predicts about  $w=1$  here, that is, whether  $w=1$  falls inside its near-optimality interval. The reading is possible because the proposition’s thresholds depend on the reward asymmetry ratio  $\alpha = |R^-|/R^+$  (not the reward scale that Section 6.1 also writes as  $\alpha$ ), along with the observation accuracy and cost. The penalty structure is asymmetric on the missed-fault side. Declaring nominal pays +2 when correct and  $-50$  when a fault is missed ( $\alpha = 25$  for that decision, the missed-fault penalty divided by the correct-nominal reward), while declaring a fault is symmetric (+5 when correct,  $-5$  on a false alarm,  $\alpha = 1$ ). Reading Proposition 2’s two-state thresholds onto this multi-state environment is a heuristic reduction, as with the two-state-reduced rows of Table 1. With that caveat in place, the missed-fault decision, where the large penalty sits, places the environment in the proposition’s large-asymmetry regime. In that regime Part (a) says that a heavy penalty for a wrong commit drives the lower threshold  $w_{\text{thresh}}$  downward (the proposition’s  $w_{\text{thresh}}^*$ , written without the star from Corollary 1 onward), and once that threshold falls below zero,  $w=1$  clears the interval’s lower endpoint. The interval’s upper endpoint, the over-observation threshold of Part (b), would then also have to exceed 1 for  $w=1$  to lie inside the interval, and we do not evaluate that threshold for Inspection. The reading is therefore that Inspection sits in the regime where Part (a) stops binding, which is the regime where the proposition predicts  $w=1$  is a reasonable untuned default, and not a computed placement of  $w=1$  inside the full interval.

## 7.6 Summary

EFE matches Planning+IG at  $w=1$  within their stated scope, and that information-unit weight lands inside the reward-maximizing bracket on three of the five swept environments without per-environment search. Where multiple observation actions matter most, as on Diagnosis and Bandit, untuned EFE Pareto-dominates

same-horizon planning. Where tuned bonuses chase success harder, they can beat EFE on accuracy at a reward cost. Across Tileworld, RockSample, and Inspection, EFE remains a competitive untuned trade-off rather than uniquely best on every metric. The advantage over  $w=0$  Planning opens at the largest scale of the Tileworld sweep (Figure 16), and it is not a general size pattern. It vanishes entirely on RockSample[11,11] despite that instance having the most rocks to check, and on Structural Inspection the accuracy advantage narrows from +18.0pp at  $N=8$  ( $|S|=256$ ) to +12.6pp at  $N=16$  ( $|S|=65,536$ ) even as its reward cost narrows in parallel.

## 8 Discussion

*The information weight as an operational shadow price.* Every claim in this section about  $w=1$  being an untuned default is conditional on one reward convention shared across every comparison in this paper. Information gain is measured in bits while task reward is measured in whatever units the designer chose, so the pragmatic-epistemic exchange rate  $w$  is necessarily convention-dependent. Conditioning on the convention is therefore the correct framing rather than a caveat. For the receding-horizon planner, Proposition 4 proves that the dependence is exact and predictable rather than merely bounded, under the tie-break and rounding qualifications of Section 4.2. Because the dependence is exact, the question “what should  $w$  be” can be replaced by “how much sensing should the agent use in expectation,” a question with a physically meaningful answer that a designer can state without knowing anything about information theory. The answer is a budget  $B$  in observation count or cost. Section 6 shows that the resulting operational shadow price  $w^*(B)$  is computable offline from a usage curve. Proposition 6 gives stationary convergence of the projected, decaying-step dual-descent estimate of that price under stated conditions, and Section 6.3 demonstrates empirically that same controller re-adapting through a reward-scale shift on nine of the ten controller seeds tested within the post-rescale window. The canonical weight induces an implicit budget  $B_{\text{EFE}} = U(w=1)$ , the mean sensing usage it happens to spend, and  $w=1$  is a member of that budget’s level set (Definition 2). Inverting the curve at  $B_{\text{EFE}}$  need not return  $w=1$  itself, since on a plateau the crossing threshold sits at the plateau’s onset, the last jump up to that level, rather than at any particular member of the level set, and at a usage floor the crossing from below is undefined, so  $w=1$  is the special case that buys whatever sensing the ambient reward scale happens to imply and need not be the price the procedure would report for that budget (Section 4.2).

The distractor-robustness result (Section 6.10) marks the boundary of what a budget alone can do, and it supplies the second instrument, relevance, which a price alone cannot provide. A sensing budget sets a target for expected sensing usage, counted in observations or in their cumulative cost, and says nothing about which uncertainty those observations resolve. Neither ordinary information gain nor the reward-aware IDS baseline we tested redirects that usage away from reward-irrelevant uncertainty on its own. Scoring information gain on the

partition of states into reward-equivalence classes does redirect it, to exactly zero reward-irrelevant spend at every swept weight, using only quantities the agent already holds. Price and relevance are complementary instruments. The budget sets how much sensing is bought, as an expected-usage target. The partition decides which uncertainty that sensing is spent on.

*When does EFE-as- $\rho$  help?* In the regimes we tested, EFE’s advantage appears where (1) the agent chooses among multiple observation actions of differing informativeness and (2) the planning horizon is long enough for recursive EFE to propagate epistemic value. These are the observed regimes, not necessary conditions. When (1) fails, as on Tiger with its single listen action, reward-only planning matches EFE at every discount factor tested except the heaviest, where a small significant difference appears (Appendix K). Condition (2) is examined below in its discount-factor form, after the discussion of tuning. Tileworld adds a third condition, (3) a reward signal too diffuse to guide scan selection on its own. Tileworld’s scaling sweep produces that condition by growing the state space, and the separation is a threshold rather than a steady growth, opening at  $8 \times 8$  where Planning collapses (Figure 16, Section 7.3). Within-episode dynamics (Appendix D.5) show the mechanism. EFE concentrates belief by partition-based narrowing, each noisy scan answer shifting belief mass toward one side of a bit-level split, and then commits once the value of committing exceeds the value of observing, a crossover that emerges automatically from  $w=1$  without tuning (Figure 17, Appendix D.1). The per-decision value-of-information audit trail (Appendix P) exposes the decomposition behind these decisions, and Appendix O scores the terminal belief itself rather than the decision it produces.

*The information-unit weight and tuning.* EFE fixes the exploration–exploitation weight at  $w=1$  rather than eliminating it (Section 3.3). The Pareto analysis (Figure 14) shows this weight sits inside the reward-maximizing tied bracket  $w_{\text{ret}}^*$  on three of the five swept environments, while the success-maximizing  $w_{\text{succ}}^*$  used for Planning+IG in the main tables lies at 10–50. The gap between  $w_{\text{ret}}^*$  and  $w_{\text{succ}}^*$  has practical implications.  $w=1$  is tuned to neither criterion. Its objective is reward plus information gain at weight one, and the fact that its measured reward sits with the reward-maximizing weights rather than with the success-maximizing ones on three of the five swept environments is an empirical finding of the sweep, not a design target. In safety-critical settings where near-certain accuracy is required, a higher weight may be appropriate, though such weights must still be tuned per-environment. EFE provides a principled starting point rather than a scale-free automatic calibration. Section 7.2 quantifies the accuracy-for-reward trade, which on Tiger and Diagnosis buys at most 1.8pp of accuracy for a reward cost that arrives entirely at the single grid step to  $w=50$ . The weight’s dependence on the reward scale and on the bit-versus-nat convention, with the per-environment consequences of the latter, is set out in the reward-to-nats calibration paragraph of Section 3, and Section 7.2 reports the measured sweeps behind those consequences.

A further complication for tuning is that adding planning depth to a weighted IG bonus amplifies the weight’s effect, so the tuning problem becomes harder with depth. The same weight buys more sensing at depth two than at depth one, so the map from weight to sensing usage shifts with the horizon. A weight tuned at one depth need not carry over to another. One measurement supports this directly. At  $w=100$  on Bandit under the canonical 5-seed protocol, Planning+IG at  $H=2$  takes 12.36 inspections against myopic Info Gain’s 11.50 (seed-level Welch  $p=0.003$ , `experiments/run_bandit_w100_depth_comparison.py`).

Condition (2) above, a sufficient planning horizon, also has a discount-factor form. In the discount sweep (Appendix K, 5 seeds), the information-unit weight’s advantage on the multi-observation environments appears at  $\gamma \geq 0.99$  and not at the two lower values tested, and the transition is sharp. On Diagnosis ( $N=4$ ), EFE’s success advantage over Planning is significant at  $\gamma \in \{0.99, 1.0\}$  and not statistically significant at  $\gamma \in \{0.95, 0.90\}$  (Table 24). The mechanism is that discounting truncates the effective planning horizon below the number of observations needed for confident diagnosis, erasing EFE’s advantage. On Tiger (single observation), EFE and Planning differ significantly only at the heaviest discounting tested ( $\gamma=0.90$ ) and are numerically identical at  $\gamma \geq 0.95$ . This is a meaningful practical limitation within the range swept. The sweep covers four discount factors on three environments, so it does not establish a general threshold, but it does show that applications whose time pressure mandates discounting as heavy as  $\gamma=0.95$  on Diagnosis-like problems should not expect EFE’s epistemic drive to pay off.

*Zero-shot weight transfer.* The tuning problem is not merely inconvenient. Whether a tuned weight survives transfer to a new environment depends on the tuning criterion. Table 12 evaluates  $w=1$ , reward-tuned weights  $w_{\text{ret}}^*$ , and success-tuned weights  $w_{\text{succ}}^*$  across environments. Success-tuned transfer harms reward at the top of the weight grid.  $w_{\text{succ}}^*=50$  is the success-tuned weight for both Tiger and Diagnosis, tuned within the Planning+IG agent class that the table evaluates. It yields +4.25 on Tiger, where it was tuned, and  $-0.38$  on Testbed, where that same weight is carried to an environment it was not tuned for, versus  $w=1$  at +5.33 and +0.37. Reward-tuned moderate weights ( $w \in \{0.5, 1\}$ ) transfer well within the shared reward scale. On Tiger and Diagnosis,  $w=1$  is tied for best, and the tie is four-way, shared with  $w=0.5$ , Testbed’s  $w_{\text{succ}}^*=10$ , and Bandit’s  $w_{\text{succ}}^*=20$ . The reason is that the reward surface on each of those two environments is flat enough at moderate weights that reward-tuned and success-tuned weights from that band, whichever environment they were tuned on, all attain the same reward there. On Bandit,  $w=1$  remains best outright without search. The honest conclusion is that moderate weights transfer robustly under a shared reward convention whatever their provenance, while success-tuned weights overfit once pushed to  $w=50$ , the top of the tested grid. In that moderate band,  $w=1$  is the variational default, not a scale-free optimum.

*When is  $w=1$  near-optimal?* Having seen that  $w=1$  transfers as a moderate default, we now ask when it is also near-optimal, first in closed form for the

two-state case and then empirically across planning horizons. Proposition 2 formalizes lower and upper thresholds for the two-state case, written  $w_{lo}^*$  and  $w_{hi}^*$  in Table 1. The lower threshold  $w_{lo}^*$  (the proposition’s  $w_{thresh}^*$ ) is the weight below which the agent commits without observing at  $H=1$ , and it depends on the reward asymmetry  $\alpha$ , the informativeness  $\eta$ , and the observation accuracy  $p$ , all three defined in the proposition. The upper threshold  $w_{hi}^*$  (the proposition’s  $w_{over}^*$ ) is the weight above which the agent, at the post-first-observation belief, buys a second observation whose instrumental value falls short of its cost, so it bounds over-observation. Part (a) of the proposition predicts that large reward asymmetry drives the lower threshold below zero, which leaves  $w=1$  inside the interval whenever the upper threshold exceeds 1. Section 3.3 read Table 1 against that prediction. The high-asymmetry environments ( $\alpha \geq 5$ , namely Tiger, Diagnosis, and Tileworld) place  $w=1$  interior to the interval, on the genuinely two-state row and, more heuristically, on the two-state-reduced rows, while Testbed places  $w=1$  near the upper threshold, consistent with over-exploration.

The proposition’s two parts live at two horizons, Part (a)’s lower threshold at  $H=1$  and Part (b)’s near-optimality interval at  $H=2$ . To ask what happens to the near-optimality of  $w=1$  as the planning horizon varies, including beyond those two cases, Appendix J reports a Monte Carlo study over 100 randomly generated two-state environments at  $H \in \{1, 2, 3\}$  (Figure 24). This study uses a different, empirical criterion of near-optimality from the closed-form interval above. The interval is stated for the two-state case at  $H=2$  only, whereas a reward-gap criterion, which asks how much reward  $w=1$  gives up relative to the best weight found by grid search, can be applied at any horizon, and the study counts  $w=1$  as near-optimal when that gap is within  $\max(5\%, 0.5)$  of the best reward. The near-optimality rate increases from 30% at  $H=1$  to 58% at  $H=2$  and 79% at  $H=3$ , consistent with multi-step planning widening the near-optimality basin, the region of environment parameters where  $w=1$  passes this test. High-asymmetry environments ( $\alpha \geq 10$ ) follow the same trend, rising from 26% to 53% to 79%. The failure rate, the fraction of environments where  $w=1$  does not pass this test, is the more useful number for a practitioner and we state it directly.  $w=1$  fails this near-optimality test in about 21% of the random family even at  $H=3$ , and in 70% of it at  $H=1$ . The criterion is also lenient by construction, since the absolute floor of 0.5 reward units does most of the work on environments whose reward range is small.

The widening with horizon of the basin in which  $w=1$  passes this test is qualitatively consistent with what we observed on Bandit. Bandit’s asymmetry is a contrast between two positive arm rewards (correct-arm reward +10 vs. small-reward-arm +1) rather than the two-state  $\alpha=|R^-|/R^+$  ratio this section’s formula assumes. The formula therefore does not apply to it directly and we do not report a numeric  $\alpha$  for it. Bandit, read loosely as a low-asymmetry case relative to the high-asymmetry environments above, is where the one-step analysis would lead one to expect the least from  $w=1$ , although we do not compute its threshold. Multi-step EFE at  $H=2$  achieves 86.9% success on it (Table 8), consistent with the aggregate direction of the random-environment study, where

the fraction of environments in which  $w=1$  meets that study’s reward-gap near-optimality criterion rises from 30% at  $H=1$  to 58% at  $H=2$ .

Many real-world decision problems have  $\alpha \gg 1$ . In medical diagnosis, fault detection, and security screening, a missed condition costs far more than another test. This is the regime where the interval analysis is most favorable to  $w=1$ , and the multi-step study shows the near-optimality basin widening with planning depth. The practical consequence is that in these domains the information-unit weight is a strong untuned starting default under the shared reward convention, not a guarantee of near-optimality. When the default’s induced sensing differs from a stated expected-usage target, the budgeted machinery of Section 4 selects, within the single-price receding-horizon family, the weight or endpoint mixture whose expected usage meets that target.

*Robustness to model misspecification.* Our main results assume exact knowledge of the generative model. Appendix L investigates robustness when the agent’s believed observation accuracy differs from the true value, over mismatches from  $-0.15$  to  $+0.10$ , where a negative mismatch means the agent believes the sensor to be less accurate than it is and a positive mismatch means the reverse. One question the analysis can answer is whether EFE’s epistemic term buffers a misspecified sensor in a way that reward-only Planning does not. On Tiger, success stays close to its zero-mismatch value at every tested level, and EFE and Planning respond identically to every tested mismatch (Table 25), so there is no measurable EFE-specific buffering effect on this environment. On Diagnosis, degradation is larger but graceful, and both agents’ behavior steps between two plateaus rather than degrading smoothly with the mismatch magnitude, a higher one with more tests and higher success and a lower one with fewer tests and lower success. Changing the mismatch moves an agent between the plateaus rather than along a gradient (Table 26). Overestimating sensor accuracy (positive mismatch) is more harmful than underestimating it, because the agent commits prematurely on insufficient evidence.

*Approximate planning with MCTS-EFE.* At each depth the tree branches over  $K$  observation actions and then, per action, over its  $|\mathcal{O}|$  observation outcomes, so the exact tree search cost is  $\mathcal{O}((K \cdot |\mathcal{O}|)^H)$ , limiting practical horizons to  $H=2-3$  with  $K \geq 3$ . To push beyond this, we implement MCTS-EFE, a UCB1 tree search whose design choices we state in turn. The tree branches over observe and commit actions and, after each observe action, enumerates its  $|\mathcal{O}|$  observation outcomes as child nodes. It adds the exact per-node information gain to the in-tree reward of each observe action, so that within the tree an observation is credited for what it reveals as well as charged for what it costs. Values propagate by max-backup, each node taking the maximum over its children’s values rather than the average of the returns sampled through it. Leaves are scored by an EFE-greedy rollout, which at each step takes the observation with the best one-step EFE value and credits the better of committing now and continuing to observe. The UCB1 exploration constant is left at its default, the commit reward range (110 on Tiger). That default is a static analog of the  $R_{hi} - R_{lo}$  rule of Silver and

Veness [55], which sizes the constant to the spread between the highest and lowest returns seen in preliminary sample runs of the search, whereas the default here takes the spread once from the commit reward table and never revisits it. On Tiger at  $H=10$ , MCTS-EFE with 500 simulations achieves  $97.7\% \pm 0.5\text{pp}$  success (23–28 ms per episode across the three horizons tested, `results_mcts_efe.csv`), compared to  $88.8\% \pm 0.2\text{pp}$  for POMCP at the same simulation count (15–19 ms per episode across horizons, seed-level SE, seed-level Welch  $p=5.2 \times 10^{-6}$ ). The comparison is simulation-matched, not compute-matched, since MCTS-EFE’s leaf evaluations are more expensive per simulation. In this configuration, EFE’s closed-form information valuation, acting as the in-tree observe reward and inside the leaf rollouts, outperforms POMCP’s semi-informed rollouts (Appendix S).

The two solvers differ in more than the leaf, so an ablation narrows where that advantage comes from. Two of the components that distinguish this search from a UCT tree (Upper Confidence bounds applied to Trees, the standard UCB1 tree search) can be turned off independently, the in-tree information gain and the max-backup, and Table 33 runs all four combinations on Tiger, Diagnosis, and Tileworld-6 $\times$ 6. Neither component shows a detectable effect on success or reward at five seeds after Holm correction (Appendix S). What remains to carry the contribution is the EFE-greedy leaf rollout together with the exact belief representation and the exact commit-child values, which the ablation holds fixed. A commit child is the node reached by a commit action, and the search sets its value to the belief-expected commit reward computed directly from the belief instead of estimating it from sampled returns.

A natural objection is that the POMCP gap reflects an untuned exploration constant rather than the solver. The observe-then-commit POMCP runs with a single untuned UCB1 exploration constant per battery, and Appendix S sweeps that constant on Tiger, Diagnosis, and Tileworld-6 $\times$ 6 (Table 34). The sweep changes the Tiger reading substantially. At its best swept constant POMCP reaches  $96.3\% \pm 0.6\text{pp}$  success, and we cannot distinguish that configuration from MCTS-EFE at its default constant on either metric, that is, we fail to reject the null hypothesis of no difference (seed-level Welch  $p=0.09$  and  $p=0.54$ ). The 88.8% figure above therefore measures an untuned constant rather than the solver. The sweep also shows MCTS-EFE’s default is not its own best, so the appendix compares the two solvers with each at a constant selected on tuning seeds disjoint from every evaluation seed, under a rule recorded before the selection was run. At those constants, MCTS-EFE stays ahead on Tiger success ( $98.2\% \pm 0.3\text{pp}$  against  $96.3\% \pm 0.6\text{pp}$ ,  $p=0.024$ , surviving Holm correction within the appendix’s six-test family) with reward not distinguishable ( $p=0.16$ ), so tuning narrows the Tiger gap sharply without removing it. Because the canonical runs had been inspected before the rule was written, that narrow Tiger result is retrospective evidence rather than a prospective confirmation, and a confirmatory claim would need fresh evaluation seeds that no one has looked at. The Diagnosis and Tileworld differences are large enough that the same caveat does not change their reading. Where the agent must choose among tests, tuning the constant leaves most of the gap in place. POMCP’s selected configuration reaches  $71.4\% \pm 1.5\text{pp}$  on Diagnosis

and  $19.7\% \pm 1.7\text{pp}$  on Tileworld- $6 \times 6$  against MCTS-EFE’s  $94.7\% \pm 0.4\text{pp}$  and  $95.4\% \pm 0.6\text{pp}$ , every comparison significant after Holm correction.

A second objection is that the gap reflects POMCP’s rollout policy. The comparison already avoids a pure-random-rollout strawman, since POMCP’s semi-informed rollouts sample observations uniformly but end in a belief-optimal commit (Appendix S). Information-gain rollouts do not close the Diagnosis and Tileworld gap either. At  $c=5$  they put Diagnosis at 71.4% against the uniform-rollout POMCP’s 68.2 at the same constant and Tileworld- $6 \times 6$  at 19.1% against 11.3, neither difference surviving Holm correction (seed-level Welch  $p=0.25$  and  $p=0.023$ ). On Tiger they change nothing at all, since a single observation action makes the informed and uniform rollout policies identical. Richer domain knowledge or a learned value function in the rollout could narrow that gap.

Taken together, the sweep and the rollout check show that the remaining gap is not a matter of the exploration constant or of the rollout policy. Where sensing is one repeated action and only its stopping time is in question, as on Tiger, tuning the constant recovers most of the difference. Where the agent must choose which of several tests to run, neither control removes more than a small part of the gap, which stays above twenty points on Diagnosis and above seventy points on Tileworld- $6 \times 6$  at every swept configuration. What the evidence supports as the explanation is MCTS-EFE’s joint configuration of EFE-greedy leaf evaluation over exact beliefs and exact commit-child values, since those three components were held fixed together in every run of the sweep, the rollout check, and the ablation. We therefore do not attribute the gap to directed observation selection in isolation. Separating the leaf evaluation from the exact belief and commit values would need a further matched-compute comparison that we have not run. The claim is not that EFE-based planning exceeds state-of-the-art POMDP solvers, but that the EFE objective supplies a principled, zero-tuning in-tree reward and leaf evaluation in place of hand-designed rollouts.

The approach also scales to multiple observation actions. For single-observation environments such as Tiger, the observation-outcome enumeration (2 outcomes per action) keeps expansion costs low, and the question is whether the same holds once the agent must choose among several tests. On Diagnosis ( $N=4$ ,  $K=2$  tests), MCTS-EFE at  $H=5$  achieves  $94.2\% \pm 0.5\text{pp}$  success compared to POMCP’s  $68.7\% \pm 1.8\text{pp}$  at the same simulation count (200 simulations, seed-level SE, seed-level Welch  $p=7.2 \times 10^{-5}$ , `results_mcts_efe.csv`). To exercise the approach on a larger multi-observation environment, we evaluate on Tileworld  $6 \times 6$  ( $|S|=36$ ,  $K=6$  scans). MCTS-EFE saturates at  $95.4\% \pm 0.6\text{pp}$  success across both simulation budgets tested (200 and 500, seed-level SE), well above the  $71.9\% \pm 1.5\text{pp}$  of Exact-EFE (the full recursive EFE agent, called EFE in the earlier tables, evaluated here within the same MCTS battery), but at a reward cost ( $-25.02$  vs. Exact-EFE’s  $-21.62$ ) from scanning substantially more (32.3 vs. 14.8 observations on average). MCTS’s sampled rollouts do not value information as precisely as the exact recursion, so it trades reward for a large success-rate gain by scanning more thoroughly. On Tiger and Diagnosis there is no such trade, since MCTS-EFE sits slightly below Exact-EFE on success and further be-

low it on reward (97.7% against 99.9% and +3.73 against +5.65 on Tiger, 94.2% against 97.1% and -3.26 against -1.44 on Diagnosis). POMCP collapses on this larger observation menu regardless of simulation budget ( $10.7\% \pm 1.1\text{pp}$  at 200 simulations,  $5.6\% \pm 1.1\text{pp}$  at 500, seed-level SE). MCTS-EFE’s 85–90pp success-rate gap over POMCP (seed-level Welch  $p < 10^{-9}$  at both budgets) therefore shows that the gap persists at this scale, even though the specific reward/success trade-off MCTS-EFE strikes here differs from the smaller environments. Semi-informed rollouts still cannot identify which of the 6 scans to perform. For larger observation spaces, more efficient tree policies, such as progressive widening or double progressive widening [5], techniques that limit how many children a node expands as a function of its visit count, would further improve scalability. We leave this to future work.

*Practical guidance.* The experiments suggest the following guidance for choosing an information weight. The numeric thresholds for reward asymmetry  $\alpha$ , horizon  $H$ , and discount factor  $\gamma$  are estimates from this benchmark suite, not established regime boundaries.

On our suite, reward asymmetry  $\alpha \geq 5$  places  $w=1$  well inside Proposition 2’s  $H=2$  near-optimality interval, because the penalty for a wrong commit far exceeds the observation cost. The random-environment study tempers this into a favorable-regime heuristic rather than a guarantee (79% of  $\alpha \geq 10$  samples pass at  $H=3$ , Appendix J).

With multiple observation actions, EFE’s joint pragmatic–epistemic objective selects which information to gather and can improve on reward-only planning at matched horizon. The observed advantage depends on the gap between greedy and information-optimal behavior and on whether the search is deep enough to represent the latter. State-space size alone does not predict this gap. On the Tileworld sweep, the separation is within sampling error at  $4 \times 4$  and  $6 \times 6$  and opens at  $8 \times 8$ , where Planning collapses (Figure 16). Diagnosis at  $N=16$  under the  $H=2$  scaling protocol shows no consistent EFE–Planning gap (Appendix G), the advantage vanishes on RockSample[11,11] (Section 7.4), and on Structural Inspection the accuracy gap narrows from  $N=8$  to  $N=16$  (Section 7.5).

When observation and navigation preserve the hidden state, the information-unit-weight equivalence also holds for interleaved observe-act settings (factored observation POMDPs, Table 2). EFE then directs both where to go and what to check (Tables 10, 11).

When a weight cannot be tuned per task, transfer performance depends on the target environment. The information-unit weight  $w=1$  ties the transferred success-tuned  $w=10$  and  $w=20$  on Tiger and Diagnosis, where only  $w=50$ , the top of the tested grid, degrades. On Bandit and Testbed, every transferred success-tuned weight costs reward relative to  $w=1$  (Table 12).

Recursive EFE propagates epistemic value across steps at horizons  $H \geq 2$ . At  $H=1$ , it reduces to myopic information gain with  $w=1$ . On the tested multi-observation environments, its advantage appears at discount factors  $\gamma \geq 0.99$ . Heavier discounting truncates the effective horizon below the number of

observations needed for confident disambiguation (Appendix K). This discount threshold is read off two environments at one horizon each and should not be treated as a general boundary.

EFE is not recommended in three situations. The first is symmetric penalties,  $\alpha \approx 1$ . The second is navigation-style POMDPs where observations arrive only as a side effect of moving, as proximity feedback at the agent’s current position, so that a greedy mover already receives informative feedback (Table 20). The third is a model that overestimates observation accuracy, the direction where measured degradation already appears at +0.05 mismatch (Appendix L).

*Limitations and future work.* The formal equivalence extends from observe-then-commit (Proposition 1) to factored observation POMDPs (Proposition 3), covering settings where observation actions preserve the hidden state. POMDPs where information-gathering actions change the hidden state (e.g., destructive testing) require an epistemic term recomputed from transition-aware predictive and posterior distributions, whose departure from the observation-only posterior the transition–observation coupling term  $\Delta_T$  diagnoses, and are not covered. Extending the formal treatment to such settings remains future work. For slowly drifting hidden states (e.g., progressive disease),  $\Delta_T$  may be nonzero, and whether it is small where it is depends on support and regularity conditions we have not established. With a perfect sensor the observation-only posterior can be a point mass, and an arbitrarily small probability of a state change then places posterior mass outside that support, so the divergence as defined is infinite rather than small. What the coupling depends on is how far the transition kernel sits from the identity on the hidden state and on the supports involved, not the transition’s entropy.  $\Delta_T$  is a diagnostic of when the reduction’s hypothesis fails. We have not shown it to be an additive correction to the EFE value or a bound on the resulting decision error, and a perturbation result of that kind is future work. Example 1 makes the distinction concrete, since its destructive test is deterministic and therefore has zero transition entropy while its coupling term is maximal. Characterizing the regime where the error remains acceptable is an open question.

On the practical side, three limitations stand out, namely the cost of tree search on multi-observation environments, the assumption of a known model, and the domain structure on which the advantage depends. The first is that scaling MCTS-EFE to multi-observation environments requires more efficient tree policies, as discussed above. The second is that the EFE agent assumes a known generative model. The misspecification analysis (Appendix L) shows graceful degradation with moderate model errors. Extending to full model learning, where AIF and Bayes-Adaptive MDPs (BAMDPs) converge on treating uncertainty about the model itself as part of the state that the agent’s actions inform, remains important future work. The third concerns where the advantage appears at all. Navigation (Appendix F) shows that scale alone does not rescue epistemic planning when the observation model is proximity-based. NavMyopic, the reward-only greedy mover that always heads toward the most likely goal cell, leads nominally at all three grid sizes tested and significantly at  $5 \times 5$ , and no

epistemic agent significantly beats it at any size (Appendix F). The practical contrast is with domains that offer explicit, choice-set observation actions (Diagnosis, Tileworld, RockSample), where EFE separates from reward-only planning once the search depth reaches the informative actions and greedy behavior has slack to lose. State-space size alone does not predict this separation, as the slack condition above and the RockSample[11,11] reading of Section 7.4 spell out. Finally, the two kinds of claim in this section carry different status. The formal EFE-to- $\rho$ -POMDP equivalences (Propositions 1 and 3) are exact, while the numeric regime guidance above is estimated from this suite. The principled derivation of exploration weights could benefit safety-critical decision-making by reducing reliance on ad-hoc tuning.

## 9 Conclusion

We treat the weight on an information-gain utility as a price rather than a hyperparameter. Recasting  $w$  as the operational shadow price of a sensing budget  $B$  (Section 4) turns the caveat that  $w=1$  is not invariant to rescaling the rewards into a theorem. The receding-horizon planner’s decisions are scale equivariant under the scale-independent tie-break the implementation uses (Proposition 4), so its observation count is unchanged and its sensing cost rescales with the costs. At a usage gap the crossing threshold is a single number that the procedure reports as a grid bracket, and the budget is attained in expectation by a per-episode mixture of the two endpoint policies (Definition 2). Closed-form onset thresholds appear as the first knot of the usage staircase (Corollary 1). The projected, decaying-step dual controller converges under stated conditions (Proposition 6) and, empirically, re-adapts through a  $\times 10$  reward rescale, roughly  $3\times$  faster on the restricted recovery time common to all ten controller seeds when its step-size clock is reset after a detected shift, a heuristic the proposition does not cover (Section 6.3). At target budgets set by a predeclared rule from the calibration curve rather than at the endogenous  $w=1$  usage, the calibrated family meets its expected-usage target on held-out seeds to within 0.13 observations and declares unattainable budgets rather than returning a fallback. Against the near-optimal constrained reference evaluated on the same held-out seeds, its reward falls short of the estimated feasible envelope on eight of the eleven distinct attainable budgets with nominal pointwise 95 percent intervals excluding zero, by amounts that are largest at each environment’s largest budget and reach about a fifth of the reference reward at Bandit’s, and the other three intervals include zero. Two of the eight, Tiger’s two smallest budgets, rest on held-out episodes in which no policy commits wrongly and would not survive the reference’s canonical wrong-commit rate. It is also below the best budget-feasible single weight wherever the budget exceeds the sensing that reward alone justifies (Section 6.9). The procedure calibrates expected usage within a family. It does not maximize reward under a cap, and the study measures how large that difference is. We also report where the budget alone does not help. A usage target fixes how much sensing is bought in expectation, not what it is bought about. Of the mechanisms we

test, only scoring information gain on the reward-equivalence partition of the state space removes reward-irrelevant sensing (Section 6.10). Naive information gain misvalues an action outright once observation changes the hidden state (Example 1).

That price curve is interpretable because of a second result. Under the reward-based recursive objective we specify, the  $\rho$ -POMDP framework and active inference converge on the same object, a belief-dependent utility that adds information gain at weight  $w=1$  to the reward. That identification is exact when the reward is itself a log score [4], conditional on the reward-based recursion otherwise, and not scale-invariant in either case. Empirically that coefficient is a robust untuned default. It lands inside the reward-maximizing tied bracket on Tiger, Diagnosis, and Bandit, falls outside it on Testbed, and is Pareto-dominated by  $w=20$  on Tileworld (Section 7.2). It is statistically equivalent to SARSOP by a fixed-margin TOST on the three core environments, under both unpaired and paired analyses (Section 6.7), and close to an estimated constrained-optimal frontier (Section 6.8). Proposition 3 carries the identity into factored observation POMDPs for their hidden-state-preserving actions, and RockSample and Structural Inspection exercise it. What predicts EFE’s advantage over reward-only planning in the regimes we tested is the search depth reaching the informative actions and greedy behavior having slack to lose, not problem size alone. Against POMCP, the gap where the agent chooses among several tests survives both a tuned exploration constant and information-gain rollouts, and we attribute it to MCTS-EFE’s joint configuration of EFE leaf evaluation over exact beliefs and exact commit-child values rather than to directed observation selection in isolation. On Tiger, where sensing is a single repeated action, tuning POMCP’s exploration constant recovers most of the difference (Section 8). Under a shared reward convention, active inference supplies an untuned information-unit weight that serves as a strong default in the tested settings; it promises neither scale-free calibration nor uniform optimality. A higher weight tuned for success can buy accuracy where near-certain commits are required, at the reward cost shown by the sweep. When expected sensing usage is the design requirement, stating a budget and calibrating the corresponding weight gives the engineer an operating point away from that default.

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**Table 10.** RockSample results. RS[5,3] and RS[7,4] run 500 episodes  $\times$  10 seeds (5,000 episodes), and RS[7,8] and RS[11,11] run 100 episodes  $\times$  5 seeds (500 episodes, reduced protocol). Greedy samples without checking. POMCP is the genuine search-tree solver at its tuned, frozen configuration (2,048 simulations, Appendix M). Rollout only runs POMCP’s approach-then-check rollout policy standalone, with no search tree. Bold marks the best mean reward per instance plus any agent not significantly different from it by a seed-level Welch  $t$ -test with Holm–Bonferroni correction, corrected within metric (complete pairwise statistics in the committed `results_rocksample_*_stats.csv` files). Reward reported as mean  $\pm$  seed-level SE (SE of the per-seed means over 10 seeds for RS[5,3] and RS[7,4] and over 5 seeds for RS[7,8] and RS[11,11]). Steps and Checks omitted here (per-instance step and check counts, and a Flat-MC(1000) reference on every instance, are in Appendix M).

| Instance  | Agent                      | Good | Bad  | Reward                              |
|-----------|----------------------------|------|------|-------------------------------------|
| RS[5,3]   | Greedy                     | 1.49 | 1.51 | $+3.71 \pm 0.26$                    |
|           | Planning ( $w=0$ , $d=3$ ) | 0.99 | 0.04 | $+14.28 \pm 0.13$                   |
|           | Plan+IG ( $w=5$ , $d=3$ )  | 1.44 | 0.01 | <b><math>+16.41 \pm 0.13</math></b> |
|           | EFE ( $w=1$ , $d=3$ )      | 1.44 | 0.03 | <b><math>+16.47 \pm 0.12</math></b> |
|           | POMCP (2048 sims)          | 0.68 | 0.03 | $+12.74 \pm 0.11$                   |
|           | Rollout only (approach)    | 1.41 | 0.07 | <b><math>+16.67 \pm 0.13</math></b> |
| RS[7,4]   | Greedy                     | 1.98 | 2.02 | $-1.38 \pm 0.36$                    |
|           | Planning ( $w=0$ , $d=4$ ) | 0.95 | 0.05 | $+13.97 \pm 0.11$                   |
|           | Plan+IG ( $w=5$ , $d=4$ )  | 1.69 | 0.01 | $+14.56 \pm 0.14$                   |
|           | EFE ( $w=1$ , $d=4$ )      | 1.44 | 0.01 | <b><math>+15.96 \pm 0.15</math></b> |
|           | POMCP (2048 sims)          | 0.66 | 0.03 | $+12.01 \pm 0.06$                   |
|           | Rollout only (approach)    | 1.88 | 0.10 | <b><math>+15.85 \pm 0.17</math></b> |
| RS[7,8]   | Greedy                     | 4.02 | 3.98 | $-5.60 \pm 1.03$                    |
|           | Planning ( $w=0$ , $d=3$ ) | 0.48 | 0.03 | $+12.31 \pm 0.35$                   |
|           | Plan+IG ( $w=5$ , $d=3$ )  | 3.40 | 0.01 | $+23.76 \pm 0.58$                   |
|           | EFE ( $w=1$ , $d=3$ )      | 2.61 | 0.06 | $+21.85 \pm 0.79$                   |
|           | POMCP (2048 sims)          | 1.71 | 0.10 | $+15.07 \pm 0.45$                   |
|           | Rollout only (approach)    | 3.83 | 0.19 | <b><math>+28.33 \pm 0.82</math></b> |
| RS[11,11] | Greedy                     | 5.48 | 5.52 | $-18.90 \pm 1.04$                   |
|           | Planning ( $w=0$ , $d=2$ ) | 0.48 | 0.03 | $+13.23 \pm 0.18$                   |
|           | Plan+IG ( $w=5$ , $d=2$ )  | 0.50 | 0.03 | $+13.18 \pm 0.15$                   |
|           | EFE ( $w=1$ , $d=2$ )      | 0.48 | 0.03 | $+13.23 \pm 0.18$                   |
|           | POMCP (2048 sims)          | 0.49 | 0.03 | $+12.94 \pm 0.24$                   |
|           | Rollout only (approach)    | 5.22 | 0.23 | <b><math>+28.66 \pm 0.83</math></b> |

**Table 11.** Structural Inspection results.  $N=8$  and  $N=16$  run 2,500 episodes each (500 per seed  $\times$  5 seeds). Reward and accuracy are reported as mean  $\pm$  seed-level SE. Greedy diagnoses without testing. Bold marks the best accuracy and the best reward per instance, plus any value within sampling error of that best (seed-level Welch  $p > 0.05$ , uncorrected). The two are not always the same agent. EFE is a competitive untuned trade-off, not uniquely best on all metrics. Accuracy is the fraction of components correctly diagnosed.

| Instance               | Agent                 | Accuracy                           | Missed | Tests | Reward                              |
|------------------------|-----------------------|------------------------------------|--------|-------|-------------------------------------|
| $N=8$<br>$ S =256$     | Greedy                | $70.7 \pm 0.5\%$                   | 2.34   | 0.0   | $-114.33 \pm 1.96$                  |
|                        | Planning ( $w=0$ )    | $73.0 \pm 0.3\%$                   | 0.07   | 12.7  | <b><math>-17.85 \pm 0.23</math></b> |
|                        | Planning+IG ( $w=5$ ) | <b><math>96.3 \pm 0.1\%</math></b> | 0.10   | 19.7  | $-27.98 \pm 0.44$                   |
|                        | EFE ( $w=1$ )         | $91.1 \pm 0.2\%$                   | 0.09   | 15.5  | $-20.95 \pm 0.30$                   |
| $N=16$<br>$ S =65,536$ | Greedy                | $70.5 \pm 0.4\%$                   | 4.72   | 0.0   | $-233.09 \pm 3.13$                  |
|                        | Planning ( $w=0$ )    | $78.2 \pm 0.2\%$                   | 0.24   | 22.4  | <b><math>-45.42 \pm 0.23</math></b> |
|                        | Planning+IG ( $w=5$ ) | <b><math>97.3 \pm 0.0\%</math></b> | 0.14   | 38.3  | $-54.74 \pm 0.40$                   |
|                        | EFE ( $w=1$ )         | $90.8 \pm 0.1\%$                   | 0.24   | 28.1  | <b><math>-45.80 \pm 0.60</math></b> |

**Table 12.** Zero-shot weight transfer (500 episodes  $\times$  5 seeds, reward reported as mean  $\pm$  seed-level SE, `results_transfer.csv`). Rows include  $w=1$ , reward-tuned weights ( $w_{\text{ret}}^*$ ), and success-tuned weights ( $w_{\text{succ}}^*$ ), the Planning+IG-class values recorded in `results_supplementary_tuned_weights.csv`, matching the evaluated agent class. Success-tuned transfer harms reward at the top of the grid, while moderate weights transfer well. Bold. Best reward per column (Tiger and Diagnosis show a four-way tie among  $w \in \{0.5, 1, 10, 20\}$ , bit-identical under the fixed pipeline, not merely within SE).

| Weight                                             | Tiger                              | Diagnosis                          | Bandit                             | Testbed                            |
|----------------------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| $w=0.5$ ( $w_{\text{ret}}^*$ )                     |                                    |                                    |                                    |                                    |
| Diag./Testbed<br>$w=1$ (EFE / $w_{\text{ret}}^*$ ) | <b><math>+5.33 \pm 0.26</math></b> | <b><math>-1.26 \pm 0.34</math></b> | $+6.05 \pm 0.11$                   | <b><math>+0.48 \pm 0.02</math></b> |
| Tiger/Bandit<br>$w=10$ ( $w_{\text{succ}}^*$ )     | <b><math>+5.33 \pm 0.26</math></b> | <b><math>-1.26 \pm 0.34</math></b> | <b><math>+6.24 \pm 0.09</math></b> | $+0.37 \pm 0.01$                   |
| Testbed<br>$w=20$ ( $w_{\text{succ}}^*$ )          | <b><math>+5.33 \pm 0.26</math></b> | <b><math>-1.26 \pm 0.34</math></b> | $+5.25 \pm 0.06$                   | $+0.01 \pm 0.01$                   |
| Bandit<br>$w=50$ ( $w_{\text{succ}}^*$ )           | <b><math>+5.33 \pm 0.26</math></b> | <b><math>-1.26 \pm 0.34</math></b> | $+5.05 \pm 0.07$                   | $-0.18 \pm 0.01$                   |
| Tiger/Diag.                                        | $+4.25 \pm 0.08$                   | $-3.68 \pm 0.14$                   | $+4.23 \pm 0.06$                   | $-0.38 \pm 0.01$                   |

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## A Proof of Proposition 1

*Proof.* We show that  $\arg \min_a \mathcal{G}(a, b, d) = \arg \max_a V_\rho(a, b, d)$  at every belief node, where  $V_\rho$  is the  $\rho$ -POMDP value function with  $\rho_{\text{EFE}}(b, a)$  as defined in the proposition.

Define  $V(a, b, d) \triangleq -\mathcal{G}(a, b, d)$ . Since negation reverses ordering,  $\arg \min_a \mathcal{G} = \arg \max_a V$ .

Case 1, commit actions. For commit action  $i$ , Equation 3 gives  $\mathcal{G}(\text{commit}_i) = -\mathbb{E}_b[R_i]$ , so  $V(\text{commit}_i, b) = \mathbb{E}_b[R_i]$ . The  $\rho$ -POMDP value is  $V_\rho(\text{commit}_i, b) = \mathbb{E}_b[R_i] + \rho_{\text{EFE}}(b, \text{commit}_i) = \mathbb{E}_b[R_i] + 0$ . These are identical.

Case 2, observation actions. For observation action  $k$ , Equation 3 gives  $\mathcal{G}(\text{obs}_k) = c_k - I_k(b) + \mathbb{E}_o[\min_{a'} \mathcal{G}(a', b'_o)]$ . Negating both sides gives  $V(\text{obs}_k, b) = -c_k + I_k(b) + \mathbb{E}_o[\max_{a'} V(a', b'_o)]$ . The  $\rho$ -POMDP Bellman backup with  $R(b, \text{obs}_k) = -c_k$  and  $\rho_{\text{EFE}}(b, \text{obs}_k) = I_k(b)$  gives  $V_\rho(\text{obs}_k, b) = -c_k + I_k(b) + \gamma \mathbb{E}_o[\max_{a'} V_\rho(a', b'_o)]$ . With  $\gamma = 1$  (undiscounted finite horizon), the two recursions are structurally identical. Since the terminal values (commit actions) agree and the recursive updates agree, by induction on the remaining horizon  $H - d$ ,  $V(a, b, d) = V_\rho(a, b, d)$  for all  $a, b, d$ . Policies therefore coincide at every belief node.

## B Full Per-Environment Results

## C Two-State Testbed

Under symmetric rewards (+1/−1), EFE gathers more information than is instrumentally optimal (+0.38 vs. Planning’s +0.48, seed-level Welch  $p=1.1 \times 10^{-5}$ ) because uncertainty reduction has intrinsic value under EFE. The success-tuned agents show the same effect more starkly. At their in-class tuned weights both drive reward to a near-zero +0.01 while reaching 99.5% success, giving up almost the entire return to buy the last few points of accuracy. This is the one environment where the posterior-vote baseline exactly ties the best agent, and on Bandit it posts the highest mean reward of any agent. Posterior-vote matches reward-only Planning bit-for-bit (+0.48, 89.8%, 3.15 observations), and both beat EFE on reward. The Epistemic-only ablation of Table 3, which sets  $\mathcal{G}(\text{commit}) = 0$ ,

**Table 13.** Tiger problem (5,000 episodes at 1,000 per seed  $\times$  5 seeds,  $H=6$ ). Reward asymmetry  $+10/-100$ . Bold marks the best-reward row and any row bit-identical with it or within sampling error of it (seed-level Welch  $p > 0.05$ , uncorrected, the conservative direction for a tie claim), here and in the Diagnosis and Bandit tables of this appendix. Printed uncertainty in these tables is pooled episode-level SE, while the bolding rule uses the seed-level test.

| Agent                       | Listens     | Success      | Reward                             |
|-----------------------------|-------------|--------------|------------------------------------|
| Myopic                      | 1.00        | 85.3%        | $-7.15 \pm 0.55$                   |
| <b>Planning</b> ( $H=6$ )   | <b>4.20</b> | <b>99.4%</b> | <b><math>+5.19 \pm 0.12</math></b> |
| <b>Info Gain</b> ( $w=20$ ) | <b>4.20</b> | <b>99.4%</b> | <b><math>+5.19 \pm 0.12</math></b> |
| Planning+IG ( $w=50$ )      | 5.66        | 99.9%        | $+4.21 \pm 0.06$                   |
| Epistemic-only              | 0.00        | 49.7%        | $-45.31 \pm 0.78$                  |
| Posterior-vote              | 2.64        | 96.8%        | $+3.79 \pm 0.28$                   |
| <b>EFE</b> ( $H=6$ )        | <b>4.20</b> | <b>99.4%</b> | <b><math>+5.19 \pm 0.12</math></b> |

**Table 14.** Diagnosis ( $N=4$ ,  $K=2$  tests,  $H=3$ , 5,000 episodes at 1,000 per seed  $\times$  5 seeds).

| Agent                  | Tests       | Success      | Reward                             |
|------------------------|-------------|--------------|------------------------------------|
| Myopic                 | 2.00        | 64.7%        | $-13.17 \pm 0.41$                  |
| Planning ( $H=3$ )     | 5.86        | 88.8%        | $-2.57 \pm 0.27$                   |
| Info Gain ( $w=100$ )  | 13.19       | 99.4%        | $-3.58 \pm 0.09$                   |
| Planning+IG ( $w=50$ ) | 13.16       | 99.2%        | $-3.66 \pm 0.10$                   |
| Epistemic-only         | 0.00        | 24.7%        | $-35.16 \pm 0.37$                  |
| Posterior-vote         | 5.83        | 88.4%        | $-2.78 \pm 0.27$                   |
| <b>EFE</b> ( $H=3$ )   | <b>9.66</b> | <b>97.2%</b> | <b><math>-1.37 \pm 0.15</math></b> |

**Table 15.** Structured bandit ( $K=4$  arms,  $H=2$ , 5,000 episodes at 1,000 per seed  $\times$  5 seeds).

| Agent                  | Inspections | Success      | Reward                             |
|------------------------|-------------|--------------|------------------------------------|
| Myopic                 | 2.06        | 62.2%        | $+5.56 \pm 0.06$                   |
| Planning ( $H=2$ )     | 3.29        | 71.0%        | $+5.75 \pm 0.06$                   |
| Info Gain ( $w=20$ )   | 9.54        | 98.8%        | $+5.12 \pm 0.04$                   |
| Planning+IG ( $w=20$ ) | 9.57        | 98.9%        | $+5.11 \pm 0.04$                   |
| Epistemic-only         | 0.00        | 24.7%        | $+3.23 \pm 0.05$                   |
| <b>Posterior-vote</b>  | <b>5.01</b> | <b>87.0%</b> | <b><math>+6.32 \pm 0.05</math></b> |
| <b>EFE</b> ( $H=2$ )   | <b>5.11</b> | <b>86.9%</b> | <b><math>+6.27 \pm 0.05</math></b> |

**Table 16.** Tileworld 6×6 (2,500 episodes at 500 per seed × 5 seeds,  $H=2$ ). This covers every agent in the Tileworld battery (`experiments/run_tileworld.py`). POMCP and MCTS-EFE were additionally run on Tileworld 6×6 under their own protocols and are reported in Table 32 and Section 8. Posterior-vote and the PyMDP-AIF consistency agent are observe-then-commit agents and were not run on the spatial environments, so they are absent here rather than omitted. Bold marks EFE and any agent not significantly different from it at the seed level (Welch  $p > 0.05$ , uncorrected), not the single best value per column. Epistemic-only commits immediately (0 scans) rather than exhausting the step cap, matching Myopic exactly (Section 7). Reward uncertainty is pooled episode-level SE, matching the other tables of this appendix.

| Agent                         | Scans        | Success      | Reward                              |
|-------------------------------|--------------|--------------|-------------------------------------|
| Myopic                        | 0.00         | 2.9%         | $-48.25 \pm 0.20$                   |
| Planning ( $H=2$ )            | 15.64        | <b>73.8%</b> | <b><math>-21.39 \pm 0.54</math></b> |
| Info Gain ( $w=1$ )           | 5.20         | 31.7%        | $-36.16 \pm 0.56$                   |
| Info Gain ( $w=100$ )         | 32.30        | 98.0%        | $-23.50 \pm 0.24$                   |
| Planning+IG ( $w=50$ )        | 32.25        | 97.3%        | $-23.85 \pm 0.26$                   |
| Epistemic-only                | 0.00         | 2.9%         | $-48.25 \pm 0.20$                   |
| <b>EFE (<math>H=2</math>)</b> | <b>14.75</b> | <b>72.0%</b> | <b><math>-21.53 \pm 0.54</math></b> |

**Table 17.** Tileworld 8×8 (2,500 episodes at 500 per seed × 5 seeds,  $H=2$ ). As with the 6×6 table, this covers every agent in the Tileworld battery. Posterior-vote and PyMDP-AIF are observe-then-commit agents and were not run here. Bold marks EFE and any agent not significantly different from it at the seed level (Welch  $p > 0.05$ , uncorrected), not the single best value per column, so the higher success rates of the two heavily scanning tuned agents are unbolded by design. EFE’s success rate here (69.8%) is nearly identical to the 69.4% reported in the independent spatial-scaling sweep (Figure 16). Likewise Planning’s 1.8% here is close to that sweep’s 1.4%. The two figures come from separate experiment scripts with different per-seed episode counts, and the small residual gaps are consistent with ordinary seed-to-seed variation. Planning, untuned Info Gain, and Epistemic-only all fail to explore because a single noisy scan over 64 states barely narrows the belief, so all three commit immediately and tie exactly with Myopic, while EFE’s epistemic bonus drives meaningful exploration. Reward uncertainty is pooled episode-level SE, matching the other tables of this appendix.

| Agent                         | Scans        | Success      | Reward                              |
|-------------------------------|--------------|--------------|-------------------------------------|
| Myopic                        | 0.00         | 1.8%         | $-48.94 \pm 0.16$                   |
| Planning ( $H=2$ )            | 0.00         | 1.8%         | $-48.94 \pm 0.16$                   |
| Info Gain ( $w=1$ )           | 0.00         | 1.8%         | $-48.94 \pm 0.16$                   |
| Info Gain ( $w=100$ )         | 39.45        | 97.2%        | $-31.15 \pm 0.25$                   |
| Planning+IG ( $w=50$ )        | 39.49        | 97.5%        | $-31.00 \pm 0.25$                   |
| Epistemic-only                | 0.00         | 1.8%         | $-48.94 \pm 0.16$                   |
| <b>EFE (<math>H=2</math>)</b> | <b>17.60</b> | <b>69.8%</b> | <b><math>-25.69 \pm 0.56</math></b> |

**Table 18.** Information-seeking testbed (5,000 episodes at 1,000 per seed  $\times$  5 seeds,  $H=4$ ). Symmetric rewards  $+1/-1$ . Each tuned agent is success-tuned in its own agent class, as `tune_info_gain_weight` requires, at  $w_{\text{succ}}^*=20$  for myopic Info Gain and  $w_{\text{succ}}^*=10$  for Planning+IG (both recorded in `results_supplementary_tuned_weights.csv`). At their respective tuned weights the two converge to the same policy and are bit-identical on all three metrics. Bold marks the best value in the success and reward columns, with exact ties bold together. Both buy near-ceiling success by scanning almost twice as much as EFE, and both give up nearly all the return to do it. Every row here is produced by `experiments/run_supplementary.py`.

| Agent                                  | Obs. | Success      | Reward                             |
|----------------------------------------|------|--------------|------------------------------------|
| Myopic                                 | 1.00 | 76.3%        | $+0.43 \pm 0.01$                   |
| Planning ( $H=4$ )                     | 3.15 | 89.8%        | <b><math>+0.48 \pm 0.01</math></b> |
| Info Gain ( $w_{\text{succ}}^*=20$ )   | 9.79 | <b>99.5%</b> | $+0.01 \pm 0.01$                   |
| Planning+IG ( $w_{\text{succ}}^*=10$ ) | 9.79 | <b>99.5%</b> | $+0.01 \pm 0.01$                   |
| Posterior-vote                         | 3.15 | 89.8%        | <b><math>+0.48 \pm 0.01</math></b> |
| EFE ( $H=4$ )                          | 5.50 | 96.3%        | $+0.38 \pm 0.01$                   |

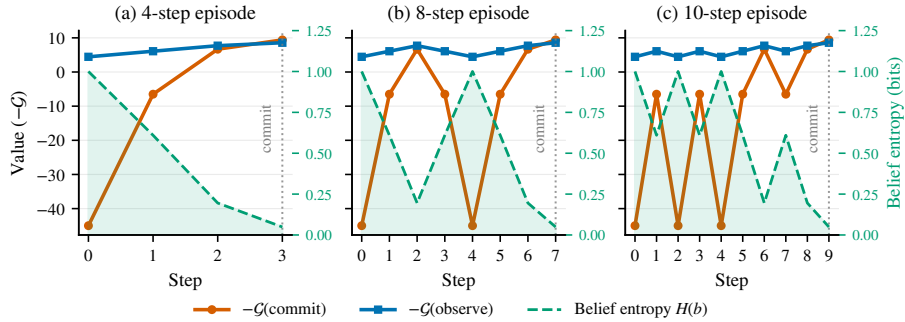
also observes 3.15 times and succeeds at 89.8% here, matching reward-only Planning exactly on both figures (`results_summary.csv`), because Testbed’s observation is cheap enough that information gain alone clears its cost. On Tiger, Diagnosis, Bandit, and Tileworld it instead commits immediately at the chance rate (Appendix B). The testbed result is informative about the conditions under which  $w=1$  is near-optimal. With symmetric  $\pm 1$  rewards a second observation at the post-first-observation belief has zero instrumental value, since the disconfirming outcome returns the belief exactly to the tie point, so its 0.1 cost buys no reward at all. Proposition 2’s over-observation ceiling is correspondingly tight here,  $w_{\text{hi}}^* \approx 1.01$  nats (Table 1), which sits below the implemented bit-denominated  $w=1$  at 1.44 nats. The reward-optimal weight therefore falls below the canonical one. In this regime, EFE assigns too much weight to information gain. Many real-world decision problems, including medical diagnosis, fault detection, and security screening, have asymmetric penalties ( $|R^-|/|R^+| \gg 1$ ), precisely the regime where  $w=1$  is well-calibrated (Tiger, Diagnosis).

## D Supplementary Figures

This appendix collects figures that support the main text but are not needed to follow it. They cover the decomposition of the EFE objective along a trajectory, how observation actions scale and how tests are selected, the reward-asymmetry sweep on Tiger, the evolution of the belief within an episode, and exploration efficiency.

### D.1 EFE Trajectory Decomposition

Figure 17 decomposes EFE within Tiger episodes. The agent commits at the crossover where commit value exceeds observe value. As belief entropy falls, the



**Fig. 17.** EFE decomposition within Tiger episodes ( $H=6$ ). The agent commits at the crossover where commit value (circles) exceeds observe value (squares). As belief entropy (dashed line with shading) decreases, the transition from exploration to exploitation occurs automatically.

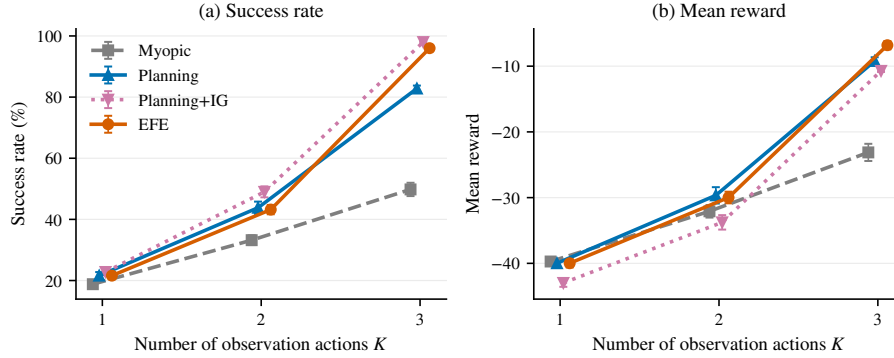
transition from exploration to exploitation happens on its own, with no stopping rule.

## D.2 Observation Action Scaling and Adaptive Test Selection

Holding  $N=8$  fixed on Diagnosis and varying  $K$  from 1 to 3 isolates the “which information” hypothesis (Figure 18). At  $K=1$  all agents face the same single test, and EFE and Planning are bit-identical. At  $K=2$  the two remain within seed-level sampling error of each other (43.2% against 43.8% success), with the multi-test evidence at that scale carried by the main Diagnosis battery (Section 7). The separation opens at  $K=3$  (96.0% against 82.8%, +13.2pp), because selecting among competing tests is what the information-gain term contributes. The pattern is a threshold rather than a steady growth. Figure 19 shows the mechanism at work over a single extended episode. Tests become differentially informative as belief concentrates, since some partitions of the state space stop mattering once the condition is narrowed down while others remain decisive. The EFE agent changes its test selection as belief evolves. It optimizes a recursive pragmatic-epistemic score, so its chosen test need not maximize the one-step information gain plotted in panel (b). This episode illustrates adaptive test selection, with the aggregate comparison reported in Section 7.

## D.3 Reward Asymmetry on Tiger

To test whether EFE’s Tiger parity holds across reward scales, we sweep the penalty magnitude from  $|R^-|=1$  to 500 (Figure 20). EFE adapts its listening on its own, from 2.7 observations at the smallest penalty to 5.8 at the largest, tracking the stakes without reconfiguration. The sweep re-tunes the myopic InfoGain-Tuned agent’s weight at every penalty, and the tuned weight comes back as  $w=20$



**Fig. 18.** Observation action scaling on Diagnosis ( $N=8$ ,  $H=3$ , 500 episodes). EFE separates from Planning only once several tests compete. The gap is zero at  $K=1$  (bit-identical policies),  $-0.6\text{pp}$  and within seed-level sampling error at  $K=2$ , and  $+13.2\text{pp}$  at  $K=3$ , where choosing among differentially informative tests is exactly what the information-gain term contributes. InfoGain-Tuned is not part of this battery, so the legend carries four agents rather than the five of Figure 20.

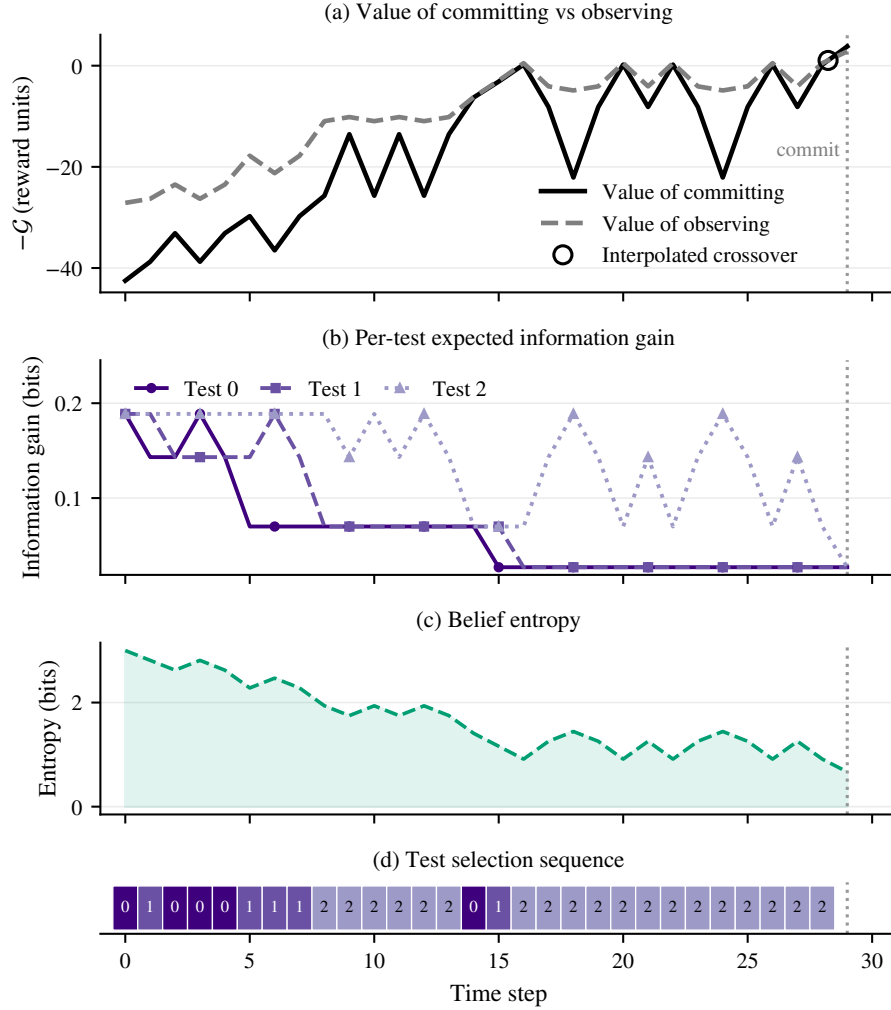
at all of them. At that fixed weight the myopic agent listens 4.37 times at every penalty, while the deeper Planning+IG rises to 5.77 observations at penalties 200 and 500. At low stakes that over-listening costs about 1.6 reward against reward-only Planning. At the largest penalty the fixed-weight myopic agent sits marginally ahead of EFE (4.61 against 4.23), so the case for adaptation is robustness across the sweep rather than uniform dominance.

#### D.4 Belief Evolution

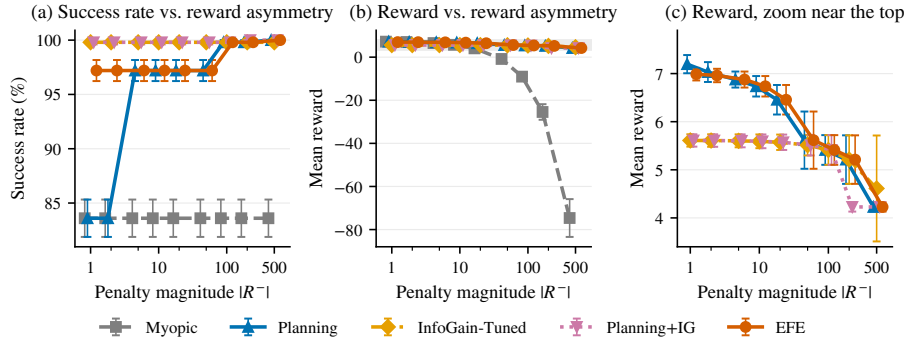
Figure 21 follows belief within one EFE episode on the  $6 \times 6$  Tileworld. Figure 22 does the same on extended Diagnosis instances ( $N=8$ ,  $K=3$ , accuracy 0.75) long enough for the plotted episode to run from fourteen to thirty-two observation steps across the three agents. EFE narrows by partition. Early tests eliminate large groups of states and later tests discriminate among what is left. Reward-only Planning commits earlier and, over the battery, less accurately (Section 7 and Figure 18), though in the plotted episode its fourteenth observation already leaves the belief concentrated on the true state. InfoGain-Tuned keeps testing past the point where further tests pay for themselves in reward.

#### D.5 Exploration Efficiency

Figure 23 averages the temporal dynamics of epistemic foraging over 300 episodes. EFE reduces entropy at a rate comparable to tuned Planning+IG but commits earlier, staying out of the diminishing-returns regime. Its survival curve falls off through steps 10 to 25 while Planning+IG’s extends further right.



**Fig. 19.** EFE decomposition over an extended Diagnosis episode ( $N=8$ ,  $K=3$ ). (a) Value of committing vs. observing. The open circle marks the value crossover interpolated between sampled steps. The dotted vertical line marks the actual discrete commit step in all four panels. (b) Per-test expected information gain. Tests become differentially informative as belief concentrates, and the agent shifts its test selection as they do, though the recursive score it optimizes is not the one-step information gain plotted here. (c) Belief entropy decay. (d) Test selection sequence showing the agent's adaptive strategy.

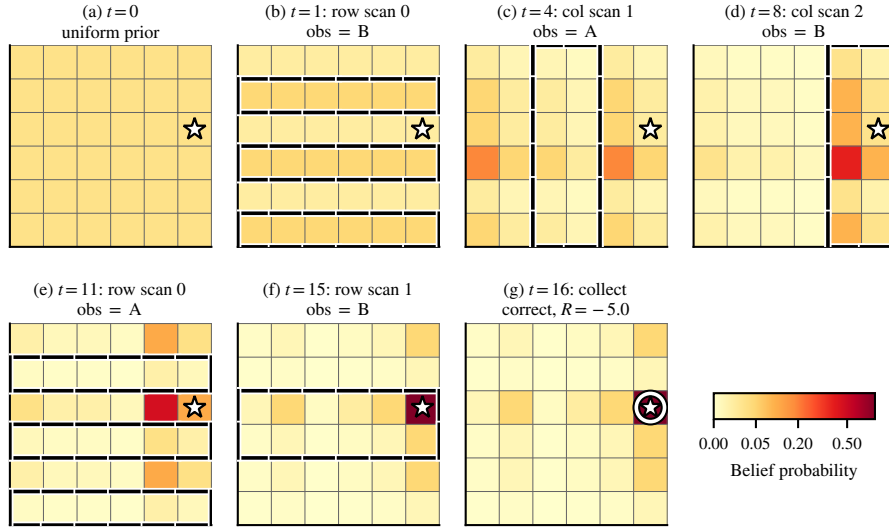


**Fig. 20.** Reward asymmetry sweep on Tiger ( $H=6$ , 500 episodes per point, 5 canonical seeds). EFE and Planning adapt their listening as the penalty grows, with EFE rising from 2.7 observations at  $|R^-|=1$  to 5.8 at 500. The myopic InfoGain-Tuned agent at the fixed tuned weight listens 4.37 times at every penalty, while Planning+IG at the same weight rises to 5.77 at the two largest penalties. The myopic agent’s fixed weight over-listens at low stakes, costing about 1.6 reward against reward-only Planning at  $|R^-|=1$ , and its reward sits marginally ahead of EFE’s only at the largest penalty, where it listens less (4.37 against 5.77).

## E Environment Specifications

RockSample check accuracy decays with distance from the rock (0.95 at range zero, halving every 4 cells, floored at 0.5). The tabulated value is the maximum. Each RockSample instance runs under an episode step cap of  $N^2 + 10K$  steps, giving 55, 89, 129, and 231 for RS[5,3], RS[7,4], RS[7,8], and RS[11,11], and the fixed rock maps are those in ROCKSAMPLE\_CONFIGS (experiments/run\_rocksample.py). On RS[11,11] the start cell (5,0) coincides with a rock while every other rock lies at Euclidean distance 3.16 or more, at or beyond the limit of the sensor’s useful range (accuracy roughly 0.55 at distance 3.16, and floored at chance from 3.7 cells). This geometry is behind the single-informative-check pattern of Section 7.4. Its per-instance tree depth was fixed a priori before the reported runs, not selected on measured agent performance, and the depth-3 check on RS[11,11] (Section 7.4) shows the reported pattern is insensitive to one extra ply. Inspection additionally pays  $-0.5$  per move, gives  $+2$  for a correct nominal call, and  $-5$  for a false alarm, with fault prior 0.3.

All environments are implemented as Gymnasium [64] environments with the standard `reset/step` API. The Diagnosis environment uses  $K = \lceil \log_2 N \rceil$  binary tests, each providing noisy information about a different partition of the state space. The Navigation environment uses proximity-based warm/cold signals with accuracy depending on Manhattan distance to the hidden goal. Tile-world projects the Diagnosis partition structure onto a 2D grid.  $K = 2 \lceil \log_2 n \rceil$  scans, for grid side  $n$ , partition the grid by bit-level splits of the row and column indices, enabling complete spatial disambiguation.



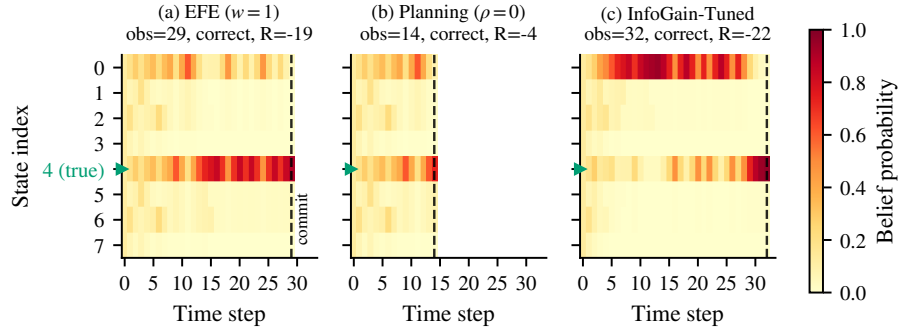
**Fig. 21.** Belief evolution within an EFE episode on the 6x6 Tileworld. Panels (a)–(g) show belief probability over the 36 grid cells on a shared square-root color scale (colorbar, darker = higher probability). Black outlines mark the cell group queried by each scan, each panel’s title reports that scan’s noisy binary readout as obs=A or obs=B naming which half of the queried group the signal favors, and the black-edged white star marks the target tile in every panel. Belief concentrates toward the target, and in the final panel a black-stroked white ring marks the committed cell.

**Table 19.** Full environment specifications for every environment reported in this paper’s results tables and appendices. Variant instances used outside the main results tables (Navigation  $5 \times 5$  and  $7 \times 7$ , Tileworld  $4 \times 4$ , Diagnosis  $N \in \{2, 8, 16\}$ ) differ from their tabulated siblings only in the stated size parameter.  $H$  is the planning horizon for observe-then-commit environments and the tree depth for the interleaved ones.  $\gamma$  is the discount used by the reporting agents. All main-text batteries are undiscounted ( $\gamma=1.0$ , the agent default), and  $\gamma$  is set to 0.999 inside the SARSOP/CPOMDP exports of Section 6.7, which require a discounted infinite-horizon model, and is otherwise varied only in the discount sweep of Appendix K and in two rows of the RockSample POMCP sensitivity battery.  $|S|$  counts hidden-state configurations only, excluding fully observable components such as the RockSample agent position. Move actions are listed under observation actions only for Navigation, where they are the sole information source. Inspection additionally has four move (navigation) actions, omitted from its observation-action count.

| Environment                                            | $ S $           | Obs.<br>actions     | Commit<br>actions | Accuracy  |
|--------------------------------------------------------|-----------------|---------------------|-------------------|-----------|
| Tiger                                                  | 2               | 1                   | 2                 | 0.85      |
| Testbed                                                | 2               | 1                   | 2                 | 0.75      |
| Diagnosis                                              |                 |                     |                   |           |
| ( $N=4$ )                                              | 4               | 2                   | 4                 | 0.80      |
| Bandit                                                 |                 |                     |                   |           |
| ( $K=4$ )                                              | 4               | 4                   | 4                 | 0.80      |
| Navigation                                             | 9               | 4 (move)            | 0 (implicit)      | var.      |
| Tileworld                                              |                 |                     |                   |           |
| ( $6 \times 6$ )                                       | 36              | 6 (scan)            | 36 (collect)      | 0.80      |
| Tileworld                                              |                 |                     |                   |           |
| ( $8 \times 8$ )                                       | 64              | 6 (scan)            | 64 (collect)      | 0.80      |
| Interleaved observe-act (Proposition 3). Undiscounted. |                 |                     |                   |           |
| RockSample                                             |                 |                     |                   | 0.95      |
| RS[5,3]                                                | $2^3=8$         | 3 (check)           | sample/exit       | decaying  |
| RockSample                                             |                 |                     |                   | 0.95      |
| RS[7,4]                                                | $2^4=16$        | 4 (check)           | sample/exit       | decaying  |
| RockSample                                             |                 |                     |                   | 0.95      |
| RS[7,8]                                                | $2^8=256$       | 8 (check)           | sample/exit       | decaying  |
| RockSample                                             |                 |                     |                   | 0.95      |
| RS[11,11]                                              | $2^{11}=2,048$  | 11 (check)          | sample/exit       | decaying  |
| Inspection                                             |                 |                     |                   |           |
| ( $N=8$ )                                              | $2^8=256$       | 2 types $\times$ 8  | diagnose          | 0.70/0.90 |
| Inspection                                             |                 |                     |                   |           |
| ( $N=16$ )                                             | $2^{16}=65,536$ | 2 types $\times$ 16 | diagnose          | 0.70/0.90 |

**Table 19.** Full environment specifications (continued). Sensing costs, commit rewards, and planning horizon or tree depth  $H$  with discount  $\gamma$ . Every RockSample action costs  $-0.5$ . Inspection additionally pays  $-0.5$  per move, and its other outcomes are specified below.

| Environment                                            | Obs. cost     | Correct    | Incorrect    | $H / \gamma$ |
|--------------------------------------------------------|---------------|------------|--------------|--------------|
| Tiger                                                  | $-1.0$        | $+10$      | $-100$       | $6 / 1.0$    |
| Testbed                                                | $-0.1$        | $+1$       | $-1$         | $4 / 1.0$    |
| Diagnosis                                              |               |            |              |              |
| ( $N=4$ )                                              | $-1.0$        | $+10$      | $-50$        | $3 / 1.0$    |
| Bandit                                                 |               |            |              |              |
| ( $K=4$ )                                              | $-0.5$        | $+10$      | $+1$         | $2 / 1.0$    |
| Navigation                                             | $-0.5$        | $+20$      | $-$          | $2 / 1.0$    |
| Tileworld                                              |               |            |              |              |
| ( $6 \times 6$ )                                       | $-1.0$        | $+10$      | $-50$        | $2 / 1.0$    |
| Tileworld                                              |               |            |              |              |
| ( $8 \times 8$ )                                       | $-1.0$        | $+10$      | $-50$        | $2 / 1.0$    |
| Interleaved observe-act (Proposition 3). Undiscounted. |               |            |              |              |
| RockSample                                             |               |            |              |              |
| RS[5,3]                                                | $-0.5$        | $+10$      | $-10$        | $3 / 1.0$    |
| RockSample                                             |               |            |              |              |
| RS[7,4]                                                | $-0.5$        | $+10$      | $-10$        | $4 / 1.0$    |
| RockSample                                             |               |            |              |              |
| RS[7,8]                                                | $-0.5$        | $+10$      | $-10$        | $3 / 1.0$    |
| RockSample                                             |               |            |              |              |
| RS[11,11]                                              | $-0.5$        | $+10$      | $-10$        | $2 / 1.0$    |
| Inspection                                             |               |            |              |              |
| ( $N=8$ )                                              | $-0.5 / -2.0$ | $+5$ fault | $-50$ missed | $3 / 1.0$    |
| Inspection                                             |               |            |              |              |
| ( $N=16$ )                                             | $-0.5 / -2.0$ | $+5$ fault | $-50$ missed | $2 / 1.0$    |

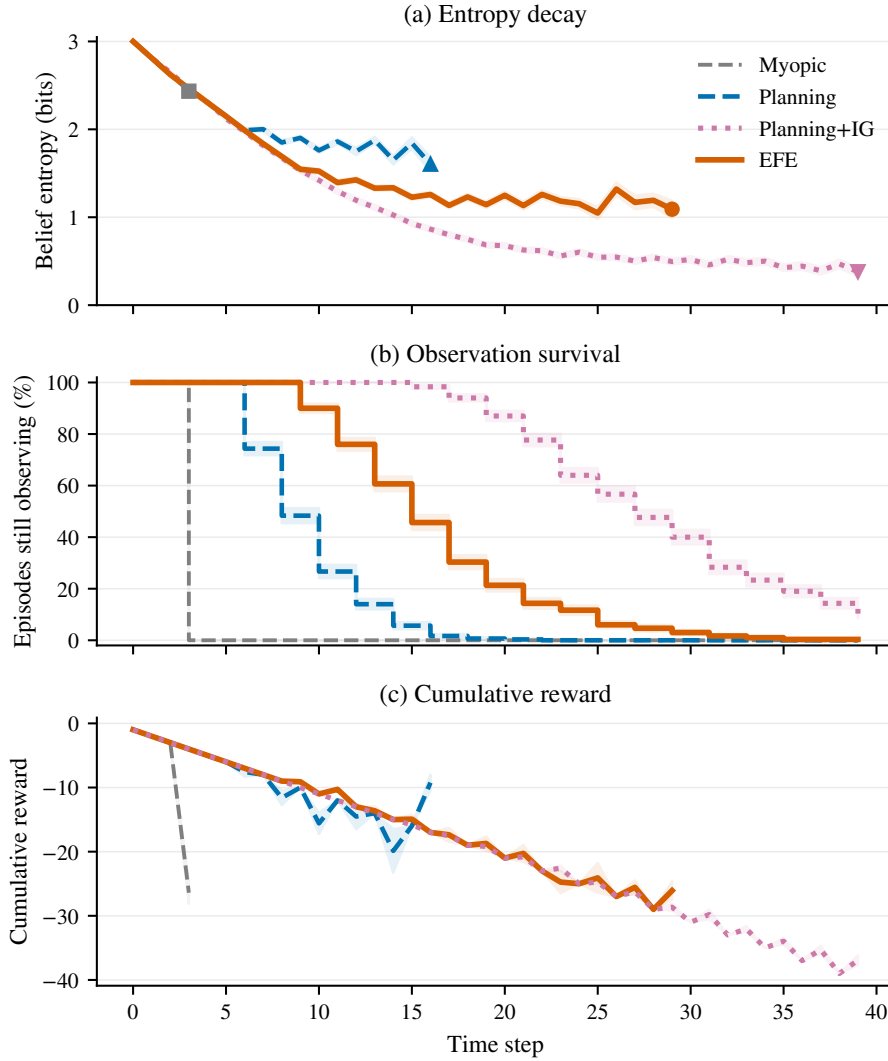


**Fig. 22.** Belief evolution within one illustrative Diagnosis episode per agent, not chosen to be typical in length ( $N=8$ ,  $K=3$ , accuracy 0.75). Darker = higher probability (shared colorbar). The green arrow in the left margin marks the true state row, the dashed line marks the commit point, and the three panels share the time axis so episode lengths compare directly. In the plotted episodes all three agents commit correctly, Planning after 14 observations on a belief already concentrated on the true state, EFE after 29, and InfoGain-Tuned after 32 with its darkest band on a wrong row for most of the episode.

## F Navigation Results

Navigation uses only move actions. The goal cell is hidden, and each move yields a proximity-based warm/cold observation (Section E). Unlike Diagnosis or Tileworld, there is no separate menu of observation actions. Information arrives as a side effect of translation, and greedy progress toward the belief mode already provides correlated evidence.

NavMyopic (move toward the highest-probability cell) is the strongest baseline at every scale, nominally at  $3 \times 3$  and  $7 \times 7$  and significantly at  $5 \times 5$ , where its lead over NavEFE holds on reward ( $p=0.002$ ), success ( $p=0.010$ ), and observation count ( $p=0.001$ , all seed-level Welch, Holm-corrected within metric at each grid size, `results_navigation_scaling_seed_stats.csv`). At the other two sizes no pairwise difference survives correction at 5 seeds. Extra steps devoted to exploratory detours incur the per-move cost without the sharp information returns seen when tests can be chosen explicitly, and at no size does an epistemic agent significantly beat the myopic rule. This is not a failure of  $w=1$  so much as a structural regime change relative to our main benchmarks. NavEFE is competitive with one-step NavInfoGain at  $7 \times 7$ , with nominally higher mean reward ( $-40.86$  against  $-44.28$ , seed-level  $p=0.040$  uncorrected, not significant after Holm) and comparable success. This is evidence that recursive epistemic valuation is not harmful once the grid is large enough that myopic information gain is already costly. The contrast with Tileworld (Figure 16) underscores the multiple-observation-actions condition of Section 8. EFE’s clearest wins appear



**Fig. 23.** Exploration efficiency on Diagnosis ( $N=8$ ,  $K=3$ , 300 episodes). (a) Belief entropy decay. (b) Percentage of episodes still observing. (c) Cumulative reward. Panels are stacked on a shared time axis, shaded bands show  $\pm 1$  SE across episodes, and in panel (a) a marker terminates each series at its final plotted step. Panels (a) and (c) average, at each step, only the episodes still running at that step. Planning+IG uses the myopic Info Gain agent’s success-tuned weight, selected on 100 tuning episodes and transferred to Planning+IG for this visualization, not the own-class tuning protocol of the main tables (Section 3). EFE stops observing later than reward-only Planning and earlier than Planning+IG, whose survival curve lies at or above EFE’s at every step (panel b).

**Table 20.** Navigation scaling (150 episodes per seed  $\times$  5 seeds = 750 episodes per agent per grid). Step budget  $3n^2$ . NavEFE uses depth-limited planning  $H=2$ . Uncertainty is seed-level SE. See `results_navigation_scaling.csv` and the companion seed-level Welch tests in `results_navigation_scaling_seed_stats.csv`.

| Grid         | $ S $ | Agent            | Success          | Mean reward       |
|--------------|-------|------------------|------------------|-------------------|
| $3 \times 3$ | 9     | NavMyopic        | $90.4 \pm 1.3\%$ | $+2.04 \pm 1.51$  |
|              |       | NavInfoGain      | $86.4 \pm 1.6\%$ | $-1.37 \pm 1.69$  |
|              |       | NavEFE ( $H=2$ ) | $87.2 \pm 1.7\%$ | $-1.29 \pm 1.58$  |
| $5 \times 5$ | 25    | NavMyopic        | $80.7 \pm 1.4\%$ | $-16.64 \pm 1.54$ |
|              |       | NavInfoGain      | $76.0 \pm 1.5\%$ | $-23.24 \pm 1.75$ |
|              |       | NavEFE ( $H=2$ ) | $74.4 \pm 1.2\%$ | $-25.76 \pm 1.14$ |
| $7 \times 7$ | 49    | NavMyopic        | $65.7 \pm 2.8\%$ | $-38.42 \pm 2.77$ |
|              |       | NavInfoGain      | $60.3 \pm 1.2\%$ | $-44.28 \pm 1.12$ |
|              |       | NavEFE ( $H=2$ ) | $62.8 \pm 0.7\%$ | $-40.86 \pm 0.77$ |

when the agent must select among differentially informative observation actions, not only where to translate next under a smooth proximity field.

## G Scaling Analysis

At  $H=2$ , EFE and Planning show no consistent gap at any  $N$  tested here (bit-identical at  $N=2$  and  $N=4$ , at  $N=8$ ,  $84.08\% \pm 0.66\text{pp}$  vs.  $83.92\% \pm 0.82\text{pp}$ , and at  $N=16$ ,  $77.12\% \pm 0.78\text{pp}$  vs.  $77.36\% \pm 0.98\text{pp}$ , seed-level SE, with no consistent direction), both substantially outperforming Myopic and untuned Info Gain. This contrasts with the  $H=3$  Diagnosis result (Table 8), where EFE outperformed Planning by  $+8.3$  pp, confirming that EFE’s advantage requires sufficient recursive depth, not merely a larger state space. EFE-over-Myopic advantage grows from  $+13.8$  pp at  $N=2$  ( $94.12\% \pm 0.50\text{pp}$  vs.  $80.28\% \pm 0.91\text{pp}$ ) to  $+36.3$  pp at  $N=16$  ( $77.12\% \pm 0.78\text{pp}$  vs.  $40.84\% \pm 1.69\text{pp}$ , seed-level SE), confirming that deeper planning is increasingly valuable as state spaces grow, independent of whether EFE or reward-only Planning supplies that depth.

## H PyMDP Consistency Check

We check our EFE computation against the `pymdp` library [26] by wrapping `pymdp`’s standard Agent class. This comparison checks consistency between implementations; it does not certify numerical equivalence, so the evidence below is stated in terms of the behaviors actually compared. The quantitative form of the check is a deployed comparison on the two-state testbed. Run as an ordinary observe-then-commit agent over the canonical protocol, `pymdp-AIF` returns 3.15 observations, 89.8% success, and  $+0.480$  reward, bit-identical to reward-only Planning on all three metrics (`results_summary.csv`). The meaningful comparator here is our own myopic Info Gain agent at  $w=1$ , which `results_summary.csv`

**Table 21.** Scaling analysis. Diagnosis  $N = 2$  to 16 (2,500 episodes at 500 per seed  $\times$  5 seeds,  $H=2$ ). Bold marks the best success and reward per  $N$  and the tests of those rows, plus any row within sampling error of it (seed-level Welch  $p > 0.05$ , uncorrected, `results_scaling_stats.csv`). At  $N=8$  and  $N=16$  the EFE and Planning rows are statistical ties.

| $N$ | Agent           | Tests        | Success      | Reward        |
|-----|-----------------|--------------|--------------|---------------|
| 2   | Myopic          | 1.00         | 80.3%        | -2.83         |
|     | <b>Planning</b> | <b>2.91</b>  | <b>94.1%</b> | <b>+3.56</b>  |
|     | Info Gain       | 1.00         | 80.3%        | -2.83         |
|     | <b>EFE</b>      | <b>2.91</b>  | <b>94.1%</b> | <b>+3.56</b>  |
| 4   | Myopic          | 2.00         | 64.5%        | -13.29        |
|     | <b>Planning</b> | <b>5.83</b>  | <b>88.8%</b> | <b>-2.58</b>  |
|     | Info Gain       | 2.00         | 64.5%        | -13.29        |
|     | <b>EFE</b>      | <b>5.83</b>  | <b>88.8%</b> | <b>-2.58</b>  |
| 8   | Myopic          | 3.00         | 51.1%        | -22.33        |
|     | <b>Planning</b> | <b>8.79</b>  | <b>83.9%</b> | <b>-8.44</b>  |
|     | Info Gain       | 3.00         | 51.1%        | -22.33        |
|     | <b>EFE</b>      | <b>8.83</b>  | <b>84.1%</b> | <b>-8.38</b>  |
| 16  | Myopic          | 4.00         | 40.8%        | -29.50        |
|     | <b>Planning</b> | <b>11.65</b> | <b>77.4%</b> | <b>-15.23</b> |
|     | Info Gain       | 4.00         | 40.8%        | -29.50        |
|     | <b>EFE</b>      | <b>11.64</b> | <b>77.1%</b> | <b>-15.37</b> |

records at the same 3.15 observations, 89.8% success, and +0.480 reward. Both agents evaluate single-step policies, and by Proposition 1 the myopic  $w=1$  agent is the myopic EFE agent. This is therefore a like-for-like agreement. Our  $H=4$  recursive EFE agent legitimately differs on this environment (5.50 observations), through planning depth rather than through any disagreement about the epistemic term. The check is therefore evidence that the two myopic implementations agree, not that recursive EFE matches pymdp. It remains weak evidence, since on this environment the myopic  $w=1$  agent is itself bit-identical to reward-only Planning, so the comparison cannot separate a correctly implemented epistemic term from its absence. Both implementations agree on observe-vs-commit decisions across all tested belief states (uniform, intermediate, and confident beliefs on the two-state environments). Information gain values computed by our recursive scheme and pymdp’s `control` module agree qualitatively, with differences attributable to our recursive multi-step evaluation vs. pymdp’s single-step policy enumeration.

## I Supplementary Statistics

Pairwise comparisons in the main results tables use independent two-sample Welch  $t$ -tests on per-seed means with Holm–Bonferroni correction for family-wise error control, and comparisons reported without that correction are marked uncorrected where they appear. The companion pooled episode-level column in the six stats files behind the main observe-then-commit and Tileworld tables is an equal-variance two-sample test rather than a Welch test, which changes no non-degenerate comparison’s outcome at  $\alpha=0.05$  across the 510 pooled comparisons in those files. The RockSample and Structural Inspection producers use Welch for both levels. The correction family is the full set of pairwise agent comparisons and metrics (reward, success/accuracy, and, where reported, observation count) reported together within a single table. For the main results tables (Table 8 and the RockSample and Structural Inspection tables), the family is scoped to one environment or instance at a time, since each table’s rows are computed and corrected together, independently of every other table. Concretely, with  $A$  agents and  $M$  corrected metrics the family size is  $M \cdot \binom{A}{2}$ , giving 108 for Tiger ( $A=9$ ,  $M=3$  over reward, success, and observations), 84 for Diagnosis and Bandit ( $A=8$ ,  $M=3$ ), and 18 for each Structural Inspection instance ( $A=4$ ,  $M=3$  over reward, accuracy, and tests). These are nominal sizes. A comparison whose seed-level  $p$ -value is undefined, because both agents’ per-seed values have zero variance, is excluded from its family, which leaves 107 defined tests for Tiger and 83 for Diagnosis. RockSample corrects within metric rather than pooling, giving 28 tests per metric per instance over its eight agents, because the RockSample table’s bolding rule treats a non-rejection as a tie with the best row and pooling metrics into one larger family would loosen exactly that rule. The exceptions are the discount-sweep appendix table (Table 24), where the family spans every environment and  $\gamma$  value in that sweep jointly (24 tests, from 3 environments  $\times$  4 discount factors  $\times$  2 metrics), and the two MCTS batteries of Appendix S, where

the ablation corrects within metric across all three environments jointly and the exploration sweep corrects within metric within each of its three comparison families. These exceptions arise because in each case those rows are accumulated and corrected together before the table is written. Bootstrap 95% confidence intervals use 10,000 resamples with fixed seed for reproducibility. Effect sizes use Cohen’s  $d$  with pooled standard deviation. Because episodes within a seed are not independent, those primary tests aggregate each seed’s episodes to a single mean before testing ( $n=5$  seeds), and the pooled episode-level columns are reported alongside them. The main-text table (Table 8) reports the seed-level SE as the primary uncertainty measure, since each answers a different question (episode-level precision versus between-seed replicability) and neither alone is a strict superset of the other. The Tiger, Diagnosis and Bandit tables of Appendix B report the pooled episode-level SE only and omit the untuned Info Gain rows that duplicate Myopic exactly, together with Tiger’s pymdp-AIF row. The two Tileworld tables report point estimates for every agent run, duplicates included. The companion `*_stats.csv` files committed alongside the main battery carry the seed-level Welch  $t$ -tests and Cohen’s  $d$  this section describes. The Tileworld scaling battery reports seed-level standard errors for reward and success without a companion Welch file.

The reward table shows that EFE vs. Planning is negligible to none on Tiger, Diagnosis, and Bandit (exactly zero on Tiger, where EFE and Planning tie). Once both agents take enough informative actions, mean returns are similar, even though success rates differ. The success-rate rows isolate the axis where EFE improves most relative to Planning on Diagnosis and Bandit. Against over-exploring tuned alternatives (InfoGain-Tuned, Planning+IG), EFE shows medium reward effects on the Testbed ( $d = 0.69$ ) and small effects on Bandit ( $d \approx 0.36$ - $0.37$ ), reflecting the cost of over-exploration at Bandit’s success-tuned weight of  $w=20$ .

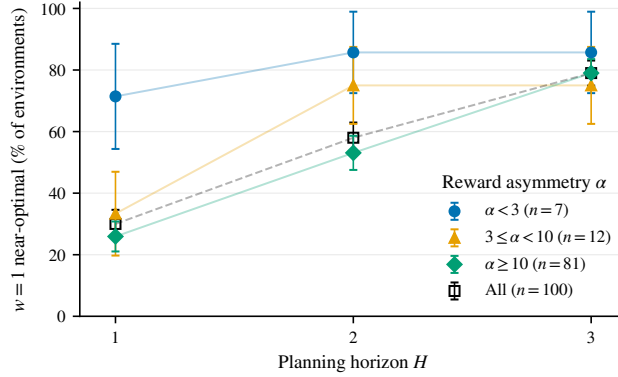
## J Near-Optimality across Planning Horizons

Proposition 2 bounds  $w_{lo}^*$  at  $H=1$  and  $w_{hi}^*$  at  $H=2$ . To check whether the resulting near-optimality claim survives recursive multi-step planning, we generated 100 random two-state environments (reward asymmetry  $\alpha \in [1, 50]$ , observation accuracy  $p \in [0.55, 0.95]$ , cost  $\in [0.1, 5]$ ). For each environment and each  $H \in \{1, 2, 3\}$ , we computed the reward-optimal weight  $w^*$  by grid search and recorded whether  $w=1$  falls within  $\max(5\%, 0.5)$  of the best achievable reward. Each (environment, weight, horizon) cell runs 50 episodes at a single seed, and  $w^*$  is selected over the six-point grid  $\{0.01, 0.5, 1, 2, 5, 50\}$  (`experiments/run_nearopt_horizon.py`). This is a deliberate deviation from the canonical five-seed protocol, because the study is a descriptive sweep over 100 synthetic environments rather than a headline comparison, and its conclusions are read as coarse rates rather than per-environment estimates. Figure 24 reports the fraction of environments where  $w=1$  passes this test at each horizon, stratified by  $\alpha$ . The near-optimality rate rises from 30% at  $H=1$  to 58% at  $H=2$  and 79%

**Table 22.** Cohen’s  $d$  for EFE vs. each baseline on mean episode reward (top) and vs. Planning on success rate (bottom). Positive  $d$  favors EFE. Interpretation.  $|d| < 0.2$  negligible, 0.2–0.5 small, 0.5–0.8 medium,  $|d| > 0.8$  large. Success-rate  $d$  uses the same pooled- $s$  convention as reward (proportions treated as per-episode Bernoulli outcomes).

| Environment                                     | Comparison             | $d$ (reward) | Interpretation         |
|-------------------------------------------------|------------------------|--------------|------------------------|
| Tiger                                           | EFE vs. Myopic         | +0.44        | small                  |
| Tiger                                           | EFE vs. Planning       | 0.00         | negligible (exact tie) |
| Tiger                                           | EFE vs. InfoGain-Tuned | 0.00         | negligible (exact tie) |
| Tiger                                           | EFE vs. Planning+IG    | +0.14        | negligible             |
| Tiger                                           | EFE vs. Posterior-vote | +0.09        | negligible             |
| Testbed                                         | EFE vs. Myopic         | −0.07        | negligible             |
| Testbed                                         | EFE vs. Planning       | −0.18        | negligible             |
| Testbed                                         | EFE vs. InfoGain-Tuned | +0.69        | medium                 |
| Testbed                                         | EFE vs. Planning+IG    | +0.69        | medium                 |
| Testbed                                         | EFE vs. Posterior-vote | −0.18        | negligible             |
| Diagnosis                                       | EFE vs. Myopic         | +0.55        | medium                 |
| Diagnosis                                       | EFE vs. Planning       | +0.08        | negligible             |
| Diagnosis                                       | EFE vs. InfoGain-Tuned | +0.25        | small                  |
| Diagnosis                                       | EFE vs. Planning+IG    | +0.25        | small                  |
| Diagnosis                                       | EFE vs. Posterior-vote | +0.09        | negligible             |
| Bandit                                          | EFE vs. Myopic         | +0.18        | negligible             |
| Bandit                                          | EFE vs. Planning       | +0.14        | negligible             |
| Bandit                                          | EFE vs. InfoGain-Tuned | +0.36        | small                  |
| Bandit                                          | EFE vs. Planning+IG    | +0.37        | small                  |
| Bandit                                          | EFE vs. Posterior-vote | −0.01        | negligible             |
| <i>Success rate (<math>d</math> favors EFE)</i> |                        |              |                        |
| Tiger                                           | EFE vs. Planning       | 0.00         | negligible (exact tie) |
| Testbed                                         | EFE vs. Planning       | +0.26        | small                  |
| Diagnosis                                       | EFE vs. Planning       | +0.33        | small                  |
| Bandit                                          | EFE vs. Planning       | +0.40        | small                  |

at  $H=3$  (26%  $\rightarrow$  53%  $\rightarrow$  79% restricted to  $\alpha \geq 10$ ), so deeper planning widens the basin in which the untuned information-unit weight is reward-competitive rather than narrowing it. This is the same trend reported for Bandit in Section 8, where the low reward asymmetry predicts only marginal  $H=1$  sufficiency yet multi-step EFE performs well in practice.



**Fig. 24.** Fraction of randomly generated two-state environments where  $w=1$  is near-optimal, as a function of planning horizon  $H$ , stratified by reward asymmetry  $\alpha$ . The near-optimality basin widens with horizon, consistent with multi-step planning amplifying the value of observation. 100 environments,  $\alpha \in [1, 50]$ ,  $p \in [0.55, 0.95]$ , cost  $\in [0.1, 5]$ . Markers show the three measured horizons  $H \in \{1, 2, 3\}$  with binomial standard-error bars, the legend reports the number of environments in each  $\alpha$  stratum, and the light guide lines connect measured points only.

*Agreement across planning horizons.* Figure 24 stratifies only by reward asymmetry  $\alpha$ . Table 23 breaks the same 100 environments down by all three sampled regime axes and adds a direct per-environment comparison of the  $H=1$  classification against  $H \geq 2$ . Eight of the nine regime strata show the same qualitative pattern, with near-optimality rarer at  $H=1$  than at  $H=3$ . The effect is sharpest for high-informativeness observations ( $p \geq 0.85$ , rising 17%  $\rightarrow$  90%). The exception is the low-informativeness stratum ( $p < 0.65$ , 25 environments), which is flat and mildly non-monotone at 80%  $\rightarrow$  84%  $\rightarrow$  80%, since weak observations leave little for a deeper horizon to exploit. Low reward asymmetry rises but less sharply ( $\alpha < 3$ , from 71% at  $H=1$  to 86% by  $H=2$ , holding there through  $H=3$ , on 7 environments). The per-environment comparison uses the producing script’s definition of agreement. An environment agrees when  $w=1$  is near-optimal at  $H=1$  and at some deeper horizon, or at neither. By that definition the two horizons agree on 44 of the 100 environments. In 30 the  $H=1$  check passes and near-optimality recurs at some deeper horizon. In 26 of those 30 it holds at every deeper horizon tested. In 14 the check fails and never recovers.

**Table 23.** Horizon map, the fraction of randomly generated two-state environments where  $w=1$  is near-optimal (within  $\max(5\%, 0.5)$  of the reward-optimal weight), stratified by each sampled regime axis and planning horizon  $H$ . Same 100 environments as Figure 24 (Appendix J). This is an empirical boundary map, not an adaptive-submodularity guarantee.

| Regime                                              | $H=1$ | $H=2$ | $H=3$ |
|-----------------------------------------------------|-------|-------|-------|
| <i><math>\alpha</math> (reward asymmetry)</i>       |       |       |       |
| $\alpha < 3$                                        | 71%   | 86%   | 86%   |
| $3 \leq \alpha < 10$                                | 33%   | 75%   | 75%   |
| $\alpha \geq 10$                                    | 26%   | 53%   | 79%   |
| <i><math>p</math> (observation informativeness)</i> |       |       |       |
| $p < 0.65$                                          | 80%   | 84%   | 80%   |
| $0.65 \leq p < 0.85$                                | 11%   | 40%   | 71%   |
| $p \geq 0.85$                                       | 17%   | 63%   | 90%   |
| <i>cost (observation cost)</i>                      |       |       |       |
| $\text{cost} < 1$                                   | 16%   | 16%   | 58%   |
| $1 \leq \text{cost} < 3$                            | 26%   | 55%   | 79%   |
| $\text{cost} \geq 3$                                | 40%   | 79%   | 88%   |

$H=1$  vs.  $H \geq 2$  agreement: 44% of 100 environments agree; myopic overclaims 0, underclaims 56.

The remaining 56 environments fail the  $H=1$  check yet turn out near-optimal at some deeper horizon (myopic underclaiming, the safe direction, a missed opportunity rather than a risk). On this random sample, zero environments show the opposite pattern (myopic overclaiming, where  $H=1$  predicts near-optimality but no deeper horizon delivers it). Four of the 30 passers fail at exactly one of the two deeper horizons (environments 2, 42, 63, and 95 in the committed CSV). A practitioner validating  $w=1$  with a cheap  $H=1$  check would therefore have been misled at one specific deeper horizon in 4 of 30 cases, and in no case toward a weight that fails at every deeper horizon. This is a property of the observed sample, not a proven guarantee. We report it as an empirical boundary map over one random family of two-state environments, not an adaptive-submodularity result. We make no claim that the  $H=1$  check is monotone or that overclaiming is impossible in general, only that it did not occur here (the per-environment data is in `results/results_nearopt_horizon.csv`, from which `experiments/build_horizon_map.py` derives the aggregate agreement counts in `results/results_horizon_map_agreement.csv`).

## K Discount Factor Sensitivity

We test EFE and Planning agents' sensitivity to discounting by sweeping  $\gamma \in \{0.9, 0.95, 0.99, 1.0\}$  across three environments, at the canonical 5-seed protocol. Including Planning at each  $\gamma$  enables comparison of the relative gap between agents under discounting.

**Table 24.** EFE and Planning agent performance across discount factors  $\gamma$  (2,500 episodes at 500 per seed  $\times$  5 seeds, lighter than Table 8’s 5,000-episode battery, so small numeric differences from that table on the  $\gamma=1.0$  row are expected).  $\Delta$ Succ. shows the success rate gap (EFE – Planning) at each  $\gamma$ , with the seed-level Welch  $p$ -value in parentheses. \* marks significance after Holm–Bonferroni correction.

| Environment | Agent    | $\gamma$ | Obs. | Success | Reward | $\Delta$ Succ. ( $p$ )   |
|-------------|----------|----------|------|---------|--------|--------------------------|
| Tiger       | EFE      | 0.90     | 4.23 | 99.6%   | +5.33  | +2.7 ( $p < 10^{-3}$ )*  |
|             | Planning | 0.90     | 2.65 | 96.9%   | +3.91  |                          |
|             | EFE      | 0.95     | 4.23 | 99.6%   | +5.33  | 0.0 ( $p=1.0$ )          |
|             | Planning | 0.95     | 4.23 | 99.6%   | +5.33  |                          |
|             | EFE      | 0.99     | 4.23 | 99.6%   | +5.33  | 0.0 ( $p=1.0$ )          |
|             | Planning | 0.99     | 4.23 | 99.6%   | +5.33  |                          |
|             | EFE      | 1.00     | 4.23 | 99.6%   | +5.33  | 0.0 ( $p=1.0$ )          |
|             | Planning | 1.00     | 4.23 | 99.6%   | +5.33  |                          |
| Diagnosis   | EFE      | 0.90     | 5.83 | 88.8%   | −2.58  | +0.0 ( $p=0.96$ )        |
|             | Planning | 0.90     | 5.86 | 88.7%   | −2.62  |                          |
|             | EFE      | 0.95     | 5.83 | 88.8%   | −2.58  | +0.0 ( $p=0.96$ )        |
|             | Planning | 0.95     | 5.86 | 88.7%   | −2.62  |                          |
|             | EFE      | 0.99     | 9.67 | 97.5%   | −1.19  | +7.9 ( $p < 10^{-4}$ )*  |
|             | Planning | 0.99     | 5.89 | 89.6%   | −2.15  |                          |
|             | EFE      | 1.00     | 9.68 | 97.4%   | −1.26  | +8.6 ( $p < 10^{-4}$ )*  |
|             | Planning | 1.00     | 5.85 | 88.8%   | −2.59  |                          |
| Bandit      | EFE      | 0.90     | 2.04 | 61.9%   | +5.55  | −0.2 ( $p=0.95$ )        |
|             | Planning | 0.90     | 2.04 | 62.0%   | +5.56  |                          |
|             | EFE      | 0.95     | 2.04 | 61.9%   | +5.55  | −0.2 ( $p=0.95$ )        |
|             | Planning | 0.95     | 2.04 | 62.0%   | +5.56  |                          |
|             | EFE      | 0.99     | 5.16 | 86.9%   | +6.24  | +22.4 ( $p < 10^{-4}$ )* |
|             | Planning | 0.99     | 2.38 | 64.5%   | +5.61  |                          |
|             | EFE      | 1.00     | 5.16 | 86.9%   | +6.24  | +17.0 ( $p < 10^{-4}$ )* |
|             | Planning | 1.00     | 3.24 | 69.9%   | +5.67  |                          |

*Sensitivity to discounting.* On Tiger (single observation action), both agents achieve  $\geq 96\%$  success across all  $\gamma$  values, with a significant but small +2.7 pp EFE advantage at the heaviest discounting tested ( $\gamma=0.9$ , seed-level Welch  $p < 10^{-3}$ ). At  $\gamma \geq 0.95$  the two agents are numerically identical ( $p=1.0$ ). On Diagnosis and Bandit, heavy discounting ( $\gamma=0.9-0.95$ ) renders both agents effectively myopic, erasing EFE’s advantage entirely. At  $\gamma=0.9-0.95$ , EFE’s epistemic term cannot propagate value through future observations because discounting truncates the effective horizon below the number of observations needed for disambiguation. Both agents observe the same number of times on Bandit ( $\sim 2$ ) and Diagnosis ( $\sim 6$ ) at these discount levels, and the residual success gaps are not statistically significant (Diagnosis diff  $\approx 0.04$ pp,  $p=0.96$ , Bandit diff  $\approx -0.16$ pp,  $p=0.95$ , Table 24). The transition between  $\gamma=0.95$  and  $\gamma=0.99$  is sharp and, with 5-seed replication, clearly significant. At  $\gamma=0.99$ , EFE’s success gap over Planning jumps to +7.9 pp on Diagnosis and +22.4 pp on Bandit (both  $p < 10^{-4}$ ), matching the undiscounted pattern (+8.6 pp Diagnosis, +17.0 pp Bandit at  $\gamma=1.0$ , both  $p < 10^{-4}$ ). This confirms that EFE’s advantage depends on the discount factor preserving a sufficient effective horizon for recursive epistemic evaluation. In practical terms, on the multi-observation environments Diagnosis and Bandit, EFE differentiates from reward-only planning on success at the two largest discount factors tested,  $\gamma \in \{0.99, 1.0\}$ , and not at  $\gamma \in \{0.90, 0.95\}$ , so the transition lies between 0.95 and 0.99 on this grid, and the 5-seed replication makes the significance of that transition a confirmed finding rather than a single-seed point estimate.

## L Model Misspecification Sensitivity

Our main results assume the agent’s generative model matches the true environment dynamics. Here we investigate robustness when the agent’s believed observation accuracy  $p_{\text{agent}}$  differs from the true accuracy  $p_{\text{true}}$ , creating a systematic model mismatch.

*Asymmetric sensitivity to sensor error.* On Tiger, success falls at most 2.7pp below the calibrated 99.6% for both EFE and Planning across the tested range of  $-0.15$  to  $+0.10$ . The degradation pattern is asymmetric. For EFE and Planning, negative mismatch (underestimating accuracy, including exact calibration) leaves observation count and success unchanged (4.23 observations, 99.6% success at every level from  $-0.15$  to  $0.00$ ), whereas positive mismatch (overestimating accuracy) sharply reduces observation count (to 2.65) and lowers success to 96.9%. Under the fixed pipeline, EFE and Planning give bit-identical rows at every mismatch level tested on Tiger (Table 25), not merely statistically indistinguishable ones. With a single observation action, the belief-update dynamics driven by the agent’s believed accuracy determine when committing becomes attractive, and information gain has no separate observation action to redirect toward. There is therefore no EFE-specific buffering effect to report here. This finding is specific to the tested environment and cannot be attributed to the epis-

**Table 25.** Model misspecification on Tiger ( $p_{\text{true}}=0.85$ ,  $H=4$ , 2,500 episodes across 5 seeds, lighter than Table 8’s 5,000-episode battery, so small numeric differences from that table’s calibrated row are expected). The mismatch column shows  $p_{\text{agent}} - p_{\text{true}}$ . Reward reported as mean  $\pm$  seed-level SE (`results_model_misspec.csv`). EFE and Planning rows are bit-identical at every level tested, not merely close, so no significance test applies to that comparison.

| Agent    | $p_{\text{agent}}$ | Mismatch   | Obs  | Success | Reward           |
|----------|--------------------|------------|------|---------|------------------|
| EFE      | 0.70               | −0.15      | 4.23 | 99.6%   | +5.33 $\pm$ 0.26 |
| EFE      | 0.75               | −0.10      | 4.23 | 99.6%   | +5.33 $\pm$ 0.26 |
| EFE      | 0.80               | −0.05      | 4.23 | 99.6%   | +5.33 $\pm$ 0.26 |
| EFE      | 0.85               | $\pm 0.00$ | 4.23 | 99.6%   | +5.33 $\pm$ 0.26 |
| EFE      | 0.90               | +0.05      | 2.65 | 96.9%   | +3.91 $\pm$ 0.38 |
| EFE      | 0.95               | +0.10      | 2.65 | 96.9%   | +3.91 $\pm$ 0.38 |
| Planning | 0.70               | −0.15      | 4.23 | 99.6%   | +5.33 $\pm$ 0.26 |
| Planning | 0.85               | $\pm 0.00$ | 4.23 | 99.6%   | +5.33 $\pm$ 0.26 |
| Planning | 0.95               | +0.10      | 2.65 | 96.9%   | +3.91 $\pm$ 0.38 |

**Table 26.** Model misspecification on Diagnosis ( $p_{\text{true}}=0.80$ ,  $H=3$ , 2,500 episodes across 5 seeds). Reward reported as mean  $\pm$  seed-level SE (`results_model_misspec.csv`).

| Agent    | $p_{\text{agent}}$ | Mismatch   | Obs  | Success | Reward           |
|----------|--------------------|------------|------|---------|------------------|
| EFE      | 0.65               | −0.15      | 5.86 | 88.7%   | −2.62 $\pm$ 0.31 |
| EFE      | 0.70               | −0.10      | 9.68 | 97.4%   | −1.26 $\pm$ 0.34 |
| EFE      | 0.75               | −0.05      | 9.68 | 97.4%   | −1.26 $\pm$ 0.34 |
| EFE      | 0.80               | $\pm 0.00$ | 9.68 | 97.4%   | −1.26 $\pm$ 0.34 |
| EFE      | 0.85               | +0.05      | 5.86 | 88.7%   | −2.62 $\pm$ 0.31 |
| EFE      | 0.90               | +0.10      | 5.86 | 88.7%   | −2.62 $\pm$ 0.31 |
| Planning | 0.65               | −0.15      | 5.84 | 89.2%   | −2.35 $\pm$ 0.51 |
| Planning | 0.70               | −0.10      | 9.71 | 97.6%   | −1.12 $\pm$ 0.26 |
| Planning | 0.75               | −0.05      | 9.68 | 97.3%   | −1.29 $\pm$ 0.32 |
| Planning | 0.80               | $\pm 0.00$ | 5.85 | 88.8%   | −2.59 $\pm$ 0.37 |
| Planning | 0.85               | +0.05      | 5.86 | 88.7%   | −2.62 $\pm$ 0.31 |
| Planning | 0.90               | +0.10      | 5.82 | 88.7%   | −2.61 $\pm$ 0.40 |

temic term. The same asymmetric degradation appears in reward-only Planning, which has no information-gain term at all.

On Diagnosis, EFE’s behavior steps between exactly two levels rather than degrading smoothly with mismatch magnitude (Table 26). The high-testing level (9.68 observations, 97.4% success) occurs at mismatch  $\in \{-0.10, -0.05, 0.00\}$ . The low-testing level (5.86 observations, 88.7% success) occurs at mismatch  $\in \{-0.15, +0.05, +0.10\}$ . This is not a “well-calibrated agent tests most, both directions of error reduce testing” pattern. Perfect calibration ( $\pm 0.00$ ) ties with two levels of underestimation rather than uniquely maximizing testing, and only the most extreme underestimate tested ( $-0.15$ ) drops to the low-testing level, matching every overestimate we tested. Overestimating accuracy (the agent believes tests are more informative than they truly are) makes each test look more decisive than it is, so the agent commits prematurely on insufficient evidence and lands in the low-testing, lower-success regime at every overestimate level tested. Underestimating accuracy makes each test look less informative than it is, but only past a threshold. Mild underestimation ( $-0.05, -0.10$ ) does not visibly change behavior from perfect calibration, while severe underestimation ( $-0.15$ ) triggers the same reduced-testing response as overestimation, producing a similar-looking success rate (88.7%) that is, as before, driven by fewer rather than more observations. Reward-only Planning shows a comparable but not identical step pattern (Table 26), so this threshold structure is not unique to EFE’s epistemic term. We report the pattern without assigning a mechanism to the precise transition point.

*Implications for learned models.* These results suggest that EFE-based agents can tolerate moderate model errors without catastrophic failure. In practice, observation models learned from data will have estimation error. The graceful degradation observed here indicates that approximate models suffice for effective EFE-driven information gathering, provided the model does not systematically overestimate sensor informativeness.

## M RockSample as an Interleaved Observe-Act Setting (Extended)

This appendix provides extended RockSample results beyond the main text (Section 7.4), including a flat one-ply Monte Carlo (Flat-MC) reference on every instance, the genuine POMCP at its tuned frozen configuration, POMCP’s approach-then-check rollout policy run standalone, the two-point Planning+IG sweep ( $w=5, 10$ ), and per-instance step and check counts. The extended table (Table 27) is read directly from the same CSVs as Table 10 (`results_rocksample_{5x3,7x4,7x8,11x11}.csv`), regenerated by `experiments/build_rocksample_tables.py`. The tuning, budget-scaling, horizon, and sensitivity numbers quoted in the POMCP discussion come from their own producer CSVs (`results_rocksample_pomcp_{tuning,budget,horizon,sensitivity}.csv`).

The Greedy agent moves to rocks and samples them without checking quality first, incurring frequent bad-rock penalties ( $1.51\text{--}5.52 \pm 0.01\text{--}0.05$ , seed-level SE, across the four instances). EFE moves toward uncertain rocks, checks them to resolve quality, then samples selectively, so bad samples stay low ( $0.01\text{--}0.06 \pm 0.001\text{--}0.013$ , seed-level Welch  $p < 5 \times 10^{-8}$  vs. Greedy at every instance). RS[11,11] is exceptional in a different sense, where every tree-search agent, including Planning at  $w=0$ , converges to the same low-activity policy, discussed in Section 7.4. The Flat-MC baseline collapses to a large negative reward on every instance ( $-9.92$  to  $-87.22$ ), but we do not read this as a directed-checking result. It should not be compared to the POMCP findings of Appendix S. Flat-MC is a one-ply Monte Carlo evaluator with no search tree and no UCB1 selection (`rho_aif/agents/rocksample_agents.py`). Because its rollout policy conditions on the rock-quality hypothesis sampled at the root, a check accrues no simulated value beyond the root action itself, so the agent structurally cannot represent a check that informs a later decision. On the two larger instances this degenerates outright. It checks almost continuously (82.4 checks/episode on RS[7,8], 165.8 on RS[11,11]) and never reaches the exit, hitting the episode step cap on 100% of episodes for both (mean steps 129.00 of a 129-step cap, and 231.00 of 231). Its returns are therefore that truncation, not evidence about information-directed planning. We report it as a weak reference point under an accurate name rather than dropping it.

A genuine POMCP for RockSample is reported here (`rho_aif/agents/rocksample_pomcp.py`). It builds a search tree over action-observation histories, selects with UCB1, backs up means as UCT does, and evaluates newly expanded leaves by Monte Carlo rollout. Three departures in the search procedure itself remain, all keyed to fully observable quantities (the belief representation and depth-cap leaf value are separate implementation choices, disclosed and sensitivity-checked below). The depth bound shrinks as the episode step cap approaches. The action set is masked to exclude wall bumps, checks on already-sampled rocks, and samples where no rock sits. The subtree under the realized history is not reused, since under a fixed depth bound statistics accumulated at depth one would be credited with a full horizon after promotion. Its configuration was selected on three tuning seeds disjoint from every evaluation seed, over a two-stage coordinate grid covering the rollout policy, the exploration constant, the planning horizon, and the root criterion (`results_rocksample_pomcp_tuning.csv`), and frozen before any evaluation number was inspected. Belief is the exact factored posterior by default, which is available here because the RockSample posterior factorizes exactly. A literal unweighted particle filter with rejection sampling is also implemented and scores within sampling error of the exact version ( $+12.93 \pm 0.25$  against  $+12.81 \pm 0.43$ ,  $p=0.81$ ), so the exact belief is not what carries the result. The depth-cap leaf value is a second implementation choice worth the same check. Replacing the frozen exit-valued leaf with a greedy-belief leaf scores  $+14.52 \pm 0.34$  against the frozen configuration's  $+12.81 \pm 0.43$  ( $p=0.015$  uncorrected), still below that battery's own standalone rollout at  $+17.16 \pm 0.44$  ( $p=0.002$  uncorrected, and neither comparison survives that battery's per-metric Holm family). The leaf

choice moves the number without changing the ordering, so we report it rather than retune around it.

At the frozen configuration POMCP reaches  $+12.74 \pm 0.11$  on RS[5,3] at the full protocol, against EFE’s  $+16.47 \pm 0.12$ , and it costs 46.6 ms per decision against EFE’s 1.76 ms. The gap holds at every instance though its ratio varies, 767.5 against 103.5 ms on RS[7,4], 75.5 against 7.2 on RS[7,8], and 87.6 against 0.46 on RS[11,11] (`mean_wallclock_ms_per_step` in the instance CSVs). Two further measurements are needed before that comparison means anything, and both cut against the baseline being taken at face value.

First, the search tree is not what produces the score. The rollout policy POMCP was tuned to use, run standalone with no tree at all, scores  $+16.67 \pm 0.13$  on the same protocol at 0.005 ms per decision, which is 3.9 reward above the full search (seed-level Welch  $p=1.9 \times 10^{-14}$  at the full protocol, significant after Holm–Bonferroni). On this domain the tree subtracts value from the policy it rolls out with. The mechanism is the position-independent exit acting as a zero-variance outside option worth +9.50 from any cell, against mean UCT backup. The forced exploration that makes UCT sound is what drags the alternatives below it, so the tree commits to exiting after 7.5 steps where its own rollout plays a 13.5-step collection tour.

Second, the frozen planning horizon handicapped it, and we say so because we measured it rather than because a referee asked. Sweeping the horizon post hoc on RS[5,3] at two simulation budgets (`results_rocksample_pomcp_horizon.csv`) shows the frozen  $H=10$  is the worst setting tested at both. At 16,384 simulations,  $H=10$  gives  $+12.52 \pm 0.26$  while  $H=15$ ,  $H=25$  and  $H=40$  give  $+15.60 \pm 0.37$ ,  $+15.24 \pm 0.39$  and  $+15.62 \pm 0.36$ . The headline 2,048-simulation budget was fixed before evaluation as double the tuning stage’s 1,024, not itself a tuned choice. The budget-scaling direction reverses with the horizon, and it is not a clean slope in either direction, since at  $H=10$  the sweep is non-monotone across the budget range (256 scores  $+11.73$ , the worst tested, and the peak sits at 1,024). At  $H=10$  more simulations beyond 1,024 make POMCP worse, from  $+13.19$  at 1,024 to  $+12.52$  at 16,384, because a longer search only sharpens a conclusion that is wrong for being truncated. At  $H=15$  more simulations help substantially, from  $+14.00$  to  $+15.60$ , which is the behavior the algorithm is supposed to show. This is a real gap in our tuning protocol and not a property of POMCP. Stage B selected the horizon at a fixed budget of 1,024 simulations, so a budget-by-horizon interaction was structurally invisible to it. The horizon sweep ran on the canonical evaluation seeds and is therefore reported as a disclosed post-hoc diagnostic, not used to re-select the frozen configuration.

The same diagnostic on RS[11,11] (`results_rocksample_pomcp_horizon_11x11.csv`, at the 2,048-simulation tier only, since the 16,384 tier is prohibitively slow at this instance’s per-step planning cost) shows the opposite pattern. Deepening the horizon does not rescue POMCP at scale. It makes it monotonically worse.  $H=10$  scores  $+12.94 \pm 0.24$ ,  $H=15$  falls to  $-7.54 \pm 1.20$ ,  $H=25$  to  $-34.65 \pm 1.12$ , and  $H=40$  to  $-54.78 \pm 0.86$ , with every step of the decline significant (seed-level Welch  $p \leq 4.3 \times 10^{-5}$ , all surviving Holm correction). The

behavioral trace says why. At  $H=10$  the tree exits after 3.3 steps with 1.6 checks. At  $H=40$  it plays 196-step episodes with 144 checks, thirteen times its own rollout’s 11, yet still collects only 3.4 good rocks against the rollout’s 5.2. The deeper tree does see the collection tour, but what it assembles from noisy distant checks is a bad tour. The  $-0.5$  cost on every one of those extra actions swamps the sampling profit. So the tuning-protocol gap the RS[5,3] sweep exposed does not transfer to the large instance. On RS[5,3] the frozen  $H=10$  was the worst measured setting. On RS[11,11] it is the best by a wide margin, and the frozen configuration was not handicapping POMCP there.

Taken together, the fair reading is that POMCP at its best measured configuration on RS[5,3] reaches roughly  $+15.6$  at 530 ms per decision. EFE run under that diagnostic’s own protocol (5 seeds  $\times$  100 episodes, `results_rocksample_efe_diagnostic.csv`) scores  $+16.90 \pm 0.45$  at 1.7 ms per decision, nominally 1.3 above POMCP’s best and not significantly so (seed-level Welch  $p=0.058$  against  $H=15$  and  $H=40$ ,  $p=0.024$  against  $H=25$ , none surviving Holm correction over the nine diagnostic rows). The tree therefore reaches statistical parity with the closed-form recursion on this instance at roughly three hundred times the per-decision cost. On RS[11,11] the frozen  $H=10$ ’s  $+12.94$  is matched but not clearly beaten by any measured alternative, the best being the 4,096-simulation budget at  $+13.14 \pm 0.15$ , a nominal gain inside sampling error. On RS[5,3], POMCP’s remaining deficit against its own rollout policy at 16,384 simulations and  $H \geq 15$  is 1.5 to 1.9 reward, 9 to 11 percent of the rollout’s return (seed-level Welch  $p=0.012$  to 0.028 uncorrected, none surviving the seed-level Holm correction over the battery’s 36 reward comparisons, though the  $H=25$  deficit does survive the pooled-level correction). There the tree therefore comes within roughly ten percent of the rollout’s return at five orders of magnitude more compute per decision. On RS[11,11] the gap does not close, since the 4,096-simulation budget still trails its own rollout’s  $+28.66 \pm 0.83$  (seed-level SE) by 15.5 reward (seed-level Welch  $p=3.2 \times 10^{-5}$ , surviving Holm correction at both levels, `results_rocksample_pomcp_budget_stats.csv`). We report the predeclared row in the main table and this diagnostic beside it, so a reader can see both what the protocol committed to and what a more generous configuration achieves.

Planning+IG at  $w=5$  and  $w=10$  lets us check whether  $w=1$  leaves performance on the table when a higher weight is allowed. On RS[5,3],  $w=5$  is not significantly different from EFE ( $w=1$ ) ( $+16.41$  vs.  $+16.47$ , seed-level Welch  $p=0.74$ ), while  $w=10$  is significantly worse ( $+14.60$  vs.  $+16.47$ ,  $p < 10^{-8}$ ). Nothing is therefore gained by tuning beyond  $w=1$  here, and pushing the weight too far past it actively hurts. On RS[7,4], EFE ( $w=1$ ) outright beats both tuned weights ( $+15.96$  vs.  $+14.56$  and  $+14.27$ , both  $p < 10^{-5}$ ). More checking here (Plan+IG’s higher check counts) does buy more good rocks sampled, but not enough to pay for the extra steps it costs, so the untuned weight is better calibrated on this instance. On RS[7,8], the pattern is neither monotone in the weight nor a simple case of “more checking helps”.  $w=5$  ( $+23.76$ ) and EFE ( $+21.85$ ) are not significantly different ( $p=0.09$  uncorrected), while  $w=10$  ( $+20.55$ ) is significantly worse than  $w=5$  ( $p=0.003$ , significant under the per-metric Holm family).

The intermediate tuned weight is therefore at worst tied with the untuned  $w=1$  and reliably ahead of the more aggressive  $w=10$ . That heavier weight’s extra checking (57.47 steps, 18.26 checks per episode, versus  $w=5$ ’s 40.27/14.19) buys 0.54 more good rocks per episode but spends 17.2 more steps at the  $-0.5$  per-step move cost to get them. The honest summary is that  $w=1$  is not uniformly best on RockSample, and neither is any fixed direction of “tune the weight higher”.  $w=1$  leads the weighted family outright on RS[7,4] and is never significantly beaten within that family on RS[5,3] and RS[7,8]. Outside the family the picture changes. On RS[7,8] the standalone approach-then-check heuristic beats  $w=1$  decisively (+28.33 vs. +21.85,  $p=4.6\times 10^{-4}$ , significant after correction). POMCP at its frozen configuration trails EFE and both tuned Plan+IG weights significantly on the three smaller instances, beats reward-only Planning on RS[7,8] (+15.07 vs. +12.31,  $p=0.002$ ), and sits within sampling error of the entire family on RS[11,11] (seed-level Welch  $p \geq 0.36$ , none significant after correction). The weight sweep bounds what tuning  $w$  can buy, not what this instance admits.

These results exercise the extension Proposition 3 already establishes for interleaved settings despite the transition–observation coupling analyzed in Section 3 (factored observation extension). The agent’s behavior mirrors the observe-then-commit pattern. It checks (gathers information), then samples or exits (commits), even though checking and acting are interleaved with movement.

## N IDS Baseline (Observe-then-Commit)

We adapt information-directed sampling [48] to the observe-then-commit setting by minimizing the information ratio  $\Delta^2/I$  over observation actions, where  $\Delta$  is the opportunity cost of observing versus committing now and  $I$  is expected information gain about the optimal commit. Two fidelity gaps against Russo and Van Roy must be stated. Their IDS minimizes the ratio over randomized action distributions  $\pi \in D(A)$ . The journal version of their work [49] gives an example in which every stationary deterministic policy incurs linear regret for some prior, which they read as randomization playing a fundamental role. Our baseline instead takes a deterministic argmin over single observation actions. And where their formulation assumes the information gain is nonzero, our implementation substitutes a raw state-entropy term when the gain about the optimal commit vanishes. Table 28 reports a matched comparison (2,500 episodes at 500 per seed  $\times$  5 seeds, the canonical seed set) against Myopic, Planning, success-tuned Planning+IG, and EFE.

On multi-observation environments, IDS reaches near-ceiling success but at a reward cost similar to success-tuned Planning+IG, whereas EFE ( $w=1$ ) retains higher mean reward. A seed-level Welch test on reward rejects the null in both cases (Diagnosis  $-1.26$  vs.  $-3.63$ ,  $p=0.0009$ , Bandit  $+6.24$  vs.  $+4.65$ ,  $p=1.5\times 10^{-6}$ , `results_ids.csv`). The adaptive information ratio does not eliminate the need for a stopping trade-off calibrated to the reward scale. Under our shared reward convention, the fixed information-unit weight remains the stronger untuned default for expected reward.

## O Proper-Scoring Calibration

The reward and success-rate metrics used throughout this paper score decisions, not beliefs. To check that the epistemic term is doing more than inflating a task-performance proxy, we separately score the terminal belief itself against the true hidden state using two proper scoring rules [22], the log score ( $\log b(s^*)$ , the same functional the epistemic value term is derived from) and the Brier score ( $\sum_s (b(s) - \mathbb{I}[s = s^*])^2$ ). These scores assess overall terminal-belief quality, which depends on both calibration and how informative the belief is. A uniform posterior before any observation and an exact point-mass posterior after a perfect observation can both be calibrated, although the latter has better expected scores. Table 29 reports mean  $\pm$  SE over 5 seeds for Planning (reward-only), EFE ( $w=1$ ), and a reference near-optimal or IG-heavy policy on the three core OTC environments plus both Structural Inspection instances (500 episodes per seed for Tiger, Diagnosis, Bandit, and Inspection-N8, 200 per seed for Inspection-N16). The reference is the point-based near-optimal solver SARSOP [32], run as a self-built alpha-vector policy evaluated in-simulator, for the three OTC environments where its belief is small enough to reach near-optimal convergence in reasonable time. For Structural Inspection, whose state space exceeds what our SARSOP build solves in reasonable time, the reference is Planning+IG at the atlas’s higher canonical budget (Appendix T.3).

Planning (reward-only) is dominated on both terminal-belief scores by every epistemic-aware agent on every instance except Tiger, where its single observation action already suffices and all three agents converge to the same policy and hence the same terminal belief. On Diagnosis, Bandit, and both Inspection instances, EFE’s terminal belief scores substantially better under both rules than reward-only Planning’s, and on the three OTC environments EFE ( $w=1$ ) lies within overlapping SE bands of SARSOP’s terminal-belief scores. This is evidence that the epistemic term produces better-informed terminal beliefs, not only better terminal actions. The two need not coincide, since an agent can commit correctly from a confidently wrong belief on environments with few reachable outcomes.

## P Per-Test Value-of-Information Audit Trail

Section 8 argues that the epistemic value term supports interpretable decisions, but nothing in the main results exposes that decomposition directly. We add an optional, backward-compatible audit record at the shared Planning+IG evaluation boundary used by `PlanningInfoGainAgent`, `EFEAgent` (at  $w=1$ ), and `InspectionTreeSearchAgent`. For every candidate action considered at the root of the decision, the record stores the sensing cost, the expected information gain in bits, the weighted information gain  $w \cdot \text{IG}$ , the expected task value, and the resulting total score, plus which action was selected. Enabling this record does not change any action selection. It is a read-only projection of quantities the agent already computes, and unit tests confirm that the recorded total score always ranks the chosen action first and that its components sum correctly.

Table 30 shows one representative decision point from a Structural Inspection-N8 episode ( $w=5$ ) where the ranking by total score disagrees with the ranking by raw information gain. `test1` carries about  $5\times$  the information gain of `test0`, yet the agent selects `test0` because `test1`’s higher sensing cost is not offset by a compensating gain in expected task value. This is the audit trail’s purpose. It lets a practitioner check, for any single decision, whether an unexpected action was chosen for an information-theoretic reason or a task-value reason, rather than only reporting an aggregate reward or success rate. We keep this as an interpretability appendix artifact. It does not change any reported headline result.

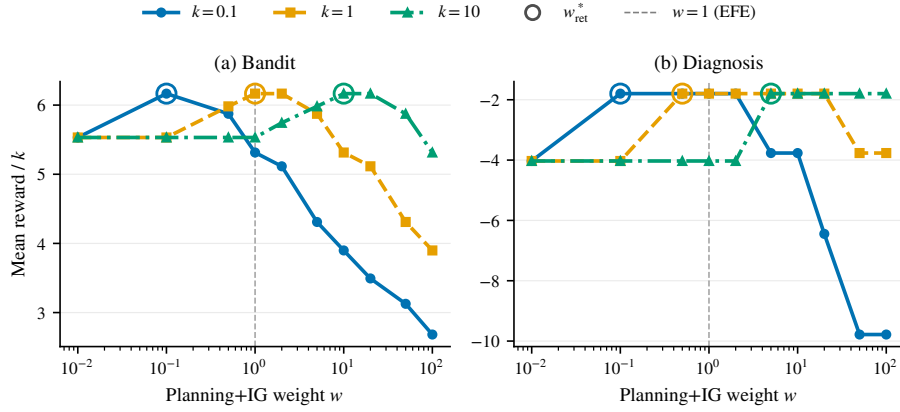
## Q Reward-Rescaling Invariance

To test the scale objection directly, we multiply all rewards and costs by  $k \in \{0.1, 1, 10\}$  (the factor written  $\alpha$  in Proposition 4, renamed here because  $\alpha$  also denotes reward asymmetry) on Diagnosis and Bandit and re-sweep Planning+IG weights. Maximizing  $\mathbb{E}[kR] + wI$  induces the same policy as maximizing  $\mathbb{E}[R] + (w/k)I$ . Matching a given unscaled policy therefore requires scaling the weight up with  $k$ , and  $w_{\text{ret}}^*(k) \approx k w_{\text{ret}}^*(1)$ . Equivalently, keeping  $w$  fixed while scaling rewards by  $k$  is indistinguishable from rescaling the weight by  $1/k$ . Figure 25 and Table 31 report the sweep (200 episodes  $\times$  3 seeds per  $(k, w)$  pair). Results confirm covariance of  $w_{\text{ret}}^*$  with  $k$ .  $w=1$  is not scale-invariant, and the paper’s near-optimality claims are conditional on the shared reward convention used throughout the suite.

## R Information Base and the Nat-Canonical Agent

The implementation measures information in bits, so the implemented  $w=1$  agent carries an effective nat-denominated weight of  $1/\ln 2 \approx 1.44$ , and the exactly nat-canonical identification of Proposition 1 corresponds to  $w = \ln 2 \approx 0.69$  in the implementation’s bit units (Section 3.3). Whether the gap between the implemented weight and the canonical one changes any reported result is a question the weight sweep of Section 7.2 can answer in part. On Tiger the reward-maximizing tied bracket is  $[0.01, 20]$  and on Diagnosis it is  $[0.5, 20]$ . Both weights, the implemented  $w=1$  and the nat-canonical 0.69 in the implementation’s bit units, therefore lie inside it, and the choice of information base changes no qualitative conclusion there. On Bandit the tied bracket’s grid endpoints are  $[1, 2]$ , which contains  $w=1$  but not 0.69. Because the sweep measures only grid weights, a weight that falls between two grid points is not directly observed, and 0.69 falls between two of them. The nearest swept weight below it,  $w=0.5$ , scores 6.05 against the tied 6.24 (`results_pareto_sweep.csv`), so the sweep grid alone cannot say which side of that boundary the nat-canonical weight falls on.

To settle the remainder directly, we ran the exactly nat-canonical agent of Proposition 1,  $w=\ln 2$  in the implementation’s bit units, together with a fresh  $w=1$  run on the same seeds, under the sweep’s own protocol (500 episodes  $\times$  5 seeds) on all five swept environments (`results_nat_canonical_check.csv`,



**Fig. 25.** Reward rescaling on Bandit (a) and Diagnosis (b). Scale-normalized mean reward (mean reward divided by  $k$ ) vs. Planning+IG weight  $w$  under global scale factors  $k$ . Normalization places the three scales on one axis, so the equivariance is read directly. The three curves share one shape, and the reward-maximizing weight (open circles,  $w_{\text{ret}}^*$ ) shifts toward larger  $w$  as  $k$  grows, consistent with  $w_{\text{ret}}^*(k) \propto k$ . The dashed rule marks  $w=1$ , the EFE weight.

experiments/run\_nat\_canonical\_check.py). On Tiger, Diagnosis, Bandit, and Testbed the two agents are bit-identical in reward, success rate, and mean observation count (5.33 and 99.6% on Tiger,  $-1.26$  and 97.4% on Diagnosis, 6.24 and 86.9% on Bandit, 0.37 and 96.4% on Testbed). On four of the five swept environments, then, the choice of information base moves none of the sweep’s reported numbers, and on Bandit in particular the nat-canonical weight lands inside the tied bracket rather than with  $w=0.5$ . Testbed’s exactly identical result carries one further remark. Proposition 2 derives, for a two-observation horizon ( $H=2$ ), a closed-form interval of near-optimal weights. On Testbed that interval’s upper endpoint falls between the nat-denominated weight  $w=1$  and the implemented bit-denominated agent’s 1.44 nats, a separation taken up in the discussion following Table 1. Under the sweep’s  $H=4$  protocol, however, the two agents behave identically, both scoring 0.37 reward against the 0.48 that the weights of the tied reward-maximizing bracket  $[0.01, 0.5]$  achieve.

Tileworld is the one swept environment where the base changes the policy. There the nat-canonical agent behaves like the sweep’s  $w=0.5$  grid point rather than like its  $w=1$  grid point ( $-20.81$  reward, 74.7% success, and 15.64 scans against  $-21.53$ , 72.0%, and 14.75, `results_nat_canonical_check.csv`). The difference is significant in observation count and within sampling error in reward and success rate. Seed-level Welch  $t$ -tests under the protocol of Appendix I give  $p=9.5 \times 10^{-5}$  for observation count, surviving Holm correction over the three metrics, against  $p=0.34$  and 0.070 for reward and success rate. The conclusion Tileworld supports is unchanged by the check. Under either unit convention  $w=1$

is Pareto-dominated there by the larger weight  $w=20$ , which beats it on both axes of the sweep, reward and success rate ( $-18.95$  and  $91.8\%$ , `results_pareto_sweep.csv`). That conclusion now rests on a direct run of the nat-canonical agent rather than on reading between two grid points.

## S POMCP Baseline Comparison

We compare against POMCP [55], a standard online POMDP solver that handles exploration through Monte Carlo tree search with UCB1 action selection. This addresses whether EFE’s explicit information gain valuation provides benefit over the implicit exploration in POMCP’s tree search. Our implementation adapts the algorithm to the observe-then-commit action structure, and three fidelity gaps against Silver and Veness must be stated before the numbers. Rollouts initialize from the agent’s root belief regardless of tree depth, ignoring the observations accumulated along the tree path. The depth-cap leaf returns the best commit reward for the sampled state, a clairvoyant quantity. And rollouts credit the belief-expected commit reward rather than the sampled-state reward. The clairvoyant leaf and the belief-expected commit credit are favorable or neutral to POMCP on these small instances. The root-belief rollout cuts the other way, since it denies observation branches credit for the in-tree observations that produced them, so the net direction of these simplifications is mixed rather than uniformly favorable. None is present in the RockSample POMCP of Appendix M, whose own departures from the published algorithm, all keyed to fully observable quantities, are stated there. Crucially, our POMCP rollout policy is semi-informed. Observation actions are selected uniformly at random, but the rollout terminates (stochastically, with 30% probability per step) with a belief-optimal commit, i.e., the commit action maximizing expected reward under the Bayesian-updated rollout belief. This is strictly stronger than a fully random rollout, which would also randomize over commit actions. Each of the five outer seeds initializes both the environment stream and the planner’s internal random stream (particle sampling and rollouts) from that seed, so the reported variability reflects genuine run-to-run differences rather than a fixed planner seed collapsing one axis of variation. On Diagnosis and Tileworld, where the agent must choose among several tests, the gap is largest, and the exploration sweep below shows that neither the exploration constant nor the rollout policy closes it. Bandit, where the agent also chooses among several inspections, shows no success gap, so a choice among tests is not by itself what produces one. EFE chooses which test to run based on information gain, while POMCP samples tests uniformly, but we do not attribute the gap to that difference in isolation, since exact EFE also differs from this POMCP in its exact beliefs, exact tree enumeration, and closed-form leaf values, and the ablation below does not separate those from one another. We sweep POMCP simulation budgets across  $\{500, 1000, 2000, 5000\}$  and report wall-clock timings for a simulation-budget scaling analysis. The comparison is simulation-matched, not compute-matched, the same distinction drawn for the MCTS-EFE comparison above. Tree-node action selection uses UCB1 with explo-

ration constant 10.0 (`rho_aif/agents/pomcp.py`), fixed across all environments. This was not tuned per environment, and because reward scales differ across our benchmarks (Section 5), the same exploration constant implies a different effective exploration–exploitation balance on each one, a caveat symmetric to the reward-scale sensitivity we report for  $w$  (Appendix Q). Tree and rollout depth are capped at the reporting agent’s planning horizon plus three steps (`experiments/run_pomcp.py`).

*POMCP comparison.* Table 32 shows POMCP underperforming both EFE and Planning on reward on every environment, and on success rate everywhere except Bandit. On Tiger, POMCP at this table’s untuned exploration constant commits prematurely (1.80 observations vs. 4.20 for EFE), yielding 89.4% success against EFE’s 99.4% (here numerically identical to Planning, per the exact tie documented in Section 7). On Diagnosis, POMCP achieves 74.1% success ( $-9.49$  reward) while EFE reaches 97.2% ( $-1.37$  reward). POMCP’s semi-informed rollout policy selects observation actions uniformly at random and so cannot evaluate which of the  $K=2$  diagnostic tests to run. It treats all observation actions equally, whereas EFE’s information gain term directly values differential informativeness. On Bandit, POMCP’s Monte Carlo exploration achieves a slightly higher success rate than EFE ( $87.7 \pm 0.5\%$  vs.  $86.9 \pm 0.3\%$  seed-level SE,  $+0.8$  pp) but with lower reward ( $+5.29 \pm 0.04$  vs. EFE’s  $+6.27 \pm 0.04$ ), buying that success with 7.20 inspections against EFE’s 5.11. On success the two are close. EFE is ahead on reward. On Tileworld ( $6 \times 6$ ,  $|S|=36$ ), POMCP collapses to 5.6% success, as the 36 commit actions and 6 scan actions create a branching factor that overwhelms 1,000 simulations.

*Simulation-budget scaling.* Increasing POMCP’s simulation budget to 5,000 does not qualitatively change the picture. On Diagnosis, POMCP(5000) reaches  $75.5 \pm 0.6\%$  success (seed-level SE), still far short of EFE’s  $97.2 \pm 0.3\%$ , while taking substantially more wall-clock time per episode (311.9 ms/ep vs. EFE’s closed-form computation). On Tiger, POMCP(5000) reaches  $90.3 \pm 0.3\%$  but at roughly  $5 \times$  the compute of the 1,000-simulation budget. On Bandit the sweep is not monotone in the middle. POMCP(2000) reaches  $90.8 \pm 0.3\%$  success at  $+5.90 \pm 0.05$  reward, ahead of POMCP(1000) on both metrics and nominally above Planning’s  $+5.75$ . POMCP(5000) then reaches  $97.2 \pm 0.2\%$  success (clearly ahead of EFE), but its reward falls to  $+5.18 \pm 0.04$ . Its wall-clock cost rises to roughly 0.93 s/episode, a  $\sim 6.9 \times$  slowdown versus the 1,000-simulation budget. The 2,000-to-5,000 step is the clearest case in our results of buying success-rate with both compute and return. These results indicate that EFE’s advantage is not merely a compute-budget artifact but reflects the benefit of closed-form information gain valuation over Monte Carlo exploration as configured here, with this rollout policy and a single untuned exploration constant, particularly on multi-observation-action environments where the agent must choose which information to gather. A stronger POMCP configuration does narrow the gap, sharply so on Tiger (Table 34), so we do not claim the comparison bounds what a best-practice POMCP can achieve.

*MCTS-EFE.* To disentangle EFE’s information valuation from the planning mechanism, we implement MCTS-EFE, the UCB1 tree search of Section 8, which uses EFE as the leaf heuristic rather than random rollouts. On Tiger at  $H=10$ , MCTS-EFE with 500 simulations achieves 97.7% success (23–28 ms per episode across the three horizons tested, `results_mcts_efe.csv`), compared to POMCP’s 88.8% at the same simulation count (15–19 ms per episode across horizons, simulation-matched, not compute-matched) and exact EFE’s 99.9% at  $H=6$  (103 ms per episode, both from the MCTS battery’s own 200-episodes-per-seed protocol rather than Table 8’s battery). In this configuration, the EFE-driven search, with exact per-node information gain as the in-tree observe reward, max-backup, exact commit-child values, and an EFE-greedy leaf, substantially outperforms POMCP’s mean-backup search with semi-informed rollouts. The two solvers differ in more than the leaf, so the two components that distinguish this search from a UCT tree are ablated below. We additionally ran a compute-matched check on this machine (`experiments/run_pomcp_compute_matched.py`), instrumented with seed-level statistics throughout. Timing MCTS-EFE(500) end to end for the full protocol (5 seeds  $\times$  200 episodes) gives 24.97s ( $97.7\% \pm 0.5\text{pp}$  success, seed-level SE). Calibrating POMCP’s wall-clock cost at 200/500/1,000 simulations on the same protocol and machine and solving for the simulation budget matching 24.97s gives 699 simulations, roughly 40% more than the simulation-matched comparison. POMCP(699) reaches  $89.0\% \pm 0.8\text{pp}$  success ( $-3.82 \pm 0.81$  reward, 24.99s), and a seed-level Welch test against the simulation-matched POMCP(500) ( $88.8\% \pm 0.2\text{pp}$  success,  $-4.03 \pm 0.21$  reward, 19.08s) confirms the two are not significantly different (success  $p=0.81$ , reward  $p=0.81$ ). Giving POMCP the same wall-clock budget as MCTS-EFE rather than the same simulation count does not close the gap at this exploration constant, and this is a direct test result rather than a bare assertion. Both POMCP configurations remain significantly behind MCTS-EFE on both metrics (seed-level Welch  $p < 0.0002$  in every comparison). On Tiger, MCTS-EFE’s success and observation count are insensitive to both horizon (6, 8, 10) and simulation budget (200, 500, 1,000) in this range, saturating at 97.7% success and 3.735 observations throughout. This reflects Tiger’s small reachable-belief set, which the tree already covers well within 200 simulations, rather than the horizon or budget parameters having no effect in general.

*Component ablation.* MCTS-EFE differs from a UCT tree in its leaf evaluation, in its exact belief representation and exact commit-child values, and in two switchable places, the in-tree information gain and the max-backup. Table 33 runs all four combinations of those two switches under this battery’s own protocol. Neither switchable component shows a detectable effect. No variant differs from the reported configuration on success or reward on any of the three environments after Holm correction within metric. The only difference surviving that correction anywhere is Tiger’s observation count, which falls from 3.73 to 3.48 when the in-tree information gain is removed (seed-level Welch  $p=0.0014$ ). At five seeds this is a failure to detect an effect rather than evidence that there is none. What remains to carry the contribution is the EFE-greedy leaf evaluation

together with the exact belief representation and the exact commit-child values, which the ablation holds fixed. That is a narrower attribution than crediting the EFE objective throughout the tree, and it is the one these data support.

*Exploration constant and informed rollouts.* Table 34 sweeps POMCP’s UCB1 exploration constant on the MCTS battery’s three environments, Tiger, Diagnosis, and Tileworld-6×6, over the seven values  $\{1, 2, 5, 10, 20, 50, R\}$ , where  $R$  is the commit reward range that MCTS-EFE takes as its own default. Bandit is not swept. Outside the sweep, the observe-then-commit POMCP runs with a single untuned constant per battery, 10.0 in Table 32 and 5.0 in the MCTS-EFE battery of `results_mcts_efe.csv`, and Table 34 writes the constant as  $c$ , a symbol distinct from the observation cost  $c$  of Proposition 2. The sweep also adds an information-gain rollout that selects the observation with the largest exact one-step information gain at every rollout step. Three findings follow. First, the constant matters a great deal on Tiger. POMCP at  $c=50$  reaches  $96.3\% \pm 0.6$  success at  $+3.21 \pm 0.62$  reward, and against MCTS-EFE at its default constant we fail to reject on either metric ( $p=0.09$  and  $p=0.54$ , a failure to reject rather than a TOST-backed equivalence claim). The 88.8% figure above therefore measures an untuned constant and not the solver. Second, MCTS-EFE’s own default constant is not its best either. At  $c=5$  on Tileworld-6×6 it earns  $-22.26 \pm 0.46$  against  $-25.02 \pm 0.25$  at the default (seed-level Welch  $p=0.0017$ ) at indistinguishable success, so comparing the two solvers at their respective defaults is not a tuned-against-tuned comparison in either direction. Taking each solver at a configuration selected on the disjoint tuning seeds  $\{11, 22, 33\}$  at 100 episodes per seed instead (highest success on those seeds, ties broken by reward, `results_pomcp_exploration_tuning.csv` and `experiments/select_pomcp_constants.py`), and reading the frozen selection off the canonical five-seed rows of Table 34, MCTS-EFE remains ahead on Tiger success ( $98.2\% \pm 0.3$  at  $c=5$  against POMCP’s  $96.3\% \pm 0.6$  at  $c=50$ , seed-level Welch  $p=0.024$ , surviving Holm correction within the stated six-test family of success and reward on the three environments) with reward not distinguishable ( $p=0.16$ ). The rule was recorded in `experiments/select_pomcp_constants.py` before the selection was run, over the same candidate grid, and the comparison family is success and reward on the three environments, six tests under Holm correction with success the primary metric. It picks  $c=5$  for MCTS-EFE on Tiger and Diagnosis and its default on Tileworld, and for POMCP  $c=50$  with uniform rollouts on Tiger,  $c=5$  with information-gain rollouts on Diagnosis, and  $c=2$  with uniform rollouts on Tileworld. On Tiger and Diagnosis these are the configurations the evaluation seeds would also have selected, and on Tileworld the evaluation seeds would have selected  $c=1$  for POMCP. The canonical runs had been inspected before the rule was written, so what the protocol removes is selection on the reported seeds, not our prior sight of them, and the narrow Tiger result counts as retrospective evidence (Section 8). Tuning the constant narrows the Tiger gap sharply and does not remove it. Third, no swept constant rescues POMCP where the agent must choose among tests. Its tuning-seed selection,  $c=5$  with information-gain rollouts on Diagnosis and  $c=2$  with uniform rollouts

on Tileworld-6×6, reaches  $71.4\% \pm 1.5$  and  $19.7\% \pm 1.7$  against MCTS-EFE’s  $94.7\% \pm 0.4$  at  $c=5$  and  $95.4\% \pm 0.6$  at its default, every comparison significant after Holm correction. The evaluation seeds’ own best POMCP configuration on Tileworld,  $c=1$  at  $22.3\% \pm 1.7$ , is not the selected one and is reported here only as a descriptive row of the sweep. On Tileworld-6×6 success falls as  $c$  rises over most of the range, from  $22.3\%$  at  $c=1$  to  $2.8\%$  at  $c=50$ , recovering slightly to  $3.5\%$  at  $c=R$ . That environment’s best therefore sits at the grid’s lower edge. A constant below 1 was not tested, so whether one would do better is not settled here. Information-gain rollouts leave both gaps open, putting Diagnosis at  $71.4\%$  against  $68.2$  and Tileworld at  $19.1\%$  against  $11.3$  at  $c=5$ , neither difference surviving Holm correction (seed-level Welch  $p=0.25$  and  $p=0.023$ ). They leave Tiger bit-identical, since with one observation action selecting the most informative observation and selecting uniformly are the same policy. POMCP’s planner stream is seeded per run from the outer seed in this battery, as it is in Table 32’s, while the MCTS battery of `results_mcts_efe.csv` left it at a fixed constant, so these POMCP rows are not bit-identical to that battery’s.

## T Budgeted $\rho$ -POMDP Supplementary Details

This appendix gives the details behind the budgeted experiments that the main text summarizes. It records how the SARSOP models were built and exported, proves the factorization property the distractor-diagnosis environment relies on, and presents the atlas of measured shadow prices.

### T.1 SARSOP Build and Export Details

Section 6.7’s reference solver is the original APPL C++ toolkit (`pomdpsol`), built from source by `tools/build_sarsop.sh` with four documented patches required for a modern clang toolchain on Apple Silicon. The patches drop unsupported `-msse2` and `-mfpmath=sse` flags, replace raw-pointer `SparseCol` iterators (modern `libc++` cannot construct `std::vector::const_iterator` from a raw pointer), convert chained comparison assertions to conjunctions, and suppress implicit-function-declaration errors in the legacy Cassandra parser. The resulting binary is not committed, but the patch script is. `experiments/run_sarsop_baseline.py` exports any discrete observe-then-commit benchmark generically from the same hidden-state, action-layout, and observation-model definitions the agents already use (an absorbing done state, discount  $\gamma=0.999$ ), solves to precision  $10^{-3}$ , parses the resulting alpha-vector policy from its XML output, and evaluates it through the same episode runner used for every other agent so that mechanics and metrics are shared exactly. Four unit tests (`tests/test_sarsop_export.py`) cover the exporter, the policy parser, and the alpha-vector argmax agent without requiring the compiled binary, so the export pipeline is checked even in environments where the solver is unavailable.

## T.2 Distractor-Diagnosis Environment and Its Factorization Guarantee

Section 6.10’s `DistractorDiagnosisEnv` joint state space is the product of a 4-way condition and a binary nuisance factor. Every observation model in the environment (condition tests, the distractor test, and the terminal commit reward) is constructed to depend on exactly one of the two factors. For a belief that itself factors as a product of independent condition and nuisance marginals, observing the distractor test therefore updates the nuisance marginal and leaves the condition marginal exactly unchanged. The same holds symmetrically for condition tests and the nuisance marginal. This is verified by 13 unit tests (`tests/test_distractor_diagnosis.py`). They include a direct check that the condition marginal is unchanged before and after a distractor observation (up to floating-point tolerance) when the prior belief is factored, which makes their KL divergence exactly zero, and a symmetric check for condition tests against the nuisance marginal. The guarantee is stated for factored beliefs specifically. A belief with residual condition–nuisance correlation (which does not arise under this environment’s independent generative process, but could under model misspecification) is outside the scope of the proof.

## T.3 The $w^*$ Atlas

Table 35 aggregates the usage-curve range, implicit EFE budget  $B_{\text{EFE}} = U(w=1)$ , and two canonical-budget crossing brackets (Definition 2) for the eight benchmark instances whose usage curves were swept, generated by `experiments/run_w_atlas.py` from the saved usage-curve CSVs of Sections 6.5 and 6.6. Brackets are grid intervals by construction. The table implies no closed-form relationship between  $B_{\text{EFE}}$  and environment parameters such as reward scale, state count, or horizon, and none should be inferred from it.

**Table 27.** RockSample extended results with the full agent roster, including a flat one-ply Monte Carlo (Flat-MC) reference, the genuine POMCP at its tuned frozen configuration, POMCP’s rollout policy run standalone (Rollout only), and Planning+IG at both  $w=5$  and  $w=10$ . Steps and checks are episode means. RS[5,3]/RS[7,4]: 500 episodes  $\times$  10 seeds. RS[7,8]/RS[11,11]: 100 episodes  $\times$  5 seeds.

| Instance  | Agent                   | Steps  | Checks | Good | Bad  | Reward            |
|-----------|-------------------------|--------|--------|------|------|-------------------|
| RS[5,3]   | Greedy                  | 12.00  | 0.00   | 1.49 | 1.51 | $+3.71 \pm 0.26$  |
|           | Flat-MC (1000)          | 51.00  | 20.69  | 1.22 | 0.03 | $-9.92 \pm 0.20$  |
|           | POMCP (2048 sims)       | 7.47   | 2.97   | 0.68 | 0.03 | $+12.74 \pm 0.11$ |
|           | Planning (d=3)          | 10.38  | 3.71   | 0.99 | 0.04 | $+14.28 \pm 0.13$ |
|           | Plan+IG $w=5$ (d=3)     | 15.81  | 5.36   | 1.44 | 0.01 | $+16.41 \pm 0.13$ |
|           | Plan+IG $w=10$ (d=3)    | 20.35  | 6.63   | 1.48 | 0.00 | $+14.60 \pm 0.11$ |
|           | EFE $w=1$ (d=3)         | 15.26  | 4.79   | 1.44 | 0.03 | $+16.47 \pm 0.12$ |
|           | Rollout only (approach) | 13.48  | 3.00   | 1.41 | 0.07 | $+16.67 \pm 0.13$ |
| RS[7,4]   | Greedy                  | 22.00  | 0.00   | 1.98 | 2.02 | $-1.38 \pm 0.36$  |
|           | Flat-MC (1000)          | 85.97  | 41.05  | 1.58 | 0.04 | $-25.51 \pm 0.20$ |
|           | POMCP (2048 sims)       | 8.67   | 3.46   | 0.66 | 0.03 | $+12.01 \pm 0.06$ |
|           | Planning (d=4)          | 10.00  | 2.00   | 0.95 | 0.05 | $+13.97 \pm 0.11$ |
|           | Plan+IG $w=5$ (d=4)     | 24.55  | 8.15   | 1.69 | 0.01 | $+14.56 \pm 0.14$ |
|           | Plan+IG $w=10$ (d=4)    | 26.82  | 8.80   | 1.78 | 0.01 | $+14.27 \pm 0.15$ |
|           | EFE $w=1$ (d=4)         | 16.81  | 5.36   | 1.44 | 0.01 | $+15.96 \pm 0.15$ |
|           | Rollout only (approach) | 23.98  | 4.00   | 1.88 | 0.10 | $+15.85 \pm 0.17$ |
| RS[7,8]   | Greedy                  | 32.00  | 0.00   | 4.02 | 3.98 | $-5.60 \pm 1.03$  |
|           | Flat-MC (1000)          | 129.00 | 82.43  | 1.51 | 0.05 | $-49.90 \pm 0.57$ |
|           | POMCP (2048 sims)       | 22.18  | 11.77  | 1.71 | 0.10 | $+15.07 \pm 0.45$ |
|           | Planning (d=3)          | 4.51   | 1.00   | 0.48 | 0.03 | $+12.31 \pm 0.35$ |
|           | Plan+IG $w=5$ (d=3)     | 40.27  | 14.19  | 3.40 | 0.01 | $+23.76 \pm 0.58$ |
|           | Plan+IG $w=10$ (d=3)    | 57.47  | 18.26  | 3.94 | 0.01 | $+20.55 \pm 0.29$ |
|           | EFE $w=1$ (d=3)         | 27.26  | 9.84   | 2.61 | 0.06 | $+21.85 \pm 0.79$ |
|           | Rollout only (approach) | 36.02  | 8.00   | 3.83 | 0.19 | $+28.33 \pm 0.82$ |
| RS[11,11] | Greedy                  | 57.00  | 0.00   | 5.48 | 5.52 | $-18.90 \pm 1.04$ |
|           | Flat-MC (1000)          | 231.00 | 165.75 | 2.98 | 0.16 | $-87.22 \pm 0.48$ |
|           | POMCP (2048 sims)       | 3.28   | 1.60   | 0.49 | 0.03 | $+12.94 \pm 0.24$ |
|           | Planning (d=2)          | 2.50   | 1.00   | 0.48 | 0.03 | $+13.23 \pm 0.18$ |
|           | Plan+IG $w=5$ (d=2)     | 3.09   | 1.56   | 0.50 | 0.03 | $+13.18 \pm 0.15$ |
|           | Plan+IG $w=10$ (d=2)    | 3.09   | 1.56   | 0.50 | 0.03 | $+13.18 \pm 0.15$ |
|           | EFE $w=1$ (d=2)         | 2.50   | 1.00   | 0.48 | 0.03 | $+13.23 \pm 0.18$ |
|           | Rollout only (approach) | 62.44  | 11.00  | 5.22 | 0.23 | $+28.66 \pm 0.83$ |

**Table 28.** IDS vs. baselines (2,500 episodes at 500 per seed  $\times$  5 seeds, lighter than Table 8’s 5,000-episode battery, so small numeric differences from that table’s EFE/Planning rows are expected). Reward reported as mean  $\pm$  seed-level SE. IDS is competitive on Tiger (exactly tying Planning and EFE there) but over-explores on Diagnosis and Bandit relative to EFE’s untuned  $w=1$ , matching success-tuned Planning+IG on Diagnosis reward and trailing even it on Bandit. Success-tuned Planning+IG runs at the Planning+IG-class weights recorded in `results_supplementary_tuned_weights.csv`.

| Environment | Agent                  | Obs.  | Success | Reward            |
|-------------|------------------------|-------|---------|-------------------|
| Tiger       | Myopic                 | 1.00  | 84.6%   | $-7.90 \pm 0.67$  |
|             | Planning               | 4.23  | 99.6%   | $+5.33 \pm 0.26$  |
|             | Planning+IG ( $w=50$ ) | 5.66  | 99.9%   | $+4.25 \pm 0.08$  |
|             | EFE                    | 4.23  | 99.6%   | $+5.33 \pm 0.26$  |
|             | IDS                    | 4.23  | 99.6%   | $+5.33 \pm 0.26$  |
| Diagnosis   | Myopic                 | 2.00  | 64.5%   | $-13.29 \pm 0.88$ |
|             | Planning               | 5.85  | 88.8%   | $-2.59 \pm 0.37$  |
|             | Planning+IG ( $w=50$ ) | 13.17 | 99.2%   | $-3.68 \pm 0.14$  |
|             | EFE                    | 9.68  | 97.4%   | $-1.26 \pm 0.34$  |
|             | IDS                    | 13.17 | 99.2%   | $-3.63 \pm 0.16$  |
| Bandit      | Myopic                 | 2.05  | 61.6%   | $+5.52 \pm 0.13$  |
|             | Planning               | 3.24  | 69.9%   | $+5.67 \pm 0.13$  |
|             | Planning+IG ( $w=20$ ) | 9.72  | 99.0%   | $+5.05 \pm 0.07$  |
|             | EFE                    | 5.16  | 86.9%   | $+6.24 \pm 0.09$  |
|             | IDS                    | 10.62 | 99.5%   | $+4.65 \pm 0.07$  |

**Table 29.** Proper-scoring calibration of the terminal belief against the true hidden state (log score in nats and Brier score, mean  $\pm$  SE over 5 seeds). Reference is SARSOP for the three OTC environments and Planning+IG at the atlas’s higher canonical budget (Appendix T.3) for Structural Inspection, where the near-optimal solver used here does not scale to the instance’s state space.

| Instance       | Agent                   | Log score          | Brier score       |
|----------------|-------------------------|--------------------|-------------------|
| Tiger          | Planning (reward-only)  | $-0.035 \pm 0.006$ | $0.011 \pm 0.002$ |
| Tiger          | EFE ( $w=1$ )           | $-0.035 \pm 0.006$ | $0.011 \pm 0.002$ |
| Tiger          | SARSOP                  | $-0.035 \pm 0.006$ | $0.011 \pm 0.002$ |
| Diagnosis      | Planning (reward-only)  | $-0.433 \pm 0.018$ | $0.199 \pm 0.011$ |
| Diagnosis      | EFE ( $w=1$ )           | $-0.134 \pm 0.004$ | $0.049 \pm 0.002$ |
| Diagnosis      | SARSOP                  | $-0.154 \pm 0.021$ | $0.058 \pm 0.010$ |
| Bandit         | Planning (reward-only)  | $-0.901 \pm 0.033$ | $0.463 \pm 0.019$ |
| Bandit         | EFE ( $w=1$ )           | $-0.525 \pm 0.027$ | $0.240 \pm 0.016$ |
| Bandit         | SARSOP                  | $-0.510 \pm 0.034$ | $0.231 \pm 0.018$ |
| Inspection-N8  | Planning (reward-only)  | $-0.417 \pm 0.004$ | $0.286 \pm 0.003$ |
| Inspection-N8  | EFE ( $w=1$ )           | $-0.245 \pm 0.004$ | $0.144 \pm 0.003$ |
| Inspection-N8  | Planning+IG ( $w=100$ ) | $-0.008 \pm 0.001$ | $0.002 \pm 0.000$ |
| Inspection-N16 | Planning (reward-only)  | $-0.382 \pm 0.005$ | $0.251 \pm 0.003$ |
| Inspection-N16 | EFE ( $w=1$ )           | $-0.257 \pm 0.003$ | $0.150 \pm 0.002$ |
| Inspection-N16 | Planning+IG ( $w=100$ ) | $-0.013 \pm 0.003$ | $0.004 \pm 0.001$ |

**Table 30.** Per-action value-of-information audit for one representative decision (Inspection-N8, info-gain weight  $w = 5$ , step 2 of the episode seeded at 42). Total score is the maximized planning objective (Section 3). Task value is net of sensing cost, so Total score = Task value +  $w \cdot \text{IG}$ , with Cost shown for reference. The chosen action need not have the largest raw information gain once sensing cost and downstream task value are accounted for.

| Action            | Cost | IG (bits) | $w \cdot \text{IG}$ | Task value | Total score |
|-------------------|------|-----------|---------------------|------------|-------------|
| test0             | 0.50 | 0.021     | 0.106               | -26.29     | -26.18      |
| test1             | 2.00 | 0.106     | 0.528               | -26.87     | -26.34      |
| diagnose(nominal) | 0.00 | 0.000     | 0.000               | -26.86     | -26.86      |
| diagnose(faulty)  | 0.00 | 0.000     | 0.000               | -31.05     | -31.05      |

**Table 31.** Reward-optimal Planning+IG weight under reward scale  $k$ . The direction of covariance is as predicted (larger scales favor larger weights). On Bandit the ratio  $w_{\text{ret}}^*/k$  is exactly constant across the three tested scales under the fixed pipeline. On Diagnosis it varies twofold (flat high- $w$  plateaus mean several grid weights tie for  $w_{\text{ret}}^*$ , and ties are broken toward the smallest weight reaching the maximum), so the table is qualitative evidence of covariance on Diagnosis and a tight quantitative match on Bandit, not a precise stability estimate on this finite grid.

| Environment | $k$  | $w_{\text{ret}}^*$ | $w_{\text{ret}}^*/k$ |
|-------------|------|--------------------|----------------------|
| Diagnosis   | 0.1  | 0.1                | 1.0                  |
| Diagnosis   | 1.0  | 0.5                | 0.5                  |
| Diagnosis   | 10.0 | 5.0                | 0.5                  |
| Bandit      | 0.1  | 0.1                | 1.0                  |
| Bandit      | 1.0  | 1.0                | 1.0                  |
| Bandit      | 10.0 | 10.0               | 1.0                  |

**Table 32.** POMCP comparison config (5,000 total episodes across 5 per-run seeds {42, 123, 456, 789, 1024} for Tiger/Diagnosis/Bandit and 1,000 for Tileworld, matching the main-table protocol on those three environments, and EFE/Planning here are re-run inside the POMCP harness rather than sourced from Table 8, which they reproduce exactly). POMCP uses 1,000 simulations per decision. Reward is mean  $\pm$  seed-level SE. Bold marks EFE and any agent bit-identical to it, the convention of Table 8, so it flags EFE’s row rather than a per-column best. On Tileworld 6 $\times$ 6 Planning is nominally ahead of EFE on both displayed metrics (73.6% against 71.9%,  $-21.41$  against  $-21.62$ ), and on Bandit POMCP(1000) leads on success (87.7% against 86.9%), both within seed-level sampling error.

| Environment               | Agent           | Obs.         | Success      | Reward                              |
|---------------------------|-----------------|--------------|--------------|-------------------------------------|
| Tiger                     | <b>Planning</b> | <b>4.20</b>  | <b>99.4%</b> | <b>+5.19 <math>\pm</math> 0.21</b>  |
|                           | POMCP (1000)    | 1.80         | 89.4%        | $-3.48 \pm 0.28$                    |
|                           | <b>EFE</b>      | <b>4.20</b>  | <b>99.4%</b> | <b>+5.19 <math>\pm</math> 0.21</b>  |
| Diagnosis                 | Planning        | 5.86         | 88.8%        | $-2.57 \pm 0.20$                    |
|                           | POMCP (1000)    | 3.96         | 74.1%        | $-9.49 \pm 0.22$                    |
|                           | <b>EFE</b>      | <b>9.66</b>  | <b>97.2%</b> | <b><math>-1.37 \pm 0.18</math></b>  |
| Bandit                    | Planning        | 3.29         | 71.0%        | $+5.75 \pm 0.08$                    |
|                           | POMCP (1000)    | 7.20         | 87.7%        | $+5.29 \pm 0.04$                    |
|                           | <b>EFE</b>      | <b>5.11</b>  | <b>86.9%</b> | <b><math>+6.27 \pm 0.04</math></b>  |
| Tileworld<br>6 $\times$ 6 | Planning        | 15.57        | 73.6%        | $-21.41 \pm 0.64$                   |
|                           | POMCP (1000)    | 1.35         | 5.6%         | $-47.99 \pm 0.72$                   |
|                           | <b>EFE</b>      | <b>14.76</b> | <b>71.9%</b> | <b><math>-21.62 \pm 0.87</math></b> |

**Table 33.** MCTS-EFE component ablation under the MCTS battery’s protocol (200 episodes per seed  $\times$  5 seeds, at the same horizons and simulation budgets as the MCTS-EFE runs of `results_mcts_efe.csv`). The first row of each block, max-backup with the in-tree information gain, is the manuscript’s MCTS-EFE. Uncertainty is seed-level SE. A dagger marks a variant whose success or reward differs from that row at seed-level Welch  $p < 0.05$  after Holm–Bonferroni within metric over the whole battery. No variant carries one, so no component shows a detectable effect on success or reward on any environment tested. See `results_mcts_efe_ablation.csv` and its `_stats` companion.

| Environment            | Backup | In-tree IG | Success          | Reward            | Obs.  |
|------------------------|--------|------------|------------------|-------------------|-------|
| Tiger                  | max    | yes        | $97.7 \pm 0.5\%$ | $+3.73 \pm 0.53$  | 3.73  |
|                        | mean   | yes        | $97.9 \pm 0.3\%$ | $+4.01 \pm 0.33$  | 3.68  |
|                        | max    | no         | $97.4 \pm 0.7\%$ | $+3.66 \pm 0.75$  | 3.48  |
|                        | mean   | no         | $97.7 \pm 0.5\%$ | $+3.95 \pm 0.60$  | 3.52  |
| Diagnosis ( $N=4$ )    | max    | yes        | $94.2 \pm 0.5\%$ | $-3.26 \pm 0.30$  | 9.78  |
|                        | mean   | yes        | $95.8 \pm 0.5\%$ | $-2.55 \pm 0.38$  | 10.03 |
|                        | max    | no         | $96.3 \pm 0.6\%$ | $-2.49 \pm 0.53$  | 10.27 |
|                        | mean   | no         | $95.7 \pm 0.8\%$ | $-2.97 \pm 0.40$  | 10.39 |
| Tileworld-6 $\times$ 6 | max    | yes        | $95.4 \pm 0.6\%$ | $-25.02 \pm 0.25$ | 32.26 |
|                        | mean   | yes        | $96.2 \pm 0.5\%$ | $-24.35 \pm 0.39$ | 32.07 |
|                        | max    | no         | $94.7 \pm 1.0\%$ | $-26.34 \pm 0.57$ | 33.16 |
|                        | mean   | no         | $95.7 \pm 0.3\%$ | $-25.29 \pm 0.22$ | 32.71 |

**Table 34.** POMCP exploration-constant sweep and informed rollouts under the MCTS battery’s protocol (200 episodes per seed  $\times$  5 seeds, at the same horizons and simulation budgets as the MCTS-EFE runs of `results_mcts_efe.csv`).  $R$  is the commit reward range (110 on Tiger, 60 elsewhere), MCTS-EFE’s default constant. The info-gain rollout takes the observation with the largest exact one-step information gain at every rollout step. POMCP’s planner stream is seeded per run from the outer seed here, which the MCTS battery left at a fixed constant, so these POMCP rows are not bit-identical to that battery’s. Uncertainty is seed-level SE. A dagger marks a row whose success or reward differs from MCTS-EFE at its default constant at seed-level Welch  $p < 0.05$  after Holm–Bonferroni within metric over the whole battery. See `results_pomcp_exploration_sweep.csv` and its `_stats` companion.

| Environment            | Solver   | $c$ | Rollout                | Success          | Reward            |
|------------------------|----------|-----|------------------------|------------------|-------------------|
| Tiger                  | POMCP    | 1   | uniform <sup>†</sup>   | $86.5 \pm 0.4\%$ | $-6.49 \pm 0.38$  |
|                        | POMCP    | 2   | uniform <sup>†</sup>   | $89.2 \pm 0.5\%$ | $-3.52 \pm 0.57$  |
|                        | POMCP    | 5   | uniform <sup>†</sup>   | $89.0 \pm 0.4\%$ | $-3.76 \pm 0.44$  |
|                        | POMCP    | 10  | uniform <sup>†</sup>   | $89.3 \pm 0.6\%$ | $-3.52 \pm 0.64$  |
|                        | POMCP    | 20  | uniform <sup>†</sup>   | $92.5 \pm 0.4\%$ | $-0.30 \pm 0.43$  |
|                        | POMCP    | 50  | uniform                | $96.3 \pm 0.6\%$ | $+3.21 \pm 0.62$  |
|                        | POMCP    | $R$ | uniform <sup>†</sup>   | $91.6 \pm 0.4\%$ | $-1.13 \pm 0.40$  |
|                        | POMCP    | 5   | info-gain <sup>†</sup> | $89.0 \pm 0.4\%$ | $-3.76 \pm 0.44$  |
|                        | POMCP    | $R$ | info-gain <sup>†</sup> | $91.6 \pm 0.4\%$ | $-1.13 \pm 0.40$  |
|                        | MCTS-EFE | $R$ | EFE                    | $97.7 \pm 0.5\%$ | $+3.73 \pm 0.53$  |
|                        | MCTS-EFE | 5   | EFE                    | $98.2 \pm 0.3\%$ | $+4.33 \pm 0.28$  |
| Diagnosis ( $N=4$ )    | POMCP    | 1   | uniform <sup>†</sup>   | $66.2 \pm 1.6\%$ | $-14.10 \pm 0.87$ |
|                        | POMCP    | 2   | uniform <sup>†</sup>   | $69.6 \pm 2.4\%$ | $-12.16 \pm 1.40$ |
|                        | POMCP    | 5   | uniform <sup>†</sup>   | $68.2 \pm 2.1\%$ | $-12.81 \pm 1.16$ |
|                        | POMCP    | 10  | uniform <sup>†</sup>   | $70.7 \pm 2.0\%$ | $-11.35 \pm 1.21$ |
|                        | POMCP    | 20  | uniform <sup>†</sup>   | $64.6 \pm 2.4\%$ | $-14.25 \pm 1.46$ |
|                        | POMCP    | 50  | uniform <sup>†</sup>   | $55.0 \pm 1.8\%$ | $-19.07 \pm 1.02$ |
|                        | POMCP    | $R$ | uniform <sup>†</sup>   | $55.5 \pm 2.4\%$ | $-18.67 \pm 1.41$ |
|                        | POMCP    | 5   | info-gain <sup>†</sup> | $71.4 \pm 1.5\%$ | $-11.23 \pm 0.86$ |
|                        | POMCP    | $R$ | info-gain <sup>†</sup> | $60.1 \pm 2.8\%$ | $-16.18 \pm 1.65$ |
|                        | MCTS-EFE | $R$ | EFE                    | $94.2 \pm 0.5\%$ | $-3.26 \pm 0.30$  |
|                        | MCTS-EFE | 5   | EFE                    | $94.7 \pm 0.4\%$ | $-2.75 \pm 0.31$  |
| Tileworld-6 $\times$ 6 | POMCP    | 1   | uniform <sup>†</sup>   | $22.3 \pm 1.7\%$ | $-43.99 \pm 0.91$ |
|                        | POMCP    | 2   | uniform <sup>†</sup>   | $19.7 \pm 1.7\%$ | $-44.78 \pm 0.97$ |
|                        | POMCP    | 5   | uniform <sup>†</sup>   | $11.3 \pm 1.1\%$ | $-46.55 \pm 0.61$ |
|                        | POMCP    | 10  | uniform <sup>†</sup>   | $7.5 \pm 0.7\%$  | $-47.43 \pm 0.46$ |
|                        | POMCP    | 20  | uniform <sup>†</sup>   | $5.5 \pm 1.0\%$  | $-47.87 \pm 0.56$ |
|                        | POMCP    | 50  | uniform <sup>†</sup>   | $2.8 \pm 0.3\%$  | $-48.41 \pm 0.21$ |
|                        | POMCP    | $R$ | uniform <sup>†</sup>   | $3.5 \pm 0.6\%$  | $-48.00 \pm 0.35$ |
|                        | POMCP    | 5   | info-gain <sup>†</sup> | $19.1 \pm 2.3\%$ | $-44.37 \pm 1.29$ |
|                        | POMCP    | $R$ | info-gain <sup>†</sup> | $6.2 \pm 0.9\%$  | $-47.25 \pm 0.45$ |
|                        | MCTS-EFE | $R$ | EFE                    | $95.4 \pm 0.6\%$ | $-25.02 \pm 0.25$ |
|                        | MCTS-EFE | 5   | EFE <sup>†</sup>       | $94.9 \pm 0.7\%$ | $-22.26 \pm 0.46$ |

**Table 35.** Atlas of operational sensing budgets across benchmark instances, giving the usage range of the Planning+IG family, the implicit EFE budget  $B_{\text{EFE}} = U(w=1)$  (mean  $\pm$  SE over seeds), and crossing brackets  $w^*(B)$  at two canonical budgets per instance. Each bracket estimates a single crossing threshold from the finite weight grid (Definition 2). No closed-form meta-model is implied.

| Instance       | $[U_{\min}, U_{\max}]$ | $B_{\text{EFE}}$ | $B_1$ : $w^*$ bracket | $B_2$ : $w^*$ bracket |
|----------------|------------------------|------------------|-----------------------|-----------------------|
| Bandit         | [3.23, 12.34]          | $5.11 \pm 0.11$  | 4.6: (0.0611, 0.139]  | 11.0: (19.3, 43.9]    |
| Diagnosis      | [5.90, 13.16]          | $9.77 \pm 0.14$  | 7.0: (0.139, 0.316]   | 12.1: (19.3, 43.9]    |
| Inspection-N16 | [22.38, 58.56]         | $28.02 \pm 0.13$ | 27.8: (0.316, 1.33]   | 53.1: (23.7, 100]     |
| Inspection-N8  | [12.70, 29.49]         | $15.56 \pm 0.09$ | 15.2: (0.316, 2.15]   | 27.0: (14.7, 100]     |
| RS[5,3]        | [3.00, 8.83]           | $4.79 \pm 0.05$  | 3.9: (0.316, 1.33]    | 8.0: (5.62, 23.7]     |
| RS[7,4]        | [2.00, 14.54]          | $5.37 \pm 0.07$  | 3.9: (0.316, 2.15]    | 12.7: (14.7, 100]     |
| Tiger          | [4.37, 5.77]           | $4.37 \pm 0.11$  | 4.6: (19.3, 43.9]     | 5.6: (19.3, 43.9]     |
| Tileworld-6x6  | [14.74, 32.73]         | $14.85 \pm 0.13$ | 17.4: (5.62, 23.7]    | 30.0: (23.7, 100]     |